Mathematical Structures of Computation

Completion, Representation, Truncation, and Approximation

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Computation is mathematics carried out under finite precision.

Computation starts in the finite world of data. To understand and analyze it mathematically, we pass to a more complete, infinite mathematical universe, where additional structure becomes available.

But computation itself remains finite. Representation through coordinates or discretization, together with truncation, brings infinite objects back to finite form where they can be stored and manipulated algorithmically.

This movement from finite data to the infinite mathematical universe and back shapes the mathematics underlying computation.

Mathematics therefore moves between idealized infinite structures and their finite realizations through the recurring processes of

- C COMPLETION (complete spaces, added structure)
- R REPRESENTATION (coordinates, discretization)
- T TRUNCATION (finite cut-offs)
- A APPROXIMATION (error analysis)

— which we call the computational cycle.

Contents

Part I

Analysis: Limits and Local Linear Approximation

C A Analysis introduces infinite structures through completion and controls their finite realization through approximation. It therefore forms the analytical core of the computational cycle.

Analysis provides the mathematical language for limits, approximation, and change. In this part, we develop the foundational concepts of convergence, continuity, differentiability, and integrability, starting from metric spaces through the calculus of elementary functions to the fundamental theorem of calculus.

Many of these concepts address questions that are fundamental whenever quantities are approximated, compared, or perturbed. Notions such as distance, continuity, and local linear behavior formalize stability, sensitivity, and approximation – ideas that recur throughout mathematical modeling, numerical computation, and data-driven analysis.

1 Convergence and Continuity

This chapter develops the concept of convergence as the central organizing principle of analysis. We begin by introducing metric spaces as an abstract framework for measuring distance and defining limits of sequences. This leads naturally to Cauchy sequences and the notion of completeness. Continuous functions map converging sequences onto converging sequences.

1.1 Metric spaces

Metric spaces provide a minimal framework for measuring distances between points. They consist of a set equipped with a function assigning to each pair of points a nonnegative number satisfying a few natural properties associated with distance.

Definition 1.1.1 (metric, metric space)**.** *Given a set $X \neq \emptyset$, we say that a mapping $d_X : X \times X \to [0, \infty)$ is a* metric *if the following three properties are satisfied:*

(M1) symmetry: *for all $x, y \in X$,*
$$d_X(x, y) = d_X(y, x),$$

(M2) triangle inequality: *for all $x, y, z \in X$,*
$$d_X(x, z) \leq d_X(x, y) + d_X(y, z),$$

(M3) definiteness: *for all $x, y \in X$,*
$$d_X(x, y) = 0 \quad \Leftrightarrow \quad x = y.$$

If d_X is a metric on X, then (X, d_X) is called a metric space*.*

If X is a metric space, then every nonempty subset S is also a metric space by restricting the metric d_X to $S \times S$.

Definition 1.1.2 (sequence)**.** *Let $X \neq \emptyset$ be an arbitrary set. A map $x : \mathbb{N} \to X$ is called a sequence in X and we denote $x_n := x(n)$, for $n \in \mathbb{N}$. The sequence is also written as $(x_n)_{n \in \mathbb{N}} \subseteq X$.*

Definition 1.1.3 (convergence)**.** *Let (X, d_X) be a metric space. A sequence $(x_n)_{n \in \mathbb{N}} \subseteq X$ is said to* converge *to $x \in X$ if*
$$\forall \epsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geq N \quad d_X(x_n, x) \leq \epsilon.$$

In this case, we write $\lim_{n \to \infty} x_n = x$ or simply $x_n \to x$ as $n \to \infty$.

The number N may depend on ϵ, and we sometimes write $N = N(\epsilon)$. Note that $x_n \to x$ if and only if $d_X(x_n, x) \to 0$.

Example 1.1.4 (real line). *The standard distance on* $\mathbb{R}$ *and* $\mathbb{C}$ *is*

$$d(x, y) := |x - y|. \tag{1.1}$$

It follows from Lemma **??** *that it is a metric. Therefore,* $\mathbb{R}$ *is a metric space. The sequence* $(2 + \frac{1}{n+1})_{n\in\mathbb{N}} \subseteq \mathbb{R}$ *converges to* 2*, because*

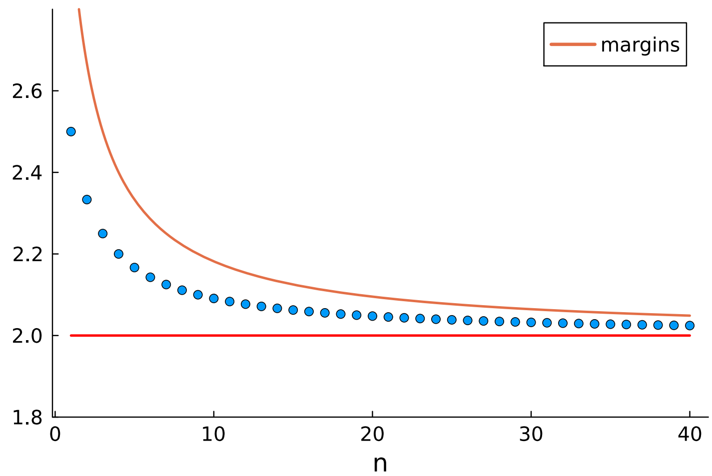
$$\left|2 + \frac{1}{n+1} - 2\right| = \left|\frac{1}{n+1}\right| \to 0.$$

Remark 1.1.5. *If we do not specify the metric on* $\mathbb{Q}$*,* $\mathbb{R}$*, or* $\mathbb{C}$*, then we implicitly refer to the modulus* (1.1)*.*

Let us denote the sequence by $(a_n)_{n\in\mathbb{N}}$, so that $2 \le a_n \le a_n + \frac{1}{n+1}$ and plot a_n with the margins 2 and $a_n + \frac{1}{n+1}$.

```
using Plots

a(n) = 2+1/(n+1)
M = 1:40
scatter(M,a.(M),ylimits=[1.8,2.8],label="")
plot!(x->2+2/(x+1),label="margins")
plot!(x->2,color="red")
```



Theorem 1.1.6 (limit is unique). *Let* (X, d_X) *be a metric space. If* $x_n \to x$ *and* $x_n \to x'$*, then* $x = x'$*.*

Proof version 1: Assume $x \neq x'$. Then $\epsilon := d_X(x, x')/3 > 0$. There exist $N_x, N_{x'} \in \mathbb{N}$ such that

$$d_X(x_n, x) \le \epsilon \quad \forall n \ge N_x,$$
$$d_X(x_n, x') \le \epsilon \quad \forall n \ge N_{x'}.$$

Hence, for $N = \max\{N_x, N_{x'}\}$, we have

$$d_X(x, x') \le d_X(x, x_N) + d_X(x_N, x') \le 2\epsilon = \frac{2}{3} d_X(x, x'),$$

which is impossible unless $d_X(x, x') = 0$. This contradicts the assumption $x \ne x'$. □

Proof version 2: The triangle inequality yields

$$d_X(x, x') \le d_X(x, x_n) + d_X(x_n, x') \to 0.$$

Thus, $d_X(x, x') = 0$, and hence $x = x'$. □

We sometimes want to quantify how quickly the sequence converges towards its limit.

Definition 1.1.7 (convergence order)**.** *Let (X, d) be a metric space and suppose that $(x_k)_{k\in\mathbb{N}} \subseteq X$ converges to $x \in X$. Then we say that the convergence is*

(i) of order $r > 1$ *if there is $c > 0$ and $k_0 \in \mathbb{N}$ such that*

$$d(x_{k+1}, x) \;\le\; c\, d(x_k, x)^r, \qquad \forall\, k \ge k_0.$$

(ii) linear *if there is $c \in (0, 1)$ and $k_0 \in \mathbb{N}$ such that*

$$d(x_{k+1}, x) \;\le\; c\, d(x_k, x), \qquad \forall\, k \ge k_0.$$

(iii) sublinear *if*

$$\limsup_{k\to\infty} \left(d(x_k, x)\right)^{\frac{1}{k}} = 1.$$

We have

$$\text{Convergence of order } r \quad \Rightarrow \quad \text{linear convergence} \quad \Rightarrow \quad \limsup_{k\to\infty} \left(\|x_k - x\|\right)^{\frac{1}{k}} < 1.$$

Example 1.1.8. *If $x_k \to x$ linearly with constant $c \in (0, 1)$, then iteration leads to*

$$d(x_{k+k_0}, x) \le c^k d(x_{k_0}, x).$$

Example 1.1.9. *The sequence*

(i) $x_k = 2^{-2^k}$ converges with order 2 towards $x = 0$ since

$$\frac{|x_{k+1} - x|}{|x_k - x|^2} = \frac{2^{-2^{k+1}}}{(2^{-2^k})^2} = \frac{2^{-2^k 2}}{(2^{-2^k})^2} = \frac{(2^{-2^k})^2}{(2^{-2^k})^2} = 1.$$

(ii) $x_k = 2^{-k}$ *converges linearly towards* $x = 0$ *since*

$$|x_{k+1} - x| = \left|2^{-(k+1)}\right| = \frac{1}{2}\left|2^{-k}\right| = \frac{1}{2}|x_k - x|.$$

(iii) $x_k = \frac{1}{k+1}$ *converges sublinearly towards* 0 *since*

$$\limsup_{k\to\infty} \left|\frac{1}{k+1}\right|^{\frac{1}{k}} = 1.$$

Definition 1.1.10. *Let* (X, d_X) *and* (Y, d_Y) *be metric spaces and* $f : X \to Y$. *Then* f *is called an isometry if*

$$d_Y(f(x), f(x')) = d_X(x, x'), \qquad \forall\, x, x' \in X.$$

Isometric maps preserve distances.

Definition 1.1.11. *Let* (X, d_X) *be a metric space. A sequence* $(x_n)_{n\in\mathbb{N}} \subseteq X$ *is called bounded if there exists* $x \in X$ *and* $C > 0$ *such that*

$$d_X(x_n, x) \le C, \qquad \forall\, n \in \mathbb{N}.$$

Analogously, we call a subset $S \subseteq X$ *bounded if there exists* $x \in X$ *and* $C > 0$ *such that*

$$d_X(s, x) \le C, \qquad \forall\, s \in S.$$

The triangle inequality shows that the anker point x is arbitrary. Using a different one just changes the constant C.

Convergent sequences are bounded.

Lemma 1.1.12. *Let* (X, d) *be a metric space. Every convergent sequence* $(x_n)_{n\in\mathbb{N}} \subseteq X$ *is bounded.*

Proof. Assume $(x_n)_{n\in\mathbb{N}}$ converges to some $x \in X$. Then there exists $N \in \mathbb{N}$ such that

$$d(x_n, x) < 1 \quad \text{for all } n \ge N.$$

Hence

$$d(x_n, x) \le \max\{\, d(x_1, x), \dots, d(x_{N-1}, x), 1\} \quad \text{for all } n \in \mathbb{N}.$$

□

Example 1.1.13 (Basic metric spaces)**.** *We list several simple metric spaces that will serve as guiding examples.*

1. Plane with taxicab distance: *On* $\mathbb{R}^2$, *define, for* $x = (x_1, x_2)$ *and* $y = (y_1, y_2)$,

$$d_1(x, y) := |x_1 - y_1| + |x_2 - y_2|.$$

Distances are measured by horizontal and vertical movement.

2. Plane with maximum distance: *On* $\mathbb{R}^2$*, define*

$$d_{\max}(x, y) := \max\{|x_1 - y_1|,\ |x_2 - y_2|\}.$$

The distance is given by the largest coordinate difference. Since

$$d_{\max}(x, y) \leq d_1(x, y) \leq 2d_{\max}(x, y),$$

sequences that converge in one also converge in the other metric.

3. Discrete metric*: Let* X *be any set. Define*

$$d(x, y) := \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

In this metric, distance is purely qualitative, and a sequence converges if and only if it is eventually constant.

Remark 1.1.14 (feature representation)**.** *In data analysis, elements of a set are often interpreted as data points, and a metric quantifies how similar or dissimilar two data points are. Points in* $\mathbb{R}^2$ *can be viewed as objects described by two numerical features, and metrics compare objects by comparing these features.*
Different metrics can emphasize different aspects of the data. For instance, the discrete metric reflects situations where only equality or inequality matters and naturally occurs when comparing categorical or symbolic data.

Lemma 1.1.15 (bounded, monotone implies convergence)**.** *If* $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ *is monotonically increasing and bounded from above, then it is convergent.*

Proof. The order completeness of $\mathbb{R}$ ensures that $a := \sup_{n\in\mathbb{N}} a_n$ exists. Assume that $a_n \not\to a$. Then

$$\exists \varepsilon > 0 \quad \forall N \in \mathbb{N} \quad \exists n_N \geq N \quad |a_{n_N} - a| > \varepsilon.$$

The monotonicity implies that

$$a - a_N > a - a_{n_N} > \varepsilon \quad \forall N \in \mathbb{N}.$$

and hence $a - \varepsilon > a_N$ for all $N \in \mathbb{N}$. This contradicts the property of the supremum to be the smallest upper bound. □

Lemma 1.1.16. *Let* $S \subseteq \mathbb{R}$ *be upper bounded. Then there exists a sequence* $(x_n)_{n\in\mathbb{N}} \subseteq S$ *such that*

$$x_n \to \sup S.$$

Proof. The order completeness of $\mathbb{R}$ ensures that $s := \sup S \in \mathbb{R}$ exists. For all positive $n \in \mathbb{N}$, there is $x_n \in S$ such that

$$x_n \geq s - \frac{1}{n}.$$

Otherwise s were not the smallest upper bound. We further deduce $\frac{1}{n} \geq s - x_n \geq 0$, which implies $x_n \to s$. □

The analogous result holds for the infimum also.

Theorem 1.1.17. *For every $x \in \mathbb{R}$ there is a sequence $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{Q}$ such that $x_n \to x$.*

We also say that $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof. Theorem **??** states that between every pair of real numbers there is a rational number. We apply this to x and $x + \frac{1}{n+1}$. Thus, there is $x_n \in \mathbb{Q}$ such that $x < x_n < x + \frac{1}{n+1}$. We derive

$$|x_n - x| = x_n - x \leq \frac{1}{n+1} \to 0, \qquad \text{for } n \to \infty,$$

which means that $(x_n)_{n\in\mathbb{N}}$ converges towards x. □

Theorem 1.1.18 (convergence respects non-negativity)**.** *Suppose that $X = \mathbb{R}$ and $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ satisfies $a_n \to a$. Then*

$$\forall n \in \mathbb{N}: \ a_n \geq 0 \quad \Rightarrow \quad a \geq 0.$$

Proof. Assume $a < 0$. Choose $\epsilon := -a/2 > 0$. Then there exists $N \in \mathbb{N}$ such that $|a_N - a| \leq \epsilon = -a/2$. Hence

$$a_N - a = |a_N - a| \leq -\frac{a}{2},$$

so $a_N \leq a/2 < 0$. This contradicts $a_N \geq 0$. □

Corollary 1.1.19 (convergence respects inequalities)**.** *Suppose that $X = \mathbb{R}$. If $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ converge to a and b respectively, and if*

$$\forall n \in \mathbb{N}: \ a_n \leq b_n,$$

then

$$a \leq b.$$

Proof. Apply Theorem 1.1.18 to $c_n := b_n - a_n$ and $c := b - a$. □

Essentials

Distance between two points is measured by a metric that lets us define convergence of sequences. Key ideas are the ϵ-N definition of convergence and the uniqueness of limits.

Every convergent sequence is bounded. You have seen that the supremum and the infimum of a bounded subset in $\mathbb{R}$ are limits of sequences from the set. The rational numbers are dense in the real numbers in the sense that each real number is the limit if a sequence of rationals.

A Convergence concepts are the basis for approximation. Every real number can be approximated by rationals.

metric, metric space, symmetry, triangular inequality, definiteness, convergence, limits

1.2 Completeness

Not every sequence that "ought to converge" actually has a limit within a given space. This section introduces Cauchy sequences and the concept of completeness, which distinguishes spaces such as $\mathbb{R}$ and $\mathbb{C}$ from incomplete ones like $\mathbb{Q}$.

Definition 1.2.1 (Cauchy). *Let (X, d_X) be a metric space. A sequence $(a_n)_{n\in\mathbb{N}} \subseteq X$ is called a* Cauchy sequence *if*

$$\forall \epsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n, m \geq N : \qquad d_X(a_n, a_m) \leq \epsilon.$$

The idea is that, in a Cauchy sequence, the elements eventually become arbitrarily close to each other, even before knowing a potential limit.

Example 1.2.2. *The sequence $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ given by $a_n = 2 - \frac{1}{n+1}$ is a Cauchy sequence. Indeed, for any $\epsilon > 0$ we may choose $N \in \mathbb{N}$ such that $1/N < \epsilon/2$. Then, for all $n, m \geq N$,*

$$|a_n - a_m| = \left|2 - \frac{1}{n+1} - \left(2 - \frac{1}{m+1}\right)\right| = \left|\frac{1}{m+1} - \frac{1}{n+1}\right| \leq \epsilon.$$

Every convergent sequence is automatically Cauchy.

Lemma 1.2.3 (convergence implies Cauchy). *Let (X, d_X) be a metric space. If $(a_n)_{n\in\mathbb{N}} \subseteq X$ converges, then it is a Cauchy sequence.*

Proof. Let $\epsilon > 0$ be arbitrary. Since $a_n \to a \in X$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$,

$$d_X(a_n, a) \leq \epsilon/2, \qquad d_X(a_m, a) \leq \epsilon/2.$$

By the triangle inequality,

$$d_X(a_n, a_m) \leq d_X(a_n, a) + d_X(a, a_m) \leq \epsilon.$$

□

The natural question is whether the converse holds. In general, it does *not*: a Cauchy sequence need not converge *inside* the metric space.

Example 1.2.4. *Consider the open interval $X = (0, 2)$ with the standard metric. The sequence $a_n = 2 - \frac{1}{n+1}$ is Cauchy, as shown above, but it converges to $2 \notin (0, 2)$. Thus, it does not converge* in *X.*

Metric spaces in which every Cauchy sequence converges are called complete.

Definition 1.2.5. *A metric space X is called* complete *if every Cauchy sequence in X is convergent in X.*

Lemma 1.2.6. *The rational numbers $\mathbb{Q}$ are not complete.*

Proof. Define a sequence $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{Q}$ inductively by $x_0 = 1$ and

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right), \qquad n \in \mathbb{N}.$$

A direct computation shows

$$x_{n+1} - \sqrt{2} = \frac{(x_n - \sqrt{2})^2}{2x_n} > 0.$$

Iterating this inequality yields

$$x_n - \sqrt{2} \leq (x_{n-1} - \sqrt{2})^2 \leq (x_{n-2} - \sqrt{2})^4 \leq \cdots \leq (x_0 - \sqrt{2})^{2^n}.$$

Another computation gives

$$\frac{x_{n+1}}{x_n} = \frac{1}{2}\left(1 + \frac{2}{x_n^2}\right) \leq 1,$$

so the sequence is decreasing and bounded below by $\sqrt{2}$. In particular, $x_n \geq \sqrt{2}$ for all $n \geq 1$. Given $\epsilon > 0$, choose N such that $(x_0 - \sqrt{2})^{2^N} < \epsilon$ (this is possible since $|x_0 - \sqrt{2}| < 1$). For $m \geq n \geq N$,

$$|x_m - x_n| = x_n - x_m < x_n - \sqrt{2} \leq (x_0 - \sqrt{2})^{2^n} \leq \epsilon.$$

Thus (x_n) is a Cauchy sequence in $\mathbb{Q}$, but it converges to $\sqrt{2} \notin \mathbb{Q}$. Therefore, $\mathbb{Q}$ is not complete. □

Theorem 1.2.7 (C). *The real numbers $\mathbb{R}$ are complete.*

We know by Theorem **??** that $\mathbb{R}$ is order-complete, meaning that every bounded set has an infimum and a supremum in $\mathbb{R}$. This is used to construct the limit for a Cauchy sequence.

Proof. Let $(x_n)_{n\in\mathbb{N}}$ be Cauchy. For $\varepsilon = 1$ there exists N such that $|x_m - x_n| \leq 1$ for all $m, n \geq N$. Hence for $n \geq N$,

$$|x_n| \leq |x_n - x_N| + |x_N| \leq 1 + |x_N|,$$

so $(x_n)_{n\in\mathbb{N}}$ is bounded.

For each n set $A_n := \{x_k : \ n \le k\}$. Then A_n is nonempty and bounded, hence

$$a_n := \inf A_n \in \mathbb{R}, \qquad b_n := \sup A_n \in \mathbb{R}$$

exist by order completeness. We have $a_n \le x_k \le b_n$ for all $k \ge n$, $(a_n)_{n\in\mathbb{N}}$ is increasing, $(b_n)_{n\in\mathbb{N}}$ is decreasing, and $a_n \le b_n$.
Fix $\varepsilon > 0$ and choose N such that $|x_m - x_k| < \varepsilon$ for all $m, k \ge N$. Then all elements of A_N lie in an interval of length ε, hence $b_N - a_N \le \varepsilon$. Since b_n decreases and a_n increases,

$$0 \le b_n - a_n \le b_N - a_N \le \varepsilon \qquad \forall\, n \ge N,$$

so $b_n - a_n \to 0$. This also implies

$$0 \le x_n - a_n \le b_n - a_n \to 0.$$

Define $a := \sup_{n\in\mathbb{N}} a_n$ and $b = \inf_{n\in\mathbb{N}} b_n$. Then we have $a \le b$ and $b_n - a_n \to 0$ yields $a = b =: \gamma$. Since $a_n \le \gamma \le b_n$, we deduce

$$0 \le \gamma - a_n \le b_n - a_n \to 0,$$

so that $a_n \to \gamma$. Finally, we have

$$|x_n - a| \le |x_n - a_n| + |a_n - a| \to 0.$$

□

We now consider complex numbers and recall that for $z = x + \mathrm{i}y \in \mathbb{C}$ with $x, y \in \mathbb{R}$, real and imaginary part are

$$\mathrm{Re}(z) = x, \qquad \mathrm{Im}(z) = y.$$

We have $|z|^2 = x^2 + y^2$ because

$$|z|^2 = z\overline{z} = (x + \mathrm{i}y)(x - \mathrm{i}y) = x^2 + \mathrm{i}xy - \mathrm{i}xy - \mathrm{i}^2y^2 = x^2 + y^2.$$

This helps transporting completeness of $\mathbb{R}$ to completeness of $\mathbb{C}$.

Theorem 1.2.8 (C). *The complex numbers $\mathbb{C}$ are complete.*

Proof. Given a Cauchy sequence $(z_n)_{n\in\mathbb{N}} \subseteq \mathbb{C}$, we may write $z_n = x_n + \mathrm{i}y_n$, where $x_n, y_n \in \mathbb{R}$. We directly compute

$$\begin{aligned} |z_n - z_m|^2 &= |x_n - x_m + \mathrm{i}(y_n - y_m)|^2 \\ &= |x_n - x_m|^2 + |y_n - y_m|^2. \end{aligned}$$

This implies that both $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ are Cauchy sequences in $\mathbb{R}$. Since $\mathbb{R}$ is complete, there are $x, y \in \mathbb{R}$ such that $x_n \to x$ and $y_n \to y$.
We again obtain

$$|z_n - (x + \mathrm{i}y)|^2 = |x_n - x + \mathrm{i}(y_n - y)|^2 = |x_n - x|^2 + |y_n - y|^2.$$

Therefore, $|z_n - (x + \mathrm{i}y)| \to 0$. □

Remark 1.2.9 (Limits of approximation)**.** *In data analysis, sequences often arise from iterative approximation procedures. Completeness guarantees that such sequences converge to a well-defined limit within the space under consideration, rather than converging to an object outside the model.*

In many situations, relevant information is contained in selected elements of a sequence rather than in its full progression. This motivates the introduction of subsequences.

Definition 1.2.10 (subsequences)**.** *A subsequence $(a_{n_j})_{j\in\mathbb{N}}$ of a sequence $(a_n)_{n\in\mathbb{N}}$ is formed by selecting terms with strictly increasing indices,*

$$n_0 < n_1 < \cdots < n_j < \cdots$$

Lemma 1.2.11 (subsequences respect convergence)**.** *If $(a_n)_{n\in\mathbb{N}}$ converges to a, then every subsequence also converges to a.*

Proof. Indeed, for $\epsilon > 0$ choose N such that $d_X(a_n, a) \leq \epsilon$ for all $n \geq N$. Since $n_j \geq j$, we obtain $d_X(a_{n_j}, a) \leq \epsilon$ for all $j \geq N$. □

Example 1.2.12. *A subsequence of $(a_n)_{n\in\mathbb{N}} = (n^2)_{n\in\mathbb{N}}$ is obtained by setting $n_j = 2j$, giving $(a_{n_j})_{j\in\mathbb{N}} = (4j^2)_{j\in\mathbb{N}}$.*

Recall that a sequence of real numbers $(a_n)_{n\in\mathbb{N}}$ is called bounded if there exist $A, B \in \mathbb{R}$ such that $A \leq a_n \leq B$ for all n. It is called increasing if $a_n \leq a_{n+1}$ for all n.

Theorem 1.2.13 (Bolzano–Weierstraß)**.** *If $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ is bounded, then it contains a convergent subsequence.*

Proof. Choose $A_0, B_0 \in \mathbb{R}$ such that $A_0 \leq a_n \leq B_0$ for all $n \in \mathbb{N}$. If we divide the interval $I_0 := [A_0, B_0]$ in the center, then either the right or the left half must contain infinitely many elements of the sequence $(a_n)_{n\in\mathbb{N}}$. We define I_1 to be the closed half that contains infinitely many elements.
Repeating this step with I_1 and so on leads to a nested family of closed intervals $(I_k)_{k\in\mathbb{N}}$,

$$I_k = [A_k, B_k] \subseteq I_{k-1}, \qquad B_k - A_k = 2^{-k}(B_0 - A_0),$$

such that each I_k contains infinitely many terms of the sequence $(a_n)_{n\in\mathbb{N}}$.
Next, we construct a subsequence $(a_{n_k})_{k\in\mathbb{N}}$ such that $a_{n_k} \in I_k$ by induction. We start by $k = 0$ and choose n_0 with $a_{n_0} \in I_0$. Now we assume that we have found $a_{n_k} \in I_k$. Since I_{k+1} contains infinitely many elements of the original sequence, there must exist $n_{k+1} > n_k$ with $a_{n_{k+1}} \in I_{k+1}$. Hence, we have constructed a subsequence $(a_{n_k})_{k\in\mathbb{N}}$ such that $a_{n_k} \in I_k$ for all $k \in \mathbb{N}$.
Given $\epsilon > 0$, choose N with $2^{-N}(B_0 - A_0) \leq \epsilon$. For $k, l \geq N$, both a_{n_k} and a_{n_l} lie in I_N, hence

$$|a_{n_k} - a_{n_l}| \leq 2^{-N}(B_0 - A_0) \leq \epsilon.$$

Thus $(a_{n_k})_{k\in\mathbb{N}}$ is Cauchy and therefore convergent because $\mathbb{R}$ is complete. □

Example 1.2.14. *The sequence* $a_n = (-1)^n + \frac{1}{n+1}$ *is bounded and has the convergent subsequence*

$$b_j = a_{2j} = 1 + \frac{1}{2j+1} \to 1.$$

Essentials

Completeness is the property that guarantees Cauchy sequences actually converge inside the space. The rationals $\mathbb{Q}$ are not complete, the reals $\mathbb{R}$ are complete. Bolzano-Weierstraß ensures that bounded sequences have convergent subsequences.

C The real and complex numbers are complete spaces.

Cauchy sequence, completeness, subsequences

1.3 Compactness

Many central results in analysis can be unified by a single structural property of sets, called *compactness*. In metric spaces, compactness admits a simple and intuitive characterization in terms of sequences.

Definition 1.3.1 (compactness)**.** *A metric space X is called* compact *if every sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ has a convergent subsequence $(x_{n_j})_{j\in\mathbb{N}}$ (with limit in X).*

In words: from every sequence in X we can extract a subsequence that converges *without leaving X*. Compactness will later be combined with continuity to obtain strong global statements about functions.

Lemma 1.3.2. *Every compact metric space X is complete.*

Proof. Let $(x_n)_{n\in\mathbb{N}} \subseteq X$ be a Cauchy sequence. Since X is compact, it has a convergent subsequence $x_{n_j} \to x \in X$ for $j \to \infty$. Given $\varepsilon > 0$, there are $N_1, N_2 \in \mathbb{N}$ such that

$$\begin{aligned} d_X(x_n, x_m) &\le \varepsilon/2, & \forall n, m \ge N_1, \\ d_X(x_{n_j}, x) &\le \varepsilon/2, & \forall j \ge N_2. \end{aligned}$$

Let $N := \max(N_1, N_2)$. Then for all $n \ge N$, we have $n_N \ge N_1$ and

$$d_X(x_n, x) \le d_X(x_n, x_{n_N}) + d_X(x_{n_N}, x) \le \varepsilon/2 + \varepsilon/2 \le \varepsilon.$$

Thus, $x_n \to x$. □

Open and closed subsets

Definition 1.3.3. *Let X be a metric space and $x \in X$.*

(i) We define a ball $B_\epsilon(x)$ of radius $\epsilon > 0$ and its interior $\mathring{B}_\epsilon(x)$ by

$$B_\epsilon(x) := \{y \in X : d_X(x,y) \leq \epsilon\},$$
$$\mathring{B}_\epsilon(x) := \{y \in X : d_X(x,y) < \epsilon\}.$$

(ii) We say a set $S \subseteq X$ is open if

$$\forall x \in S \quad \exists \epsilon > 0 \quad B_\epsilon(x) \subseteq S.$$

(iii) We say that $S \subseteq X$ is closed if its complement $X \setminus S$ is open.

(iv) We call $S \subseteq X$ a neighborhood of x if there is $\epsilon > 0$ such that $B_\epsilon(x) \subseteq S$.

Clearly, every ball $B_\epsilon(x)$ is a neighborhood of x. Both, $\emptyset$ and X are open and closed at the same time.

Example 1.3.4. *The interior $\mathring{B}_\epsilon(x)$ is open. Indeed, take an arbitrary $y \in \mathring{B}_\epsilon(x)$. Then we may define $\delta := \frac{1}{2}(\epsilon - d_X(y,x)) > 0$. The triangle inequality ensures that $B_\delta(y) \subseteq \mathring{B}_\epsilon(x)$.*

Example 1.3.5. *Every open interval $(a,b) \subseteq \mathbb{R}$ is open and every closed interval $[a,b] \subseteq \mathbb{R}$ is closed. For instance, let $x \in (a,b)$. Then the choice $\epsilon = \frac{1}{2}\min(x-a, b-x) > 0$ yields $B_\epsilon(x) = [x-\epsilon, x+\epsilon] \subseteq (a,b)$.*

Lemma 1.3.6. *Let X be a metric space and $S \subseteq X$ a nonempty subset. Then S is closed if and only if all sequences $(x_n)_{n\in\mathbb{N}} \subseteq S$ with $x_n \to x \in X$ satisfy $x \in S$.*

In other words, S is closed if and only if it contains all limits of its converging sequences.

Proof. Let S be closed. Proof by contradiction, so assume that $(x_n)_{n\in\mathbb{N}} \subseteq S$ and $x_n \to x \in X \setminus S$. Since $X \setminus S$ is open, there is $\epsilon > 0$ such that $B_\epsilon(x) \subseteq X \setminus S$. Convergence implies that there is $N \in \mathbb{N}$ such that

$$d_X(x_n, x) \leq \varepsilon \quad \forall n \geq N.$$

This yields $x_N \in B_\epsilon(x) \subseteq X \setminus S$, which is a contradiction to $(x_n)_{n\in\mathbb{N}} \subseteq S$.
We prove the reverse implication by contraposition. Let S be not closed. Then $X \setminus S$ is not open. Hence, there is $x \in X \setminus S$ such that for all $\varepsilon > 0$ the neighborhood $B_\varepsilon(x)$ is not contained in $X \setminus S$. Thus, there is $y_\varepsilon \in B_\varepsilon(x) \cap S$. We may now choose $\varepsilon = \frac{1}{n}$, so that for all positive $n \in \mathbb{N}$, there is $y_n \in B_{1/n}(x) \cap S$. This means that $y_n \to x$ and since $y_n \in S$ we have found a sequence contained in S that converges towards $x \in X \setminus S$. □

Lemma 1.3.7. *Let X be a metric space. If $S \subseteq X$ is compact, then S is bounded and closed.*

Proof. Assume that S is unbounded. Fix $x_0 \in X$. Then for every $n \in \mathbb{N}$ there exists $x_n \in S$ such that

$$d(x_n, x_0) \geq n. \tag{1.2}$$

Thus we obtain a sequence $(x_n)_{n\in\mathbb{N}} \subseteq S$.
Due to compactness, there exists a subsequence $(x_{n_j})_{j\in\mathbb{N}}$ that converges to some $x \in S$. This subsequence is bounded by Lemma 1.1.12. This contradicts (1.2). Therefore, S must be bounded.
To verify that S is closed, let $(x_n)_{n\in\mathbb{N}} \subseteq S$ satisfy $x_n \to x \in X$. We aim to show that x is contained in S, so that Lemma 1.3.6 then yields that S is closed. Since S is compact, there is a converging subsequence $x_{n_j} \to x$ for $j \to \infty$. This implies that $x \in S$. □

Example 1.3.8. *The closed interval $[a, b] \subseteq \mathbb{R}$ is compact. Indeed, every sequence in $[a, b]$ is bounded and therefore has a convergent subsequence by the Bolzano-Weierstraß theorem. Since $[a, b]$ is closed, the limit belongs to $[a, b]$.*

Example 1.3.9. *The open interval $(0, 1)$ is not compact. The sequence $x_n = 1/n$ satisfies $x_n \in (0, 1)$ for all n, but converges to $0 \notin (0, 1)$. Hence none of its subsequences converges in $(0, 1)$.*

Remark 1.3.10 (no escape of data)**.** *Compactness expresses that sequences cannot drift off or oscillate without producing a convergent subsequence. In data analysis, this corresponds to working with data sets or models where relevant information remains controlled and does not escape from the model.*

In $\mathbb{R}$, the reverse implication of Lemma 1.3.7 also holds.

Theorem 1.3.11 (Heine-Borel theorem in $\mathbb{R}$)**.** *Suppose that $\mathbb{R}$ is equipped with the standard metric. A set $K \subseteq \mathbb{R}$ is compact if and only if it is closed and bounded.*

Proof. Suppose that $K \subseteq \mathbb{R}$ is compact but unbounded. Thus, there is a sequence $(x_n)_{n\in\mathbb{N}} \subseteq K$ such that $|x_n| > n$, for $n \in \mathbb{N}$. This sequence must have a monotone subsequence in K that is also unbounded. However, every subsequence of this subsequence must also be unbounded, hence, cannot converge. This violates compactness of K since every sequence must have a convergent subsequence. Hence, K must be bounded.
Now suppose that $K \subseteq \mathbb{R}$ is compact but not closed. Thus, the complement is not open and there must be some $x \in \mathbb{R}\backslash K$ such that $B_\epsilon(x) \not\subseteq \mathbb{R}\backslash K$ for all $\epsilon > 0$. In particular, for all $\epsilon = \frac{1}{n+1}$ for $n \in \mathbb{N}$. Therefore, there is $x_n \in B_{\frac{1}{n+1}}(x) \cap K$. This sequence $(x_n)_{n\in\mathbb{N}} \subseteq K$ converges towards $x \notin K$. Hence, none of its subsequences can converge in K. Hence, K must be closed.
Conversely, if K is closed and bounded, then every sequence in K is bounded and hence admits a convergent subsequence by Bolzano–Weierstraß. Closedness ensures that the limit lies in K, so K is compact. □

In regards to convergence the behavior of the metric $d_X(x, y)$ for $x \approx y$ is important. The value when x and y are far from each other has no influence.

Lemma 1.3.12 (Truncating the metric preserves convergence properties)**.** *Let (X, d) be a metric space and define*

$$d_1(x, y) := \min\{1, d(x, y)\}, \qquad x, y \in X.$$

Then the following hold:

(i) d_1 *is a metric.*

(ii) For a sequence $(x_n)_{n\in\mathbb{N}} \subset X$ *and* $x \in X$,

$$x_n \to x \text{ with respect to } d \iff x_n \to x \text{ with respect to } d_1.$$

(iii) $K \subseteq X$ *is compact with respect to* d *if and only if* K *is compact with respect to* d_1.

Proof. (i) To verify that d_1 is a metric, we first observe that (M1) and (M3) are obviously satisfied. For the triangle inequality (M2), we compute

$$\min\{1, d(x, z)\} \leq \min\{1, d(x, y) + d(y, z)\} \leq \min\{1, d(x, y)\} + \min\{1, d(y, z)\},$$

so that d_1 is a metric.
(ii,iii) The convergence claim is obvious because $d_1(x_n, x) \to 0$ if and only if $d(x_n, x) \to 0$. Compactness is defined via converging subsequences. □

Essentials

Every compact metric space is complete. A metric space also admits open and closed sets. Every compact set is closed and bounded. In the real numbers, the Heine-Borel theorem gives the converse: every closed and bounded set in the reals is compact. Hence, the compact intervals in $\mathbb{R}$ are precisely the closed and bounded intervals.

Ⓒ Compact metric spaces are complete.

open, closed, compact, Heine-Borel

1.4 Continuity in metric spaces

Continuity expresses preservation of convergence. A function is continuous precisely when it maps convergent sequences to convergent sequences. In this way, continuous maps respect approximation, ensure robustness under perturbations in data-oriented settings, preserve compactness, and occupy a central position in the computational cycle.

We formalize preservation of convergence for mappings between metric spaces.

Definition 1.4.1 (continuity)**.** *Given two metric spaces* (X, d_X) *and* (Y, d_Y) *and a function* $f : X \to Y$, *we say that* f *is* continuous *in* $x \in X$ *if for every sequence* $(x_n)_{n\in\mathbb{N}} \subseteq X$ *with* $x_n \to x$ *we have*

$$f(x_n) \to f(x).$$

Thus, continuity means

$$\lim_{n\to\infty} f(x_n) = f\big(\lim_{n\to\infty} x_n\big) \qquad \forall x_n \to x.$$

Considering limits over all possible sequences approaching x is often written as $\lim_{z\to x}$, so continuity in x means

$$\lim_{z\to x} f(y) = f(x).$$

We say that f is *continuous* if it is continuous in every $x \in X$.

Lemma 1.4.2 (composition)**.** *Suppose that $f : X \to Y$ and $g : Y \to Z$ are continuous. Then the composition $g \circ f : X \to Z$ is continuous.*

Proof. Fix $x \in X$ and let $x_n \to x$. Since f is continuous, we have $f(x_n) \to f(x)$. By continuity of g,

$$g(f(x_n)) \to g(f(x)).$$

Hence $g \circ f$ is continuous in x, and since x was arbitrary, $g \circ f$ is continuous on X. □

Example 1.4.3 (sums and products of continuous functions are continuous)**.** *Let $f, g : \mathbb{R} \to \mathbb{R}$ be continuous. Then $f + g$ and $f \cdot g$ are continuous because, for $x_n \to x$,*

$$(f+g)(x_n) = f(x_n) + g(x_n) \to f(x) + g(x)$$
$$(f \cdot g)(x_n) = f(x_n)g(x_n) \to f(x)g(x).$$

Example 1.4.4 (polynomial functions are continuous)**.** *For $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x^2 + 3$, consider an arbitrary sequence $x_n \to 2$. Then $x_n^2 \to 2^2 = 4$ and hence*

$$f(x_n) = 2x_n^2 + 3 \to 2 \cdot 2^2 + 3 = 11 = f(2).$$

Thus f is continuous in 2. Since polynomials can be written as finite sums and products of the identity and constant functions, and these are continuous, all polynomial functions

$$p(x) = \sum_{n=0}^{N} a_n x^n$$

are continuous on $\mathbb{R}$.

Example 1.4.5 (piecewise-defined functions)**.** *Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by*

$$f(x) = \begin{cases} 2x + 1, & x \geq 0, \\ 4x^2 + 2, & x < 0. \end{cases}$$

This function is not *continuous at 0 because for $x_n = -1/n$ we obtain*

$$f(x_n) = 4x_n^2 + 2 \to 2 \neq 1 = f(0).$$

It is continuous at every $x \in \mathbb{R} \setminus \{0\}$.

Lemma 1.4.6. *Let (X, d) be a metric space and fix $y \in X$. Then the function*

$$\phi_y : X \to \mathbb{R}, \qquad \phi_y(x) = d(x, y),$$

is continuous.

Proof. Let $(x_n)_{n\in\mathbb{N}} \subseteq X$ converge towards $x \in X$. By the triangle inequality,

$$d(x_n, y) \leq d(x_n, x) + d(x, y) \quad \Rightarrow \quad d(x_n, y) - d(x, y) \leq d(x_n, x).$$

Swapping x_n and x gives

$$d(x, y) - d(x_n, y) \leq d(x, x_n).$$

Hence,

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x) \to 0.$$

□

Theorem 1.4.7. *Let X and Y be metric spaces. Suppose that X is compact and $f : X \to \mathbb{R}$ is continuous. Then the range $f(X)$ is compact.*

Proof. Let $(y_n)_{n\in\mathbb{N}} \subseteq f(X)$ be an arbitrary sequence. Then there is $(x_n)_{n\in\mathbb{N}} \subseteq X$ such that $f(x_n) = y_n$, for all $n \in \mathbb{N}$. Since X is compact, there is a subsequence $(x_{n_j})_{j\in\mathbb{N}}$ that converges towards some $x \in X$. Hence, $(y_{n_j})_{j\in\mathbb{N}}$ is a subsequence of $(y_n)_{n\in\mathbb{N}}$ and satisfies

$$y_{n_j} = f(x_{n_j}) \xrightarrow{j\to\infty} f(x) \in f(X).$$

Thus, every sequence in $f(X)$ has a subsequence that converges in $f(X)$. □

The interaction between compactness and continuity leads to strong global statements about functions.

Theorem 1.4.8 (maximum and minimum are attained)**.** *Suppose that X is a compact metric space and $f : X \to \mathbb{R}$ is continuous. Then there exist points $p, q \in X$ such that*

$$f(p) = \max_{x\in X} f(x), \qquad f(q) = \min_{x\in X} f(x).$$

Proof. We prove the statement for the maximum. The argument for the minimum is analogous. Let

$$M := \sup_{x\in X} f(x).$$

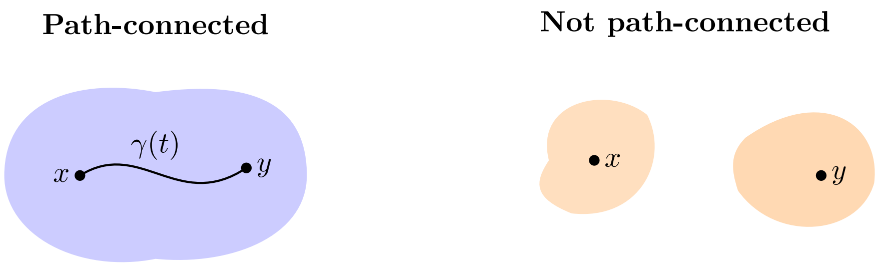
By Lemma 1.1.16, there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ such that $f(x_n) \to M$. Since X is compact, there is a convergent subsequence $(x_{n_j})_{j\in\mathbb{N}}$ with limit $x \in X$. By continuity of f,

$$\lim_{j\to\infty} f(x_{n_j}) = f(x).$$

On the other hand, $(f(x_{n_j}))_{j\in\mathbb{N}}$ is a subsequence of $(f(x_n))_{n\in\mathbb{N}}$ and therefore also converges to M. Hence $f(x) = M$, and the maximum is attained at $p := x$. □

Remark 1.4.9 (existence of extrema). *The combination of compactness and continuity guarantees that extreme values are actually attained. In data-related settings, this ensures that optimal values are not merely approached but realized within the model.*

Definition 1.4.10. *A metric space X is called path-connected if for all $x, y \in X$ there is a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.*
Such a function γ is called a path from x to y.



A path from x to y No path between x and y

Lemma 1.4.11. *Let X and Y be metric spaces. If X is path-connected and $f : X \to Y$ is continuous, then $f(X)$ is path-connected.*

Proof. Let $a_0, a_1 \in f(X)$ be arbitrary. Then there are $x_0, x_1 \in X$ such that $f(x_0) = a_0$ and $f(x_1) = a_1$. Since X is path-connected, there is a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. The function $f \circ \gamma : [0,1] \to f(X)$ is also continuous, because f is assumed to be continuous. It satisfies $f \circ \gamma(0) = f(x_0) = a_0$ and $f \circ \gamma(1) = f(x_1) = a_1$. $\square$

Theorem 1.4.12. *Let $X \subseteq \mathbb{R}$. Then X is path-connected if and only if it is an interval.*

Proof. Let X be an interval. For every pair $x_0, x_1 \in X$, we may define $\gamma : [0,1] \to X$ by $\gamma(t) = (1-t)x_0 + tx_1$. It is obviously continuous with $\gamma(0) = x_0$ and $\gamma(1) = x_1$, and its other values lie between x_0 and x_1, so that it does map into the interval X. This shows that X is path-connected.
Now, suppose that X is path-connected. Choose two arbitrary points $x_0, x_1 \in X$ such that $x_0 < x_1$. Let $y \in \mathbb{R}$ be an arbitrary point in between, $x_0 < y < x_1$. We aim to show that y is also in X.
Since X is path-connected, there is a continuous function $\gamma : [0,1] \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. We define

$$Q = \{t \in [0,1] : \gamma(t) \leq y\} \subseteq \mathbb{R}\,.$$

Then Q is obviously bounded, hence by the order completeness of $\mathbb{R}$, the supremum exists $t_0 = \sup Q$. By Lemma 1.1.16 there is a sequence $(q_n)_{n\in\mathbb{N}} \subseteq Q$ such that $q_n \to t_0$.
Since $\gamma(q_n) \leq y$, we also have $\gamma(t_0) \leq y$. If $\gamma(t_0) < y$ were true, then we may define $\varepsilon := |\gamma(t_0) - y| > 0$ and there is $\delta > 0$ such that $|\gamma(t) - \gamma(t_0)| \leq \varepsilon/2$ for all $|t - t_0| \leq \delta$. This

implies that $\gamma(t_0 + \delta) \leq y$, which contradicts that t_0 is an upper bound. Hence, $\gamma(t_0) = y$. Since $\gamma([0,1]) \subseteq X$, we must have $y \in X$. This shows that X is an interval. □

Corollary 1.4.13. *Let $K \subseteq \mathbb{R}$. Then K is compact and path-connected if and only if it is a closed, bounded interval.*

Proof. This is a consequence of the Heine Borel Theorem 1.3.11 and Theorem 1.4.12. □

Next, we are more restrictive and assume that $K \subseteq \mathbb{R}$ is a closed interval.

Theorem 1.4.14 (intermediate value theorem)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous, then $f([a,b])$ is a closed, bounded interval. In particular, if $f(a) < 0$ and $f(b) > 0$, then there exists $\xi \in (a,b)$ such that*

$$f(\xi) = 0.$$

Proof. The interval $[a,b]$ is compact by the Heine Borel Theorem 1.3.11. Theorem 1.4.7 implies the image under f is compact. The range $f([a,b])$ is path-connected by Theorem 1.4.12. Hence, $f([a,b])$ is a closed interval and we have

$$0 \in [f(a), f(b)] \subseteq f([a,b]).$$

This means there is $\xi \in [a,b]$ such that $f(\xi) = 0$. □

RTA **Bisection method:** For $f : [a,b] \to \mathbb{R}$ continuous with $f(a) > 0$ and $f(b) < 0$, find $\hat{x} \in \mathbb{R}$ such that

$$f(\hat{x}) = 0.$$

Define $J_0 := [a_0, b_0] := [a,b]$ and denote $J_n := [a_n, b_n]$, for $n \in \mathbb{N}$, and

$$x_n = \frac{a_n + b_n}{2}.$$

If $f(x_n) = 0$, then simply $\hat{x} = x_n$. We consider

$$J_{n+1} := [a_{n+1}, b_{n+1}] := \begin{cases} [a_n, x_n], & f(a_n) f(x_n) < 0, \\ [x_n, b_n], & \text{otherwise.} \end{cases}$$

We only evaluate f on subsequent interval centers, which is a discretization. Truncation is a derived by some stopping criterion, and approximation needs an error bound.

Theorem 1.4.15 (RTA Bisection method)**.** *Suppose that $f : [a,b] \to \mathbb{R}$ is continuous with $f(a) > 0$ and $f(b) < 0$. If $f(\frac{a_n+b_n}{2}) \neq 0$, for all $n \in \mathbb{N}$, then*

$$\{\hat{x}\} = \bigcap_{n \in \mathbb{N}} J_n, \qquad f(\hat{x}) = 0,$$

with the error estimate

$$|x_n - \hat{x}| \leq \frac{1}{2^n}|b-a|. \tag{1.3}$$

Compare the bound (1.3) with Example 1.1.8 that contains a consequence of linear convergence.

Proof. In both cases, the diameter satisfies

$$\operatorname{diam}(J_{n+1}) = b_n - \frac{a_n+b_n}{2} = \frac{1}{2}|b_n - a_n| = \frac{1}{2}\operatorname{diam}(J_n) = \ldots = \frac{1}{2^{n+1}}\underbrace{\operatorname{diam}(J_0)}_{|b-a|}.$$

Therefore, $\{\hat{x}\} = \bigcap_{n\in\mathbb{N}} J_n$ and $a_n, x_n, b_n \to \hat{x}$. Continuity of f implies

$$0 \leq f(a_n) \to f(\hat{x}), \qquad 0 \geq f(b_n) \to f(\hat{x}).$$

Hence, $f(\hat{x}) = 0$. Since $a_n, x_n, b_n, \hat{x} \in J_n$, we have

$$|a_n - \hat{x}|, |x_n - \hat{x}|, |b_n - \hat{x}| \leq \operatorname{diam}(J_n) \leq \frac{1}{2^n}|b-a|.$$

□

Consider $f : [0,2] \to \mathbb{R}$ with $f(x) = 2 - x^2$.

```
function myBisection(f,a,b; tol=1e-4,maxiter=50)
    x = Vector{Float64}(undef, maxiter)
    for k in 1:maxiter
        x[k] = (a+b)/2
        if abs(a-b)<tol || abs(f(x[k])) < 2*eps()    # stopping criteria
            return x[1:k]
        end
        if f(a)*f(x[k])<0
            b = x[k]
        else
            a = x[k]
        end
    end
    println("no convergence")
    return x
end

f(z)=z^2-2
myBisection(f,0,2)
```

```
16-element Vector{Float64}:
 1.0
 1.5
 1.25
 1.375
 1.4375
 1.40625
 1.421875
 1.4140625
 1.41796875
```

```
1.416015625
1.4150390625
1.41455078125
1.414306640625
1.4141845703125
1.41424560546875
1.414215087890625
```

The exact root is $\sqrt{2} \approx 1.41421356\ldots$.

Remark 1.4.16 (Solving nonlinear equations)**.** *In applied contexts, the intermediate value theorem provides a theoretical foundation for the existence of solutions of nonlinear equations $f(x) = 0$. The latter is equivalent to a fixed point of $F(x) = f(x) + x$.*

We now put the computational cycle into action. The Banach Fixed Point Theorem illustrates it in a particularly transparent way.

Theorem 1.4.17 (CRTA Banach Fixed Point Theorem)**.** *Let (X, d) be a complete metric space and let $f : X \to X$ be a contraction, i.e. there exists a constant $q \in [0, 1)$ such that*

$$d\big(f(x), f(y)\big) \le q\, d(x, y) \qquad \text{for all } x, y \in X.$$

Then:

1. *f admits a unique fixed point $x \in X$, i.e. $f(x) = x$.*
2. *For every $x_0 \in X$, the iterates $x_{n+1} := f(x_n)$ converge linearly to the fixed point x.*
3. *One has the approximation estimates, for all $n \ge 0$,*

$$d(x_n, x) \le \frac{q^n}{1 - q}\, d(x_1, x_0).$$

CRTA Begin with X_0 and a mapping $f_0 \colon X_0 \to X_0$. Suppose we seek a fixed point, that is, a point x such that $f_0(x) = x$.

1. Completion: Embed X_0 into a complete metric space X and extend f_0 to a mapping $f \colon X \to X$ such that $f|_{X_0} = f_0$.
2. Structure in the infinite regime: Within the complete space X, contractivity,
$$\exists\, q \in (0, 1) \quad \text{such that} \quad d_X\big(f(x), f(y)\big) \le q\, d_X(x, y) \quad \forall\, x, y \in X,$$
implies the existence and uniqueness of a fixed point $x \in X$.
3. Representation: Starting from an arbitrary $x_0 \in X$, the recursively defined sequence
$$x_{n+1} = f(x_n), \qquad n \in \mathbb{N},$$
forms a countable approximation process that converges to the fixed point x.

4. Truncation and approximation: For computation, the infinite iteration is truncated after n steps. The approximation error is controlled by the quantitative estimate

$$d_X(x_n, x) \leq \frac{q^n}{1-q} d_X(x_1, x_0). \tag{1.4}$$

Quantitative bounds such as the contraction constant q may themselves require further analytical or numerical estimation, so that the computational cycle can reappear at a secondary level within the same problem.

Example 1.4.18. *Start with* $f_0 : (0,1) \to (0,1)$ $x \mapsto \frac{1}{4}x^2$*. Following completion, we switch to* $f : [0,1] \to [0,1]$ *and observe*

$$|f(x) - f(y)| = \frac{1}{4}|x^2 - y^2| = \frac{1}{4}|x+y||x-y| \leq \frac{1}{2}|x-y|, \qquad \forall\, x, y \in [0,1].$$

For $x_0 = 1$*, we define the recursive sequence* $x_{n+1} = f(x_n)$*, for* $n \in \mathbb{N}$*. It converges towards a fixed point* $x \in [0,1]$*, so that* $x = f(x) = \frac{1}{4}x^2$*. This implies*

$$0 = \frac{1}{4}x^2 - x = \frac{1}{4}x(x-4).$$

The fixed point is $0 = x \in [0,1]$*. Truncation after* n *steps with* $q = \frac{1}{2}$ *translates the approximation bound* (1.4) *into*

$$|x_n| \leq \frac{q^n}{1-q}|\frac{1}{4} - 1| = 2^{-n+1}\frac{3}{4} = 2^{-(n+1)}3.$$

In reality we are even doing better. One may verify by induction that $x_n = 2^{-2^{n+1}}4$.

Proof. Fix an arbitrary $x_0 \in X$ and define the Picard iteration $x_{n+1} := f(x_n)$ for $n \geq 0$. By the contraction property,

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq q\, d(x_n, x_{n-1}).$$

Iterating this inequality yields, for all $n \geq 0$,

$$d(x_{n+1}, x_n) \leq q^n\, d(x_1, x_0).$$

To verify that $(x_n)_{n\in\mathbb{N}}$ is Cauchy, we consider $m > n$ and deduce

$$\begin{aligned} d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) &\leq \sum_{k=n}^{m-1} q^k d(x_1, x_0) \\ &\leq d(x_1, x_0) \sum_{k=n}^{\infty} q^k \\ &= d(x_1, x_0)\, \frac{q^n}{1-q}. \end{aligned}$$

Since $q \in [0,1)$, the right-hand side tends to 0 as $n \to \infty$, hence $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in X.

Because X is complete, there exists $x \in X$ such that $x_n \to x$. To verify $f(x) = x$, we use the triangle inequality and the contraction property,

$$d(f(x), x) \leq d(f(x), f(x_n)) + d(f(x_n), x) \leq q\, d(x, x_n) + d(x_{n+1}, x).$$

Letting $n \to \infty$ gives $d(f(x), x) = 0$, hence $f(x) = x$.
If y is another fixed point, then

$$d(x, y) = d(f(x), f(y)) \leq q\, d(x, y).$$

Since $q < 1$, this implies $d(x, y) = 0$ and hence $x = y$.
The convergence is linear because

$$d(x_{n+1}, x) = d(f(x_n), f(x)) \leq q d(x_n, x)$$

and $q \in (0, 1)$.
To derive the bound on the approximation error, we send $m \to \infty$ in our Cauchy estimates and obtain, for all $n \geq 0$,

$$d(x_n, x) \leq \frac{q^n}{1-q}\, d(x_1, x_0),$$

which in particular implies $x_n \to x$. □

Remark 1.4.19. *Continuous maps are remarkable. They preserve convergence and thus respect approximation. They preserve structure such as compactness and path-connectedness. Continuity also preserves topological features, but in retrospect, as the preimage of every open set is open. It is therefore no coincidence that nearly all computation is anchored, in one way or another, in continuous mappings.*

Uniform continuity and uniform convergence

Uniform notions sharpen pointwise control to joint control, as realized in uniform continuity and uniform convergence.

(ϵ, δ)-definition of continuity

We first derive a few equivalent notions of continuity.

Lemma 1.4.1 ((ϵ, δ)-definition of continuity)**.** *Let X and Y be metric spaces. The function $f : X \to Y$ is continuous in $x \in X$ if and only if*

$$\forall \epsilon > 0\ \exists\, \delta > 0 : \ d_Y(f(x), f(x')) < \epsilon \quad \forall\, x' \in X \ \textit{with}\ d_X(x, x') < \delta. \tag{1.5}$$

Proof. Suppose that (1.5) holds. Let $(x_n)_{n\in\mathbb{N}} \subseteq X$ converge towards $x \in X$, and let $\epsilon > 0$. Then there is $\delta > 0$ such that

$$d_Y\big(f(x), f(x')\big) \leq \epsilon \qquad \forall\, x' \in X \text{ with } d_X(x, x') < \delta.$$

Since $x_n \to x$, there exists $N = N(\delta) \in \mathbb{N}$ such that

$$d_X(x, x_n) \leq \delta \qquad \forall n \geq N.$$

Thus, we have

$$d_Y(f(x), f(x_n)) \leq \epsilon \qquad \forall n \geq N,$$

which means that $f(x_n) \to f(x)$. We have shown that (ϵ, δ)-continuity implies sequence-continuity.
For the converse implication, we assume that (1.5) does not hold, and we then verify that sequence-continuity is violated. For all $n \in \mathbb{N}$ there is $x_n \in X$ such that

$$d_X(x, x_n) \leq \frac{1}{n} \qquad \text{and} \qquad d_Y(f(x), f(x_n)) > \epsilon.$$

This means that $x_n \to x$ but $f(x_n) \not\to f(x)$. $\square$

Remark 1.4.2 (stability under perturbations)**.** *Continuity as stated in Lemma 1.4.1 formalizes the idea that small changes in the input lead to small changes in the output. In data analysis, this corresponds to robustness with respect to noise or small measurement errors in the data.*

Theorem 1.4.3. *Let X and Y be metric spaces and $f : X \to Y$. Then f is continuous if and only if $f^{-1}(U)$ is open for all open subsets $U \subseteq Y$.*

Proof. We first note that the (ϵ, δ)-definition of continuity yields that f is continuous in x if and only if for all $\epsilon > 0$ there is $\delta > 0$ such that

$$f(B_\delta(x)) \subseteq B_\epsilon(f(x)).$$

(a) Let f be continuous and $U \subseteq Y$ an open set. If $x \in f^{-1}(U)$, then there is $\epsilon > 0$ such that $B_\epsilon(f(x)) \subseteq U$. Since f is continuous, there is $\delta > 0$ such that

$$f(B_\delta(x)) \subseteq B_\epsilon(f(x)) \subseteq U.$$

Hence, $B_\delta(x) \subseteq f^{-1}(U)$.
(b) Assume that $f^{-1}(U)$ is open for all open subsets $U \subseteq Y$. We choose an arbitrary $x \in X$ and an arbitrary $\epsilon > 0$. Since $\mathring{B}_\epsilon(f(x))$ is open, the pre-image $f^{-1}(\mathring{B}_\epsilon(f(x)))$ is open. The point x is contained in this pre-image, hence, there is $\delta > 0$ such that

$$B_\delta(x) \subseteq f^{-1}(\mathring{B}_\epsilon(f(x))).$$

This implies that $f(B_\delta(x)) \subseteq \mathring{B}_\epsilon(f(x))$, which shows that f is continuous in x. $\square$

Lemma 1.4.4. *Let X be a metric space. If $f : X \to \mathbb{R}$ is continuous in $x \in X$ and $f(x) > 0$, then there is a neighborhood of x, on which f is positive.*

Proof. The interval $(0,\infty)$ is open, hence, $f^{-1}((0,\infty))$ is open and contains x. Thus, it contains a neighborhood of x that is then mapped into $(0,\infty)$. □

Uniform continuity

We now sharpen the notion of continuity.

Definition 1.4.5 (uniform continuity)**.** *Let X and Y be metric spaces. The function $f : X \to Y$ is called uniformly continuous if*

$$\forall \epsilon > 0\ \exists\, \delta > 0 : \ d_Y(f(x_1), f(x_2)) \le \epsilon \quad \forall\, x_1, x_2 \in X \text{ with } d_X(x_1, x_2) \le \delta.$$

Since f is continuous at every x_1, there is $\delta = \delta(x_1)$ such that

$$d_Y(f(x_1), f(x_2)) \le \epsilon, \qquad \forall\, x_2 \in X \text{ with } d_X(x_1, x_2) \le \delta.$$

The point of uniform continuity is that δ does not depend on x_1.

Theorem 1.4.6 (uniformly continuous on compact sets)**.** *Let X and Y be metric spaces. If X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous.*

Proof. Assume that f is not uniformly continuous. Then there is $\epsilon > 0$ and points $x_n, z_n \in X$ such that

$$d_X(x_n, z_n) \le \frac{1}{n}, \qquad d_Y\left(f(x_n), f(z_n)\right) \ge \epsilon.$$

Since X is compact, there is a convergent subsequence $(x_{n_j})_{j\in\mathbb{N}} \to x \in X$. The sequence $(z_{n_j})_{j\in\mathbb{N}}$ has a subsequence converging towards $(z_{n_{j_i}})_{i\in\mathbb{N}} \to z \in X$. Hence, to simplify notation, we may assume that we have chosen a common subsequence right from the beginning such that $(x_{n_j})_{j\in\mathbb{N}} \to x$ and $(z_{n_j})_{j\in\mathbb{N}} \to z$. By design we obtain

$$d_X(x, z) \le d_X(x, x_n) + d_X(x_n, z_n) + d_X(z_n, z) \to 0.$$

Hence, $x = z$ and continuity of f yields

$$d_Y\left(f(x_{n_j}), f(z_{n_j})\right) \le d_Y\left(f(x_{n_j}), f(x)\right) + d_Y\left(f(z), f(z_{n_j})\right) \to 0.$$

This is a contradiction to $d_Y\left(f(x_{n_j}), f(z_{n_j})\right) \ge \epsilon$, for all $j \in \mathbb{N}$. □

Uniform convergence

We now consider sequences of functions and investigate their convergence behavior.

Theorem 1.4.7 (uniform convergence)**.** *Let X and Y be metric spaces. For $n \in \mathbb{N}$, let $f_n : X \to Y$ be continuous and $f : X \to Y$ be another function such that*

$$\sup_{x\in X} d_Y(f_n(x), f(x)) \to 0, \quad \text{for } n \to \infty. \tag{1.6}$$

Then f is continuous.

Obviously, if (1.6) is satisfied, then $f_n(x) \to f(x)$ for all $x \in X$. The latter is referred to as pointwise convergence while (1.6) is called uniform convergence.

Proof. Suppose $(x_m)_{m\in\mathbb{N}} \subseteq X$ such that $x_m \to x \in X$. To verify that $f(x_m) \to f(x)$, we estimate

$$\begin{aligned} d_Y(f(x_m), f(x)) &\leq d_Y(f(x_m), f_n(x_m)) + d_Y(f_n(x_m), f_n(x)) + d_Y(f_n(x), f(x)) \\ &\leq d_\infty(f_n, f) + d_Y(f_n(x_m), f_n(x)) + d_\infty(f_n, f) \\ &= 2d_\infty(f_n, f) + d_Y(f_n(x_m), f_n(x)). \end{aligned}$$

This chain of inequalities holds for all $n \in \mathbb{N}$. Taking the limit $m \to \infty$ and using that f_n is continuous, we obtain

$$0 \leq \lim_{m\to\infty} d_Y(f(x_m), f(x)) \leq 2d_\infty(f_n, f), \qquad \forall n \in \mathbb{N}.$$

The limit $n \to \infty$ leads to

$$0 \leq \lim_{m\to\infty} d_Y(f(x_m), f(x)) \leq \lim_{n\to\infty} 2d_\infty(f_n, f) = 0.$$

Hence, $f(x_m) \to f(x)$ and f is continuous. □

Example 1.4.8. *The functions $f_n : [0,1] \to \mathbb{R}$ defined by $f_n(x) = \frac{x}{n+1} + 2$ satisfy*

$$\sup_{x\in[0,1]} |f_n(x) - 2| = \sup_{x\in[0,1]} |\frac{x}{n+1}| \leq \frac{1}{n+1} \to 0.$$

Thus, $f_n \to 2$ uniformly. The limit function $f(x) = 2$ for $x \in [0,1]$ is constant and hence continuous.

Example 1.4.9. *The functions $f_n : [0,1] \to \mathbb{R}$ defined by $f_n(x) = x^n$ satisfy the pointwise convergence*

$$f_n(x) \to \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1. \end{cases}$$

Since the limit function is not continuous in 1, the convergence cannot be uniform.

Example 1.4.10. *The functions $f_n : [0,1] \to \mathbb{R}$ defined by $f_n = 1_{[\frac{1}{n},1]}$ satisfy the pointwise convergence*

$$f_n(x) \to \begin{cases} 1, & x \in (0,1], \\ 0, & x = 0. \end{cases}$$

Since the limit function is not continuous in 0, the convergence cannot be uniform.

We now define a metric space that consists of functions. Fix two metric spaces (X, d_X) and (Y, d_Y) and consider the set of bounded, continuous functions

$$\mathcal{C}_b(X,Y) = \{f : X \to Y \quad | \quad f \text{ bounded and continuous}\}.$$

For $f, g \in \mathcal{C}_b(X,Y)$, define

$$d_\infty(f,g) := \sup_{x\in X} d_Y(f(x), g(x)). \tag{1.7}$$

Since f and g are bounded, this is well-defined. It satisfies the three conditions of a metric, so that $(\mathcal{C}_b(X,Y), d_\infty)$ is a metric space.

Theorem 1.4.11. *Let X and Y be metric spaces. If Y is complete, then $\mathcal{C}_b(X,Y)$ is complete.*

Proof. Let $(f_n)_{n\in\mathbb{N}} \subseteq \mathcal{C}_b(X,Y)$ be a Cauchy sequence. For every $x \in X$, the sequence $(f_n(x))_{n\in\mathbb{N}} \subseteq Y$ is Cauchy. Since Y is complete, there exists a limit in Y, say $f(x)$. This way we define a function $f : X \to Y$.
To verify that f is bounded, we notice that the Cauchy property of $(f_n)_{n\in\mathbb{N}}$ ensures that there is $N \in \mathbb{N}$ such that

$$d_\infty(f_n, f_N) \leq 1 \qquad \text{for all } n \geq N.$$

Fix $x \in X$ and let $n \geq N$. By the triangle inequality,

$$d_Y\big(f_n(x), y\big) \leq d_Y\big(f_n(x), f_N(x)\big) + d_Y\big(f_N(x), y\big) \leq 1 + d_Y\big(f_N(x), y\big).$$

Letting $n \to \infty$ yields

$$d_Y\big(f(x), y\big) \leq 1 + d_Y\big(f_N(x), y\big).$$

Taking the supremum over $x \in X$ and using that f_N is bounded, we obtain

$$\sup_{x\in X} d_Y\big(f(x), y\big) \leq 1 + \sup_{x\in X} d_Y\big(f_N(x), y\big) < \infty.$$

Hence f is bounded.
To verify that f is continuous, we show that $(f_n)_{n\in\mathbb{N}}$ converges uniformly towards f. Let $\varepsilon > 0$. Choose N such that $d_\infty(f_n, f_m) \leq \varepsilon$ for all $n, m \geq N$. Fix $n \geq N$ and $x \in X$. Letting $m \to \infty$ in

$$d_Y\big(f_n(x), f_m(x)\big) \leq d_\infty(f_n, f_m) < \varepsilon$$

gives $d_Y\big(f_n(x), f(x)\big) \leq \varepsilon$. Taking the supremum over x yields $d_\infty(f_n, f) \leq \varepsilon$ for all $n \geq N$. Hence $f_n \to f$ in d_∞. The uniform convergence implies that f is continuous, hence, $f \in \mathcal{C}_b(X,Y)$. $\square$

Essentials

Continuity expresses that small changes in the input produce small changes in the output. Continuous functions map compact sets to compact sets and path-connected sets to path-connected sets. In particular, the image of a closed and bounded interval is again a closed and bounded interval, so maxima and minima are attained. The intermediate value theorem and the Banach fixed point theorem ensure the existence of solutions to nonlinear equations. Continuity admits an ϵ-δ-characterization, and f is continuous if and only if the pre-image of open sets is open. Continuous functions are uniformly continuous on compact sets. Continuous functions that converge uniformly have a continuous limit function.

CRTA Continuous functions preserve approximation, so that approximating inputs yield approximating outputs. The entire computational cycle is present in the Banach fixed point theorem for contractive maps. The uniform limit of continuous functions is continuous.

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continuous, polynomial functions, sums, products, composition, intermediate value theorem, maxima, minima, complex conjugation, intermediate value theorem, Banach fixed point theorem, uniform continuity, uniform convergence

2 Elementary Functions

In this chapter, we restrict to $X = Y = \mathbb{R}$ and define function values as limits of special sequences. From another viewpoint, we consider sequences of functions $f_n : \mathbb{R} \to \mathbb{R}$ that converge to a limit function f. This perspective reflects the notion of uniform convergence introduced in the previous chapter.

We first study infinite series as limits of partial sums and use them to construct functions via power series. A central example is the exponential function. We then define and analyze further special functions, including logarithmic and trigonometric functions.

2.1 Series

Infinite series arise as limits of special sequences whose elements are built from partial sums. They provide a fundamental tool for the construction of special functions. We study criteria for convergence, absolute convergence, and reordering.

Given a sequence $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{C}$, we consider the sequence of finite sums

$$S_N := \sum_{n=0}^{N} a_n.$$

We then ask whether the sequence $(S_N)_{N\in\mathbb{N}}$ converges. If it does, we define the *series*

$$\sum_{n=0}^{\infty} a_n := \lim_{N\to\infty} \sum_{n=0}^{N} a_n.$$

Equivalently, we say that the infinite sum $\sum_{n=0}^{\infty} a_n$ *exists* or *converges.*

A first necessary condition for convergence is the following.

Lemma 2.1.1. *If $\sum_{n=0}^{\infty} a_n$ converges, then $a_n \to 0$.*

Proof. Since the sequence of partial sums $(S_N)_{N\in\mathbb{N}}$ converges, it is a Cauchy sequence. Let $\epsilon > 0$. There exists $N_0 \in \mathbb{N}$ such that

$$|S_N - S_M| \le \epsilon \qquad \forall N, M \ge N_0.$$

In particular, for all $N \ge N_0$,

$$|a_{N+1}| = \left| \sum_{n=0}^{N+1} a_n - \sum_{n=0}^{N} a_n \right| = |S_{N+1} - S_N| \le \epsilon.$$

Thus $|a_n| \le \epsilon$ for all $n \ge N_0 + 1$, and so $a_n \to 0$. $\square$

Remark 2.1.2 (aggregating information). *Infinite series naturally arise from adding many small contributions. In data analysis, similar aggregation processes are used to combine information from multiple sources or features. Convergence ensures that this aggregation leads to a stable and meaningful result.*

Example 2.1.3 (telescope sum). *Consider the series* $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. *Since* $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, *we can rewrite the partial sums by*

$$\begin{aligned}\sum_{n=1}^{N} \frac{1}{n(n+1)} &= \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots + \left(\frac{1}{N} - \frac{1}{N+1}\right) \\ &= 1 - \frac{1}{N+1} \to 1.\end{aligned}$$

Hence, $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$.

For the special choice $a_k = x^k$, the partial sums $S_N = \sum_{k=0}^{N} x^k$ are the geometric sum.

Example 2.1.4 (geometric series). *We have verified in Theorem* **??** *that*

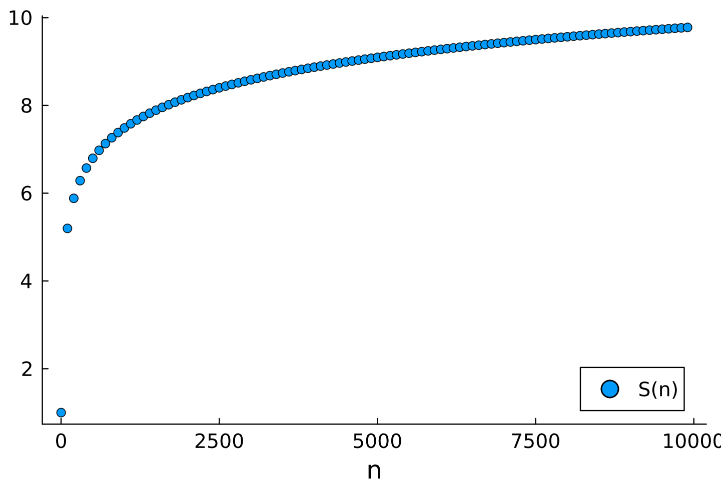
$$\sum_{k=0}^{N} x^k = (1 - x^{N+1})(1 - x)^{-1}, \quad x \neq 1.$$

In Example **??** *we observed that this formula holds in every ring with identity* 1 *as long as* $1 - x$ *has a multiplicative inverse. It therefore holds for* $x \in \mathbb{C}$ *with* $x \neq 0$. *If* $|x| < 1$, *then* $x^{N+1} \to 0$, *and the geometric series converges,*

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \qquad \forall\, x \in \mathbb{C},\ |x| < 1.$$

At first sight, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ may seem to converge.

```
S(n) = sum(1/k for k in 1:n)
n = 1:100:10000
scatter(n,S.(n), label="S(n)", xlabel="n")
```



The harmonic series does not converge; it diverges, albeit very slowly.

Example 2.1.5 (harmonic series)**.** *The condition $a_n \to 0$ does not necessarily imply that $\sum_{n=0}^{\infty} a_n$ converges. For instance, the harmonic series satisfies*

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + &\underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq \frac{2}{4} = \frac{1}{2}} \\ &+ \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}}_{\geq \frac{4}{8} = \frac{1}{2}} \\ &+ \underbrace{\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}}_{\geq \frac{8}{16} = \frac{1}{2}} \\ &+ \underbrace{\frac{1}{17} + \ldots + \frac{1}{32}}_{\frac{16}{32} = \frac{1}{2}} \\ &\vdots\end{aligned}$$

This means that

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \sum_{k=0}^{\infty} \underbrace{\sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n}}_{\geq \frac{2^k}{2^{k+1}} = \frac{1}{2}} = \infty.$$

We next discuss convergence of general series.

Definition 2.1.6 (absolute convergence)**.** *A series $\sum_{n=0}^{\infty} a_n$ is called* absolutely convergent *if $\sum_{n=0}^{\infty} |a_n|$ converges.*

Lemma 2.1.7 (absolutely convergent series converge)**.** *If* $\sum_{n=0}^{\infty} |a_n|$ *converges, then* $\sum_{n=0}^{\infty} a_n$ *also converges.*

Proof. Since the partial sums of $\sum_{n=0}^{N} |a_n|$ form a Cauchy sequence, there exists N_0 such that

$$\left| \sum_{n=0}^{N} |a_n| - \sum_{n=0}^{M} |a_n| \right| \leq \epsilon \qquad \forall N, M \geq N_0.$$

For $N \geq M \geq N_0$,

$$\begin{aligned} \left| \sum_{n=0}^{N} a_n - \sum_{n=0}^{M} a_n \right| &= \left| \sum_{n=M+1}^{N} a_n \right| \leq \sum_{n=M+1}^{N} |a_n| \\ &= \left| \sum_{n=0}^{N} |a_n| - \sum_{n=0}^{M} |a_n| \right| \leq \epsilon. \end{aligned}$$

Thus the partial sums of $\sum a_n$ form a Cauchy sequence. □

Lemma 2.1.8. *If* $\sum_{n=0}^{\infty} a_n$ *is absolutely convergent, then any reordering of the terms converges to the same limit.*

Proof. Define $A := \sum_{n=0}^{\infty} a_n$ and let $\tau : \mathbb{N} \to \mathbb{N}$ be a bijective mapping. For $\epsilon > 0$ there is $N_0 \in \mathbb{N}$ such that

$$\left| \sum_{n=0}^{\infty} |a_n| - \sum_{n=0}^{N_0} |a_n| \right| = \sum_{n=N_0+1}^{\infty} |a_n| \leq \epsilon/2.$$

This also implies

$$\left| \sum_{n=0}^{\infty} a_n - \sum_{n=0}^{N_0} a_n \right| = \left| \sum_{n=N_0+1} a_n \right| \leq \sum_{n=N_0+1}^{\infty} |a_n| \leq \epsilon/2.$$

There is $N \in \mathbb{N}$ such that

$$\{0, 1, \ldots, N_0\} \subseteq \{\tau(0), \ldots, \tau(N)\}.$$

Therefore, we obtain for $M \geq N$,

$$\left| \sum_{n=0}^{M} a_{\tau(n)} - \sum_{n=0}^{N_0} a_n \right| \leq \sum_{n=N_0+1}^{\infty} |a_n| \leq \epsilon/2.$$

This leads to

$$\left| \sum_{n=0}^{M} a_{\tau(n)} - \sum_{n=0}^{\infty} a_n \right| \leq \left| \sum_{n=0}^{M} a_{\tau(n)} - \sum_{n=0}^{N_0} a_n \right| + \left| \sum_{n=0}^{N_0} a_n - \sum_{n=0}^{\infty} a_n \right| \leq \epsilon.$$

Thus, $\sum_{n=0}^{M} a_{\tau(n)} \to \sum_{n=0}^{\infty} a_n$ as $M \to \infty$. □

Remark 2.1.9 (robust aggregation)**.** *For absolutely convergent series, the value is independent of the order of the terms. This is crucial in computational settings, where the order of summation may change due to implementation details or optimization. Absolute convergence guarantees that such changes do not affect the result.*

Decimal expansions provide a concrete example of converging series.

Lemma 2.1.10 (decimal number representation)**.** *Let $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ be such that $(a_n)_{n=1}^\infty \subseteq \{0,1,2,\dots,9\}$. The series*

$$\sum_{n=0}^{\infty} a_n \left(\frac{1}{10}\right)^n$$

converges absolutely.

Proof. We estimate

$$\begin{aligned}\sum_{n=0}^{\infty} \left| a_n \left(\frac{1}{10}\right)^n \right| &= |a_0| + \sum_{n=1}^{\infty} \left| a_n \left(\frac{1}{10}\right)^n \right| \\ &\le |a_0| + 9 \sum_{n=1}^{\infty} \left(\frac{1}{10}\right)^n \\ &= |a_0| + 9 \left(\frac{1}{1-\frac{1}{10}} - 1 \right) \\ &= |a_0| + 1.\end{aligned}$$

Thus, the series is absolutely convergent. □

Example 2.1.11. *For $a_0 = 0$ and $a_n = 9$ for all $n = 1, 2, \dots$, we obtain*

$$0,999\ldots = \sum_{n=0}^{\infty} a_n \left(\frac{1}{10}\right)^n = 9 \sum_{n=1}^{\infty} \left(\frac{1}{10}\right)^n = 9 \left(\frac{1}{1-\frac{1}{10}} - 1 \right) = 1.$$

```
decimals(a) = sum(a[n]*0.1^n for n=1:length(a))
fill(9, 3), decimals(fill(9,16)), decimals(fill(9,17))
```

```
([9, 9, 9], 0.9999999999999999, 1.0)
```

The following tests are standard tools for establishing convergence.

Theorem 2.1.12 (convergence criteria for series)**.** *Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences of complex numbers.*

(1) Comparison test: *If $|a_n| \le |b_n|$ for all n and $\sum_{n=0}^\infty |b_n|$ converges, then $\sum_{n=0}^\infty |a_n|$ converges.*

(2) Ratio test (quotient criterion): *If there exists* $0 < \theta < 1$ *and* $N \in \mathbb{N}$ *such that*

$$\left|\frac{a_{n+1}}{a_n}\right| \leq \theta \qquad \forall n \geq N, \tag{2.1}$$

then $\sum_{n=0}^{\infty} |a_n|$ *converges.*

(3) Leibniz criterion: *If* $(a_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ *is nonnegative, monotonically decreasing, and* $a_n \to 0$*, then the alternating series* $\sum(-1)^n a_n$ *converges.*

Remark 2.1.13. *If the ratios converge towards*

$$\left|\frac{a_{n+1}}{a_n}\right| \to q < 1, \quad n \to \infty,$$

then the quotient criterion is satisfied for every number $q < \theta < 1$*.*

Proof. (1) Since $0 \leq \sum_{n=0}^{N} |a_n| \leq \sum_{n=0}^{N} |b_n|$, the sequence of partial sums of $\sum |a_n|$ is increasing and bounded above, hence convergent.
(2) Since convergence is independent of the first N elements, we may assume that (2.1) holds for all $n \in \mathbb{N}$. Repeated application gives

$$|a_n| \leq \theta |a_{n-1}| \leq \theta^2 |a_{n-2}| \leq \cdots \leq \theta^n |a_0|.$$

The comparison test together with the geometric series shows that $\sum |a_n|$ converges.
(3) We define $s_k := \sum_{n=0}^{k}(-1)^n a_n$ and directly observe

$$\begin{aligned} s_{2k+2} - s_{2k} &= a_{2k+2} - a_{2k+1} \leq 0 \\ s_{2k+3} - s_{2k+1} &= a_{2k+2} - a_{2k+3} \geq 0. \end{aligned}$$

Thus, the sequence of partial sums satisfies

$$\begin{aligned} s_{2k} &\geq s_{2k+2} \\ s_{2k+1} &\leq s_{2k+3}. \end{aligned}$$

Since $s_{2k+1} - s_{2k} = -a_{2k+1} \leq 0$, we have $s_{2k+1} \leq s_{2k}$. Thus, the sequence $(s_{2k})_{k\in\mathbb{N}}$ is monotonically decreasing and lower bounded, hence, convergent, say $s_{2k} \to S^+ \in \mathbb{R}$. Analogously, the sequence $(s_{2k+1})_{k\in\mathbb{N}}$ is monotonically increasing and upper bounded, hence, convergent, say $s_{2k+1} \to S^- \in \mathbb{R}$. To verify that $S^+ = S^-$ and that the entire sequence $(s_k)_{k\in\mathbb{N}}$ converges, we first observe that

$$S^+ - S^- = \lim_{k\to\infty}(s_{2k} - s_{2k+1}) = \lim_{k\to\infty} a_{2k+1} = 0.$$

If the two subsequences $s_{2k}, s_{2k+1} \to S := S^+ = S^-$, then also the entire sequence converges towards S.

$\square$

Example 2.1.14. *Let $a_n = 2^{-n}$. Then $a_n > 0$, a_n is decreasing, and $a_n \to 0$. By the Leibniz criterion, the alternating series converges. Its limit is*

$$\sum_{n=0}^{\infty}(-1)^n 2^{-n} = \sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}.$$

Example 2.1.15. *Consider the series*

$$\sum_{n=1}^{\infty}\frac{n^2}{3^n}.$$

With $a_n = \frac{n^2}{3^n}$, we have $\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{3n^2}$. Since $\lim_{n\to\infty}\frac{(n+1)^2}{3n^2} = \frac{1}{3} < 1$, the ratio test implies that the series converges.

Example 2.1.16. *Let $a_n = \frac{2^n}{n!}$. To check on*

$$\sum_{n=1}^{\infty}\frac{2^n}{n!},$$

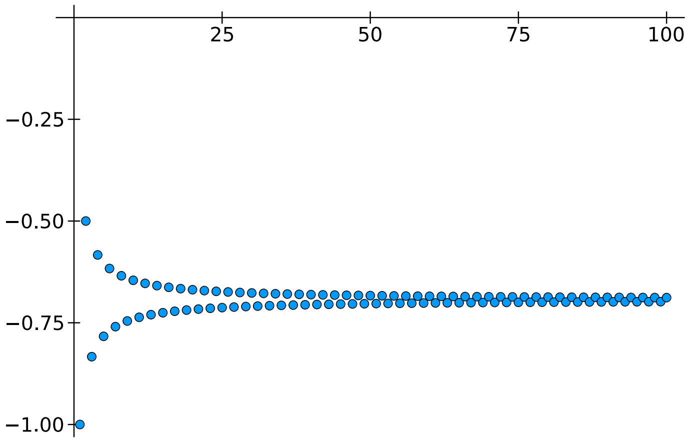
we observe $\frac{a_{n+1}}{a_n} = \frac{2}{n+1} \to 0$. Since the ratio tends to a value less than 1, the series converges by the ratio test.

Example 2.1.17. *Consider the series*

$$\sum_{n=1}^{\infty}(-1)^n\frac{1}{n}.$$

Let $b_n = \frac{1}{n}$. Then $b_n \geq 0$, $b_{n+1} \leq b_n$, and $b_n \to 0$. Thus, by the Leibniz criterion, the alternating series converges. The series does not converge absolutely, because the harmonic series does not converge.

```
alterHarmonic(n) = sum((-1)^k/k for k=1:n)
M = 1:100
scatter(M,alterHarmonic.(M))
```



Example 2.1.18. *For the harmonic series* $\sum_{n=1}^{\infty} \frac{1}{n}$, *the quotient satisfies*

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \leq 1.$$

This shows that the existence of a number $\theta < 1$ *in Theorem 2.1.12 is critical.*

Essentials

Infinite series are limits of partial sums, and there are tools to decide convergence (comparison, ratio test, Leibniz criterion). Absolute convergence has a central role due to its invariance under reordering.

C Series are limits of partial sums.

series, partial sum, telescope sum, geometric series, harmonic series, absolute convergence, decimal numbers, comparison test, ratio test, Leibniz criterion

2.2 The exponential function

Many fundamental functions arise as limits of series. We now turn to one of the most widely used functions in mathematics: the exponential function, whose significance throughout science can hardly be overstated.

Definition 2.2.1 (exponential function)**.** *For* $z \in \mathbb{C}$, *the exponential series is defined by*

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

and Euler's number e *is*

$$e := \exp(1) = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

The series indeed converges for every $z \in \mathbb{C}$, because

$$\frac{\frac{z^{n+1}}{(n+1)!}}{\frac{z^n}{n!}} = \frac{z}{n+1}.$$

For fixed z, choose $N_0 \in \mathbb{N}$ such that $|z| < N_0$. Then, for all $n \geq N_0$,

$$\frac{|z|}{n+1} \leq \frac{|z|}{N_0+1} =: \theta < 1.$$

Hence the ratio test (quotient criterion) implies that $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ converges.

Theorem 2.2.2 (remainder of exp)**.** *For every $z \in \mathbb{C}$ and $N \in \mathbb{N}$ we have*

$$\exp(z) = \sum_{n=0}^{N} \frac{z^n}{n!} + R_{N+1}(z),$$

where the remainder term R_{N+1} satisfies

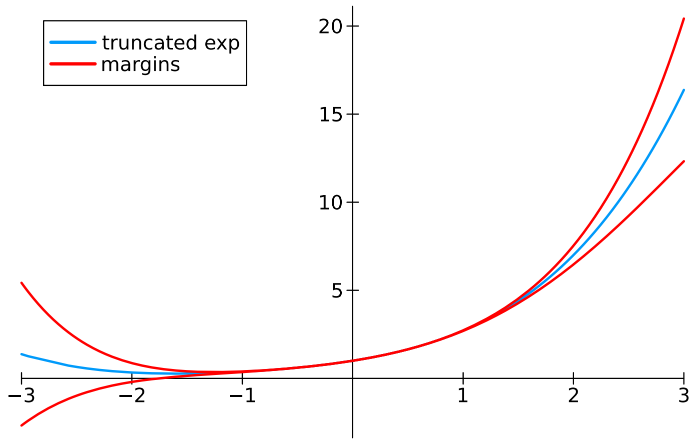
$$|R_{N+1}(z)| \leq 2\,\frac{|z|^{N+1}}{(N+1)!}, \qquad \forall\, |z| \leq (N+2)/2.$$

Proof. We estimate the tail of the series:

$$\begin{aligned}
|R_{N+1}(z)| &\leq \sum_{n=N+1}^{\infty} \frac{|z|^n}{n!} \\
&= \frac{|z|^{N+1}}{(N+1)!}\left(1 + \frac{|z|}{N+2} + \frac{|z|^2}{(N+2)(N+3)} + \dots\right) \\
&\leq \frac{|z|^{N+1}}{(N+1)!}\left(1 + \frac{|z|}{N+2} + \frac{|z|^2}{(N+2)^2} + \dots\right) \\
&\leq \frac{|z|^{N+1}}{(N+1)!}\left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right) \\
&= \frac{|z|^{N+1}}{(N+1)!} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \\
&= 2\,\frac{|z|^{N+1}}{(N+1)!}.
\end{aligned}$$

□

```
myExp(x,N) = sum(x^n/factorial(big(n)) for n=0:N)
R(x,N) = 2*abs(x)^(N+1)/factorial(big(N+1))
N = 4
plot(x->myExp(x,N),xlimits=[-(N+2)/2,(N+2)/2],label="truncated exp")
plot!(x->myExp(x,N)+R(x,N),color="red",label="margins")
plot!(x->myExp(x,N)-R(x,N),color="red")
```



Remark 2.2.3 (constructing functions). *Series provide a way to construct functions from elementary building blocks. In data science, complex models are often built by combining simple components, and convergence guarantees that the resulting function is well-defined.*

Example 2.2.4. *Let $f_N : [-1,1] \to \mathbb{R}$ be defined by the partial sum $f_N(x) = \sum_{n=0}^{N} \frac{x^n}{n!}$. Each f_N is continuous. It is also bounded by*

$$|f_N(x)| \leq \sum_{n=0}^{N} \frac{1}{n!}.$$

For $N \geq 1$, the remainder term yields

$$\sup_{x\in[-1,1]} |f_N(x) - \exp(x)| \leq \sup_{x\in[-1,1]} \frac{2|x|^{N+1}}{(N+1)!} \leq \frac{2}{(N+1)!} \to 0.$$

Thus, $(f_N)_{N\in\mathbb{N}}$ converges uniformly towards $\exp$ on $[-1,1]$. Theorem 1.4.7 implies that $\exp|_{[-1,1]}$ is continuous.

We now compute the product of two absolutely convergent series.

Lemma 2.2.5 (Cauchy product). *Assume that $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \subseteq \mathbb{C}$ are such that $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are absolutely convergent. Define a new sequence $(c_n)_{n\in\mathbb{N}}$ by*

$$c_n := \sum_{k=0}^{n} a_k b_{n-k}.$$

Then the series $\sum_{n=0}^{\infty} c_n$ is also absolutely convergent and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} a_n\right) \cdot \left(\sum_{n=0}^{\infty} b_n\right).$$

Proof. For even $N \in \mathbb{N}$, we define

$$\Box_N := \{(k,l) \in \mathbb{N}^2 : 0 \le k,l \le N\}$$
$$\Delta_N := \{(k,l) \in \mathbb{N}^2 : k + l \le N\}.$$

The partial sums are denoted by

$$A := \sum_{n=0}^{\infty} a_n, \quad A_N := \sum_{n=0}^{N} a_n, \qquad B := \sum_{n=0}^{\infty} b_n, \quad B_N := \sum_{n=0}^{N} b_n, \qquad C_N := \sum_{n\in\mathbb{N}} c_n.$$

We also have

$$A_N B_N = \sum_{(k,l)\in\Box_N} a_k b_l, \qquad C_N := \sum_{(k,l)\in\Delta_N} a_k b_l.$$

Since $\Delta_N \subseteq \Box_N$, we may write

$$A_N B_N - C_N = \sum_{(k,l)\in\Box_N\setminus\Delta_N} a_k b_l,$$

so that $\Box_{N/2} \subseteq \Delta_N$ leads to

$$\begin{aligned} |A_N B_N - C_N| &\le \sum_{(k,l)\in\Box_N\setminus\Delta_N} |a_k|\,|b_l| \\ &\le \sum_{(k,l)\in\Box_N\setminus\Box_{N/2}} |a_k|\,|b_l| \\ &= \left(\sum_{n=0}^{N} |a_n|\right)\left(\sum_{n=0}^{N} |b_n|\right) - \left(\sum_{n=0}^{N/2} |a_n|\right)\left(\sum_{n=0}^{N/2} |b_n|\right). \end{aligned}$$

Since the series converge, the product is a Cauchy sequence and therefore $|A_N B_N - C_N| \to 0$ as $N \to \infty$. Hence, not only $A_n B_N \to AB$ but also $C_N \to AB$.
Absolute convergence follows by replacing a_n and b_n by $|a_n|$ and $|b_n|$ and

$$|c_n| \le \sum_{k=0}^{n} |a_k|\,|b_{n-k}|.$$

□

Theorem 2.2.6 (functional equation of exp)**.** *The exponential function satisfies*

$$\exp(x+y) = \exp(x)\,\exp(y), \qquad x, y \in \mathbb{C}.$$

Proof. Both series $\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and $\exp(y) = \sum_{n=0}^{\infty} \frac{y^n}{n!}$ are absolutely convergent. By

Lemma 2.2.5 and $a_n = \frac{x^n}{n!}$ and $b_n = \frac{y^n}{n!}$, their Cauchy product is

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} = \sum_{k=0}^{n} \frac{x^k}{k!} \frac{y^{n-k}}{(n-k)!}$$
$$= \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = \frac{(x+y)^n}{n!}.$$

Thus

$$\exp(x)\exp(y) = \sum_{n=0}^{\infty} c_n = \sum_{n=0}^{\infty} \frac{(x+y)^n}{n!} = \exp(x+y).$$

□

Corollary 2.2.7. *The exponential function* $\exp : \mathbb{C} \to \mathbb{C}$ *is continuous and satisfies the following:*

(i) For all $z \in \mathbb{C}$*:* $\exp(-z) = \dfrac{1}{\exp(z)}$.

(ii) For all $x \in \mathbb{R}$*:* $\exp(x) > 0$.

(iii) For all $k \in \mathbb{Z}$*:* $\exp(k) = e^k$.

Proof. The remainder estimate with $N = 0$ in Theorem 2.2.2 yields

$$|\exp(z) - 1| \leq 2|z|, \qquad \forall |z| \leq 1.$$

Hence $\lim_{z\to 0} \exp(z) = 1 = \exp(0)$ and exp is continuous at 0.
Now let $z \in \mathbb{C}$ be arbitrary and $z_n \to z$. Using the functional equation,

$$\exp(z_n) = \exp(z)\exp(z_n - z) \to \exp(z)\exp(0) = \exp(z),$$

so exp is continuous everywhere.
(i) By Theorem 2.2.6,

$$\exp(z)\exp(-z) = \exp(z - z) = \exp(0) = 1,$$

so $\exp(-z) = 1/\exp(z)$.
(ii) For $x \geq 0$, the series expansion gives

$$\exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \cdots \geq 1 > 0.$$

For $x < 0$, we have $-x > 0$ and therefore $\exp(-x) > 0$. Using part (i),

$$\exp(x) = \frac{1}{\exp(-x)} > 0.$$

(iii) For $k = 0$, we have $\exp(0) = 1 = \exp(1)^0$. Assume $\exp(k) = \exp(1)^k$ for some $k \in \mathbb{N}$. Then

$$\exp(k+1) = \exp(k)\exp(1) = \exp(1)^k \exp(1) = \exp(1)^{k+1}.$$

Hence $\exp(k) = e^k$ for all $k \in \mathbb{N}$.
For negative integers, using (i),

$$\exp(-k) = \frac{1}{\exp(k)} = \frac{1}{e^k} = e^{-k}.$$

Thus $\exp(k) = e^k$ for all $k \in \mathbb{Z}$. $\square$

Essentials

The exponential function exp is defined by a power series and satisfies $\exp(x + y) = \exp(x)\exp(y)$.

RTA Many special functions are represented by power series. Truncating the power series of the exponential function yields a finite representation. Error bounds quantify the resulting approximation.

exponential function exp, Cauchy product, functional equation, e

2.3 Logarithm and general exponents

Proposition 2.3.1. *Let $\mathcal{I}, \mathcal{J} \subseteq \mathbb{R}$ be intervals. Suppose that $f : \mathcal{I} \to \mathcal{J}$ is continuous and surjective.*

(i) If f is strictly monotone, then f is bijective and its inverse function $f^{-1} : \mathcal{J} \to \mathcal{I}$ is also continuous.

(ii) If f is bijective, then f is strictly monotone and its inverse function $f^{-1} : \mathcal{J} \to \mathcal{I}$ is also continuous.

Proof. (i) Strict monotonicity implies that f is injective, hence, bijective and the inverse exists. Suppose that f is strictly increasing. To verify that f^{-1} is continuous in $y \in \mathcal{J}$, not a boundary point, let $\epsilon > 0$. Then $x = f^{-1}(y) \in \mathcal{I}$ is not a boundary point and we may assume that ϵ is sufficiently small such that $[x - \epsilon, x + \epsilon] \subseteq \mathcal{I}$.
Since f is strictly increasing, we have

$$f([x - \epsilon, x + \epsilon]) = [f(x - \epsilon), f(x + \epsilon)]. \tag{2.2}$$

Define $\delta = \min(y - f(x - \epsilon), f(x + \epsilon) - y) > 0$. Then $[y - \delta, y + \delta] \subseteq [f(x - \epsilon), f(x + \epsilon)]$, and the relation (2.2) implies

$$f^{-1}((y - \delta, y + \delta)) \subseteq [x - \epsilon, x + \epsilon].$$

Thus, f^{-1} is continuous in y.
The case when y is a boundary point is treated similarly and so is the case when f is strictly decreasing.
(ii) Suppose that f is bijective but not strictly monotone. Take $x_1, x_2, x_3 \in \mathcal{I}$ such that $x_1 < x_2 < x_3$ and assume that

$$f(x_1) \leq f(x_2) \geq f(x_3).$$

Injectivity of f yields $f(x_1) < f(x_2) > f(x_3)$. If $f(x_3) \geq f(x_1)$, then $f(x_3) \in f([x_1, x_2])$, which contradicts the injectivity. If $f(x_1) \geq f(x_3)$, then $f(x_1) \in f([x_2, x_3])$, and this also contradicts the injectivity.
The assumption $f(x_1) \geq f(x_2) \leq f(x_3)$ similarly leads to a contradiction. Altogether it implies that f is strictly monotone, and part (i) then yields that f^{-1} is continuous. □

Example 2.3.2 (k-th roots)**.** *Let $k \in \mathbb{N}$ with $k \geq 1$.*

$$f : [0, \infty) \to [0, \infty), \quad x \mapsto x^k$$

is continuous and strictly monotonically increasing. Hence, its inverse function

$$f^{-1} : [0, \infty) \to [0, \infty), \quad x \mapsto \sqrt[k]{x}$$

is well-defined and continuous.

Theorem 2.3.3 (logarithm)**.** *The exponential function* $\exp : \mathbb{R} \to (0, \infty)$ *is continuous and strictly monotonically increasing. Its inverse function*

$$\log : (0, \infty) \to \mathbb{R}$$

is continuous and satisfies

$$\log(xy) = \log(x) + \log(y), \qquad \forall x, y \in (0, \infty).$$

Proof. (a) $\exp : \mathbb{R} \to (0, \infty)$ *is strictly monotonically increasing.* For $\xi > 0$,

$$\exp(\xi) = 1 + \xi + \frac{\xi^2}{2} + \ldots > 1.$$

If $x < y$, put $\xi = y - x > 0$ and compute

$$\exp(y) = \exp(x + \xi) = \exp(x) \exp(\xi) > \exp(x).$$

(b) *The image of* $\exp$ *is* $(0, \infty)$. For $n \in \mathbb{N}$, $\exp(n) \geq 1 + n$ and

$$\exp(-n) = \frac{1}{\exp(n)} \leq \frac{1}{1 + n}.$$

Thus the range of exp is $(0, \infty)$.
(c) *Functional equation for* $\log$. For $a = \log(x)$ and $b = \log(y)$, we have

$$\exp(a + b) = \exp(a) \exp(b) = xy.$$

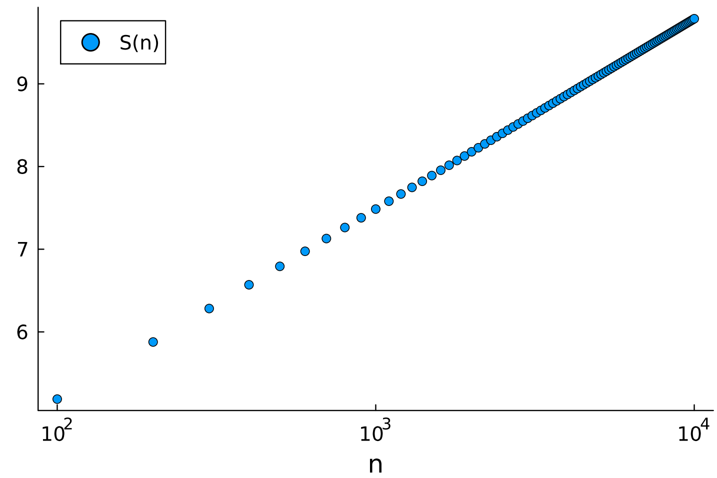
Therefore,

$$\log(xy) = a + b = \log(x) + \log(y).$$

□

We now plot the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ with the n-axis spaced uniformly on a logarithmic scale.

```
S(n) = sum(1/k for k in 1:n)
n = 100:100:10000
scatter(n,S.(n), xaxis=:log, label="S(n)", xlabel="n")
```



Its divergence now becomes obvious.

We have already defined the expressions a^x for $a > 0$ and $x \in \mathbb{R}$ in Section **??** via a limit process: rational exponents are well defined, and irrational exponents are obtained by approximation with rationals. Although this approach could be extended analogously to complex exponents, we take a different route here and define a^z for $a > 0$ and $z \in \mathbb{C}$ by a single expression.

Definition 2.3.4 (general exponents). *For $a > 0$ and $z \in \mathbb{C}$, we define*

$$a^z := \exp(z \log a).$$

Lemma 2.3.5 (rules for general exponents). *For $a > 0$, the function $z \mapsto a^z$ is continuous on $\mathbb{C}$ and the following properties hold:*

1. $a^{x+y} = a^x a^y$, *for all $x, y \in \mathbb{C}$.*
2. $\log(a^x) = x \log a$, *for all $x \in \mathbb{R}$.*
3. $\left(a^x\right)^y = a^{xy}$, *for all $x \in \mathbb{R}$ $y \in \mathbb{C}$.*
4. $(ab)^x = a^x b^x$, *for all $a, b > 0$ and all $x \in \mathbb{C}$.*
5. $a^{-x} = 1/a^x$, *for all $x \in \mathbb{C}$.*
6. $a^n = \underbrace{a \cdots a}_{n\text{-times}}$ *for all $n \in \mathbb{N}$.*
7. *for all $n \in \mathbb{N}$, $a^{1/n} = \sqrt[n]{a}$ is the unique positive number defined in Proposition **??**, whose n-th power is a.*
8. $e^x = \exp(x)$, *for all $x \in \mathbb{C}$.*

Proof. 1. By definition and the functional equation of exp, we obtain

$$\begin{aligned} a^{x+y} &= \exp((x+y)\log a) \\ &= \exp(x\log a + y\log a) \\ &= \exp(x\log a)\exp(y\log a) \\ &= a^x a^y. \end{aligned}$$

2. $\log(a^x) = \log(\exp(x\log a)) = x\log a$. Since log is the inverse of exp on $\mathbb{R}$, we require $x\log a \in (0,\infty)$ and thus assume $x \in \mathbb{R}$.

3. $\left(a^x\right)^y = \exp\left(y\log(a^x)\right) = \exp(xy\log a) = a^{xy}$. To evaluate log, we require $a^x \in (0,\infty)$ and thus assume $x \in \mathbb{R}$.

4. $(ab)^x = \exp(x\log(ab)) = \exp(x\log a + x\log b) = \exp(x\log a)\exp(x\log b) = a^x b^x$.

5. $a^{-x} = \exp(-x\log a) = 1/\exp(x\log a) = 1/a^x$.

6. We compute

$$\begin{aligned} a^n &= \exp(n\log a) \\ &= \exp(\underbrace{\log a + \ldots + \log a}_{n\text{-times}}) \\ &= \exp(\log a)\ \cdots\ \exp(\log a) \\ &= \underbrace{a\cdots a}_{n\text{-times}}. \end{aligned}$$

7. Obviously $\left(a^{1/n}\right)^n = \exp\left(\frac{1}{n}\log a\right)^n = \exp(\log a) = a$ and $a^{1/n} = \exp\left(\frac{1}{n}\log a\right) > 0$. Thus, $a^{1/n}$ is the unique positive real number $x > 0$ such that $x^n = a$, see Proposition **??**.

8. For $x \in \mathbb{R}$, we have $e^x = \exp(x\log(e)) = \exp(x)$.

□

For $z \in \mathbb{C}$ with $z = a + \mathrm{i}b$ and $a, b \in \mathbb{R}$, complex conjugation yields $\overline{z} = a - \mathrm{i}b$. The real part is a and the imaginary part is b. They satisfy

$$\operatorname{Re} z = \tfrac{1}{2}(z + \overline{z}), \qquad \operatorname{Im} z = \tfrac{1}{2\mathrm{i}}(z - \overline{z}).$$

Lemma 2.3.6. *Complex conjugation is continuous on $\mathbb{C}$, i.e., for $z \in \mathbb{C}$,*

$$\lim_{\xi\to z} \overline{\xi} = \overline{z}.$$

Proof. Let $z_n \to z$. For any $\epsilon > 0$, there exists N such that $|z_n - z| \le \epsilon$ for all $n \ge N$. We now obtain

$$|\overline{z_n} - \overline{z}| = |\overline{z_n - z}| = |z_n - z| \le \epsilon,$$

which proves the claim. □

Essentials

The inverse of a continuous function is again continuous. It led to the logarithm as the inverse of the exponential function and enables us to define powers for complex exponents.

logarithm log, general exponents, k-th roots

2.4 Trigonometric functions

Trigonometric functions are introduced via the complex exponential, revealing their deep connection to analysis and complex numbers. This approach yields power series representations, fundamental identities, and periodicity properties in a unified way.

Definition 2.4.1 (sine and cosine). *For $x \in \mathbb{R}$, we define the trigonometric functions by*

$$\cos(x) := \mathrm{Re}(\exp(\mathrm{i}x)), \qquad \sin(x) := \mathrm{Im}(\exp(\mathrm{i}x)).$$

Thus, cosine and sine arise naturally as the real and imaginary parts of the exponential function evaluated on the imaginary axis. In particular,

$$\exp(\mathrm{i}x) = \cos x + \mathrm{i}\sin x,$$

which is known as *Euler's formula*.

Theorem 2.4.2 (series expansion of sine and cosine). *For every $x \in \mathbb{R}$, the following series expansions hold:*

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}, \qquad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

Proof. We split the exponential series into even and odd powers:

$$\begin{aligned}\exp(\mathrm{i}x) &= \sum_{n=0}^{\infty} \frac{(\mathrm{i}x)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{(\mathrm{i}x)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(\mathrm{i}x)^{2k+1}}{(2k+1)!}.\end{aligned}$$

Using $\mathrm{i}^{2k} = (-1)^k$ and $\mathrm{i}^{2k+1} = \mathrm{i}(-1)^k$, we obtain

$$\exp(\mathrm{i}x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} + \mathrm{i}\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

Comparing real and imaginary parts yields the formulas. $\square$

Lemma 2.4.3. *Both* cos *and* sin *are continuous and satisfy*

$$\cos x = \tfrac{1}{2}\big(e^{\mathrm{i}x} + e^{-\mathrm{i}x}\big), \qquad \sin x = \tfrac{1}{2\mathrm{i}}\big(e^{\mathrm{i}x} - e^{-\mathrm{i}x}\big), \qquad x \in \mathbb{R}\,.$$

Proof. For $z \in \mathbb{C}$, the real and imaginary parts satisfy

$$\operatorname{Re} z = \tfrac{1}{2}(z + \overline{z}), \qquad \operatorname{Im} z = \tfrac{1}{2\mathrm{i}}(z - \overline{z}).$$

Applying this to Euler's formula gives, for all $x \in \mathbb{R}$,

$$\cos x = \tfrac{1}{2}\big(e^{\mathrm{i}x} + \overline{e^{\mathrm{i}x}}\big), \qquad \sin x = \tfrac{1}{2\mathrm{i}}\big(e^{\mathrm{i}x} - \overline{e^{\mathrm{i}x}}\big).$$

Since complex conjugation is continuous, we have

$$\overline{e^{\mathrm{i}x}} = e^{\overline{\mathrm{i}x}} = e^{-\mathrm{i}x}.$$

□

Theorem 2.4.4 (trigonometric identities)**.** *For all $x \in \mathbb{R}$,*

$$|e^{\mathrm{i}x}| = 1, \qquad \cos^2 x + \sin^2 x = 1.$$

Proof. We compute $|e^{\mathrm{i}x}|^2 = e^{\mathrm{i}x}e^{-\mathrm{i}x} = 1$. The identity then follows from $|z|^2 = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2$. □

Lemma 2.4.5 (addition formulas)**.** *For all $x, y \in \mathbb{R}$,*

$$\cos(x + y) = \cos x \cos y - \sin x \sin y,$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y.$$

Proof. Using Euler's formula and the functional equation,

$$\begin{aligned} \exp(\mathrm{i}(x + y)) &= \exp(\mathrm{i}x)\exp(\mathrm{i}y) \\ &= (\cos x + \mathrm{i}\sin x)(\cos y + \mathrm{i}\sin y) \\ &= (\cos x \cos y - \sin x \sin y) + \mathrm{i}\,(\sin x \cos y + \cos x \sin y)\,. \end{aligned}$$

Comparing real and imaginary parts yields the identities. □

Lemma 2.4.6. *The function* cos *has a zero in* $(0, 2)$.

Proof. The series expansion $\cos x = \sum_{k=0}^{\infty}(-1)^k \frac{x^{2k}}{(2k)!}$ yields $\cos 0 = 1$. We now evaluate at

$x = 2$. Consecutive summands satisfy

$$\frac{2^{2(k+1)}}{(2(k+1))!} \frac{(2k)!}{2^{2k}} = \frac{4}{(2k+1)(2k+2)} < 1, \qquad k = 1, 2, \dots,$$

hence, are decaying. This implies

$$\begin{aligned}
\cos 2 &= \sum_{k=0}^{\infty} (-1)^k \frac{2^{2k}}{(2k)!} \\
&= 1 - \frac{4}{2} + \frac{16}{4!} \underbrace{- \frac{64}{6!} + \frac{2^8}{8!}}_{<0} \underbrace{- \frac{2^{10}}{10!} + \frac{2^{12}}{12!}}_{<0} - \dots \\
&\le 1 - \frac{4}{2} + \frac{16}{4!} = 1 - 2 + 2/3 = -1/3.
\end{aligned}$$

Since cos is continuous on $[0, 2]$ and $\cos(0) = 1$ and $\cos(2) < -\frac{1}{3}$, the intermediate value theorem ensures that there is $\xi \in (0, 2)$ such that $\cos \xi = 0$. ☐

Definition 2.4.7 (π). *We define $\pi/2 := \xi$, where ξ is the smallest positive zero of* cos.

Lemma 2.4.8. *For all $n \in \mathbb{N}$,*

$$e^{\mathrm{i}n\pi/2} = \mathrm{i}^n.$$

Proof. Since $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, Euler's formula gives

$$e^{\mathrm{i}\pi/2} = \mathrm{i}.$$

Raising both sides to the n-th power yields the claim. ☐

Theorem 2.4.9 (periodicity). *For all $x \in \mathbb{R}$ and $k \in \mathbb{Z}$,*

$$e^{\mathrm{i}(x+2\pi k)} = e^{\mathrm{i}x}, \qquad \cos(x + 2\pi k) = \cos x, \qquad \sin(x + 2\pi k) = \sin x.$$

Proof. By Lemma 2.4.8, we have $e^{\mathrm{i}2\pi} = 1$. For $k \in \mathbb{N}$, this also yields

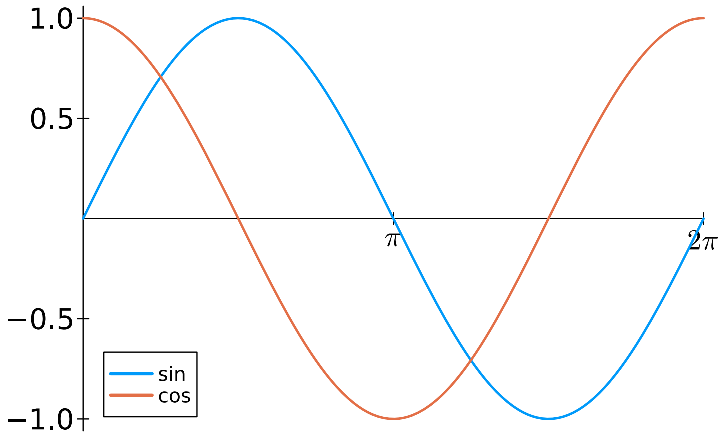
$$e^{\mathrm{i}2\pi k} = \left(e^{\mathrm{i}2\pi}\right)^k = 1,$$

and negative k lead to the same. Thus,

$$e^{\mathrm{i}(x+2\pi k)} = e^{\mathrm{i}x} e^{\mathrm{i}2\pi k} = e^{\mathrm{i}x}.$$

Comparing real and imaginary parts concludes the proof. ☐

```
plot(sin,xlimits=[0,2π],label="sin")
plot!(cos,label="cos")
```



Remark 2.4.10 (periodic patterns)**.** *Trigonometric functions provide canonical models for periodic and oscillatory behavior. In data analysis, such functions are used to represent repeating patterns and cyclic effects, for example in time-dependent or spatial data that are analyzed within the mathematical field of Fourier series.*

Lemma 2.4.11. *Sine and cosine satisfy*

$$\lim_{h\to 0}\frac{\cos h - 1}{h} = 0, \qquad \lim_{h\to 0}\frac{\sin h}{h} = 1.$$

Proof. The series expansion leads to

$$\begin{aligned}\frac{\cos h - 1}{h} &= \frac{1}{h}\left(\sum_{k=0}^{\infty}(-1)^k\frac{h^{2k}}{(2k)!} - 1\right)\\ &= \frac{1}{h}\sum_{k=1}^{\infty}(-1)^k\frac{h^{2k}}{(2k)!}\\ &= h\sum_{k=1}^{\infty}(-1)^k\frac{h^{2k-2}}{(2k)!}.\end{aligned}$$

Taking the absolute value for $|h| \le 1$ yields

$$\left|\frac{\cos h - 1}{h}\right| \le |h|\sum_{k=1}^{\infty}\frac{1}{(2k)!}.$$

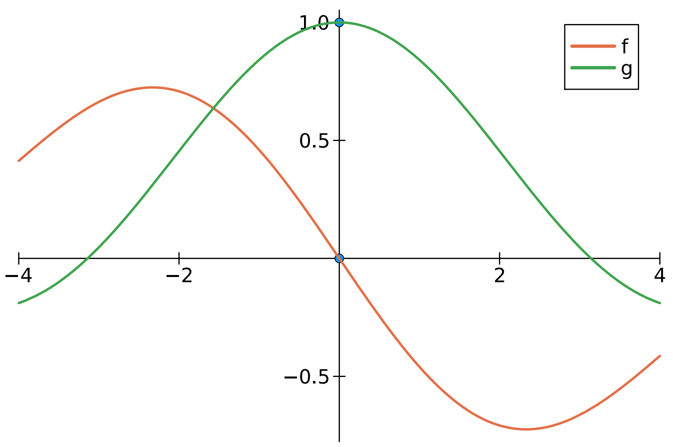
The quotient criterion can be used to verify that the series $\sum_{k=1}^{\infty}\frac{1}{(2k)!}$ converges and hence $\frac{\cos h - 1}{h} \to 0$ for $h \to 0$.

The proof for sine is similar. Its series expansion yields

$$\begin{aligned}\frac{\sin h}{h} - 1 &= \frac{1}{h}\sum_{k=0}^{\infty}(-1)^k\frac{h^{2k+1}}{(2k+1)!} - 1\\ &= \sum_{k=0}^{\infty}(-1)^k\frac{h^{2k}}{(2k+1)!} - 1\\ &= \sum_{k=1}^{\infty}(-1)^k\frac{h^{2k}}{(2k+1)!}.\end{aligned}$$

As above, we can now extract a factor of h to verify that $\frac{\sin h}{h} - 1 \to 0$ for $h \to 0$. □

```
f(h) = (cos(h)-1)/h
g(h) = sin(h)/h

scatter([0,0],[1,0])
plot!(f,label="f",xlimits=[-4,4])
plot!(g,label="g")
```



Essentials

Trigonometric functions are defined using the complex exponential, leading naturally to Euler's formula $\exp(\mathrm{i}x) = \cos x + \mathrm{i}\sin x$, for $x \in \mathbb{R}$. Sine and cosine have power series expansions, satisfy addition formulas, and are periodic. The number π is introduced analytically as a distinct zero of the cosine function.

R T A As with the exponential function, the series representations of the trigonometric functions allow for truncation. Suitable truncations of trigonometric series approximate the exact values.

sine, cosine, trigonometric identities, addition formulas, Euler's formula, π, periodicity, arcsine, arccosine

3 Differentiation and Integration

Differentiability refines the notion of continuity by capturing the idea of local linear approximation. In this chapter, we define derivatives via difference quotients and show how this induces a local, linear approximation of the function. We compute derivatives of elementary functions and establish the fundamental differentiation rules, including linearity, product, and chain rules. These tools are then applied to inverse functions, leading to derivatives of the logarithm and inverse trigonometric functions.

We also introduce integration as a limit of finite sums that determine the area under the curve. Building on continuity and uniform approximation by step functions, we define the integral and study its fundamental properties. The central result is the Fundamental Theorem of Calculus, which connects integration and differentiation.

The Taylor polynomial is a local polynomial approximation whose coefficients are determined by the derivatives at the expansion point, and whose approximation error can be quantified through an integral remainder term.

3.1 Differentiation

Differentiability is defined through the existence of a limit of difference quotients. We begin with the formal definition.

Definition 3.1.1 (differentiability). *Let $U \subseteq \mathbb{R}$ be an open interval and $f : U \to \mathbb{R}$. We say that f is* differentiable at $x \in U$ *if the limit*

$$\lim_{y \to x} \frac{f(y) - f(x)}{y - x}$$

exists. The value of this limit is denoted by $f'(x)$ and is called the derivative of f at x*. Equivalently, by writing $h = y - x$, the function f is differentiable at x if and only if*

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

exists.

Example 3.1.2 (constants are differentiable). *If $f(x) = c$ is constant, then $f'(x) = 0$ for all $x \in \mathbb{R}$, since*

$$\frac{f(x+h) - f(x)}{h} = \frac{c - c}{h} = 0.$$

Example 3.1.3 (linear functions are differentiable). *If $f(x) = ax$ with $a \in \mathbb{R}$, then $f'(x) = a$*

for all $x \in \mathbb{R}$, because

$$\frac{a(x+h)-ax}{h} = \frac{ah}{h} = a.$$

Example 3.1.4 (squares are differentiable)**.** *If $f(x) = x^2$, then $f'(x) = 2x$. Indeed,*

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \to 0} 2x.$$

Lemma 3.1.5 (exp is differentiable)**.** *The exponential function satisfies*

$$\exp' x = \exp x.$$

Proof. We compute the difference quotient by the functional equation,

$$\frac{\exp(x+h) - \exp(x)}{h} = \exp(x)\frac{\exp(h)-1}{h}.$$

Using the expansion from Theorem 2.2.2 with $N = 1$, we have

$$\exp(h) = 1 + h + R_2(h),$$

where $|R_2(h)| \leq |h|^2$ for $|h| \leq 3/2$. Hence

$$\frac{\exp(h)-1}{h} = 1 + \frac{R_2(h)}{h},$$

and since $|R_2(h)/h| \leq |h| \to 0$ as $h \to 0$, we obtain

$$\frac{\exp(x+h) - \exp(x)}{h} \to \exp(x).$$

□

Lemma 3.1.6 (sine and cosine are differentiable)**.** *The trigonometric functions satisfy*

$$\sin' x = \cos x, \qquad \cos' x = -\sin x.$$

Proof. Using the addition formulas,

$$\frac{\sin(x+h) - \sin x}{h} = \sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h}.$$

As $h \to 0$, we have $\frac{\cos h - 1}{h} \to 0$ and $\frac{\sin h}{h} \to 1$, see Lemma 2.4.11. Hence $\sin' x = \cos x$. Similarly,

$$\frac{\cos(x+h) - \cos x}{h} = \cos x \frac{\cos h - 1}{h} - \sin x \frac{\sin h}{h},$$

which converges to $-\sin x$.

□

We say that a function is continuously differentiable if it is differentiable and its derivative is continuous. Hence, exp, cos, and sin are continuously differentiable. In fact, they are arbitrarily often differentiable.

Remark 3.1.7 (local change). *Differentiability captures how a function changes locally. In data-related models, derivatives quantify the sensitivity of the output with respect to small changes in the input.*

Theorem 3.1.8 (A differentiability as linear approximability). *Let $f : U \to \mathbb{R}$ and $x_0 \in U$. The function f is differentiable at x_0 if and only if there exist $c \in \mathbb{R}$ and a function φ such that*

$$f(x) = f(x_0) + c(x - x_0) + \varphi(x),$$

where

$$\lim_{x \to x_0} \left| \frac{\varphi(x)}{x - x_0} \right| = 0.$$

In this case, $c = f'(x_0)$.

For a function φ satisfying $\lim_{x\to x_0} \left|\frac{\varphi(x)}{x-x_0}\right| = 0$ we simply write $o(|x - x_0|)$. The linear approximability reads as

$$f(x) = f(x_0) + c(x - x_0) + o(|x - x_0|).$$

Proof. If f is differentiable at x_0, set $c := f'(x_0)$ and define

$$\varphi(x) = f(x) - f(x_0) - c(x - x_0).$$

Then

$$\frac{\varphi(x)}{x - x_0} = \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \xrightarrow{x \to x_0} 0.$$

Conversely, if such a representation exists, then

$$\frac{f(x) - f(x_0)}{x - x_0} = c + \frac{\varphi(x)}{x - x_0} \xrightarrow{x \to x_0} c,$$

so f is differentiable at x_0 with derivative c. $\square$

The mapping

$$x \mapsto f(x_0) + f'(x_0)(x - x_0)$$

is a polynomial of degree 1 in x, hence, an affine linear map. It provides a good local approximation of f near x_0, because the deviation is captured by φ that quickly decays to zero in proximity to x_0.

Remark 3.1.9 (Local linear approximation). *Derivatives provide local linear models of nonlinear functions. This allows complex relations to be analyzed via simpler linear behavior.*

Differentiability is a refinement of continuity.

Corollary 3.1.10 (differentiability implies continuity). *If f is differentiable at x_0, then f is continuous at x_0.*

Proof. Using the linear approximation,

$$f(x) = f(x_0) + \underbrace{f'(x_0)(x - x_0)}_{\to 0} + \underbrace{\varphi(x)}_{\to 0} \xrightarrow{x \to x_0} f(x_0).$$

□

Remark 3.1.11. *There are also alternative notations of the derivative $f'(x)$ that are useful, in particular, when the variable is not obvious from the context. We also write*

$$f'(x) = \left(\frac{df}{dx}\right)(x) = (\partial_x f)(x).$$

Direct computation of derivatives by the difference quotient quickly becomes impractical. This section develops general differentiation rules such as linearity, product rule, chain rule, and applies them to compositions and inverse functions, including logarithmic and inverse trigonometric functions.

Throughout this section, let $U \subseteq \mathbb{R}$ be an open interval.

Lemma 3.1.12 (linearity of differentiation). *Let $f, g : U \to \mathbb{R}$ be differentiable at $x \in U$ and let $\alpha, \beta \in \mathbb{R}$. Then the function $\alpha f + \beta g$ is differentiable at x, and*

$$(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x).$$

Proof. We compute the difference quotient:

$$\begin{aligned}
\frac{(\alpha f + \beta g)(x+h) - (\alpha f + \beta g)(x)}{h} &= \frac{\alpha f(x+h) - \alpha f(x) + \beta g(x+h) - \beta g(x)}{h} \\
&= \alpha \frac{f(x+h) - f(x)}{h} + \beta \frac{g(x+h) - g(x)}{h}.
\end{aligned}$$

Taking the limit $h \to 0$ and using differentiability of f and g leads to $\alpha f'(x) + \beta g'(x)$. □

Theorem 3.1.13 (product rule). *Let $f, g : U \to \mathbb{R}$ be differentiable at $x \in U$. Then the product fg is differentiable at x, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Proof. By subtracting and adding $f(x)g(x+h)$, we obtain

$$\begin{aligned}\frac{f(x+h)g(x+h)-f(x)g(x)}{h} &= \frac{f(x+h)g(x+h)-f(x)g(x+h)}{h} \\ &\quad + \frac{f(x)g(x+h)-f(x)g(x)}{h} \\ &= g(x+h)\frac{f(x+h)-f(x)}{h} + f(x)\frac{g(x+h)-g(x)}{h}.\end{aligned}$$

Since differentiability implies continuity, we have $g(x+h) \to g(x)$ as $h \to 0$. Taking the limit yields $g(x)f'(x) + f(x)g'(x)$. □

Remark 3.1.14 (polynomial functions are differentiable)**.** *As a consequence of linearity and the product rule, all polynomial functions are differentiable, and their derivatives can be computed term by term.*

Example 3.1.15. *Let $f(x) = x$ and $g(x) = x^2$, so that $f(x)g(x) = x^3$. Then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) = x^2 + x\,2x = 3x^2.$$

Theorem 3.1.16 (chain rule)**.** *Let $f : U \to \mathbb{R}$ be differentiable at $x \in U$, and let $g : V \to \mathbb{R}$ be differentiable at $f(x)$, where $f(U) \subseteq V$. Then the composition $g \circ f$ is differentiable at x, and*

$$(g \circ f)'(x) = g'(f(x))\, f'(x).$$

Proof. Provided that $f(x+h) \neq f(x)$, we may write

$$\frac{g(f(x+h)) - g(f(x))}{h} = \frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} \cdot \frac{f(x+h) - f(x)}{h},$$

Since f is differentiable at x, we have $f(x+h) \to f(x)$ as $h \to 0$. Thus, by differentiability of g at $f(x)$,

$$\frac{g(f(x+h)) - g(f(x))}{f(x+h) - f(x)} \to g'(f(x)).$$

At the same time,

$$\frac{f(x+h) - f(x)}{h} \to f'(x).$$

Taking limits gives

$$(g \circ f)'(x) = g'(f(x))\, f'(x).$$

For the case when $f(x+h) \neq f(x)$ is violated, we need to define an additional function. Fix $x \in U$. Since g is differentiable at $f(x)$, we define $\eta : V \to \mathbb{R}$ by

$$\eta(y) := \begin{cases} \dfrac{g(y) - g(f(x))}{y - f(x)} - g'(f(x)), & y \neq f(x), \\ 0, & y = f(x). \end{cases}$$

By the definition of the derivative, we have

$$\lim_{y \to f(x)} \eta(y) \to 0.$$

Moreover, for every $y \in V$ we can rewrite

$$g(y) - g(f(x)) = g'(f(x))(y - f(x)) + \eta(y)\,(y - f(x)).$$

Since f is differentiable at x, we have $f(x+h) \to f(x)$ as $h \to 0$. Hence, for h sufficiently small, $f(x+h) \in V$ and thus

$$(g \circ f)(x+h) - (g \circ f)(x) = g'(f(x))\big(f(x+h) - f(x)\big) + \eta\big(f(x+h)\big)\big(f(x+h) - f(x)\big).$$

Dividing by $h \neq 0$ yields

$$\frac{(g \circ f)(x+h) - (g \circ f)(x)}{h} = g'(f(x))\frac{f(x+h) - f(x)}{h} + \eta\big(f(x+h)\big)\frac{f(x+h) - f(x)}{h}.$$

Letting $h \to 0$, the first term converges to $g'(f(x))f'(x)$. For the second term we use

$$\eta\big(f(x+h)\big) \to 0 \quad \text{and} \quad \frac{f(x+h) - f(x)}{h} \to f'(x),$$

so their product tends to 0. Therefore,

$$\lim_{h \to 0} \frac{(g \circ f)(x+h) - (g \circ f)(x)}{h} = g'(f(x))f'(x),$$

and hence $g \circ f$ is differentiable at x with $(g \circ f)'(x) = g'(f(x))f'(x)$.

□

Example 3.1.17. *Let us compute the derivative of a polynomial function in three different ways.*

1. *directly:* $f(x) = x^2 + 5x^4$

$$f'(x) = 2x + 20x^3.$$

2. *product rule:* $f(x) = x^2(1 + 5x^2)$

$$f'(x) = 2x(1 + 5x^2) + x^2 10x = 2x + 10x^3 + 10x^3 = 2x + 20x^3.$$

3. *chain rule:* $f(x) = g(x^2)$, *where* $g(x) = x + 5x^2$ *and* $g'(x) = 1 + 10x$

$$f'(x) = g'(x^2)2x = (1 + 10x^2)2x = 2x + 20x^3.$$

Example 3.1.18. *Let* $f(x) = x^2$ *and* $g(y) = \sin y$*. Then*

$$(g \circ f)(x) = \sin(x^2),$$

and by the chain rule,

$$(g \circ f)'(x) = \cos(x^2) \cdot 2x.$$

Example 3.1.19. *Let $f(x) = \exp(x)$ and $g(y) = y^3$. Then*

$$(g \circ f)(x) = \exp(x)^3 = e^{3x},$$

and

$$(g \circ f)'(x) = 3\exp(x)^2 \exp(x) = 3e^{3x}.$$

Remark 3.1.20 (compositional structure)**.** *Differentiation rules explain how small changes behave when functions are added, multiplied, or composed.*
In data models built from multiple components, they allow us to understand how changes in the input affect the output values.

We now derive a general formula for the derivative of an inverse function and apply it to important examples.

Theorem 3.1.21 (Derivative of the inverse function)**.** *Let $f : [a, b] \to [A, B]$ be continuous and bijective. Moreover, assume that f is differentiable at $x_0 \in (a, b)$ with*

$$f'(x_0) \neq 0.$$

Then the inverse function $f^{-1} : [A, B] \to [a, b]$ is differentiable at $y_0 := f(x_0)$, and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} = \frac{1}{f'(f^{-1}(y_0))}.$$

Direct proof. Let $y = f(x)$ and $x = f^{-1}(y)$. For $y \neq y_0$ we must have $f(x) \neq f(x_0)$, and we write

$$\begin{aligned}\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} &= \frac{x - x_0}{f(x) - f(x_0)} \\ &= \left(\frac{f(x) - f(x_0)}{x - x_0}\right)^{-1}.\end{aligned}$$

Since f is continuous and bijective, its inverse is also continuous by Proposition 2.3.1. Hence, $x \to x_0$ is equivalent to $y \to y_0$. The differentiability of f and taking limits lead to

$$\lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{1}{f'(x_0)}.$$

Therefore f^{-1} is differentiable at y_0 with the stated derivative. □

Proof via chain rule. Suppose we know that f^{-1} is differentiable in y_0. Since $(f^{-1} \circ f)(x) = x$, the derivative at x_0 is $(f^{-1} \circ f)'(x_0) = 1$, and the chain rule leads to

$$1 = (f^{-1} \circ f)'(x_0) = (f^{-1})'(f(x_0))f'(x_0).$$

Division by $f'(x_0)$ yields the claim.

□

Recall that the exponential function $\exp : \mathbb{R} \to (0, \infty)$ is continuous, strictly increasing, and differentiable with $\exp'(x) = \exp(x)$. Its inverse function is the logarithm $\log : (0, \infty) \to \mathbb{R}$.

Corollary 3.1.22 (*log* is differentiable)**.** *For all $y > 0$, the logarithm is differentiable and satisfies*

$$\log'(y) = \frac{1}{y}.$$

Proof. Let $y_0 > 0$ and set $x_0 := \log(y_0)$, so that $\exp(x_0) = y_0$. Since $\exp'(x_0) = \exp(x_0) = y_0 \neq 0$, Theorem 3.1.21 applies and yields

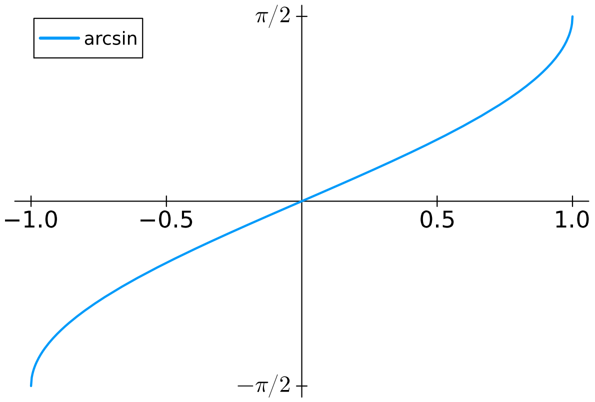
$$\log'(y_0) = \frac{1}{\exp'(x_0)} = \frac{1}{y_0}.$$

□

Remark 3.1.23 (inverse sensitivity)**.** *The derivative of an inverse function describes how small changes in the output affect the input. This will help us later understand how errors in data propagate through a model.*

Recall that $\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \to [-1, 1]$ is continuous, strictly increasing, and differentiable with $\sin'(x) = \cos x$. Its inverse function is $\arcsin : [-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$.

```
plot(asin,label="arcsin")
```



Corollary 3.1.24 (arcsin is differentiable)**.** *For all $y \in (-1, 1)$,*

$$\arcsin'(y) = \frac{1}{\sqrt{1 - y^2}}.$$

Proof. Let $y \in (-1, 1)$ and set $x := \arcsin(y)$, so that $\sin x = y$ and $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Since $\cos x > 0$ on this interval, we have $\sin'(x) = \cos x \neq 0$. By Theorem 3.1.21,

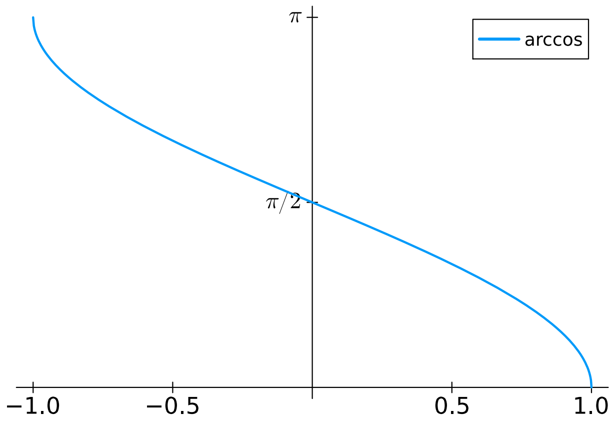
$$\arcsin'(y) = \frac{1}{\cos x}.$$

Using $\sin x = y$, $\cos^2 x = 1 - \sin^2 x$, and $\cos(x) \geq 0$, we obtain $\cos x = \sqrt{1 - y^2}$, and hence

$\arcsin'(y) = \frac{1}{\sqrt{1-y^2}}$. □

Recall that $\cos : [0, \pi] \to [-1, 1]$ is continuous, strictly decreasing, and differentiable with $\cos'(x) = -\sin x$. Its inverse function is $\arccos : [-1, 1] \to [0, \pi]$.

```
plot(acos,label="arccos")
```



Corollary 3.1.25 (arccos is differentiable). *For all $y \in (-1, 1)$,*

$$\arccos'(y) = -\frac{1}{\sqrt{1-y^2}}.$$

Proof. Let $y \in (-1, 1)$ and set $x := \arccos(y)$, so that $\cos x = y$ and $x \in (0, \pi)$. Then $\sin x > 0$ and $\cos'(x) = -\sin x \neq 0$. By Theorem 3.1.21,

$$\arccos'(y) = \frac{1}{-\sin x}.$$

Since $\sin^2 x = 1 - \cos^2 x = 1 - y^2$ and the sine is positive in $(0, \pi)$, we have $\sin x = \sqrt{1 - y^2}$, and thus $\arccos'(y) = -\frac{1}{\sqrt{1-y^2}}$. □

We now compute the derivatives of the functions $x \mapsto a^x$ and $x \mapsto x^a$. Recall that for $a > 0$ and $x \in \mathbb{R}$ we defined $a^x := \exp(x \log a)$, so that the chain rule is available.

Lemma 3.1.26 (a^x is differentiable). *Let $a > 0$. The function $f : \mathbb{R} \to (0, \infty)$ defined by $f(x) = a^x$ is differentiable on $\mathbb{R}$, and*

$$f'(x) = a^x \log a.$$

Proof. Using the definition and the chain rule with $\exp' = \exp$, we compute

$$\begin{aligned} f'(x) &= \exp(x \log a) \cdot (\log a) \\ &= a^x \log a. \end{aligned}$$

□

Example 3.1.27. *For $a = e$, we have $\log e = 1$, so that we recover $\exp'(x) = e^x$.*

For general but fixed $a \in \mathbb{R}$, the function $x \mapsto x^a$ is only defined for $x > 0$, and we therefore restrict attention to the interval $(0, \infty)$. Recall that for $x > 0$, $x^a := \exp(a \log x)$.

Lemma 3.1.28 (x^a is differentiable)**.** *Let $a \in \mathbb{R}$. The function $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) = x^a$ is differentiable on $(0, \infty)$, and*

$$f'(x) = ax^{a-1}.$$

Proof. Using the definition and the chain rule, we compute

$$\begin{aligned} f'(x) &= \exp(a \log x)\, \frac{a}{x} \\ &= x^a\, \frac{a}{x} \\ &= a\, x^{a-1}. \end{aligned}$$

□

Remark 3.1.29. *For $f(x) = x^a$, the formula $f'(x) = ax^{a-1}$ is valid for all real exponents a, provided $x > 0$. For integer exponents, it coincides with the classical power rule.*

Example 3.1.30. *For $a = \frac{1}{2}$, the derivative of the square root $f(x) = \sqrt{x}$ is*

$$f'(x) = \frac{1}{2\sqrt{x}}, \qquad x > 0.$$

Example 3.1.31 (quotient rule)**.** *The derivative of $u(x) = \frac{1}{x}$ for $x \neq 0$ is $u'(x) = -\frac{1}{x^2}$. Suppose that $G(x) = \frac{1}{g(x)}$ is well-defined and g is differentiable. Then the chain rule yields*

$$G'(x) = -\frac{1}{g^2(x)} g'(x) = -\frac{g'(x)}{g^2(x)}.$$

If $F(x) = \frac{f(x)}{g(x)}$ is well-defined and both f and g are differentiable, then the product rule yields

$$F'(x) = \frac{f'(x)}{g(x)} - f(x)\frac{g'(x)}{g^2(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}.$$

Essentials

The derivative is defined by the limit of a difference quotient. It encodes local linear approximation. Differentiability of the elementary functions exp, sin, and cos is established, including the principle that differentiability implies continuity. The main tools of calculus are the product, chain, and quotient rules, and guides to the differentiation of the compositions and inverse functions. These rules allow efficient differentiation of complicated expressions and justify formulas for derivatives of logarithms and inverse trigonometric functions.

A The first derivative gives a local linear approximation.

differentiable, difference quotient, linear approximation, product rule, chain rule, inverse function, derivatives of log, arcsin, arccos

3.2 Local extrema

Local extrema describe where a function cannot be improved by small perturbations. They can be characterized by conditions on first and second order derivatives. These local conditions form the analytical foundation of optimization and underlie numerical and learning algorithms that search for optimal states.

Let $U \subseteq \mathbb{R}$.

Definition 3.2.1 (local maximum and minimum)**.** *Let $U \subseteq \mathbb{R}$ be an open interval, $f : U \to \mathbb{R}$, and let $x_0 \in U$.*

(a) f has a local maximum *at x_0 if there exists $\delta > 0$ such that*

$$f(x) \leq f(x_0) \quad \textit{for all } x \in (x_0 - \delta, x_0 + \delta).$$

A strict *local maximum refers to a strict inequality when $x \neq x_0$.*

(b) f has a local minimum *at x_0 if there exists $\delta > 0$ such that*

$$f(x) \geq f(x_0) \quad \textit{for all } x \in (x_0 - \delta, x_0 + \delta).$$

A strict *local minimum refers to a strict inequality when $x \neq x_0$.*

If x_0 is either a (strict) local minimum or a (strict) local maximum, it is called a (strict) local extremum*.*

We now show that differentiability severely restricts where local extrema can occur.

Theorem 3.2.2 (local extrema)**.** *Let $f : (a, b) \to \mathbb{R}$ be differentiable at $x_0 \in (a, b)$. If f has a local extremum at x_0, then*

$$f'(x_0) = 0.$$

Proof. Assume that x_0 is a local maximum (the minimum case is analogous). Then there exists $\delta > 0$ such that

$$f(x) \leq f(x_0) \quad \text{for } |x - x_0| < \delta.$$

For $h > 0$ sufficiently small,

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0,$$

while for $h < 0$ sufficiently small,

$$\frac{f(x_0 + h) - f(x_0)}{h} \geq 0.$$

Since f is differentiable at x_0, both one-sided limits must coincide, hence

$$f'(x_0) = 0.$$

□

Example 3.2.3. *The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = (x-1)^2$ has a local minimum at $x_0 = 1$. Its first derivative is $f'(x) = 2(x-1)$, so that $f'(1) = 0$.*

The vanishing of the derivative is a necessary but not sufficient condition for a local extremum.

Remark 3.2.4. *The converse is false: $f'(x_0) = 0$ does not imply a local extremum. For example, $f(x) = x^3$ satisfies $f'(0) = 0$ but has no extremum at 0.*

A local extremum describes behavior in a neighborhood of a point and should be distinguished from global extrema, which depend on the entire domain.

Definition 3.2.5 (global extremum)**.** *A function $f : U \to \mathbb{R}$ has a* global maximum *at $x_0 \in U$ if*

$$f(x) \leq f(x_0) \quad \text{for all } x \in U.$$

Global minima are defined analogously.

Corollary 3.2.6 (global extrema)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous and differentiable on (a,b), then every global extremum of f occurs either at a boundary point or at a point $x_0 \in (a,b)$ with $f'(x_0) = 0$.*

Proof. By compactness of $[a,b]$, the image $f([a,b])$ is compact in $\mathbb{R}$. Hence it is closed and bounded, and therefore contains its supremum and infimum. If an extremum occurs in (a,b), then Theorem 3.2.2 implies the existence of $x_0 \in (a,b)$ such that $f'(x_0) = 0$. □

Example 3.2.7. *The function $f : [0,2] \to \mathbb{R}$, $f(x) = x$ has its global minimum and maximum at the boundary.*

Example 3.2.8. *The function $f : [-1,1] \to \mathbb{R}$, $f(x) = x - x^3$ satisfies $f(-1) = f(1) = 0$ and has its global minimum and maximum within $(-1,1)$.*

Theorem 3.2.9 (Theorem of Rolle)**.** *Assume that $f : [a,b] \to \mathbb{R}$ is continuous and $f(a) = f(b)$. If f is differentiable on (a,b), then there is $\xi \in (a,b)$ such that $f'(\xi) = 0$.*

Proof. The claim is true for constant f, so we may now assume that f is not constant. Hence, there is $x_0 \in (a,b)$ such that $f(x_0) > f(a)$ or $f(x_0) < f(a)$. In the first case, the maximum of f is attained in (a,b). The second case corresponds to the minimum. In each case, Theorem 3.2.2 leads to $\xi \in (a,b)$ such that $f'(\xi) = 0$. □

Theorem 3.2.10 (mean value theorem of differentiation)**.** *If $f : [a,b] \to \mathbb{R}$ is continuous and differentiable on (a,b), then there is $\xi \in (a,b)$ such that*

$$\frac{f(b) - f(a)}{b - a} = f'(\xi).$$

Proof. Let us define $F : [a,b] \to \mathbb{R}$ by

$$F(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a),$$

so that its derivative satisfies

$$F'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Moreover, $F(a) = f(a) = F(b)$, and the Theorem of Rolle implies that there is $\xi \in (a,b)$ such that $F'(\xi) = 0$. □

The mean value theorem of differentiation leads to a sufficient condition for being contractive and hence for Banach's Fixed Point Theorem.

Corollary 3.2.11. *Let $f : [a,b] \to \mathbb{R}$ be continuously differentiable and suppose that there is $x \in (a,b)$ such that $f(x) = x$. If*

$$|f'(x)| < 1,$$

then there is $\varepsilon > 0$ such that $f : B_\varepsilon(x) \to B_\varepsilon(x)$ and f is contractive on $B_\varepsilon(x)$.

Proof. Let $\delta := 1 - |f'(x)|$. Continuity of f' implies that there is $\varepsilon > 0$ such that

$$|f'(y)| \le 1 - \delta/2, \qquad \forall\, y \in B_\varepsilon(x).$$

The mean value theorem of differentiation yields

$$\frac{|f(y) - f(z)|}{|y - z|} \le (1 - \delta/2), \qquad \forall\, y, z \in B_\varepsilon(x).$$

Hence, f is contractive on $B_\varepsilon(x)$. To check that $f(B_\varepsilon(x)) \subseteq B_\varepsilon(x)$, we derive, for $y \in B_\varepsilon(x)$,

$$|f(y) - x| = |f(y) - f(x)| \le (1 - \delta/2)|y - x|.$$

Thus, $f : B_\varepsilon(x) \to B_\varepsilon(x)$. □

Banach's Fixed Point Theorem 1.4.17 then ensures that the iteration

$$x_{n+1} := f(x_n)$$

converges linearly towards x for all initial points $x_0 \in B_\varepsilon(x)$.

Lemma 3.2.12 (monotonicity)**.** *Let $f : (a, b) \to \mathbb{R}$ be differentiable.*

(a) If $f' \geq 0$ on (a, b), then f is increasing.

(b) If $f' > 0$ on (a, b), then f is strictly increasing.

The analogous statements hold for $f' < 0$ or $f' \leq 0$ and f (strictly) decreasing.

Proof. For $x, y \in (a, b)$, such that $x < y$, the mean value theorem of differentiation yields that there is $\xi \in (x, y)$ such that

$$\frac{f(y) - f(x)}{y - x} = f'(\xi).$$

The conditions on f' imply the respective claims. □

So far, finding a zero of the derivative is only a necessary condition for a local extremum. Next, we obtain sufficient conditions that also tell us if it is a local maximum or minimum.

Lemma 3.2.13 (first derivative test)**.** *Let $f : (a, b) \to \mathbb{R}$ be differentiable and let $x_0 \in (a, b)$ with $f'(x_0) = 0$.*

(a) If f' is increasing in a neighborhood of x_0, then f has a local minimum at x_0.

(b) If f' is strictly increasing in a neighborhood of x_0, then f has a strict local minimum at x_0.

The analogous statements hold for f' (strictly) decreasing and (strict) local maxima.

Proof. We only prove Part (b). The other parts are verified analogously.
(b) Let f' be strictly increasing in a neighborhood of x_0. Since $f'(x_0) = 0$, f' is negative to the left of x_0 and positive to the right of x_0. Lemma 3.2.12 implies that f is strictly decreasing to the left of x_0 and strictly increasing to the right of x_0. Hence, there is a strict local minimum in x_0. □

Example 3.2.14. *The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^4$ satisfies $f'(x) = 4x^3$, so that $f'(0) = 0$ and we set $x_0 = 0$. The first derivative is negative to the left of x_0 and positive to the right of x_0. Hence, x_0 is a strict local minimum.*

If the derivative f' of f is again differentiable, then we call f twice differentiable and can compute the derivative of the derivative $f'' := (f')'$, called the second derivative of f. It provides a sufficient condition for a local extrema.

Theorem 3.2.15 (second derivative test)**.** *Let $f : (a, b) \to \mathbb{R}$ be twice continuously differentiable and let $x_0 \in (a, b)$ satisfy $f'(x_0) = 0$.*

(a) If $f''(x_0) > 0$, then f has a strict local minimum at x_0.

(b) If $f''(x_0) < 0$, then f has a strict local maximum at x_0.

Proof. Assume $f''(x_0) > 0$. By continuity of f'', there exists $\delta > 0$ such that

$$f''(x) > 0 \quad \text{for } |x - x_0| < \delta.$$

Hence f' is strictly increasing on $(x_0 - \delta, x_0 + \delta)$. Since $f'(x_0) = 0$, we obtain

$$f'(x) < 0 \text{ for } x < x_0, \qquad f'(x) > 0 \text{ for } x > x_0,$$

and the claim follows from the first derivative test. The maximum case is analogous. □

Example 3.2.16. *The function $f : \mathbb{R} \to \mathbb{R}$,*

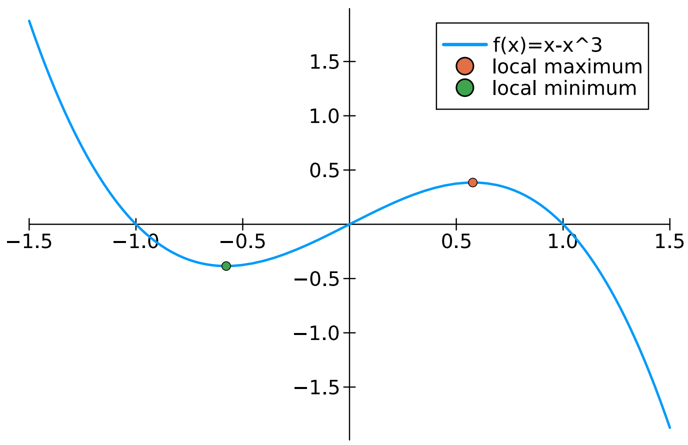
$$f(x) = x - x^3$$

satisfies $f'(x) = 1 - 3x^2$. The condition $1 - 3x^2 = 0$ yields $x^2 = 1/3$ and hence $x_1 = 1/\sqrt{3}$ and $x_2 = -1/\sqrt{3}$ are zeros of f'. The second derivative is

$$f''(x) = -6x,$$

so that $f''(1/\sqrt{3}) < 0$ and $f''(-1/\sqrt{3}) > 0$. Thus, f has a strict local maximum in $1/\sqrt{3}$ and a strict local minimum in $-1/\sqrt{3}$.

```
f(x) = x-x^3
plot(f,xlimits=[-3/2,3/2],label="f(x)=x-x^3")
x1 = 1/sqrt(3)
x2 = -1/sqrt(3)
scatter!([x1],[f(x1)],label="local maximum")
scatter!([x2],[f(x2)],label="local minimum")
```



If $f''(x_0) = 0$, the test is inconclusive. Then we either need to involve higher-order derivatives or we check f'' in a neighborhood of x_0.

Theorem 3.2.17 (second derivative test)**.** *Let $f : (a,b) \to \mathbb{R}$ be twice continuously differentiable and let $x_0 \in (a,b)$ satisfy $f'(x_0) = 0$.*

(a) If $f'' \geq 0$ in a neighborhood of x_0, then f has a local minimum at x_0.

(b) If $f'' \leq 0$ in a neighborhood of x_0, then f has a local maximum at x_0.

Proof. This is verified analogous to the proof of Theorem 3.2.15 by relying on the Lemmas 3.2.13 and 3.2.12. □

Remark 3.2.18 (local optimization)**.** *Derivatives characterize local extrema via linear approximation. In data-related optimization problems, this explains why stationary points ($f'(\xi) = 0$) are natural candidates for optimal solutions, even when global optimality is not guaranteed.*

Convex functions

In general, a local minimum of a function need not be global. Convexity is a structural condition that excludes this behavior: for convex functions, every local minimum is global. Convex functions therefore play a central role in data analysis and optimization.

Definition 3.2.19 (convex function)**.** *A function $f : \mathbb{R} \to \mathbb{R}$ is called* convex *if for all $x, y \in \mathbb{R}$ and all $t \in [0,1]$,*

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

Lemma 3.2.20 (Convexity means increasing derivative)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Then f is convex if and only if f' is increasing.*

Proof. ($\Rightarrow$) Fix $x < u < y$. Set $t := \frac{u-x}{y-x} \in (0,1)$, so $u = ty + (1-t)x$. By convexity,

$$f(u) \leq tf(y) + (1-t)f(x).$$

According to $(1-t)f(x) = f(x) - tf(x)$, we bring $f(x)$ to the left-hand-side and use $t = \frac{u-x}{y-x}$ to obtain

$$\frac{f(u) - f(x)}{u - x} \leq \frac{f(y) - f(x)}{y - x}.$$

Analogously, we can derive

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(y) - f(u)}{y - u}.$$

Hence, for fixed x and y, the functions

$$\phi_x(u) := \frac{f(u) - f(x)}{u - x}, \qquad \phi_y(u) := \frac{f(y) - f(u)}{y - u}, \qquad x < u < y,$$

satisfy the inequalities and identity

$$\phi_x(u) \leq \phi_x(y) = \phi_y(x) \leq \phi_y(u), \qquad x < u < y.$$

Taking limits yields

$$f'(x) = \lim_{u \to x} \phi_x(u) \le \phi_x(y) = \phi_y(x) \le \lim_{u \to y} \phi_y(u) = f'(y).$$

Hence f' is increasing.
($\Leftarrow$) For $x < y$ and $u = tx + (1-t)y$, $t \in (0,1)$, the mean value theorem yields that there exists $\xi_1 \in (x,u)$ and $\xi_2 \in (u,y)$ such that

$$\frac{f(u) - f(x)}{u - x} = f'(\xi_1) \le f'(\xi_2) = \frac{f(y) - f(u)}{y - u}.$$

The relations $u - x = (t-1)x + (1-t)y = (1-t)(y-x)$ and $y - u = t(y-x)$ lead to

$$\frac{f(u) - f(x)}{(1-t)} \le \frac{f(y) - f(u)}{t}.$$

This implies

$$f(u) \le tf(x) + (1-t)f(y),$$

which is the definition of convexity. $\square$

Theorem 3.2.21 (convexity means positive second derivative)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable. Then f is convex if and only if*

$$f''(t) \ge 0 \qquad \textit{for all } t \in \mathbb{R}\,.$$

Proof. Since f' is continuously differentiable, f' is increasing if and only if $f'' \ge 0$. $\square$

Theorem 3.2.22. *If $f : \mathbb{R} \to \mathbb{R}$ is convex and twice continuously differentiable, then x_0 is a global minimizer if and only if $f'(x_0) = 0$.*

Many loss functions in data science are convex, which ensures that optimization algorithms solve problems with globally optimal solutions.

Proof. If x_0 is a global minimum, then it is also a local minimum and hence $f'(x_0) = 0$ by Theorem 3.2.2.
Now assume that $f'(x_0) = 0$. Since f is convex, its second derivative satisfies $f'' \ge 0$. Hence, f' is increasing. By Lemma 3.2.13, x_0 is a local minimum. Since f' is increasing, there are no other local minima, hence, x_0 is a global minimizer. $\square$

Example 3.2.23. *The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^4$ satisfies $f'(x) = 4x^3$ and $f''(x) = 12x^2$, so that $f'' \ge 0$ and therefore f is convex. Thus, $x_0 = 0$ is a global minimum.*

Essentials

Differentiability imposes strong restrictions on the local behavior of functions. In particular, local extrema can only occur at points where the derivative vanishes. This provides a fundamental tool for analyzing local optimality. The first and second derivatives combined lead to a sufficient condition of local maxima and minima. Every local minimum of a convex function is a global one.

local maximum, local minimum, local extremum, theorem of Rolle, mean value theorem, convex function

3.3 Integration

The idea of integration is to approximate the area under the curve by summing area of rectangles on small subintervals.

Definition 3.3.1 (Partition)**.** *A* partition *of* $[a, b]$ *is a finite set*

$$\mathcal{P} = \{x_0, x_1, \ldots, x_n\} \quad \textit{with} \quad a = x_0 < x_1 < \cdots < x_n = b.$$

Recall that $f : [a, b] \to \mathbb{R}$ is called bounded if there is $C > 0$ such that $|f(x)| \leq C$ for all $x \in [a, b]$.

Definition 3.3.2 (upper and lower Darboux sums)**.** *Let* $f : [a, b] \to \mathbb{R}$ *be bounded and let* $\mathcal{P} = \{x_0, \ldots, x_n\}$ *be a partition. For* $k = 1, \ldots, n$ *define*

$$m_k := \inf_{x \in [x_{k-1}, x_k]} f(x), \qquad M_k := \sup_{x \in [x_{k-1}, x_k]} f(x).$$

The lower *and* upper Darboux sums *are*

$$L(f, \mathcal{P}) := \sum_{k=1}^{n} m_k (x_k - x_{k-1}), \qquad U(f, \mathcal{P}) := \sum_{k=1}^{n} M_k (x_k - x_{k-1}).$$

The lower Darboux sum accumulates the area of rectangles of side lengths m_k and $x_k - x_{k-1}$. The rectangles of the upper Darboux sum have side lengths M_k and $x_k - x_{k-1}$.

Example 3.3.3. *Consider* $f : [0, 1] \to \mathbb{R}$ *given by* $f(x) = 2$. *For every partition* $\mathcal{P} = \{x_0, \ldots, x_n\}$ *of the interval* $[0, 1]$, *we observe*

$$L(f, \mathcal{P}) = \sum_{k=1}^{n} m_k (x_k - x_{k-1}) = 2 \sum_{k=1}^{n} (x_k - x_{k-1}) = 2(x_n - x_0) = 2$$
$$U(f, \mathcal{P}) = \sum_{k=1}^{n} M_k (x_k - x_{k-1}) = 2 \sum_{k=1}^{n} (x_k - x_{k-1}) = 2(x_n - x_0) = 2.$$

Thus, the lower and upper Darboux sums coincide for every partition in this case.

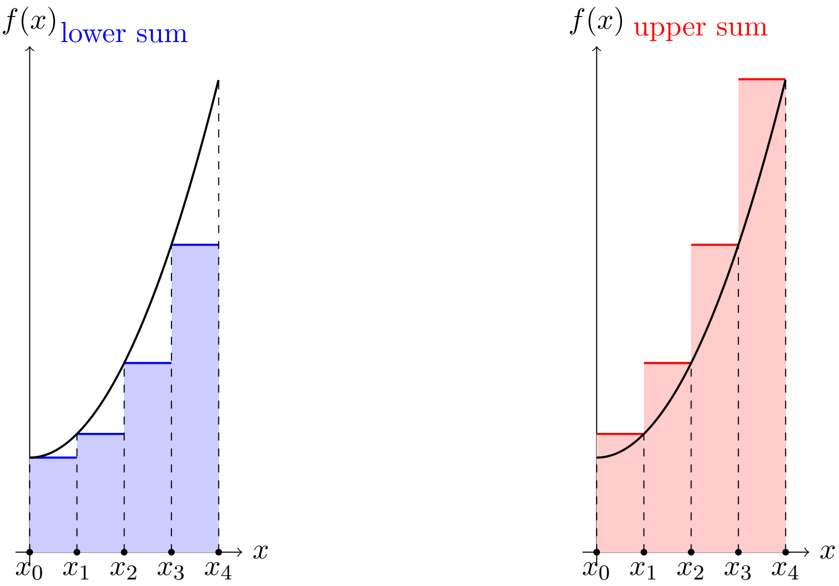


Figure 3.1: Lower and upper Darboux sums.

We say that a partition $\mathcal{P}'$ is a refinement of $\mathcal{P}$ if $\mathcal{P} \subseteq \mathcal{P}'$.

Lemma 3.3.4 (refinement 1)**.** *If $\mathcal{P}'$ is a refinement of $\mathcal{P}$, then*

$$L(f, \mathcal{P}) \le L(f, \mathcal{P}'), \qquad U(f, \mathcal{P}') \le U(f, \mathcal{P}).$$

In particular, $L(f, \mathcal{P}) \le U(f, \mathcal{P})$.

Proof. It suffices to consider the case where $\mathcal{P}'$ is obtained from $\mathcal{P}$ by inserting a single point $y \in (x_{j-1}, x_j)$. On $[x_{j-1}, x_j]$ we split into $[x_{j-1}, y]$ and $[y, x_j]$. Since both subintervals are contained in $[x_{j-1}, x_j]$, we have

$$\inf_{[x_{j-1},y]} f \ge \inf_{[x_{j-1},x_j]} f, \qquad \inf_{[y,x_j]} f \ge \inf_{[x_{j-1},x_j]} f,$$

and therefore the contribution to the lower sum does not decrease. The argument for upper sums follows analogously using suprema. Repeated insertion of points yields the general case. ☐

Lemma 3.3.5 (refinement 2)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and $\mathcal{P}$ and $\mathcal{Q}$ are partitions of $[a, b]$, then*

$$L(f, \mathcal{P}) \le U(f, \mathcal{Q}).$$

Proof. Let $\mathcal{R}$ be a common refinement, i.e., $\mathcal{P}, \mathcal{Q} \subseteq \mathcal{R}$. By the first refinement lemma,

$$L(f, \mathcal{P}) \le L(f, \mathcal{R}) \le U(f, \mathcal{R}) \le U(f, \mathcal{Q}).$$

☐

Definition 3.3.6 (© Upper and lower integral). *Let $f : [a,b] \to \mathbb{R}$ be bounded. Define*

$$\underline{\int_a^b} f(x)\,dx := \sup_{\mathcal{P}} L(f,\mathcal{P}), \qquad \overline{\int_a^b} f(x)\,dx := \inf_{\mathcal{P}} U(f,\mathcal{P}),$$

where the supremum and infimum are taken over all partitions of $[a,b]$.

Lemma 3.3.7. *For every bounded function $f : [a,b] \to \mathbb{R}$,*

$$\underline{\int_a^b} f(x)\,dx \le \overline{\int_a^b} f(x)\,dx.$$

Proof. The inequalities of the second refinement lemma are preserved by taking the supremum over $\mathcal{P}$ and the infimum over $\mathcal{Q}$. □

We have successfully approximated the area under the curve if the lower and the upper integral coincide.

Definition 3.3.8 (© Integrability). *A bounded function $f : [a,b] \to \mathbb{R}$ is called* (Riemann-Darboux) integrable *if*

$$\underline{\int_a^b} f(x)\,dx = \overline{\int_a^b} f(x)\,dx.$$

The common value is denoted by $\int_a^b f(x)\,dx$.

Remark 3.3.9. *Darboux sums formalize the idea of approximating an integral by discretization. In data analysis, integrals are typically approximated by finite sums over bins, grids, or samples. Refining a partition corresponds to increasing resolution or sample size, and integrability guarantees that these approximations converge as the discretization becomes finer.*

Example 3.3.10. *Consider $f : [0,1] \to \mathbb{R}$ given by $f(x) = 2x$ and consider the equidistant partition $\mathcal{P}_n = \{x_0, \dots, x_n\}$ of the interval $[0,1]$, i.e.,*

$$x_k = \frac{k}{n}, \qquad k = 0, \dots, n.$$

The lower Darboux sum is

$$\begin{aligned} L(f,\mathcal{P}_n) &= 2\sum_{k=1}^{n} x_{k-1}(x_k - x_{k-1}) = 2\sum_{k=1}^{n} \frac{k-1}{n}\left(\frac{k}{n} - \frac{k-1}{n}\right) \\ &= \frac{2}{n^2}\sum_{k=1}^{n}(k-1) = \frac{2}{n^2}\sum_{k=0}^{n-1} k \\ &= \frac{2}{n^2}\frac{n(n-1)}{2} = \frac{n-1}{n}. \end{aligned}$$

The upper Darboux sum is

$$U(f,\mathcal{P}_n) = 2\sum_{k=1}^{n} x_k(x_k - x_{k-1}) = 2\sum_{k=1}^{n} \frac{k}{n}\left(\frac{k}{n} - \frac{k-1}{n}\right)$$
$$= \frac{2}{n^2}\frac{n(n+1)}{2} = \frac{n+1}{n}.$$

Thus, $L(f,\mathcal{P}_n) \to 1$ *and* $U(f,\mathcal{P}_n) \to 1$ *and therefore* $f(x) = 2x$ *is integrable with* $\int_0^1 2x\,dx = 1$.

Lemma 3.3.11 (Integrability criterion)**.** *A bounded function* $f : [a,b] \to \mathbb{R}$ *is integrable if and only if*

$$\forall \varepsilon > 0\ \exists \mathcal{P} : \quad U(f,\mathcal{P}) - L(f,\mathcal{P}) \le \varepsilon.$$

Proof. ($\Rightarrow$) Let $I := \int_a^b f(x)\,dx$. Choose partitions $\mathcal{P}_1, \mathcal{P}_2$ such that

$$U(f,\mathcal{P}_1) \le I + \varepsilon/2, \qquad L(f,\mathcal{P}_2) \ge I - \varepsilon/2.$$

Let $\mathcal{P}$ be a common refinement. Then

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) \le (I + \varepsilon/2) - (I - \varepsilon/2) = \varepsilon.$$

($\Leftarrow$) Assume the condition holds. Then for every $\varepsilon > 0$,

$$0 \le \overline{\int_a^b} f(x)\,dx - \underline{\int_a^b} f(x)\,dx \le U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon.$$

Hence upper and lower integrals coincide. $\square$

Theorem 3.3.12. *Every continuous function* $f : [a,b] \to \mathbb{R}$ *is integrable.*

Proof. Since $[a,b]$ is compact, f is uniformly continuous. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$|x - y| \le \delta \quad \Rightarrow \quad |f(x) - f(y)| \le \frac{\varepsilon}{b-a}.$$

Let $\mathcal{P} = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$ such that $x_k - x_{k-1} \le \delta$. Under the notation of Definition 3.3.2, on each subinterval $[x_{k-1}, x_k]$ we have

$$M_k - m_k \le \frac{\varepsilon}{b-a}.$$

Hence, the upper and lower Darboux sums satisfy

$$\begin{aligned} U(f,\mathcal{P}) - L(f,\mathcal{P}) &\leq \sum_{k=1}^{n} \frac{\varepsilon}{b-a}(x_k - x_{k-1}) \\ &= \frac{\varepsilon}{b-a}\sum_{k=1}^{n}(x_k - x_{k-1}) \\ &= \frac{\varepsilon}{b-a}(x_n - x_0) = \varepsilon, \end{aligned}$$

where $\sum_{k=1}^{n}(x_k - x_{k-1}) = x_n - x_0 = b - a$ is a telescope sum. □

Lemma 3.3.13 (linearity)**.** *Suppose that $f, g : [a,b] \to \mathbb{R}$ are integrable.*

(i) (linear) For $\alpha, \beta \in \mathbb{R}$, the function $\alpha f + \beta g$ is integrable and

$$\int_a^b (\alpha f(x) + \beta g(x))\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

(ii) (monotone) If $f(x) \leq g(x)$ for all $x \in [a,b]$, then

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

(iii) (additive) For $q \in (a,b)$, f is also integrable on $[a,q]$ and $[q,b]$ and we have

$$\int_a^b f(x)\,dx = \int_a^q f(x)\,dx + \int_q^b f(x)\,dx.$$

Proof. We omit the direct proof of (i) and (iii).
(ii) For every partition $\mathcal{P}$,

$$L(f,\mathcal{P}) \leq L(g,\mathcal{P}), \qquad U(f,\mathcal{P}) \leq U(g,\mathcal{P}).$$

Passing to supremum and infimum proves the statement. □

Definition 3.3.14 (degenerate integration domains)**.** *Consistent with the trivial partition of a single point, we define*

$$\int_a^a f(x)\,dx := 0,$$

and, for $a \leq b$, we put

$$\int_b^a f(x)\,dx := -\int_a^b f(x)\,dx.$$

We now relate integration to differentiation that leads to methods for computing integrals.

Theorem 3.3.15 (mean value theorem of integration)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then there exists $\xi \in [a, b]$ such that*

$$\frac{1}{b-a}\int_a^b f(x)\,dx = f(\xi).$$

Moreover, if $\phi : [a, b] \to [0, \infty)$ is integrable, then there exists $\xi \in [a, b]$ such that

$$\int_a^b f(x)\phi(x)\,dx = f(\xi)\int_a^b \phi(x)\,dx.$$

Proof. Since f is continuous, it attains its minimum and maximum and the intermediate value theorem leads to $f([a, b]) = [m, M]$, see Theorem 1.4.14. The monotonicity of the integral yields

$$m\int_a^b \phi(x)\,dx = \int_a^b m\phi(x)\,dx \le \int_a^b f(x)\phi(x)\,dx \le \int_a^b M\phi(x)dx = M\int_a^b \phi(x)\,dx,$$

so that

$$\int_a^b f(x)\phi(x)\,dx \in [m, M]\int_a^b \phi(x)\,dx = f([a, b])\int_a^b \phi(x)\,dx.$$

This means a pre-image $\xi \in [a, b]$ exists.
The first identity in the theorem follows from the choice $\phi \equiv 1$. □

Lemma 3.3.16. *Let $f : [a, b] \to \mathbb{R}$ be continuous and define, for $x \in [a, b]$,*

$$F(x) := \int_a^x f(t)\,dt.$$

Then F is differentiable and

$$F'(x) = f(x), \qquad x \in [a, b].$$

This lemma is the fundamental link between integration and differentiation.

Proof. For $h \neq 0$ sufficiently small, so that $x + h \in [a, b]$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt.$$

The mean value theorem of integration implies that there exists $\xi_h \in [x, x + h]$ such that

$$\frac{1}{h}\int_x^{x+h} f(t)\,dt = f(\xi_h).$$

For $h \to 0$, we have $\xi_h \to x$, so that continuity of f yields

$$F'(x) = \lim_{h\to 0}\frac{F(x+h) - F(x)}{h} = \lim_{\xi_h \to x} f(\xi_h) = f(x).$$

□

We now formulate the Fundamental Theorem of Calculus (FTC).

Theorem 3.3.17 (FTC)**.** *If $F : [a, b] \to \mathbb{R}$ is differentiable with continuous derivative F', then*

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

If the primitive of the integrand is known, the FTC enables us to determine the value of the integral.

Proof. The function $G : [a, b] \to \mathbb{R}$,

$$G(x) := \int_a^x F'(t)\,dt$$

satisfies $G(b) = \int_a^x F'(t)\,dt$ and $G(a) = 0$. Lemma 3.3.16 leads to $G'(x) = F'(x)$. Hence $F - G$ is constant on $[a, b]$, say $F(x) = G(x) + c$ for all $x \in [a, b]$. Evaluating at a and b yields

$$F(b) - F(a) = G(b) + c - (G(a) + c) = G(b) = \int_a^b F'(t)\,dt.$$

□

Often the primitive of the integrand is not obvious and the FTC cannot be applied directly. Integration methods exploit structural properties of functions to simplify the evaluation of integrals. We derive two fundamental methods, one is based on the product rule, the other on the chain rule.

Theorem 3.3.18 (integration by parts)**.** *Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuously differentiable. Then we have*

$$\int_a^b f'(x)g(x)\,dx = f(b)g(b) - f(a)g(a) - \int_a^b f(x)g'(x)\,dx.$$

Proof. This is a direct consequence of the product rule

$$\int_a^b \big(f'(x)g(x) + f(x)g'(x)\big)\,dx = f(b)g(b) - f(a)g(a)$$

and the linearity of the integral. □

We illustrate the integration by parts formula with a few typical examples.

Example 3.3.19. *To compute $\int_a^b xe^x\,dx$, we choose $f'(x) = e^x$ and $g(x) = x$. The integration by parts formula yields*

$$\int_a^b e^x x\,dx = e^b b - e^a a - \int_a^b e^x\,dx = e^b b - e^a a - (e^b - e^a).$$

Example 3.3.20 (partial integration). *For $x > 1$, we use $f(t) = 1$ with $F(t) = t$ and $g(t) = \log t$ to obtain*

$$\int_1^x \log t \, dt = x \log x - 1 \log 1 - \int_1^x t \frac{1}{t} \, dt = x \log x - (x - 1).$$

Thus, a primitive of the logarithm is $x \mapsto x \log x - x + 1$.

The substitution rule for integration is based on the chain rule for differentiation.

Theorem 3.3.21 (substitution rule). *Suppose that $u : [a, b] \to \mathbb{R}$ is continuously differentiable with $u(a) \leq u(b)$, and $f : u([a, b]) \to \mathbb{R}$ is continuous. Then we have*

$$\int_a^b f(u(x))u'(x) \, dx = \int_{u(a)}^{u(b)} f(y) \, dy.$$

Proof. Suppose that F is a primitive of f. According to the chain rule, the derivative of the composition $F \circ u$ is

$$(F \circ u)'(x) = F'(u(x))u'(x) = f(u(x))u'(x).$$

The FTC yields

$$\int_a^b f(u(x))u'(x) \, dx = F(u(b)) - F(u(a)) = \int_{u(a)}^{u(b)} f(y) \, dy.$$

□

We give a few examples illustrating the substitution rule and note that the derivative of the function u is usually denoted by $\frac{du}{dx} = u'$.

Example 3.3.22. *To compute $\int_a^b 2x \cos(x^2) \, dx$, we substitute $u(x) = x^2$, so that $\frac{du}{dx}(x) = 2x$. The integral becomes*

$$\int_a^b 2x \cos(x^2) \, dx = \int_a^b \cos(u(x))u'(x) \, dx = \int_{a^2}^{b^2} \cos(y) \, dy.$$

If we denote the variable in the last integral by u instead of y and we replace u' by its equivalent notation $\frac{du}{dx}$, then we observe

$$\int_a^b \cos(u(x)) \frac{du}{dx}(x) \, dx = \int_{a^2}^{b^2} \cos(u) \, du.$$

Thus, the term $\frac{du}{dx} \, dx$ transforms into du as if dx were canceled by its occurrence in the denominator.

Example 3.3.23. *To compute* $\int_0^2 xe^{x^2}\,dx$*, we set* $u(x) = x^2$*, so that* $\frac{du}{dx}(x) = 2x$*. Since* $u(0) = 0$ *and* $u(2) = 4$*, we derive*

$$\int_0^2 xe^{x^2}\,dx = \int_0^2 \frac{1}{2}\frac{du}{dx}(x)e^{u(x)}\,dx = \frac{1}{2}\int_0^4 e^u\,du = \frac{1}{2}(e^4 - 1).$$

Remark 3.3.24. *The substitution rule and integration by parts are basic tools for rewriting integrals into more convenient forms. In data analysis and data science, such transformations are routinely used to simplify expressions, change coordinates, or isolate dominant contributions.*
However, in many real world scenarios, neither the substitution rule nor integration by parts reveal an integral that can be determined exactly. The function f *may not even be known explicitly but only given in terms of sample values* $f(x_0), \dots, f(x_n)$*. It is then standard to approximate integrals from those sample values. Classical rules are the mid-point and the trapezoidal rule.*

mid-point rule

Instead of using the infimum or supremum over bins for the Darboux sums, we simply evaluate f at the mid-point $\frac{x_{k-1}+x_k}{2}$ of the bin $[x_{k-1}, x_k]$. Given $f : [0,1] \to \mathbb{R}$ and $n \in \mathbb{N}$, the equidistant partition $\mathcal{P}_n$ is

$$\mathcal{P}_n = \{x_0, \dots, x_n\} = \Big\{\frac{k}{n} : k = 0, \dots, n\Big\}.$$

The mid-point rule is the quantity

$$M_n(f) = \frac{1}{n}\sum_{k=1}^{n} f\big(\frac{x_{k-1}+x_k}{2}\big) = \frac{1}{n}\sum_{k=1}^{n} f\big(\frac{k}{n} - \frac{1}{2n}\big)$$

```
function myMidPointRule(f,a,b; n=100)
    x = range(a,b,n+1)
    h = (b-a)/n
    y = x .+ h/2
    return sum(f(y[k]) for k=1:n)*h
end
```

Analogously, we can define a left-point rule by simply evaluating f at x_{k-1} for the bin $[x_{k-1}, x_k]$ or a right-point rule by evaluating f at x_k for the bin $[x_{k-1}, x_k]$.

trapezoidal rule

Instead of using rectangles to approximate the area under the curve, we use a trapezoid with the vertices x_{k-1}, x_k, $f(x_{k-1})$, and $f(x_k)$. Given $f : [0,1] \to \mathbb{R}$ and $n \in \mathbb{N}$, the equidistant partition $\mathcal{P}_n$ is

$$\mathcal{P}_n = \{x_0, \dots, x_n\} = \Big\{\frac{k}{n} : k = 0, \dots, n\Big\}.$$

The trapezoidal rule is the quantity

$$T_n(f) = \frac{1}{n}\sum_{k=1}^{n} \frac{f(x_{k-1}) + f(x_k)}{2}.$$

```
function myTrapezoidalRule(f,a,b; n=100)
    x = range(a,b,n+1)
    h = (b-a)/n
    return sum( (f(x[k])+f(x[k+1]))/2 for k=1:n)*h
end
```

Improper integrals

We extend the notion of integration beyond finite intervals and bounded integrands. We introduce improper integrals as limits of ordinary integrals and show how they naturally connect integration with questions of convergence for infinite series.

Definition 3.3.25 (improper integrals). *Given $f : [a, b) \to \mathbb{R}$ and suppose that for every sequence $(b_n)_{n\in\mathbb{N}} \subseteq (a, b)$ with $b_n \to b$ the function f is integrable on $[a, b_n]$. If the sequence of integrals $\left(\int_a^{b_n} f(x)\,dx\right)_{n\in\mathbb{N}}$ converges and the limit is independent of the chosen sequence $(b_n)_{n\in\mathbb{N}}$, then we define*

$$\int_a^b f(x)\,dx := \lim_{n\to\infty} \int_a^{b_n} f(x)\,dx = \lim_{\epsilon\searrow 0} \int_a^{b-\epsilon} f(x)\,dx.$$

We extend this definition to $b = \infty$ and make the analogous conventions for the lower integration bound.

Example 3.3.26 (improper integrals 1). *Given $f : (0, 1] \to \mathbb{R}$ by $f(x) = 1/\sqrt{x}$, we compute the integral*

$$\begin{aligned}\int_0^1 f(x)\,dx &= \lim_{a\searrow 0} \int_a^1 \frac{1}{\sqrt{x}}\,dx \\ &= \lim_{a\searrow 0} 2\sqrt{x}\Big|_a^1 \\ &= 2 \lim_{a\searrow 0} (1 - \sqrt{a}) \\ &= 2.\end{aligned}$$

Example 3.3.27 (improper integrals 2). *For $s > 1$, consider $f : [0, \infty) \to [0, \infty)$ given by*

$f(x) = \frac{1}{(1+x)^s}$, we compute the integral

$$\begin{aligned}\int_0^\infty \frac{1}{(1+x)^s}\,dx &= \lim_{b\to\infty}\int_0^b (1+x)^{-s}\,dx\\ &= \lim_{b\to\infty}\frac{1}{1-s}(1+x)^{1-s}\Big|_0^b\\ &= \frac{1}{1-s}\lim_{b\to\infty}\left((1+b)^{1-s}-1\right)\Big|_0^b\\ &= -\frac{1}{1-s} = \frac{1}{s-1}.\end{aligned}$$

The following result makes the connection between integrals and series precise. For nonnegative, monotonically decreasing functions, convergence of the integral controls convergence of the associated series and yields an explicit upper bound. This principle is a continuous analogue of the comparison test for series.

Theorem 3.3.28 (comparing series to integrals)**.** *Let $f : [0,\infty) \to [0,\infty)$ be monotonically decreasing and suppose that $\int_0^\infty f(x)\,dx$ exists. Then the series $\sum_{n=1}^\infty f(n)$ converges with*

$$\sum_{n=1}^\infty f(n) \le \int_0^\infty f(x)\,dx.$$

Proof. For $n \ge 1$, the monotonicity yields $f(n) \le \int_{n-1}^n f(x)\,dx$. We repeat this N times to obtain

$$\begin{aligned}\sum_{n=1}^N f(n) &\le \sum_{n=1}^N \int_{n-1}^n f(x)\,dx\\ &= \int_0^N f(x)\,dx\\ &\le \int_0^\infty f(x)\,dx.\end{aligned}$$

□

Example 3.3.29. *For $s > 1$, we aim to verify that the series $\sum_{n=1}^\infty \frac{1}{n^s}$ converges. Shifting the summation yields*

$$\sum_{n=1}^N \frac{1}{n^s} = 1 + \sum_{n=2}^N \frac{1}{n^s} = 1 + \sum_{n=1}^{N-1}\frac{1}{(1+n)^s}.$$

As in Example 3.3.28, define $f : [0,\infty) \to [0,\infty)$ by $f(x) = \frac{1}{(1+x)^s}$. This function is monotonically decreasing and the integral exists with $\int_0^\infty \frac{1}{(1+x)^s}\,dx = \frac{1}{s-1}$. Therefore, we have

$$\sum_{n=1}^\infty \frac{1}{n^s} = 1 + \sum_{n=1}^\infty \frac{1}{(1+n)^s} \le 1 + \frac{1}{s-1} = \frac{s}{s-1}.$$

Essentials

Integration is defined as a limit of finite sums and provides a rigorous notion of area under the curve. Continuity guarantees integrability. The Fundamental Theorem of Calculus connects integration with differentiation. The product rule and the chain rule lead to integration by parts and the substitution rule that can then help evaluating integrals. Improper integrals are defined as limits of integrals over expanding domains or approaching singularities. Monotonicity and integrability allow one to compare series with integrals, providing a practical convergence criterion and explicit upper bounds. This illustrates how continuous models can be used to control discrete sums.

CA On a structural level, integration can be considered as a completion of summation. As a limit of sums the integration is inherently an approximative process. The mid-point and trapezoidal rule yield numerical methods for approximating integrals. Improper integrals are inherently defined as approximations of proper integrals.

partition, Darboux sum, integral, Fundamental Theorem of Calculus, integration by parts, substitution rule, mid-point and trapezoidal rule,Improper integral, monotone function, integral test

3.4 Taylor expansion

Taylor expansions provide higher-order local approximations of functions. They refine the linear approximation given by the derivative by incorporating higher derivatives and quantify the approximation error by a remainder term.

We denote the k-th derivative of f by $f^{(k)}$.

Theorem 3.4.1 (Taylor expansion)**.** *Assume $f : [a,b] \to \mathbb{R}$ is $n+1$-times continuously differentiable and let $x_0 \in (a,b)$. Then we have, for all $x \in [a,b]$,*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + R_{n+1}(x), \tag{3.1}$$

where the remainder is

$$R_{n+1}(x) = \frac{1}{n!}\int_{x_0}^{x} (x-t)^n f^{(n+1)}(t)\,dt. \tag{3.2}$$

The mapping $x \mapsto \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ is a polynomial function of degree n, often referred to as the Taylor polynomial. It represents a local approximation of f near x_0.

Proof. The proof is by induction. For $n = 0$, we have

$$f(x) = f(x_0) + \int_{x_0}^{x} f'(t)\,dt,$$

which follows from the FTC. The term $\int_{x_0}^{x} f'(t)\,dt$ coincides with $R_1(x)$ as in (3.2).
We now assume the claim holds for $n-1$ with R_n in (3.2) and define R_{n+1} by (3.5). We use

the induction claim for $n-1$ and obtain

$$\begin{aligned}
R_{n+1}(x) &= f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k \\
&= \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + R_n(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k \\
&= R_n(x) - \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n \\
&= \frac{1}{(n-1)!}\int_{x_0}^{x} (x-t)^{n-1} f^{(n)}(t)\,dt - \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n \\
&= -\int_{x_0}^{x} f^{(n)}(t)\,\frac{d}{dt}\left(\frac{(x-t)^n}{n!}\right)\,dt - \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n \\
&= -f^{(n)}(t)\,\frac{(x-t)^n}{n!}\Big|_{t=x_0}^{t=x} + \int_{x_0}^{x} f^{(n+1)}(t)\frac{(x-t)^n}{n!}dt - \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n \\
&= \int_{x_0}^{x} f^{(n+1)}(t)\frac{(x-t)^n}{n!}dt.
\end{aligned}$$

□

Example 3.4.2. *For $n = 1$, the Taylor expansion yields*

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + R_2(x).$$

and the remainder R_2 coincides with φ in Theorem 3.1.8, where f is locally approximated by a linear map. Since $\lim_{x\to x_0} \frac{\varphi(x)}{x-x_0} = 0$, we also have

$$\frac{R_2(x)}{x-x_0} = \frac{\int_{x_0}^{x}(x-t)f''(t)\,dt}{x-x_0} \longrightarrow 0 \quad \text{as } x \to x_0. \tag{3.3}$$

Example 3.4.3. *For $n = 2$, the Taylor expansion yields*

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2 + R_3(x).$$

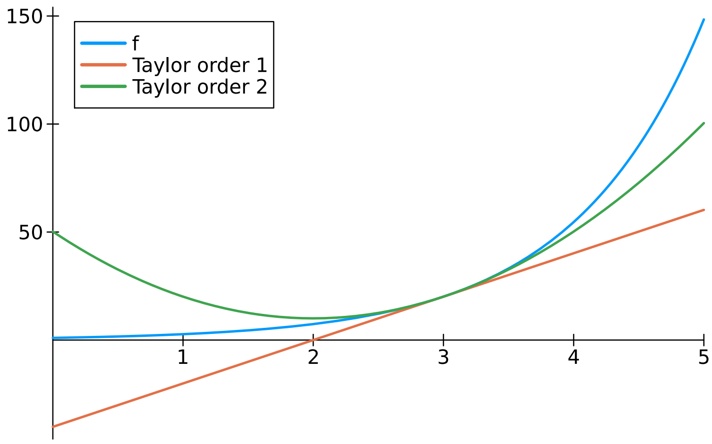
The mapping

$$x \mapsto f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2}(x-x_0)^2$$

is a polynomial function of degree 2 that locally approximates f near x_0. To estimate the quality of the approximation, we need bounds on the remainder R_3.

```
using Calculus, Plots

function myTaylor(f,x0::Float64;n=2)
    f1 = derivative(f)
    if n == 1
        return x -> f(x0) + f1(x0)*(x-x0)
```

```
    end
    f2 = second_derivative(f)
    if n == 2
        return x -> f(x0) + f1(x0)*(x-x0)+f2(x0)*(x-x0)^2 /2
    end
end

f(x) = exp(x)
x0 = 3.
T1f = myTaylor(f,x0;n=1)
T2f = myTaylor(f,x0;n=2)

plot(f,xlimits=[1,5],label="f")
plot!(T1f,label="Taylor order 1")
plot!(T2f,label="Taylor order 2")
```



To get a better grip on the size of the remainder R_{n+1}, we derive an alternative representation.

Lemma 3.4.4 (alternative form of the remainder)**.** *Assume $f : [a,b] \to \mathbb{R}$ is $n+1$-times continuously differentiable and let $x_0 \in (a,b)$. Then there is $\xi = \xi(x_0,x) \in [x_0,x] \cup [x,x_0]$ such that*

$$R_{n+1}(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}. \tag{3.4}$$

Proof. Without loss of generality, assume that $x > x_0$. According to the mean value theorem of integration there exists $\xi \in [x_0,x]$ such that

$$\begin{aligned}
R_{n+1}(x) &= \frac{1}{n!}\int_{x_0}^{x}(x-t)^n f^{(n+1)}(t)\,dt \\
&= f^{(n+1)}(\xi)\frac{1}{n!}\int_{x_0}^{x}(x-t)^n\,dt \\
&= f^{(n+1)}(\xi)\Big[\frac{-1}{(n+1)!}(x-t)^{n+1}\Big]_{x_0}^{x} \\
&= \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}.
\end{aligned}$$

□

The advantage of the form (3.4) for R_{n+1} is the explicit factor $(x-x_0)^{n+1}$. Since $f^{(n+1)}$ is assumed to be continuous and $x \to x_0$ yields $\xi \to x_0$, we now observe

$$\lim_{x\to x_0} \frac{R_{n+1}(x)}{(x-x_0)^{n+1}} = \frac{f^{(n+1)}(x_0)}{(n+1)!},$$

which extends the error for the linear approximation (3.3) to approximations by polynomial functions of degree n.

Given a function f, its Taylor polynomial of degree n at x_0 is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k.$$

Let us compute T_n for the exponential function at $x_0 = 0$.

Example 3.4.5 (Taylor expansion of exp). *To determine the Taylor polynomial of degree n of $f(x) = e^x$ at $x_0 = 0$, we first observe that* $\exp^{(k)} = \exp$. *Hence, we obtain* $\exp^{(k)}(0) = 1$, *for all $k \in \mathbb{N}$. The Taylor polynomial of degree n at $x_0 = 0$ is*

$$T_n(x) = \sum_{k=0}^{n} \frac{1}{k!}x^k.$$

These are exactly the first terms of the series expansion of exp.

We now derive the best bound on the remainder term so far. We only assume n times continuous differentiability and use the Taylor polynomial of degree n.

Theorem 3.4.6 (Taylor asymptotic). *Assume $f : [a,b] \to \mathbb{R}$ is n-times continuously differentiable and let $x_0 \in (a,b)$. Then we have, for all $x \in [a,b]$,*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + o(|x-x_0|^n). \tag{3.5}$$

The symbol $o(|x-x_0|^n)$ stands for a function φ such that

$$\lim_{x\to x_0} \frac{\varphi(x)}{|x-x_0|^n} = 0.$$

Thus, φ must decay faster than $|x-x_0|^n$.

Proof. We use the alternative form of the remainder for the Taylor polynomial of order $n-1$.

This yields

$$\begin{aligned} f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k &= R_n(x) \\ &= \frac{f^{(n)}(\xi)}{n!}(x-x_0)^n \\ &= \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n)}(\xi) - f^{(n)}(x_0)}{n!}(x-x_0)^n. \end{aligned}$$

Reordering these terms leads to

$$f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k = \frac{f^{(n)}(\xi) - f^{(n)}(x_0)}{n!}(x-x_0)^n.$$

Since $f^{(n)}$ is continuous and $x \to x_0$ enforces $\xi \to x_0$, we see that the right-hand-side is $o(|x-x_0|^n)$. □

Corollary 3.4.7. *Assume $f : [a,b] \to \mathbb{R}$ is n-times continuously differentiable and let $x \in (a,b)$ be such that $f(x) = x$. If $n \geq 2$ and*

$$0 = f^{(1)}(x) = \cdots = f^{(n-1)}(x),$$

then the iteration $x_{k+1} = f(x_k)$ converges towards x with order n for all initial points x_0 in a neighborhood of x.

Convergence of order n means that there is $c > 0$ such that

$$|x_{k+1} - x| \leq c|x_k - x|^n, \qquad k \in \mathbb{N},$$

see Definition 1.1.7.

Proof. Since $f'(x) = 0$, Corollary 3.2.11 implies contractivity in a neighborhood of x. The Banach fixed point theorem leads to linear convergence. To verify convergence order n, we consider the Taylor expansion

$$f(x_k) = \sum_{i=0}^{n} \frac{f^{(i)}(x)}{i!}(x_k - x)^i + o(|x_k - x|^n).$$

There is $\xi_k \in \mathbb{R}$ such that $\xi_k \to 0$ and

$$x_{k+1} = f(x_k) = x + \frac{f^{(n)}(x)}{n!}(x_k - x)^n + \xi_k |x_k - x|^n.$$

We bring x to the left-hand-side and estimate

$$|x_{k+1} - x| \leq (|\frac{f^{(n)}(x)}{n!}| + \xi_k)|x_k - x|^n.$$

Since $\xi_k \to 0$ and hence bounded, the convergence is of order n. □

We are now prepared to verify Newton's method to determine solutions x of $f(x) = 0$.

Theorem 3.4.8 (Newton's method). *Assume $f : [a,b] \to \mathbb{R}$ is 2-times continuously differentiable and let $x \in (a,b)$ be such that $f(x) = 0$ and $f'(x) \neq 0$. Then the iteration*

$$x_{k+1} := x_k - \frac{f(x_k)}{f'(x_k)}$$

converges towards x with order 2 for all initial points x_0 in a neighborhood of x.

Proof. The derivative f' is nonzero in a neighborhood of x, so that we may define $F(y) = y - \frac{f(y)}{f'(y)}$ in a neighborhood of x. Then $F(x) = x$ and

$$F'(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = 0.$$

Corollary 3.4.7 implies convergence order 2. □

In comparison to the bisection method, Newton's method converges faster. However, the theorem requires that the starting point x_0 is close to x to begin with.

Consider $f : [0,2] \to \mathbb{R}$ with $f(x) = x^2 - 2$.

```
function myNewton(f, ∂f, x0; tol=1e-5, maxiter=50)
    x = Vector{Float64}(undef,maxiter)
    x[1] = copy(x0)
    for k in 1:maxiter-1
        if abs(f(x[k]))<tol return x[1:k] end
        x[k+1] = x[k]-f(x[k])/∂f(x[k])
    end
    println("no convergence")
    return x
end

f(x) = x^2-2
myNewton(f,x->2x,1;tol=4eps())
```

```
6-element Vector{Float64}:
 1.0
 1.5
 1.4166666666666667
 1.4142156862745099
 1.4142135623746899
 1.4142135623730951
```

The exact root is $\sqrt{2} \approx 1.4142135623730951\ldots$.

Remark 3.4.9 (local models). *Taylor expansions formalize the idea that smooth functions can be locally approximated by simple polynomial models. In data-related settings such approximations are fundamental when complicated behavior needs to be captured in a controlled way.*

Essentials

Taylor expansions provide a systematic way to approximate smooth functions by polynomial functions near a given point. Derivatives determine the coefficients of the Taylor polynomial and the remainder term quantifies the approximation error.

RTA A function is represented by its Taylor series that is truncated to the Taylor polynomial. The remainder term quantifies the approximation error.

Taylor expansion, Taylor polynomial, remainder term, local approximation

To proceed toward multivariate phenomena, approximative linear processes, and computation, limits must be incorporated into linear algebra and linear structure into analysis. The analytical results developed in the present part therefore achieved the passage from finite algebraic structures to infinite models governed by limits, continuity, and completeness. The following part deepens this completion through infinite-dimensional linear structures such as Banach and Hilbert spaces, in which analytical phenomena acquire geometric and operator-theoretic form. Only after this structural completion does the transition to computation begin, where discretization, truncation, and numerical approximation realize these infinite models under finite precision. The movement from analysis to computation is thus not a change of subject, but a continuation of the same mathematical structures.

Part II

From Infinite-Dimensions to Computation

Normed and inner product spaces are completed to Banach and Hilbert spaces. Choosing bases or frames yields representation in coordinates. Truncating coordinates produces finite-dimensional representations suitable for computation. The Fréchet derivative quantifies the conditioning of a numerical problem, and algorithms compute this problem in finite-precision arithmetics.

So far, linear algebra has been concerned with finite linear combinations and, more generally, with finite sums. Analysis, on the other hand, has shown that infinite series can converge and thereby give rise to well-defined objects.

We now enrich linear algebra by incorporating convergence and allowing infinite linear combinations whenever they converge. Completion then transforms finite algebraic structures into infinite analytic models. Banach and Hilbert spaces arise from this passage and provide the natural setting in which limits, convergence, and continuity govern linear structure.

Conversely, analysis has so far dealt primarily with functions of a single real variable. Linear algebra introduces the language required for multivariate phenomena and thereby extends analytic concepts to higher dimensions. In finite dimensions, differentiability and local extrema are governed by linear maps represented by Jacobian matrices.

Fréchet differentiability lifts the principle of linearization to infinite-dimensional normed spaces, where nonlinear mappings are locally approximated by bounded linear operators. The sensitivity of this linearization to perturbations is captured by the concept of conditioning. Its concrete evaluation is carried out by algorithms operating in finite precision. Approximation, representation, and finite precision thus arise from the internal logic of analysis itself, and numerical computation becomes its unavoidable continuation.

Mathematics formulates problems as mappings between ideal spaces. Their conditioning describes the intrinsic sensitivity of the exact solution with respect to perturbations of the data and thus belongs to the structure of the problem itself. An algorithm provides a finite-precision realization of this mapping through representation in coordinates, truncation, and approximation. Its stability measures how faithfully this realization reflects the conditioning of the underlying problem. In this way, numerical computation does not stand outside analysis but continues it under the constraints of finite precision. The passage from problem to computation therefore follows the principles of the computational cycle.

4 Linear Algebra meets Analysis

We combine linear algebra concepts with principles from analysis such as limits and completeness.

4.1 Continuous linear maps

We start with finite-dimensional spaces and then proceed to infinite-dimensional normed spaces.

Theorem 4.1.1 (finite dimensions)**.** *Let V be a finite-dimensional normed vector space. Then the following statements hold:*

(i) V is complete.

(ii) If $\|\cdot\|_a$ and $\|\cdot\|_b$ are norms on V, then there exist constants $c, C > 0$ such that

$$c\|v\|_a \leq \|v\|_b \leq C\|v\|_a \qquad \text{for all } v \in V.$$

(iii) If $f : V \to V$ is linear, then f is continuous.

Part (iii) also holds for $f : V \to W$ for two normed linear spaces V and W as long as V is finite-dimensional.

Proof. (ii) Fix a basis $B = \{b_1, \dots, b_n\}$ of V and recall the coordinate map $\chi_B : \mathbb{F}^n \to V$ given by $\chi_B(x) = \sum_{i=1}^n x_i b_i$. Define the coordinate norm

$$\|v\|_{B,2} := \|\chi_B^{-1}(v)\|_2, \qquad v \in V,$$

where $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{F}^n$.
Upper bound. Let $M := \max_{1\leq i\leq n} \|b_i\|_a$. For $v = \sum_{i=1}^n x_i b_i$ we obtain

$$\|v\|_a \leq \sum_{i=1}^n |x_i|\, \|b_i\|_a \leq M \sum_{i=1}^n |x_i| \leq M\sqrt{n}\, \|x\|_2 = M\sqrt{n}\, \|v\|_{B,2}.$$

Lower bound. Let $\mathbb{S}^{n-1} := \{x \in \mathbb{F}^n : \|x\|_2 = 1\}$ and

$$S_B := \{v \in V : \|v\|_{B,2} = 1\} = \chi_B(\mathbb{S}^{n-1}).$$

The bound

$$\|\chi_B(x)\|_a \leq M\sqrt{n}\, \|x\|_2 \qquad \text{for all } x \in \mathbb{F}^n,$$

yields that χ_B is continuous as a map from $\mathbb{F}^n$ to $(V, \|\cdot\|_a)$. Since $\mathbb{S}^{n-1}$ is compact, the set S_B is compact in $(V, \|\cdot\|_a)$. The function $v \mapsto \|v\|_a$ is continuous on $(V, \|\cdot\|_a)$ and strictly positive on S_B, hence attains a minimum $m > 0$. For $v \neq 0$ we write $v = \|v\|_{B,2} w$ with $w \in S_B$ and conclude

$$\|v\|_a = \|w\|_a\, \|v\|_{B,2} \geq m\|v\|_{B,2}.$$

Thus there exist $\alpha, \beta > 0$ such that

$$\alpha \|v\|_{B,2} \leq \|v\|_a \leq \beta \|v\|_{B,2} \qquad \text{for all } v \in V.$$

Applying the same argument to $\|\cdot\|_b$ yields that there exist $\alpha', \beta' > 0$ such that

$$\alpha' \|v\|_{B,2} \leq \|v\|_b \leq \beta' \|v\|_{B,2} \qquad \text{for all } v \in V.$$

We combine these inequalities and obtain

$$\frac{\alpha'}{\beta} \|v\|_a \leq \|v\|_b \leq \frac{\beta'}{\alpha} \|v\|_a.$$

(i) Fix a basis $B = \{b_1, \ldots, b_n\}$ of V and recall the coordinate map $\chi_B : \mathbb{R}^n \to V$ given by $\chi_B(x) = \sum_{i=1}^n x_i b_i$. Define the coordinate norm

$$\|v\|_{B,1} := \|\chi_B^{-1}(v)\|_1, \qquad v \in V,$$

where $\|\cdot\|_1$ denotes the Euclidean 1-norm on $\mathbb{R}^n$.
Let $(v_k)_{k\in\mathbb{N}}$ be a Cauchy sequence in V. By (ii), $(v_k)_{k\in\mathbb{N}}$ is Cauchy with respect to $\|\cdot\|_{B,1}$. Setting $x_k = \chi_B^{-1}(v_k)$, we obtain a Cauchy sequence in $\mathbb{R}^n$, which converges since $\mathbb{R}^n$ is complete. Hence $v_k \to \chi_B(x) \in V$ with respect to $\|\cdot\|_{B,1}$, and therefore with respect to the original norm by (ii).

(iii) Set

$$C := \max_{1\leq i\leq n} \|f b_i\|.$$

For $v = \sum_{i=1}^n x_i b_i$ we have

$$\|fv\| \leq \sum_{i=1}^n |x_i| \, \|f b_i\| \leq C \sum_{i=1}^n |x_i| = C \, \|v\|_{B,1}.$$

By (ii), $\|v\|_{B,1} \leq C_1 \|v\|$ for some $C_1 > 0$, hence

$$\|f(v_n - v)\| \leq C C_1 \, \|v_n - v\| \to 0,$$

which proves continuity of f. □

Example 4.1.2. *For $x \in \mathbb{C}^n$,*

(i) sum norm: $\|x\|_1 = \sum_{k=1}^n |x_k|$

(ii) maximum/infinity norm: $\|x\|_\infty = \|x\|_{\max} = \max_{k=1,\ldots,n} |x_k|$

(iii) Euclidean norm: $\|x\|_2 = \left(\sum_{k=1}^n |x_k|^2\right)^{1/2}$

Example 4.1.3. *If $A \in \mathbb{C}^{n\times n}_{sym}$ is positive definite, i.e., invertible and*

$$x^* A x \geq 0, \qquad x \in \mathbb{C}^n,$$

then it induces a norm and inner product on $\mathbb{C}^n$ *by*

$$\|x\|_A := \sqrt{x^* A x}, \qquad \langle x, y\rangle_A := \langle x, Ay\rangle.$$

All norms on $\mathbb{C}^n$ *are equivalent but sometimes the constants matter.*

Theorem 4.1.4 (Boundedness and continuity of linear maps)**.** *Let* V *and* W *be normed vector spaces and let* $f : V \to W$ *be linear. The following are equivalent:*

(i) f *is continuous.*

(ii) f *is continuous at* 0.

(iii) There exists $C \geq 0$ *such that*

$$\|fv\| \leq C\|v\| \qquad \textit{for all } v \in V.$$

Proof. (i)⇒(ii) is clear.
(ii)⇒(iii) Let $\varepsilon = 1$. By continuity at 0 there exists $\delta > 0$ such that $\|v\| < \delta$ implies $\|fv\| < 1$. For $v \neq 0$ set $w := \frac{\delta}{2\|v\|}v$, so that $\|w\| = \delta/2 < \delta$ and hence $\|fw\| < 1$. By linearity,

$$\|fv\| = \frac{2\|v\|}{\delta}\,\|fw\| \leq \frac{2}{\delta}\|v\|.$$

Thus (iii) holds with $C = 2/\delta$.
(iii)⇒(i) If $\|fv\| \leq C\|v\|$, then for $v, u \in V$,

$$\|fv - fu\| = \|f(v-u)\| \leq C\|v - u\|.$$

Thus, $v \to u$ implies $fv \to fu$ and f is continuous. □

We now define the operator norm of a linear map that is continuous.

Definition 4.1.5. *Let* V *and* W *be normed vector spaces and let* $f : V \to W$ *be linear and continuous. Then the operator norm of* f *is*

$$\|f\|_{\mathrm{op}} := \sup_{0 \neq v \in V} \frac{\|fv\|}{\|v\|}.$$

The operator norm of f is the smallest constant C such that (iii) in Theorem 4.1.4 holds, in particular,

$$\|fv\| \leq \|f\|_{\mathrm{op}}\,\|v\| \qquad \text{for all } v \in V.$$

Equivalent expressions for the operator norm are

$$\|f\|_{\mathrm{op}} = \sup_{\substack{v \in V \\ \|v\| \leq 1}} \|fv\| = \sup_{\substack{v \in V \\ \|v\| = 1}} \|fv\|.$$

The linear map f is continuous if and only if its operator norm is finite. Therefore, we often say f is bounded linear when we mean a linear continuous map.

Example 4.1.6. *For $A \in \mathbb{R}^{m\times n}$ or $A \in \mathbb{C}^{m\times n}$, we have the following operator norms:*

(i) The column sum norm is the operator norm wrt. the sum norm,

$$\|A\|_1 := \sup_{\|x\|_1=1} \|Ax\|_1 = \max_{j=1,\dots,n} \sum_{i=1}^{m} |a_{i,j}|.$$

(ii) The row sum norm is the operator norm wrt. the maximum norm,

$$\|A\|_\infty := \sup_{\|x\|_\infty=1} \|Ax\|_\infty = \max_{i=1,\dots,m} \sum_{j=1}^{n} |a_{i,j}|.$$

```
opnorm(I,1),  opnorm(I,Inf),
```

```
(1.0, 1.0)
```

```
A = [1 2; 3 4]
opnorm(A,1), opnorm(A,Inf)
```

```
(6.0, 7.0)
```

Definition 4.1.7 (spectrum, spectral radius)**.** *The* spectrum *of $A \in \mathbb{C}^{n\times n}$ is*

$$\sigma(A) := \{\lambda \in \mathbb{C} : \lambda \text{ eigenvalue of } A\}.$$

The spectral radius *is*

$$\rho(A) := \max_{\lambda\in\sigma(A)} |\lambda|.$$

Lemma 4.1.8. *Let $A \in \mathbb{C}^{n\times n}$ be invertible. If $\|\cdot\|$ is a norm on $\mathbb{C}^n$, then the operator norm of A^{-1} is*

$$\|A^{-1}\| = \frac{1}{\min_{\|x\|=1} \|Ax\|}.$$

Proof. In the following, the bijectivity allows us to substitute $y = Ax$ to derive

$$\|A^{-1}\| = \max_{y\neq 0} \frac{\|A^{-1}y\|}{\|y\|} = \max_{x\neq 0} \frac{\|x\|}{\|Ax\|} = \max_{\|x\|=1} \frac{1}{\|Ax\|} = \frac{1}{\min_{\|x\|=1} \|Ax\|}$$

□

We define

$$\mathbb{C}^{n\times n}_{sym} := \{A \in \mathbb{C}^{n\times n} : A^* = A\}.$$

Lemma 4.1.9. *For $A \in \mathbb{C}^{n\times n}_{sym}$, we have*

$$\|A\|_2 = \rho(A).$$

Proof. There is an orthonormal basis of eigenvectors $\{b_1, \dots, b_n\}$. Pick $x \in \mathbb{C}^n$ with $\|A\|_2 = \|Ax\|_2$ and $\|x\|_2 = 1$. We derive

$$\|Ax\|_2^2 = \|\sum_{j=1}^n \langle x, b_j\rangle \lambda_j b_j\|_2^2 = \sum_{j=1}^n |\langle x, b_j\rangle|^2 |\lambda_j|^2 \leq \sum_{j=1}^n |\langle x, b_j\rangle|^2 \rho(A)^2 = \rho(A)^2.$$

This implies $\|A\|_2 \leq \rho(A)$. For the reverse inequality, select $\lambda \in \sigma(A)$ with $|\lambda| = \rho(A)$ and let x be the associated eigenvector with $\|x\|_2 = 1$. Then we have

$$\|A\|_2 = \max_{\|y\|_2 = 1} \|Ay\|_2 \geq \|Ax\|_2 = \|\lambda x\|_2 = \rho(A).$$

$\square$

Let V, W be normed vector spaces. We define

$$\begin{aligned} \mathcal{L}(V, W) &:= \{f : V \to W \quad : \quad f \text{ linear and continuous}\} \\ \mathcal{L}(V) &:= \mathcal{L}(V, V). \end{aligned}$$

Direct calculations reveal that the operator norm is a norm on $\mathcal{L}(V, W)$.

Lemma 4.1.10 (C). *If W is a Banach space, then the operator norm on $\mathcal{L}(V, W)$ makes it a Banach space.*

Proof. Let $(f_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(V, W)$ be a Cauchy sequence. Step 1: Pointwise limits. Fix $x \in V$. For all $n, m \in \mathbb{N}$ we have

$$\|f_n(x) - f_m(x)\|_W \leq \|f_n - f_m\| \, \|x\|_V.$$

Since (f_n) is Cauchy, the right-hand side tends to 0 as $n, m \to \infty$. Hence $(f_n(x))_{n\in\mathbb{N}}$ is a Cauchy sequence in W. Because W is complete, the limit exists. Define

$$f(x) := \lim_{n\to\infty} f_n(x) \in W, \qquad x \in V.$$

This defines a map $f : V \to W$. Direct calculations reveal that f is linear. To verify that f is also bounded, we recall that every Cauchy sequence is bounded. Hence, there is $C > 0$ such that $\|f_n\| \leq C$, and this implies that, for every $x \in V$,

$$\|f(x)\|_W = \left\|\lim_{n\to\infty} f_n(x)\right\|_W = \lim_{n\to\infty} \|f_n(x)\|_W \leq C\|x\|_V.$$

Thus f is bounded and $f \in \mathcal{L}(V, W)$.

Step 2: Convergence in operator norm. Fix $n \in \mathbb{N}$ and let $x \in V$ with $\|x\|_V \leq 1$. Then

$$\|(f_n - f)(x)\|_W = \left\|f_n(x) - \lim_{m\to\infty} f_m(x)\right\|_W = \lim_{m\to\infty} \|(f_n - f_m)(x)\|_W \leq \limsup_{m\to\infty} \|f_n - f_m\|.$$

Taking the supremum over all $\|x\|_V \leq 1$ yields

$$\|f_n - f\| \leq \limsup_{m\to\infty} \|f_n - f_m\|.$$

Since (f_n) is Cauchy, the right-hand side tends to 0 as $n \to \infty$. Therefore $\|f_n - f\| \to 0$ and $f_n \to f$ in $\mathcal{L}(V, W)$.

We have shown that every Cauchy sequence in $\mathcal{L}(V, W)$ converges in $\mathcal{L}(V, W)$. Hence $\mathcal{L}(V, W)$ is complete. $\square$

Theorem 4.1.11 (A von Neumann series). *Let B be a Banach space. If $f \in \mathcal{L}(B)$ such that there is $\theta < 1$ and*

$$\|\operatorname{id}_B - f\|_{\mathrm{op}} \leq \theta,$$

then f is invertible, $\|f^{-1}\|_{\mathrm{op}} \leq \frac{1}{1-\theta}$ and

$$f^{-1} = \sum_{k=0}^{\infty} (\operatorname{id}_B - f)^k.$$

Note that $(\operatorname{id}_B - f)^0 = \operatorname{id}_B$ by definition.

```
function myNeumann(f; n=20)
    # opnorm(I-f)<1 || throw(ArgumentError("assumption q<1 not satisfied"))
    return sum((I-f)^k for k in 0:n)
end
```

Proof. For $n \in \mathbb{N}$ define the partial sums

$$s_n := \sum_{k=0}^{n} (\operatorname{id}_B - f)^k \in \mathcal{L}(B).$$

Step 1: $(s_n)_{n\in\mathbb{N}}$ is Cauchy. Let $m > n$. Then

$$s_m - s_n = \sum_{k=n+1}^{m} (\operatorname{id}_B - f)^k.$$

By the triangle inequality and submultiplicativity,

$$\begin{aligned}
\|s_m - s_n\| \leq \sum_{k=n+1}^{m} \|(\operatorname{id}_B - f)^k\| &\leq \sum_{k=n+1}^{m} \|\operatorname{id}_B - f\|^k \\
&\leq \sum_{k=n+1}^{\infty} \|\operatorname{id}_B - f\|^k \\
&= \frac{1}{1 - \|\operatorname{id}_B - f\|} - \frac{1 - \|\operatorname{id}_B - f\|^{n+1}}{1 - \|\operatorname{id}_B - f\|} \\
&= \frac{\|\operatorname{id}_B - f\|^{n+1}}{1 - \|\operatorname{id}_B - f\|}.
\end{aligned}$$

Since $\|\operatorname{id}_B - f\| < 1$, the right-hand side tends to 0 as $n \to \infty$, hence (s_n) is Cauchy. Because B is complete, there exists $s \in \mathcal{L}(B)$ with $s_n \to s$.

Step 2: s is the inverse of f. Using the identity $f = \operatorname{id}_B - (\operatorname{id}_B - f)$ and a telescoping sum, we compute for every n:

$$\begin{aligned}
f\, s_n &= \big(\operatorname{id}_B - (\operatorname{id}_B - f)\big) \sum_{k=0}^{n} (\operatorname{id}_B - f)^k \\
&= \sum_{k=0}^{n} (\operatorname{id}_B - f)^k - \sum_{k=0}^{n} (\operatorname{id}_B - f)^{k+1} \\
&= \operatorname{id}_B - (\operatorname{id}_B - f)^{n+1}.
\end{aligned}$$

Similarly, $s_n f = \mathrm{id}_B - (\mathrm{id}_B - f)^{n+1}$.
Moreover, by submultiplicativity,

$$\|(\mathrm{id}_B - f)^{n+1}\| \leq \|\mathrm{id}_B - f\|^{n+1} \xrightarrow[n\to\infty]{} 0.$$

Passing to the limit and using continuity of multiplication,

$$fs = \lim_{n\to\infty} f s_n = \mathrm{id}_B, \qquad sf = \lim_{n\to\infty} s_n f = \mathrm{id}_B\,.$$

Thus $s = f^{-1}$. Since $s_n \to s$, we have

$$f^{-1} = s = \sum_{k=0}^{\infty} (\mathrm{id}_B - f)^k.$$

$\square$

Essentials

In finite dimensions all norms are equivalent and all linear maps are continuous. The space of bounded linear operators into a Banach space equipped with the operator norm is a Banach space. The von Neumann series of f converges in the operator norm towards the inverse provided that f is sufficiently close to the identity.

RA For Banach spaces, the von Neumann series represents and converges towards the inverse.

column sum norm, row sum norm, spectral radius, operator norm, sub-multiplicative, von Neumann series

4.2 Completions

Complete spaces are important for approximative processes such as Banach's fixed point theorem.

We first derive the classical completion of a metric space via Cauchy sequences modulo an equivalence relation. Recall that an isometry between metric spaces (X, d_X) and (Y, d_Y) is a map $J : X \to Y$ such that

$$d(J(x_1), J(x_2)) = d(x_1, x_2), \qquad \forall\, x1, x_2 \in X.$$

Every isometry is injective.

Theorem 4.2.1 (Completion of a metric space)**.** *Let (X_0, d_0) be a metric space. Then there exist a complete metric space (X, d) and an isometric embedding $J : X_0 \to X$ such that $J(X_0)$ is dense in X.*

Our focus lies on the structural ideas; technical details are deferred to the literature.

Proof. (1) Define the metric space (X, d): Let Cauchy(X_0) be the set of all Cauchy sequences in X_0. Define

$$(x_n)_{n\in\mathbb{N}} \sim (y_n)_{n\in\mathbb{N}} \quad :\Longleftrightarrow \quad d_0(x_n, y_n) \to 0,$$

and set $X := \text{Cauchy}(X_0)/\sim$, with equivalence classes $[(x_n)_{n\in\mathbb{N}}]$. It turns out that we may define

$$d\big([(x_n)_{n\in\mathbb{N}}], [(y_n)_{n\in\mathbb{N}}]\big) := \lim_{n\to\infty} d_0(x_n, y_n),$$

because $d(x_n, y_n)$ indeed converges and is independent of the specific representatives. It is a metric by passing to the limit in the triangle inequality.
(2) Density: Define $J : X_0 \to X$ by mapping x onto its constant sequence $J(x) = [(x)_{n\in\mathbb{N}}]$. Then $d(J(x), J(y)) = d(x, y)$, so J is an isometry. Moreover, if $a = [(x_n)_{n\in\mathbb{N}}] \in X$, then

$$d(J(x_n), a) = \lim_{m\to\infty} d_0(x_n, x_m) \xrightarrow[n\to\infty]{} 0,$$

so $J(X_0)$ is dense.
(3) Completeness: To prove completeness, let $(a_k)_{k\in\mathbb{N}}$ be Cauchy in X. Choose a subsequence $(a_{k_j})_{j\in\mathbb{N}}$ with $d(a_{k_{j+1}}, a_{k_j}) \le 2^{-j}$. Pick representatives $a_{k_j} = [(x_n^{(j)})_{n\in\mathbb{N}}]$ and indices n_j such that

$$d_0\Big(x_{n_j}^{(j)}, x_{n_{j+1}}^{(j+1)}\Big) \le 2^{-j}.$$

Set $z_j := x_{n_j}^{(j)} \in X_0$. Then $(z_j)_{j\in\mathbb{N}}$ is Cauchy in X_0 since for $m > j$,

$$d_0(z_j, z_m) \le \sum_{\ell=j}^{m-1} d_0(z_\ell, z_{\ell+1}) \le \sum_{\ell=j}^{m-1} 2^{-\ell} \le 2^{-(j-1)}.$$

Let $a := [(z_j)_{j\in\mathbb{N}}] \in X$. For each j,

$$d(a_{k_j}, a) \le \sum_{\ell=j}^{\infty} d(a_{k_\ell}, a_{k_{\ell+1}}) \le \sum_{\ell=j}^{\infty} 2^{-\ell} \to 0,$$

hence $a_{k_j} \to a$. Since $(a_k)_{k\in\mathbb{N}}$ is Cauchy and has a convergent subsequence, it converges to the same limit a. Thus X is complete. $\square$

Uniformly continuous maps on metric spaces can be uniquely lifted to the completions.

Theorem 4.2.2 (Extension to the completion)**.** *Let X_0 and Y_0 be metric spaces and let X and Y be their completions. Let $f_0 : X_0 \to Y_0$ be a map. Then the following are equivalent:*

1. *The map f_0 is uniformly continuous on X_0.*
2. *There exists a unique uniformly continuous map $f : X \to Y$ such that $f|_{X_0} = f_0$.*

In this case, the extension f is unique.

Proof. (1)⇒(2). Assume that f_0 is uniformly continuous. For $x \in X$, choose a sequence $(x_n)_{n\in\mathbb{N}} \subseteq X_0$ with $x_n \to x$ in X (which is possible since X_0 is dense in X). Since (x_n) is Cauchy in X_0 and f_0 is uniformly continuous, the sequence $\big(f_0(x_n)\big)$ is Cauchy in Y_0. As Y

is complete and Y_0 is dense in Y, the sequence $\big(f_0(x_n)\big)$ converges in Y. Define

$$f(x) := \lim_{n\to\infty} f_0(x_n) \in Y.$$

Well-definedness. Let (x_n) and (x'_n) be sequences in X_0 with $x_n \to x$ and $x'_n \to x$ in X. Then $d_X(x_n, x'_n) \to 0$. By uniform continuity, for every $\eta > 0$ there exists $\delta > 0$ such that $d_X(u,v) < \delta$ implies $d_Y\big(f_0(u), f_0(v)\big) < \eta$ for all $u, v \in X_0$. Hence $d_Y\big(f_0(x_n), f_0(x'_n)\big) \to 0$, so both sequences have the same limit in Y. Thus f is well-defined.
Uniform continuity of f. Fix $x \in X$ and let $\eta > 0$. Choose $\delta > 0$ from uniform continuity of f_0 corresponding to $\eta/3$. Let $x_n \in X_0$ with $x_n \to x$. Then there exists n_0 such that $d_X(x, x_{n_0}) < \delta/3$ and $d_Y\big(f(x), f_0(x_{n_0})\big) < \eta/3$. If $x' \in X$ satisfies $d_X(x, x') < \delta/3$, choose $x'_{n_0} \in X_0$ with $d_X(x', x'_{n_0}) < \delta/3$ and $d_Y\big(f(x'), f_0(x'_{n_0})\big) < \eta/3$. Then

$$d_X(x_{n_0}, x'_{n_0}) \le d_X(x_{n_0}, x) + d_X(x, x') + d_X(x', x'_{n_0}) < \delta,$$

so $d_Y\big(f_0(x_{n_0}), f_0(x'_{n_0})\big) < \eta/3$. By the triangle inequality,

$$d_Y\big(f(x), f(x')\big) \le d_Y\big(f(x), f_0(x_{n_0})\big) + d_Y\big(f_0(x_{n_0}), f_0(x'_{n_0})\big) + d_Y\big(f_0(x'_{n_0}), f(x')\big) < \eta.$$

Thus f is continuous at x, and hence continuous on X. Since δ is derived from uniform continuity of f_0, it does not depend on x and, hence, f is uniformly continuous.
Moreover, if $x \in X_0$ and we choose the constant sequence $x_n = x$, then $f(x) = \lim_n f_0(x) = f_0(x)$, so $f|_{X_0} = f_0$.
(2)⇒(1). Assume that there exists a uniformly continuous extension $f : X \to Y$ with $f|_{X_0} = f_0$. Let eps > 0. Since f is uniformly continuous on X, there exists $\delta > 0$ such that

$$d_X(x,y) < \delta \quad\Longrightarrow\quad d_Y\big(f(x), f(y)\big) < \mathrm{eps}$$

for all $x, y \in X$. In particular, for $u, v \in X_0$ with $d_X(u,v) < \delta$, we have

$$d_Y\big(f_0(u), f_0(v)\big) = d_Y\big(f(u), f(v)\big) < \mathrm{eps}\,.$$

Hence f_0 is uniformly continuous on X_0.
Uniqueness. If $g : X \to Y$ is another uniformly continuous extension with $g|_{X_0} = f_0$, then f and g agree on the dense subset $X_0 \subseteq X$. Let $x \in X$ and choose $(x_n) \subseteq X_0$ with $x_n \to x$. By continuity,

$$f(x) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} g(x_n) = g(x).$$

Thus $f = g$. □

A normed linear space that is complete is called a Banach space. We complete normed spaces to Banach space.

Corollary 4.2.3 (Completion of normed spaces)**.** *Let $(X_0, \|\cdot\|_0)$ be a normed space. Then there exist a Banach space $(X, \|\cdot\|)$ and a linear isometry $J : X_0 \to X$ with dense range.*

Proof. Apply the metric completion in Theorem 4.2.1 to $d_0(x,y) := \|x - y\|_0$. Realize X as Cauchy sequences in X_0 modulo $\sim$. Define $[(x_n)_{n\in\mathbb{N}}] + [(y_n)_{n\in\mathbb{N}}] := [(x_n + y_n)_{n\in\mathbb{N}}]$ and $\alpha[(x_n)_{n\in\mathbb{N}}] := [(\alpha x_n)_{n\in\mathbb{N}}]$; these are well-defined because $\|\cdot\|_0$ is subadditive and homogeneous.

Then X becomes a normed space with

$$\|[(x_n)_{n\in\mathbb{N}}]\| := \lim_{n\to\infty} \|x_n\|_0,$$

and completeness follows from completeness of the metric completion. $\square$

A continuous linear map is also uniformly continuous. Therefore, every continuous linear map lifts uniquely to the completions.

Theorem 4.2.4 (Extension of bounded linear maps to the completion)**.** *Let X_0 and Y_0 be normed vector spaces, and let X and Y be their completions. Let $f_0 : X_0 \to Y_0$ be bounded linear. Then there exists a unique bounded linear extension $f : X \to Y$ such that $f|_{X_0} = f_0$, and it satisfies $\|f\| = \|f_0\|$.*

Proof. Assume that f_0 is bounded linear. For $x \in X$, choose a sequence $(x_n)_{n\in\mathbb{N}} \subseteq X_0$ with $x_n \to x$ in X (possible since X_0 is dense). Then (x_n) is Cauchy in X_0, hence

$$\|f_0(x_n) - f_0(x_m)\| = \|f_0(x_n - x_m)\| \le \|f_0\| \, \|x_n - x_m\| \xrightarrow[n,m\to\infty]{} 0,$$

so $(f_0(x_n))$ is Cauchy in Y_0 and therefore converges in the completion Y. Define

$$f(x) := \lim_{n\to\infty} f_0(x_n) \in Y.$$

Well-definedness. If $(x'_n) \subseteq X_0$ is another sequence with $x'_n \to x$, then $x_n - x'_n \to 0$, and

$$\|f_0(x_n) - f_0(x'_n)\| = \|f_0(x_n - x'_n)\| \le \|f_0\| \, \|x_n - x'_n\| \to 0,$$

so both sequences have the same limit in Y. Hence f is well-defined.
Linearity. Let $x, y \in X$ and $\alpha, \beta \in \mathbb{K}$. Choose $x_n \to x$ and $y_n \to y$ in X_0. Then $\alpha x_n + \beta y_n \to \alpha x + \beta y$ and by linearity of f_0,

$$f(\alpha x + \beta y) = \lim_{n\to\infty} f_0(\alpha x_n + \beta y_n) = \alpha \lim_{n\to\infty} f_0(x_n) + \beta \lim_{n\to\infty} f_0(y_n) = \alpha f(x) + \beta f(y).$$

Boundedness and norm identity. For $x \in X$, choose $x_n \to x$ in X_0. Then

$$\|f(x)\| = \left\| \lim_{n\to\infty} f_0(x_n) \right\| = \lim_{n\to\infty} \|f_0(x_n)\| \le \|f_0\| \lim_{n\to\infty} \|x_n\| = \|f_0\| \, \|x\|.$$

Thus $\|f\| \le \|f_0\|$. On the other hand, since $f|_{X_0} = f_0$ and $X_0 \subseteq X$,

$$\|f_0\| = \sup_{x\in X_0\setminus\{0\}} \frac{\|f(x)\|}{\|x\|} \le \|f\|.$$

Hence $\|f\| = \|f_0\|$.
Moreover, if $x \in X_0$ and $x_n = x$, then

$$f(x) = \lim_{n\to\infty} f_0(x) = f_0(x),$$

so $f|_{X_0} = f_0$.
Uniqueness. Let $g : X \to Y$ be another bounded linear extension of f_0. For $x \in X$, choose $x_n \in X_0$ with $x_n \to x$. Since f and g are continuous and agree on X_0,

$$g(x) = \lim_{n\to\infty} g(x_n) = \lim_{n\to\infty} f_0(x_n) = \lim_{n\to\infty} f(x_n) = f(x).$$

Hence $g = f$. $\square$

The space $\ell^p(\mathbb{N})$ are

$$\ell^p(\mathbb{N}) = \{(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{C} : \sum_{n\in\mathbb{N}} |x_n|^p < \infty\}, \quad \|x\|_p = \Big(\sum_{n\in\mathbb{N}} |x_n|^p\Big)^{1/p}, \qquad 1 \le p < \infty,$$
$$\ell^\infty(\mathbb{N}) = \{(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{C} : \sup_{n\in\mathbb{N}} |x_n| < \infty\}, \quad \|x\|_\infty = \sup_{n\in\mathbb{N}} |x_n|.$$

Theorem 4.2.5. *For $1 \le p \le \infty$, the space $\ell^p(\mathbb{N})$ is complete.*

Proof. Let $(x^{(k)})$ be a Cauchy sequence in $\ell^p(\mathbb{N})$, $x^{(k)} = (x_n^{(k)})_n$.
We first assume $1 \le p < \infty$. For fixed n,

$$|x_n^{(k)} - x_n^{(m)}| \le \|x^{(k)} - x^{(m)}\|_p,$$

so $(x_n^{(k)})_k$ converges. Define $x_n := \lim_k x_n^{(k)}$ and $x = (x_n)_n$.
Since $(x^{(k)})$ is Cauchy, it is bounded: $\|x^{(k)}\|_p \le M$. For each N,

$$\sum_{n=1}^N |x_n|^p = \lim_{k\to\infty} \sum_{n=1}^N |x_n^{(k)}|^p \le M^p,$$

hence $x \in \ell^p$.
Letting $m \to \infty$ in $\|x^{(k)} - x^{(m)}\|_p < \varepsilon$ yields $\|x^{(k)} - x\|_p \le \varepsilon$ for k large. Thus $x^{(k)} \to x$ in ℓ^p.
For the case $p = \infty$, we observe that, for each n,

$$|x_n^{(k)} - x_n^{(m)}| \le \|x^{(k)} - x^{(m)}\|_\infty,$$

so $x_n^{(k)}$ converges; define $x_n = \lim_k x_n^{(k)}$.
Since $(x^{(k)})$ is Cauchy, it is bounded: $\|x^{(k)}\|_\infty \le M$, hence $|x_n| \le M$ and $x \in \ell^\infty(\mathbb{N})$.
Finally,

$$\|x^{(k)} - x\|_\infty = \sup_n |x_n^{(k)} - x_n| \le \varepsilon$$

for k large, so $x^{(k)} \to x$ in $\ell^\infty(\mathbb{N})$. □

An inner product space that is complete is called a Hilbert space.

Corollary 4.2.6 (Completion of inner product spaces)**.** *Let $(H_0, \langle\cdot,\cdot\rangle_0)$ be an inner product space. Then there exist a Hilbert space $(H, \langle\cdot,\cdot\rangle)$ and a linear isometry $J : H_0 \to H$ with dense range.*

Proof. Apply the previous corollary to the induced norm $\|x\|_0 = \sqrt{\langle x, x\rangle_0}$. The resulting Banach space H satisfies the parallelogram law (by continuity, since it holds on the dense subspace $J(H_0)$). Hence the polarization identity defines an inner product on H that induces $\|\cdot\|$, making H a Hilbert space, and J preserves the inner product. □

In general completions yield equivalence classes. Only in special constructions we recover genuine functions as we shall see next.

- isometry for norms?

- polarization
- orthogonal complements, projections

Essentials

Metric spaces, normed spaces, and inner product spaces can be completed to form complete metric spaces, Banach spaces, and Hilbert spaces. This completion formally consists of equivalence classes of Cauchy sequences. The original space embeds isometrically into its completion.

C Every metric, normed, or inner product space can be completed.

polarization, orthogonal complements

4.3 Hilbert spaces

Corollary 4.3.1. *The space $\ell^2(\mathbb{N})$ is a Hilbert space.*

Proof. We have seen that it is complete and straight-forward calculations reveal that

$$\langle x, y\rangle = \sum_{n\in\mathbb{N}} x_n \overline{y_n}, \qquad x, y \in \ell^2(\mathbb{N}),$$

is an inner product. $\square$

Definition 4.3.2 (Orthonormal basis)**.** *Let H be a Hilbert space. A family $(\phi_n)_{n\in\mathbb{N}} \subset H$ is called an* orthonormal basis *of H if*

$$\langle \phi_i, \phi_j\rangle = \delta_{ij} \quad \textit{for all } i, j \in I,$$

and

$$\overline{\operatorname{span}}\{\phi_n : n \in \mathbb{N}\} = H.$$

Theorem 4.3.3 (Expansion in an orthonormal basis)**.** *Let H be an infinite-dimensional Hilbert space and let $(\phi_n)_{n\in\mathbb{N}}$ be an orthonormal basis of H. Then for every $x \in H$ we have*

$$x = \sum_{n\in\mathbb{N}} \langle x, \phi_n\rangle \phi_n,$$

where the series converges in the norm of H.

Proof. Fix $x \in H$ and define the partial sums

$$s_N := \sum_{n=1}^{N} \langle x, \phi_n\rangle \phi_n \qquad (N \in \mathbb{N}).$$

For $m \le N$ we compute, using orthonormality,

$$\langle x - s_N, \phi_m \rangle = \langle x, \phi_m \rangle - \sum_{n=1}^{N} \langle x, \phi_n \rangle \langle \phi_n, \phi_m \rangle = \langle x, \phi_m \rangle - \langle x, \phi_m \rangle = 0.$$

Hence $x - s_N \perp \operatorname{span}\{\phi_1, \ldots, \phi_N\}$, and by the Pythagorean theorem,

$$\|x\|^2 = \|x - s_N\|^2 + \|s_N\|^2. \tag{4.1}$$

Moreover,

$$\|s_N\|^2 = \left\langle \sum_{n=1}^{N} \langle x, \phi_n \rangle \phi_n, \ \sum_{m=1}^{N} \langle x, \phi_m \rangle \phi_m \right\rangle = \sum_{n=1}^{N} |\langle x, \phi_n \rangle|^2.$$

Therefore (4.1) yields

$$\|x - s_N\|^2 = \|x\|^2 - \sum_{n=1}^{N} |\langle x, \phi_n \rangle|^2 \ge 0. \tag{4.2}$$

Since $(\phi_n)_{n \in \mathbb{N}}$ is an orthonormal *basis*, the subspace $\operatorname{span}\{\phi_n : n \in \mathbb{N}\}$ is dense in H. Thus for every $\varepsilon > 0$ there exists y in this span with $\|x - y\| < \varepsilon$. Writing $y = \sum_{n=1}^{N} a_n \phi_n$ and using orthonormality, we get

$$\|x - s_N\| \le \|x - y\| < \varepsilon,$$

because s_N is the orthogonal projection of x onto $\operatorname{span}\{\phi_1, \ldots, \phi_N\}$ (equivalently, it is the unique element of that subspace minimizing the distance to x). Hence $\|x - s_N\| \to 0$ as $N \to \infty$, which proves

$$x = \lim_{N \to \infty} s_N = \sum_{n=1}^{\infty} \langle x, \phi_n \rangle \phi_n.$$

□

Example 4.3.4 (Orthonormal basis of $\ell^2(\mathbb{N})$)**.** *Let*

$$\ell^2(\mathbb{N}) = \left\{ x = (x_n)_{n \in \mathbb{N}} : \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\}$$

with inner product

$$\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \overline{y_n}.$$

For each $k \in \mathbb{N}$ define the sequence $e_k \in \ell^2(\mathbb{N})$ by

$$e_k = (0, 0, \ldots, 0, 1, 0, \ldots),$$

where the entry 1 appears in the k-th position.
Then $(e_k)_{k \in \mathbb{N}}$ is an orthonormal system, since

$$\langle e_k, e_m \rangle = \delta_{km}.$$

Moreover, for any $x = (x_n)_{n \in \mathbb{N}} \in \ell^2(\mathbb{N})$ the partial sums

$$s_N = \sum_{k=1}^{N} \langle x, e_k \rangle e_k = \sum_{k=1}^{N} x_k e_k = (x_1, \ldots, x_N, 0, 0, \ldots)$$

satisfy

$$\|x - s_N\|^2 = \sum_{n>N} |x_n|^2 \longrightarrow 0 \quad (N \to \infty).$$

Hence

$$x = \sum_{k=1}^{\infty} x_k e_k,$$

so $(e_k)_{k\in\mathbb{N}}$ *is an orthonormal basis of* $\ell^2(\mathbb{N})$.

Lemma 4.3.5 (Orthogonal decomposition)**.** *Let* H *be a Hilbert space and let* $M \subseteq H$ *be a closed linear subspace. Then*

$$H = M \oplus M^\perp.$$

Proof. Let $x \in H$ and set

$$d := \inf_{m\in M} \|x - m\|.$$

Choose $(m_n) \subseteq M$ with $\|x - m_n\| \to d$. By the parallelogram identity,

$$\|m_n - m_k\|^2 = 2\|x - m_n\|^2 + 2\|x - m_k\|^2 - 4\left\|x - \frac{m_n + m_k}{2}\right\|^2.$$

Since $(m_n + m_k)/2 \in M$, we have

$$\left\|x - \frac{m_n + m_k}{2}\right\| \geq d.$$

Hence

$$\|m_n - m_k\|^2 \leq 2\|x - m_n\|^2 + 2\|x - m_k\|^2 - 4d^2 \to 0.$$

Thus (m_n) is Cauchy. Since M is closed in the Hilbert space H, there exists $m \in M$ with $m_n \to m$. By continuity of the norm,

$$\|x - m\| = d.$$

Set $u := x - m$. We show that $u \in M^\perp$. For $v \in M$ and $t \in \mathbb{R}$, the minimizing property of m gives

$$\|u\|^2 \leq \|u - tv\|^2 = \|u\|^2 - 2t\langle u, v\rangle + t^2\|v\|^2.$$

Thus

$$0 \leq -2t\langle u, v\rangle + t^2\|v\|^2 \qquad \forall t \in \mathbb{R}.$$

This implies $\langle u, v\rangle = 0$. Since $v \in M$ was arbitrary, $u \in M^\perp$. Therefore

$$x = m + u \in M + M^\perp.$$

Finally, if $x = m_1 + u_1 = m_2 + u_2$ with $m_i \in M$ and $u_i \in M^\perp$, then

$$m_1 - m_2 = u_2 - u_1 \in M \cap M^\perp.$$

But $M \cap M^\perp = \{0\}$, hence $m_1 = m_2$ and $u_1 = u_2$. Thus the decomposition is unique, and

$$H = M \oplus M^\perp.$$

□

Theorem 4.3.6 (Riesz representation theorem (real Hilbert spaces)). *Let H be a real Hilbert space. For every bounded linear functional $\varphi \in H^*$ there exists a unique $y \in H$ such that*

$$\varphi(x) = \langle x, y \rangle \qquad \forall x \in H.$$

Moreover, $\|\varphi\| = \|y\|$.

Proof. If $\varphi = 0$, take $y = 0$. Assume $\varphi \neq 0$ and set $M := \ker(\varphi)$. Then M is a linear subspace. It is closed since φ is continuous and $M = \varphi^{-1}(\{0\})$ with $\{0\}$ closed in $\mathbb{R}$.
Since M is a closed subspace of the Hilbert space H, we have the orthogonal decomposition

$$H = M \oplus M^{\perp},$$

i.e. each $x \in H$ can be written uniquely as $x = m + u$ with $m \in M$ and $u \in M^{\perp}$.
We first show that $\dim(M^{\perp}) = 1$. Pick $u \in M^{\perp} \setminus \{0\}$. Then $\varphi(u) \neq 0$ (otherwise $u \in M \cap M^{\perp} = \{0\}$). Let $v \in M^{\perp}$ be arbitrary and set

$$w := v - \frac{\varphi(v)}{\varphi(u)} u.$$

Then $w \in M^{\perp}$ and $\varphi(w) = 0$, hence $w \in M$. Thus $w \in M \cap M^{\perp} = \{0\}$, so $v = \frac{\varphi(v)}{\varphi(u)} u$.
Therefore $M^{\perp} = \operatorname{span}\{u\}$.
Let $e := u/\|u\|$, so $\|e\| = 1$ and $M^{\perp} = \operatorname{span}\{e\}$. Given $x \in H$, write $x = m + \alpha e$ with $m \in M$.
Then

$$\alpha = \langle x, e \rangle \quad \text{since} \quad \langle x, e \rangle = \langle m, e \rangle + \alpha \langle e, e \rangle = 0 + \alpha,$$

and hence

$$\varphi(x) = \varphi(m) + \alpha\varphi(e) = \alpha\varphi(e) = \varphi(e)\,\langle x, e \rangle.$$

Define $y := \varphi(e)\, e$. Then $\varphi(x) = \langle x, y \rangle$ for all x.
Uniqueness: if $\langle x, y \rangle = \langle x, z \rangle$ for all x, then $\langle x, y - z \rangle = 0$ for all x. Taking $x = y - z$ yields $\|y - z\|^2 = 0$, so $y = z$.
Finally, by Cauchy–Schwarz,

$$|\varphi(x)| = |\langle x, y \rangle| \leq \|x\|\,\|y\| \quad \Rightarrow \quad \|\varphi\| \leq \|y\|.$$

If $y \neq 0$, choose $x = y/\|y\|$ to get $|\varphi(x)| = \|y\|$, hence $\|\varphi\| \geq \|y\|$. Therefore $\|\varphi\| = \|y\|$. $\square$

Theorem 4.3.7 (Adjoint operator (real Hilbert spaces)). *Let H, K be real Hilbert spaces and let $A \in \mathcal{L}(H, K)$ be bounded and linear. Then there exists a unique bounded linear operator $A^* \in \mathcal{L}(K, H)$ such that*

$$\langle Ax, y \rangle_K = \langle x, A^*y \rangle_H \qquad \forall x \in H,\ \forall y \in K.$$

Moreover, $\|A^*\| = \|A\|$ and $(A^*)^* = A$.

Proof. Step 1: definition via Riesz. Fix $y \in K$ and define $\varphi_y \in H^*$ by $\varphi_y(x) := \langle Ax, y \rangle_K$.

Then φ_y is linear and bounded, since for all $x \in H$,

$$|\varphi_y(x)| \leq \|Ax\|_K \, \|y\|_K \leq \|A\| \, \|x\|_H \, \|y\|_K,$$

hence $\|\varphi_y\| \leq \|A\| \, \|y\|_K$. By Theorem 4.3.6, there exists a unique vector $z \in H$ such that

$$\langle Ax, y\rangle_K = \varphi_y(x) = \langle x, z\rangle_H \qquad \forall x \in H.$$

Define $A^*y := z$. This gives the adjoint identity.
Step 2: linearity and boundedness. For $\alpha, \beta \in \mathbb{R}$ and $y_1, y_2 \in K$ we have $\varphi_{\alpha y_1 + \beta y_2} = \alpha\varphi_{y_1} + \beta\varphi_{y_2}$. By uniqueness of the Riesz representative,

$$A^*(\alpha y_1 + \beta y_2) = \alpha A^* y_1 + \beta A^* y_2,$$

so A^* is linear. Moreover, by Theorem 4.3.6,

$$\|A^*y\|_H = \|\varphi_y\| \leq \|A\| \, \|y\|_K,$$

hence A^* is bounded and $\|A^*\| \leq \|A\|$.
Step 3: norm equality. For $x \in H$,

$$\|Ax\|_K = \sup_{\|y\|_K = 1} \langle Ax, y\rangle_K$$

(since $\|v\| = \sup_{\|y\|=1}\langle v, y\rangle$ in a real Hilbert space, with equality attained at $y = v/\|v\|$ if $v \neq 0$). Using the adjoint identity,

$$\|Ax\|_K = \sup_{\|y\|_K=1} \langle x, A^*y\rangle_H \leq \sup_{\|y\|_K=1} \|x\|_H \, \|A^*y\|_H \leq \|x\|_H \, \|A^*\|.$$

Taking the supremum over $\|x\|_H = 1$ gives $\|A\| \leq \|A^*\|$. Together with $\|A^*\| \leq \|A\|$, we obtain $\|A^*\| = \|A\|$.
Step 4: involution. Fix $x \in H$ and define $\psi_x \in K^*$ by $\psi_x(y) := \langle Ax, y\rangle_K$. By Theorem 4.3.6 in K, the unique representing vector of ψ_x is Ax, i.e.

$$\psi_x(y) = \langle Ax, y\rangle_K \qquad \forall y \in K.$$

On the other hand, by the adjoint identity,

$$\psi_x(y) = \langle x, A^*y\rangle_H \qquad \forall y \in K.$$

Applying Theorem 4.3.6 to the bounded functional $y \mapsto \langle x, A^*y\rangle_H$ on K (i.e. applying the construction of the adjoint to A^*) shows that its representing vector is $(A^*)^*x$. By uniqueness of the Riesz representative in K, we must have $(A^*)^*x = Ax$ for all x, hence $(A^*)^* = A$.
Uniqueness of A^* is immediate: if $B : K \to H$ also satisfies $\langle Ax, y\rangle_K = \langle x, By\rangle_H$ for all x, y, then for each y we have $\langle x, (A^* - B)y\rangle_H = 0$ for all x, hence $(A^* - B)y = 0$, so $A^* = B$. $\square$

Reproducing kernel Hilbert spaces

We introduce a special class of Hilbert spaces that do not arise as abstract completions of inner product spaces but whose completion consists of functions and not equivalence classes. While classical in origin, this completion is harder to find in the literature executed in full detail, particularly because semi-definite kernels are admitted.

Let $\mathbb{F}$ be either the real or complex numbers.

Definition 4.3.1 (RKHS). *Given $\emptyset \neq X$, a Hilbert space $H(X)$ of $\mathbb{F}$-valued functions on X, $f : X \to \mathbb{F}$, is called a* reproducing kernel Hilbert space (RKHS) *if there is $K : X \times X \to \mathbb{F}$ satisfying*

(a) $K_x \in H(X)$, *where* $K_x : X \to \mathbb{F}, \quad y \mapsto K(x, y)$,

(b) $f(x) = \langle f, K_x \rangle_{H(X)}, \ \forall x \in X, \ f \in H(X)$.

This K is called the reproducing kernel of $H(X)$.

Note that (b) implies

$$K(x, y) = \langle K_x, K_y \rangle_{H(X)}, \qquad x, y \in X.$$

Example 4.3.2. *For $k \in \mathbb{Z}$, the function*

$$x \mapsto \mathrm{e}_k(x) := e^{2\pi \mathrm{i} k x}$$

is $\mathbb{Z}$ periodic, hence well-defined on the torus $\mathbb{T} := \mathbb{R} / \mathbb{Z}$. We therefore consider $\mathrm{e}_k : \mathbb{T} \to \mathbb{C}$ and observe that

$$\{\mathrm{e}_k : |k| \leq n\}$$

is an orthonormal system with respect to the inner product

$$\langle f, g \rangle_{L^2(\mathbb{T})} := \int_{-1/2}^{1/2} f(x) \overline{g(x)} \mathrm{d}x.$$

The Dirichlet kernel

$$K(x, y) = \sum_{|k| \leq n} \mathrm{e}_k(x - y) = \sum_{|k| \leq n} \overline{\mathrm{e}_k(x)} \mathrm{e}_k(y)$$

reproduces the trigonometric polynomials $\Pi_n = \operatorname{span}\{\mathrm{e}_k : |k| \leq n\}$.

For $x \in X$, let $\delta_x : H(X) \to \mathbb{K}$, $f \mapsto f(x)$ denote the point evaluation.

Theorem 4.3.3 (RKHS). *Given $\emptyset \neq X$, let $H(X)$ be a Hilbert space of $\mathbb{F}$-valued functions on X. Then we have*

$$H(X) \text{ is a RKHS} \qquad \Leftrightarrow \qquad \delta_x \text{ continuous} \quad \forall x \in X.$$

Proof. ($\Leftarrow$) If δ_x is continuous, then the Riesz representation theorem yields that there is a unique $K_x \in H(X)$ such that

$$f(x) = \delta_x(f) = \langle f, K_x \rangle_H.$$

Define now $K(x, y) := K_x(y)$.

($\Rightarrow$) If $H(X)$ is a RKHS, then

$$|\delta_x(f)| = |\langle f, K_x\rangle_H| \leq \|K_x\|_H \|f\|_H \quad \forall f \in H(X).$$

□

Proposition 4.3.4. *If $H(X)$ is a RKHS, then*

$$f_n \to f \qquad \Rightarrow \qquad f_n(x) \to f(x) \quad \forall x \in X.$$

Proof. For $f_n \to f$, we have $|f_n(x) - f(x)| = |\delta_x(f_n - f)| \leq \|K_x\|_H \|f_n - f\|_H \to 0$. □

Proposition 4.3.5. *If $H(X)$ is a RKHS, then*

(a) its reproducing kernel K is unique,

(b) $\operatorname{span}\{K_x : x \in X\} \subseteq H(X)$ *is dense,*

(c) the inner product satisfies

$$\left\langle \sum_{i=1}^n a_i K_{x_i}, \sum_{j=1}^m b_j K_{y_j} \right\rangle = \sum_{i=1}^n \sum_{j=1}^m a_i \overline{b_j} K(x_i, y_j). \tag{4.3}$$

(d) $K(x,y) = \overline{K(y,x)}$ $\forall x, y \in X$,

(e) for any $n \in \mathbb{N}$ and $(x_j)_{j=1}^n \subseteq X$,

$$\mathbb{K}^{n\times n} \ni (K(x_i, x_j))_{i,j=1}^n \succeq 0.$$

Proof. (a) Let K and L be reproducing kernels of $H(X)$. Then we have

$$\begin{aligned}\|K_x - L_x\|^2 &= \langle K_x - L_x, K_x - L_x\rangle = \langle K_x - L_x, K_x\rangle - \langle K_x - L_x, L_x\rangle \\ &= K(x,x) - L(x,x) - K(x,x) + L(x,x) = 0.\end{aligned}$$

From $K_x = L_x$ in $H(X)$, for all $x \in X$, we deduce

$$K(x,y) = \delta_y(K_x) = \delta_y(L_x) = L(x,y), \quad \forall x, y \in X.$$

(b) If $H(X) \ni f \perp K_x$, for all $x \in X$, then

$$0 = \langle f, K_x\rangle = f(x), \quad \forall x \in X.$$

(c) The identity (4.3) follows from $\langle K_{x_i}, K_{y_j}\rangle = K(x_i, y_j)$.

(d) We have $K(x,y) = K_x(y) = \langle K_x, K_y\rangle = \overline{\langle K_y, K_x\rangle} = \overline{K_y(x)} = \overline{K(y,x)}$.

(e) For $(c_i)_{i=1}^n \subset \mathbb{K}$ and $(x_i)_{i=1}^n \subset X$, we have

$$\sum_{i,j=1}^n c_i\overline{c_j}K(x_i,x_j) = \sum_{i,j=1}^n c_i\overline{c_j}\langle K_{x_i}, K_{x_j}\rangle = \left\langle \sum_{i=1}^n c_i K_{x_i}, \sum_{j=1}^n c_j K_{x_j} \right\rangle$$
$$= \left\| \sum_{i=1}^n c_i K_{x_i} \right\|^2 \geq 0.$$

□

We now start with K.

Definition 4.3.6. *We call $K\colon X \times X \to \mathbb{F}$ a* positive semi-definite kernel *if*

- $K(x,y) = \overline{K(y,x)} \quad \forall x,y \in X,$
- *for any $n \in \mathbb{N}$ and $(x_j)_{j=1}^n \subset X$,*

$$\mathbb{K}^{n\times n} \ni (K(x_i,x_j))_{i,j=1}^n \succeq 0.$$

Lemma 4.3.7. *Let H be an inner product space and $(f_n)_{n\in\mathbb{N}} \subseteq H$ a Cauchy sequence. If there is $f \in H$ such that $\langle f_n, g\rangle \to \langle f, g\rangle$ for all g in a dense subset $H_0 \subseteq H$, then $f_n \to f \in H$.*

Proof. We split the proof into two steps. We first verify that $f_n \to f$ weakly and then argue that Cauchy sequences that converge weakly also converge strongly.
a) Any Cauchy sequence is bounded, so that there is $M > 0$ such that $\|f_n - f\| \leq M$. Let $g \in H$ be arbitrary. There is $g_n \in H_0$ such that $\|g - g_n\| \leq \frac{1}{n}$. Given $\varepsilon > 0$, choose m such that $\frac{M}{m} \leq \varepsilon/2$. We compute

$$\begin{aligned}|\langle f_n - f, g\rangle| &\leq |\langle f_n - f, g - g_m\rangle| + |\langle f_n - f, g_m\rangle| \leq \|f_n - f\|\|g - g_m\| + |\langle f_n - f, g_m\rangle| \\ &\leq \frac{M}{m} + |\langle f_n - f, g_m\rangle| \\ &\leq \varepsilon/2 + |\langle f_n - f, g_m\rangle|.\end{aligned}$$

We observe $\lim_{n\to\infty} |\langle f_n - f, g_m\rangle| = 0$, so that $|\langle f_n - f, g\rangle| \leq \varepsilon$ for sufficiently large n.
b) We now have a Cauchy sequence $(f_n)_{n\in\mathbb{N}} \subseteq H$ that converges weakly towards $f \in H$. Given $\varepsilon > 0$, $\|f_n - f_m\| \leq \varepsilon$ for sufficiently large n and m. We compute

$$\begin{aligned}\|f_n - f\|^2 &= \langle f_n - f, f_n - f\rangle \\ &\leq |\langle f_n - f, f_n - f_m\rangle| + |\langle f_n - f, f_m - f\rangle| \\ &\leq \|f_n - f\|\|f_n - f_m\| + |\langle f_n - f, f_m - f\rangle| \\ &\leq M\|f_n - f_m\| + |\langle f_n - f, f_m - f\rangle| \\ &\leq M\varepsilon + |\langle f_n - f, f_m - f\rangle|.\end{aligned}$$

This also holds for $m \to \infty$, so that $\lim_{m\to\infty} |\langle f_n - f, f_m - f\rangle| = 0$ implies

$$\|f_n - f\|^2 \leq M\varepsilon.$$

This leads to $f_n \to f$. □

Theorem 4.3.8. *If $K : X \times X \to \mathbb{F}$ is a positive semi-definite kernel, then there is a RKHS $H_K(X)$ such that K is its reproducing kernel.*

Our assumptions here are slightly weaker than those in [22, Theorem 10.7] as we do not ask for definiteness but just semi-definiteness. Moreover, we define the space explicitly, in a way that does not require any abstract completion, nor equivalence classes.

Proof. The proof is split into three steps. We first construct a suitable pre-Hilbert space. The second step is to derive a completion by functions and not just any abstract completion. The third step is dedicated to verifying that the space defined in the second step is indeed complete.
a) We define $H := \operatorname{span}\{K_x : x \in X\}$ with inner product

$$\left\langle \sum_{i=1}^{n} a_i K_{x_i}, \sum_{j=1}^{m} b_j K_{y_j} \right\rangle_H := \sum_{i=1}^{n}\sum_{j=1}^{m} a_i \overline{b_j} K(x_i, y_j). \tag{4.4}$$

We first check that (4.4) is well-defined. For $f = \sum_{i=1}^{n} a_i K_{x_i}$ and $g = \sum_{j=1}^{m} b_j K_{y_j}$, we observe

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_i \overline{b_j} K(x_i, y_j) = \sum_{j=1}^{m} \overline{b_j} f(y_j),$$

so that there is no dependency on the particular choice of a_i and x_i to represent f. The analogous fact holds for g by observing

$$\sum_{i=1}^{n}\sum_{j=1}^{m} a_i \overline{b_j} K(x_i, y_j) = \overline{\sum_{i=1}^{n}\sum_{j=1}^{m} \overline{a_i} b_j K(y_j, x_i)} = \overline{\sum_{i=1}^{n} \overline{a_i} g(x_i)} = \sum_{i=1}^{n} a_i \overline{g(x_i)}.$$

Since K is a positive semi-definite kernel, (4.4) is a positive semi-definite sesquilinear form. To check for definiteness, for $f = \sum_{i=1}^{n} a_i K_{x_i}$, assume

$$0 = \langle f, f \rangle_H.$$

We aim to verify $f(x) = 0 \ \forall x \in X$. The reproducing property holds in H since

$$\langle f, K_x \rangle_H = \sum_{i=1}^{n} a_i \langle K_{x_i}, K_x \rangle = \sum_{i=1}^{n} a_i K(x_i, x) = f(x).$$

The Cauchy-Schwartz inequality also holds for positive semi-definite sesquilinear forms, so that

$$|f(x)| = |\langle f, K_x \rangle_H| \leq \|f\|_H \|K_x\|_H = 0.$$

Thus, (**??**) is a positive definite sesquilinear form. We also observe that the point evaluation in H is continuous.
b) We now derive the completion of H by functions on X. For $f = \sum_{i=1}^{n} a_i K_{x_i} \in H$, we observe

$$\langle f, K_x \rangle_H = \sum_{i=1}^{n} a_i \langle K_{x_i}, K_x \rangle_H = \sum_{i=1}^{n} a_i K(x_i, x) = f(x).$$

Let $(f_n)_{n\in\mathbb{N}} \subset H$ be a Cauchy sequence. Cauchy-Schwartz yields

$$\begin{aligned}|f_n(x) - f_m(x)| &= |\langle f_n, K_x\rangle_H - \langle f_m, K_x\rangle_H| \\ &= |\langle f_n - f_m, K_x\rangle_H| \\ &\le \|f_n - f_m\|_H \|K_x\|_H.\end{aligned}$$

Thus, $(f_n(x))_{n\in\mathbb{N}} \subset \mathbb{K}$ is Cauchy, so that there is $f : X \to \mathbb{K}$ with $f_n(x) \to f(x)$. We collect all such f into

$$H_K(X) := \{f : X \to \mathbb{K} \mid \exists (f_n)_{n\in\mathbb{N}} \subset H \text{ Cauchy sequence}, f_n(x) \to f(x)\, \forall x \in X\}$$

We observe $H \subseteq H_K(X)$ by taking the constant sequence $f_n = f \in H$. For $f, g \in H_K(X)$, we define

$$\langle f, g\rangle := \lim_{n\to\infty} \langle f_n, g_n\rangle_H. \tag{4.5}$$

Indeed, there is convergence in (4.5) since subtracting and adding $\langle f_n, g_m\rangle$ leads to

$$\begin{aligned}|\langle f_n, g_n\rangle - \langle f_m, g_m\rangle| &= |\langle f_n, g_n - g_m\rangle + \langle f_n - f_m, g_m\rangle| \\ &\le |\langle f_n, g_n - g_m\rangle| + |\langle f_n - f_m, g_m\rangle| \\ &\le \|f_n\|\|g_n - g_m\| + \|f_n - f_m\|\|g_m\|.\end{aligned}$$

The Cauchy property yields that $\|f_n\|$ and $\|g_m\|$ are bounded and $\|g_n - g_m\|, \|f_n - f_m\| \to 0$. To verify that (4.5) is independent of the particular choice of Cauchy sequences, we consider the associated norm. The statement for the inner product then follows from the polarization formula.

Now let f_n and $\tilde{f}_n$ be Cauchy in H, so that $f_n(x), \tilde{f}_n(x) \to f(x)$. This implies that the difference $g_n := f_n - \tilde{f}_n$ is Cauchy in H and $g_n(x) \to 0$ for all $x \in X$. Lemma 4.3.7 implies that $g_n \to 0$. We derive

$$\Big| \|f_n\|_H - \|\tilde{f}_n\|_H \Big| \le \|f_n - \tilde{f}_n\|_H \to 0,$$

so that the limits of $\|f_n\|_H$ and $\|\tilde{f}_n\|_H$ coincide. Therefore, (4.5) is independent of the particular choice of Cauchy sequences.

To check that (4.5) is positive definite, we start with $0 = \|f\| = \lim_{n\to\infty} \|f_n\|_H$, so that $f_n \to 0$ in H. Continuity of the point evaluation in H yields

$$|f(x)| = \lim_{n\to\infty} |f_n(x)| = 0, \qquad x \in X,$$

so that (4.5) is indeed positive definite and $H_K(X)$ is an inner product space.

Note that (4.5) implies

$$\|f - f_n\|^2 = \langle f - f_n, f - f_n\rangle = \|f\|^2 + \underbrace{\|f_n\|^2}_{\to\|f\|^2} - \underbrace{\langle f, f_n\rangle}_{\to\|f\|^2} - \underbrace{\langle f_n, f\rangle}_{\to\|f\|^2} \to 0,$$

so that H is dense in $H_K(X)$.

The reproducing property for $f \in H_K(X)$ is also satisfied as

$$\langle f, K_x\rangle = \lim_{n\to\infty} \langle f_n, K_x\rangle_H = \lim_{n\to\infty} f_n(x) = f(x).$$

c) To check that $H_K(X)$ is complete, take a Cauchy sequence $(f_n)_{n\in\mathbb{N}} \subseteq H_K(X)$. Cauchy Schwartz implies that $(f_n(x) = \langle f_n, K_x\rangle)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{K}$, hence, convergent

towards some $f(x)$. For each n, there is a Cauchy sequence $(f_{n,m})_{m\in\mathbb{N}} \subseteq H$, such that $\lim_{m\to\infty} f_{n,m}(x) = f_n(x)$. Thus, $\langle f_{n,m} - f_n, g\rangle \to 0$ for $m \to \infty$ and all $g \in H$. Since $(f_{n,m})_{m\in\mathbb{N}}$ is a Cauchy sequence, it is bounded and the density of $H \subseteq H_K(X)$ leads to weak convergence of $f_{n,m} \to f_n$ in $H_K(X)$.
By Lemma 4.3.7, applied to the Cauchy sequence $(f_{n,m})_m \subset H_K(X)$ with weak limit f_n, we obtain $f_{n,m} \to f_n$ strongly.
For each $n \in \mathbb{N}$, there is m_n such that $g_n := f_{n,m_n}$ satisfies $\|f_n - g_n\| \le \frac{1}{n}$. Obviously, $g_n \in H$ and

$$\|g_n - g_m\| \le \|g_n - f_n\| + \|f_n - f_m\| + \|f_m - g_m\|,$$

so that $(g_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in H. To check that $g_n(x) \to f(x)$, we estimate

$$\begin{aligned}|g_n(x) - f(x)| &\le |g_n(x) - f_n(x)| + |f_n(x) - f(x)| \\ &\le |\langle g_n - f_n, K_x\rangle| + |f_n(x) - f(x)| \\ &\le \|g_n - f_n\|\|K_x\| + |f_n(x) - f(x)| \\ &\le \frac{1}{n}\|K_x\| + |f_n(x) - f(x)| \to 0.\end{aligned}$$

Therefore, we have verified that $f \in H_K(X)$. For every $g = \sum_{j=1}^m b_j K_{y_j} \in H$, the reproducing property yields

$$\langle f_n - f, g\rangle = \sum_{j=1}^m \overline{b_j}\,(f_n(y_j) - f(y_j)) \to 0.$$

Since H is dense in $H_K(X)$, this is weak convergence on a dense subspace. Thus, Lemma 4.3.7 implies strong convergence $f_n \to f \in H_K(X)$. □

Corollary 4.3.9. *If $K : X \times X \to \mathbb{F}$ is a continuous, positive semi-definite kernel such that*

$$\sup_{x\in X} K(x,x) < \infty,$$

then $H_K(X) \hookrightarrow \mathcal{C}_b(X)$.

Proof. Let $H := \operatorname{span}\{K_x : x \in X\}$ be as in the proof of Theorem 4.3.8. For $f \in H_K(X)$, there is $f_n \in H \subseteq \mathcal{C}(X)$ with $\|f_n - f\| \to 0$. We obtain

$$|f(x) - f_n(x)| = |\langle f - f_n, K_x\rangle| \le \|f - f_n\| \underbrace{\|K_x\|}_{\sqrt{K(x,x)}} \le \|f - f_n\|\sqrt{M},$$

where $M = \sup_{x\in X} |K(x,x)|$. Uniform convergence implies $f \in \mathcal{C}(X)$. Analogously, we derive

$$|f(x)| = |\langle f, K_x\rangle|\|f\|\sqrt{K(x,x)} \le \|f\|M.$$ □

Proposition 4.3.10. *Given a Hilbert space H and $F : X \to H$, then*

$$K(x,y) := \langle F(x), F(y)\rangle_H \tag{4.6}$$

is a positive semi-definite kernel.
If K is a positive semi-definite kernel on X, then there is some Hilbert space H and map $F : X \to H$ such that (4.6) *holds.*

The function F is the above proposition is often called feature map.

Proof. It is obvious that K defined by (4.6) is a positive semi-definite kernel.
For the reverse direction, we may define $F(x) := K_x$, so that Theorem 4.3.8 concludes the proof. $\square$

Approximation

Lemma 4.3.11. *If K is a positive definite kernel on Ω and $\emptyset \neq X \subseteq \Omega$ and $\Pi_X := \overline{\operatorname{span}}\{K_x : x \in X\}$, then the following holds:*

(a) The orthogonal complement of Π_X is given by

$$(\Pi_X)^\perp = \{f \in H_K(\Omega) : f(x) = 0 \quad \forall x \in X\}.$$

(b) The orthogonal projection $P_X f$ of f onto Π_X interpolates f, i.e.,

$$(P_X f)(x) = f(x), \qquad \forall x \in X.$$

Proof. (a) If $f \in (\Pi_X)^\perp$, then, for all $x \in X$,

$$0 = \langle f, K_x \rangle = f(x).$$

For the reverse inclusion, assume that $f \in H_K(\Omega)$ satisfies $f(x) = 0$ for all $x \in X$. We directly observe

$$\langle f, K_x \rangle = f(x) = 0.$$

Hence, f is also orthogonal to each finite linear combinations and therefore $f \in \operatorname{span}\{K_x : x \in X\}^\perp = (\Pi_X)^\perp$.
(b) The decomposition $H_K(\Omega) = \Pi_X \oplus (\Pi_X)^\perp$ yields that every $f \in H_K(\Omega)$ may be written as $f = P_X f + g$ with $g \in (\Pi_X)^\perp$. This implies

$$f(x) = \langle f, K_x \rangle = \langle P_X f, K_x \rangle + \langle g, K_x \rangle = (P_X f)(x) + g(x) = (P_X f)(x), \quad \forall x \in X.$$

$\square$

Given a metric space Ω and a closed subset $X \subseteq \Omega$, the covering radius ρ_X is defined by

$$\rho_X = \sup_{y \in \Omega} \operatorname{dist}(y, X)$$

and refers to the radius of the largest hole in X with respect to Ω.

Theorem 4.3.12. *Assume that K is a continuous, positive definite kernel, and, for $n \in \mathbb{N}$,* *the closed subsets* $X_n \subseteq \Omega$ *satisfy* $\lim_{n\to\infty} \rho_{X_n} = 0$. *Then*

$$\lim_{n\to\infty} \|f - P_{X_n} f\|_K = 0, \qquad \forall f \in H_K(\Omega).$$

Proof. It is sufficient to verify the claim on a dense subset. Therefore, we choose $f = \sum_{j=1}^{N} c_j K_{y_j}$, for $y_j \in \Omega$. The definition of the covering radius ensures that $\operatorname{dist}(y_j, X_n) \le \rho_{X_n}$. Thus, there are $x_n^{(j)} \in X_n$ such that $\operatorname{dist}(y_j, x_n^{(j)}) \le \rho_{X_n}$. Since $g_n := \sum_{j=1}^{N} c_j K_{x_n^{(j)}} \in \Pi_{X_n}$, we derive

$$\begin{aligned} \|f - P_{X_n} f\|_K &\le \|f - g\|_K \\ &= \Big\|\sum_{j=1}^{N} c_j (K_{y_j} - K_{x_n^{(j)}})\Big\|_K \\ &\le \sum_{j=1}^{N} |c_j| \|K_{y_j} - K_{x_n^{(j)}}\|_K. \end{aligned}$$

The continuity of K and the convergence $\lim_{n\to\infty} x_n^{(j)} = y_j$ lead to $\|f - P_{X_n} f\|_K \to 0$. □

If K additionally satisfies $\sup_{x\in\Omega} K(x,x) < \infty$, then Corollary 4.3.9 yields that convergence also holds in the supremum norm.

Essentials

A Hilbert space can be decomposed into orthogonal complements, and every orthonormal basis provides a series expansion. According to the Riesz representation theorem, every bounded linear functional on a Hilbert space arises from the inner product. Adjoint operators therefore exist even in infinite-dimensional Hilbert spaces. A reproducing kernel Hilbert space (RKHS) is a Hilbert space of real- or complex-valued functions in which point evaluation is continuous. Unlike Hilbert spaces obtained through abstract completion via equivalence classes, a RKHS consists of genuinely pointwise defined functions. The functions in the space are concrete objects, not merely representatives. Every positive semi-definite kernel determines such a space uniquely. The kernel encodes the geometry of the space and controls how functions are evaluated and compared. The interpolation problem for finitely many data points is solved by orthogonal projection onto the finite-dimensional subspace generated by the corresponding kernel sections.

C R T A The collection of kernel sections K_x is dense in the RKHS. Samples provide representations, and restricting to finitely many sampling points produces a finite-dimensional model. Interpolation then is a finite linear algebra problem and leads to a finite-dimensional approximation.

Kernel, positive semi-definite, reproducing kernel Hilbert space (RKHS), interpolation, covering radius

We now move from global Hilbert space geometry to the local behavior of nonlinear maps. Fréchet differentiability provides the linearization principle that governs sensitivity and prepares the transition from analytic structure to computation.

4.4 Fréchet differentiability

We begin by formulating a notion of differentiability for functions between normed vector spaces that captures linear approximation in a coordinate-free way. Fréchet differentiability provides a natural extension of the classical derivative and leads to a clean chain rule and a necessary condition for local extrema.

Recall that a function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a point $x \in \mathbb{R}$ if

$$f(x+h) = f(x) + f'(x)\,h + o(|h|), \qquad h \in \mathbb{R}\,.$$

The derivative $f'(x)$ is interpreted as a linear multiplication

$$f'(x) : \mathbb{R} \to \mathbb{R}, \qquad h \mapsto f'(x)\,h,$$

and f is approximated in the proximity of $z \approx x$ by the linear map $z \mapsto f(x) + f'(x)(z-x)$. This view allows us to extend the notion of differentiability from $\mathbb{R}$ to general normed vector spaces.

Definition 4.4.1 (Fréchet differentiability)**.** *Let V and W be normed vector spaces. A function $f : V \to W$ is called* Fréchet differentiable *at $x \in V$ if there exists a continuous linear map $f'(x) : V \to W$ such that*

$$f(x+h) = f(x) + f'(x)h + o(\|h\|), \qquad h \in V.$$

Since $f'(x)$ is required to be continuous, every Fréchet differentiable function is continuous.

Remark 4.4.2 (local linearization)**.** *Differentiability formalizes the idea that a nonlinear model can be approximated locally by a linear map. It also ensures that small perturbations in the input lead to small changes in the output.*

By rearranging terms and using the definition of $o(\|h\|)$, equivalently, f is Fréchet differentiable at x if

$$\lim_{h\to 0} \frac{f(x+h) - f(x) - f'(x)h}{\|h\|} = 0.$$

Example 4.4.3. *Let $f : V \to W$ be a continuous linear map.*

(a) Then f is Fréchet differentiable with $f'(x) = f$ since $f(x+h) = fx + fh$.

(b) The affine linear function $g(x) = f(x) - b$ for some fixed $b \in W$ is Fréchet differentiable with $g'(x) = f$, because

$$g(x+h) = f(x+h) - b = fx + fh - b = g(x) + fh.$$

Example 4.4.4. *Let H be an inner product space. The map $f : H \to \mathbb{R}$ defined by $f(x) =$*

$\|x\|^2$ *is Fréchet differentiable with* $f'(x) = 2\langle x, \cdot\rangle$, *because*

$$f(x+h) = \|x+h\|^2 = \|x\|^2 + 2\langle x, h\rangle + \|h\|^2,$$

and $\lim_{h\to 0} \|h\|^2/\|h\| = 0$.

Note that in the definition of Fréchet differentiability we require the derivative $f'(x)$ to be not only linear but also continuous.

Theorem 4.4.5 (chain rule)**.** *Let V, W, W' be normed vector spaces. If $f : V \to W$ and $g : W \to W'$ are Fréchet differentiable, then the composition $g \circ f : V \to W'$ is Fréchet differentiable and*

$$(g\circ f)'(x) = g'(f(x))\, f'(x).$$

Here $f'(x) : V \to W$ and $g'(f(x)) : W \to W'$ are continuous linear maps, and their composition is again a continuous linear map from V to W'.

Proof. Define $\varphi(h)$ by

$$g(f(x+h)) = g(f(x)) + g'(f(x))\, f'(x)h + \varphi(h).$$

We must show that $\varphi(h)/\|h\| \to 0$ as $h \to 0$.
Since f is Fréchet differentiable at x, there exists a remainder $\psi(h)$ such that

$$f(x+h) = f(x) + f'(x)h + \psi(h), \qquad \frac{\|\psi(h)\|}{\|h\|} \to 0 \quad (h \to 0).$$

Substituting into g yields

$$\frac{\varphi(h)}{\|h\|} = \frac{g(f(x) + f'(x)h + \psi(h)) - g(f(x)) - g'(f(x))\, f'(x)h}{\|h\|}.$$

Set $\varepsilon_h := f'(x)h + \psi(h)$. Then

$$\frac{\varphi(h)}{\|h\|} = \frac{g(f(x)+\varepsilon_h) - g(f(x)) - g'(f(x))\,\varepsilon_h}{\|h\|} + \frac{g'(f(x))\,\psi(h)}{\|h\|}.$$

The second term tends to zero since $g'(f(x))$ is continuous and $\psi(h)/\|h\| \to 0$.
For the first term we write

$$\frac{g(f(x)+\varepsilon_h) - g(f(x)) - g'(f(x))\,\varepsilon_h}{\|h\|} = \frac{\|\varepsilon_h\|}{\|h\|} \cdot \frac{g(f(x)+\varepsilon_h) - g(f(x)) - g'(f(x))\,\varepsilon_h}{\|\varepsilon_h\|}.$$

Since $f'(x)$ is continuous, $\varepsilon_h \to 0$ as $h \to 0$, and the second factor tends to zero by the Fréchet differentiability of g. Moreover,

$$\frac{\|\varepsilon_h\|}{\|h\|} \le \|f'(x)\| + \frac{\|\psi(h)\|}{\|h\|},$$

which is bounded for small h. Hence $\varphi(h)/\|h\| \to 0$, proving the claim. □

Example 4.4.6. *If $f : V \to W$ and $g : W \to W'$ are continuous linear maps, then $g \circ f$ is also continuous and linear, hence Fréchet differentiable with*

$$(g \circ f)'(x) = g \circ f.$$

This is consistent with the chain rule since $g'(f(x)) = g$ and $f'(x) = f$.

Fréchet differentiability allows us to take directional derivatives showing how a function changes in distinct directions.

Lemma 4.4.7 (directional derivative). *If $f : V \to W$ is Fréchet differentiable at x, then*

$$f'(x)u = \lim_{\lambda \to 0} \frac{f(x + \lambda u) - f(x)}{\lambda}, \qquad u \in V.$$

Proof. Let $u \in V$ and set $h = \lambda u$. If $u = 0$, the claim is trivial. For $u \neq 0$, Fréchet differentiability gives

$$\lim_{\lambda \to 0} \frac{f(x + \lambda u) - f(x) - f'(x)(\lambda u)}{\|\lambda u\|} = 0.$$

Since $\|\lambda u\| = |\lambda| \, \|u\|$ and $f'(x)$ is linear, multiplying by $\frac{|\lambda|\|u\|}{\lambda}$ (for $\lambda \neq 0$) yields

$$\lim_{\lambda \to 0} \frac{f(x + \lambda u) - f(x) - \lambda f'(x)u}{\lambda} = 0.$$

This can be rewritten as $\lim_{\lambda \to 0} \frac{f(x+\lambda u) - f(x)}{\lambda} = f'(x)u$. □

Example 4.4.8. *Every linear map $f : V \to W$ is Fréchet differentiable with $f'(x) = f$. We now compute*

$$\lim_{\lambda \to 0} \frac{f(x + \lambda u) - f(x)}{\lambda} = \lim_{\lambda \to 0} \frac{\lambda f(u)}{\lambda} = f(u) = f'(x)u.$$

Example 4.4.9. *Consider a real inner product space H and $f : H \to \mathbb{R}$, $f(x) = \|x\|^2$. The Fréchet derivative satisfies*

$$\begin{aligned} f'(x)u &= \lim_{t \to 0} \frac{\|x + tu\|^2 - \|x\|^2}{t} \\ &= \lim_{t \to 0} \frac{\|x\|^2 + \langle x, tu \rangle + \langle tu, x \rangle + \|tu\|^2 - \|x\|^2}{t} \\ &= \lim_{t \to 0} \frac{2\langle x, tu \rangle + t^2 \|u\|^2}{t} \\ &= 2\langle x, u \rangle, \end{aligned}$$

which is consistent with Example 4.4.4.

Lemma 4.4.10. *For Hilbert spaces H_0 and H, let $g : H \to \mathbb{R}$, $g(x) = \|x\|^2$ and $f : H_0 \to H$ given by $f(x) = Ax - b$, for $A : H_0 \to H$ linear continuous and $b \in H$ fixed. Then the map*

$$g \circ f : H_0 \to \mathbb{R}, \qquad (g \circ f)(x) = \|Ax - b\|^2$$

is Fréchet differentiable and

$$(g \circ f)'(x) = 2\langle A^*(Ax - b), \cdot \rangle.$$

Proof. We notice that $g'(x) = 2\langle x, \cdot \rangle$ and $f'(x) = A$. The chain rule leads to

$$\begin{aligned}(g \circ f)'(x) &= g'(f(x))f'(x)\\ &= 2\langle f(x), f'(x) \cdot \rangle\\ &= 2\langle f(x), A \cdot \rangle\\ &= 2\langle A^* f(x), \cdot \rangle.\end{aligned}$$

□

Lemma 4.4.11 (Mean value inequality for C^1 curves)**.** *Let X be a Banach space and let $g : [0, 1] \to X$ be continuously Fréchet differentiable. If*

$$\|g'(t)\| \leq M \qquad \text{for all } t \in [0, 1],$$

then

$$\|g(1) - g(0)\| \leq M.$$

Proof. Since g' is continuous on $[0, 1]$, it is uniformly continuous. Fix $\varepsilon > 0$. Then there exists $\delta > 0$ such that

$$|s - t| < \delta \quad \Longrightarrow \quad \|g'(s) - g'(t)\| < \varepsilon.$$

We first show a uniform first-order estimate. By differentiability of g at each t, we have

$$g(t + h) - g(t) = g'(t)h + r(t, h), \quad \text{with} \quad \frac{\|r(t, h)\|}{|h|} \to 0 \quad (h \to 0).$$

Uniform continuity of g' implies that this convergence is uniform for $|h| < \delta$. Hence, possibly shrinking δ, we may assume

$$\|g(t + h) - g(t) - g'(t)h\| \leq \varepsilon|h| \quad \text{whenever } |h| < \delta. \tag{$*$}$$

Now choose a partition

$$0 = t_0 < t_1 < \cdots < t_N = 1$$

with mesh size $t_{i+1} - t_i < \delta$. Applying $(*)$ with $t = t_i$ and $h = t_{i+1} - t_i$ gives

$$g(t_{i+1}) - g(t_i) = g'(t_i)(t_{i+1} - t_i) + r_i, \qquad \|r_i\| \leq \varepsilon(t_{i+1} - t_i).$$

Summing over i yields

$$g(1) - g(0) = \sum_{i=0}^{N-1} g'(t_i)(t_{i+1} - t_i) + \sum_{i=0}^{N-1} r_i.$$

Taking norms and using $\|g'(t_i)\| \leq M$ gives

$$\|g(1) - g(0)\| \leq M \sum_{i=0}^{N-1} (t_{i+1} - t_i) + \varepsilon \sum_{i=0}^{N-1} (t_{i+1} - t_i) = M + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the result follows. $\square$

Theorem 4.4.12. *Let X be a Banach space and $\Omega \subseteq X$ be open. Assume that $f : \Omega \to X$ is continuously Fréchet differentiable. Suppose that there is $x \in \Omega$ such that*

$$f(x) = x, \qquad \|f'(x)\| < 1.$$

Then there is a neighborhood $B_\varepsilon(x) \subseteq \Omega$ such that $f : B_\varepsilon(x) \to B_\varepsilon(x)$ and f is a contraction on $B_\varepsilon(x)$.

Proof. Let $\delta := 1 - \|f'(x)\|$. By continuity of $f' : \Omega \to \mathcal{L}(X)$, there is $\varepsilon > 0$ such that

$$\|f'(y)\| \leq 1 - \delta/2, \qquad \forall\, y \in B_\varepsilon(x).$$

Let $y, z \in B_\varepsilon(x)$ and define $g : [0, 1] \to X$ by

$$g(t) := f\big(z + t(y - z)\big).$$

Since $B_\varepsilon(x)$ is convex, $z + t(y - z) \in B_\varepsilon(x)$ for all $t \in [0, 1]$. By the chain rule,

$$g'(t) = f'\big(z + t(y - z)\big)(y - z),$$

hence $\|g'(t)\| \leq (1 - \delta/2)\|y - z\|$ for all $t \in [0, 1]$. Applying Lemma 4.4.11 yields

$$\|f(y) - f(z)\| = \|g(1) - g(0)\| \leq (1 - \delta/2)\|y - z\|.$$

Thus f is a contraction on $B_\varepsilon(x)$ and for $z = x$, we observe that $f(B_\varepsilon(x)) \subseteq B_\varepsilon(f(x))$. $\square$

Theorem 4.4.12 shows that Banach's fixed point theorem is available for such f on $B_\varepsilon(x)$ provided that $f(x) = x$.

Necessary conditions of local extrema

For $W = \mathbb{R}$, we now investigate local extrema of $f : V \to \mathbb{R}$.

Definition 4.4.1. *Let $U \subseteq V$ be open and let $f : U \to \mathbb{R}$. A point $x_0 \in U$ is called*

(i) *a* local minimum *if there exists $\varepsilon > 0$ such that*

$$f(x_0) \leq f(x) \quad \textit{for all } x \in B_\varepsilon(x_0),$$

a strict local minimum refers to a strict inequality for $x \in B_\varepsilon(x_0) \setminus \{x_0\}$, *(ii)* *a* local maximum *if there exists* $\varepsilon > 0$ *such that*

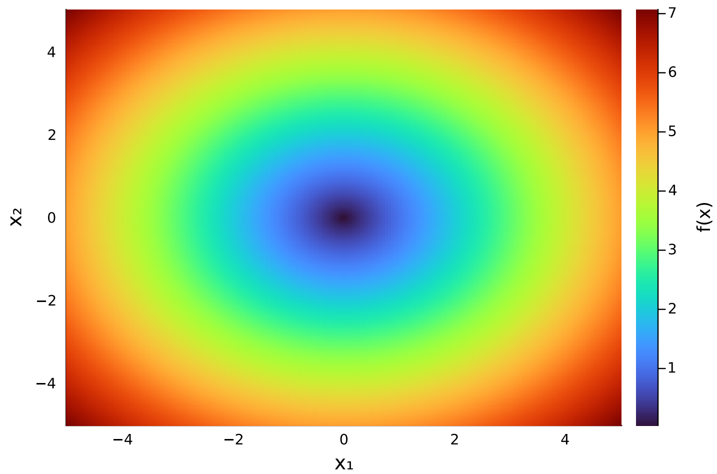
$$f(x_0) \geq f(x) \quad \text{for all } x \in B_\varepsilon(x_0),$$

a strict local maximum refers to a strict inequality for $x \in B_\varepsilon(x_0) \setminus \{x_0\}$*,*

If x_0 *is either a (strict) local minimum or a (strict) local maximum, it is called a* (strict) local extremum.

Example 4.4.2. *Let* V *be a normed space. The function* $f : V \to \mathbb{R}$ *given by* $f(x) = \|x\|$ *has a strict local minimum in* $x_0 = 0$.

```
f(x) = norm(x)

x1 = range(-5, 5, length=200)
x2 = range(-5, 5, length=200)

Z = [f([x;y]) for y in x2, x in x1]
heatmap(x1, x2, Z, xlabel="𝑥₁", ylabel="𝑥₂", colorbar_title="f(x)")
```



Analogous to univariate functions $f : \mathbb{R} \to \mathbb{R}$, the derivative vanishes in local extrema.

Theorem 4.4.3 (necessary conditions of local extrema)**.** *Let* V *be a normed vector space,* $U \subseteq V$ *an open subset, and let* $f : U \to \mathbb{R}$ *be Fréchet differentiable in* $x_0 \in U$*. If* $x_0 \in U$ *is a local extremum of* f*, then*

$$f'(x_0) = 0.$$

Since $f'(x_0)$ is a linear map, the condition $f'(x_0) = 0$ means that

$$f'(x_0)v = 0 \quad \text{for all } v \in V.$$

The proof reduces the infinite-dimensional problem to a one-dimensional calculus argument by restricting f to lines through x_0.

Proof. Let $x_0 \in U$ be a local extremum. Since U is open, there exists $\varepsilon > 0$ such that

$$x_0 + tu \in U \qquad \text{for all } u \in V \text{ with } \|u\| = 1 \text{ and all } |t| < \varepsilon.$$

Fix $u \in V$ and consider the function

$$\varphi(t) := f(x_0 + tu), \qquad |t| < \varepsilon.$$

Since f is Fréchet differentiable at x_0, the function φ is differentiable at 0 by

$$\varphi'(0) = \lim_{t \to 0} \frac{f(x_0 + tu) - f(x_0)}{t} = f'(x_0)u.$$

Since f has a local extremum at x_0, the function $\varphi : \mathbb{R} \to \mathbb{R}$ has a local extremum at $t = 0$. Thus, $\varphi'(0) = 0$ and this leads to $f'(x_0)u = 0$. Since $u \in V$ was arbitrary, we conclude that

$$f'(x_0)u = 0 \quad \text{for all } u \in V,$$

and hence $f'(x_0) = 0$. □

Example 4.4.4. *Consider a real inner product space H and the map $f : H \to \mathbb{R}$, $f(x) = \|x\|^2$. Since $f \geq 0$ and $f(0) = 0$, it has a strict local minimum at $x = 0$. The Fréchet derivative satisfies*

$$\begin{aligned} f'(x)u &= \lim_{t \to 0} \frac{\|x + tu\|^2 - \|x\|^2}{t} \\ &= \lim_{t \to 0} \frac{\|x\|^2 + \langle x, tu\rangle + \langle tu, x\rangle + \|tu\|^2 - \|x\|^2}{t} \\ &= \lim_{t \to 0} \frac{2\langle x, tu\rangle + t^2\|u\|^2}{t} \\ &= 2\langle x, u\rangle. \end{aligned}$$

The derivative is $f'(x) = 2\langle x, \cdot\rangle$ and is zero as a linear map if and only if $x = 0$.

Example 4.4.5. *For linear bounded A, the Fréchet derivative of $f(x) = \|Ax-b\|^2$ is $f'(x) = 2\langle A^*(Ax-b), \cdot\rangle$ by Lemma 4.4.10. Thus, a local minimum x_0 must satisfy $A^*(Ax_0-b) = 0$.*

Definition 4.4.6. *Points satisfying $f'(x_0) = 0$ are called* critical points. *A critical point is a* saddle point *if for all $\varepsilon > 0$ there are $x_1, x_2 \in B_\varepsilon(x_0)$ such that*

$$f(x_0) > f(x_1) \quad \text{and} \quad f(x_0) < f(x_2).$$

Example 4.4.7. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by*

$$f(x_1, x_2) = x_1 x_2.$$

To determine its derivative, we compute

$$f(x_1+h_1, x_2+h_2) = (x_1+h_1)(x_2+h_2) = x_1x_2 + x_1h_2 + x_2h_1 + h_1h_2$$
$$= f(x) + (x_2\,,\, x_1)\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} + h_1h_2,$$

and observe that $h_1h_2/\|h\| \le h_1 \to 0$*. Thus,* f *is Fréchet differentiable,* $f'(x) = (x_2\,,\, x_1)$*, and this linear map is the zero map if and only if* $x_1 = x_2 = 0$*. The critical point* $x = 0$ *is a saddle point, because for all* $\varepsilon > 0$*,*

$$f(\varepsilon, \varepsilon) = \varepsilon^2 > 0, \qquad f(\varepsilon, -\varepsilon) = -\varepsilon^2 < 0.$$

Convexity

In general, a local minimum need not be global. Convexity rules this out: for convex functions, every local minimum is global. Convex functions are therefore fundamental in data analysis and optimization, since many loss functions are convex and admit globally optimal solutions.

Definition 4.4.1 (Convex function)**.** *A function* $f : V \to \mathbb{R}$ *is called* convex *if for all* $x, y \in V$ *and all* $t \in [0, 1]$*,*

$$f(tx + (1-t)y) \le tf(x) + (1-t)f(y).$$

Lemma 4.4.2 (First-order characterization of convexity)**.** *Let* $f : V \to \mathbb{R}$ *be Fréchet differentiable. Then* f *is convex if and only if*

$$f(x) - f(y) \le f'(x)\,(x - y), \qquad x, y \in V. \tag{4.7}$$

Proof. $\Rightarrow$: Assume f is convex. For fixed $x, y \in V$, convexity implies

$$f((x + t(y-x)) \le (1-t)f(x) + tf(y).$$

We bring $f(x)$ to the left-hand-side and divide by t to obtain

$$\frac{f(x + t(y-x)) - f(x)}{t} \le f(y) - f(x) \qquad t \in (0, 1].$$

Taking the limit $t \to 0$ and using Fréchet differentiability gives

$$f'(x)(y - x) \le f(y) - f(x).$$

$\Leftarrow$: Let $z = tx + (1-t)y$. Applying $f(u) \ge f(z) + f'(z)(u - z)$ with $u = x$ and $u = y$, multiplying by t and $1 - t$, and adding gives

$$tf(x) + (1-t)f(y) \ge f(z) + f'(z)(t(x - z) + (1-t)(y - z)).$$

A brief calculations leads to $t(x - z) + (1-t)(y - z) = 0$, so that we obtain

$$f(z) \le tf(x) + (1-t)f(y).$$

The definition $z = tx + (1-t)y$ yields convexity. $\square$

Example 4.4.3. *Let H be an inner product space and $f : H \to \mathbb{R}$ given by $f(x) = \|x\|^2$. We have seen in Example 4.4.4 that $f'(x) = 2\langle x, \cdot\rangle$. Convexity follows from*

$$0 \leq \|y - x\|^2 = \|y\|^2 - 2\langle x, y\rangle + \|x\|^2.$$

This implies the convexity condition

$$\|y\|^2 \geq 2\langle x, y\rangle - \|x\|^2 = \|x\|^2 + 2\langle x, y - x\rangle.$$

Example 4.4.4. *Let H be an inner product space and $f : H \to \mathbb{R}$ given by $f(x) = \|Ax-b\|^2$. We have seen in Lemma 4.4.10 that $f'(x) = 2\langle A^*(Ax - b), \cdot\rangle$. Convexity follows from*

$$0 \leq \|Ay - b - (Ax - b)\|^2 = \|Ay - b\|^2 - 2\langle Ay - b, Ax - b\rangle + \|Ax - b\|^2.$$

This implies the convexity condition

$$\begin{aligned}\|Ay - b\|^2 &\geq 2\langle Ax - b, Ay - b\rangle - \|Ax - b\|^2\\ &= \|Ax - b\|^2 + 2\langle Ax - b, Ay - b - (Ax - b)\rangle\\ &= \|Ax - b\|^2 + 2\langle Ax - b, A(y - x)\rangle\\ &= \|Ax - b\|^2 + 2\langle A^*(Ax - b), y - x\rangle.\end{aligned}$$

Theorem 4.4.5. *Let $f : V \to \mathbb{R}$ be convex.*

(i) Every local minimum is a global minimum.

(ii) Let f be Fréchet differentiable. Then $x_0 \in V$ is a global minimum if and only if $f'(x_0) = 0$.

Proof. (i) Assume x_0 is a local minimizer. Then there exists $\delta > 0$ such that

$$f(x_0) \leq f(x) \qquad \text{for all } x \in B_\delta(x_0).$$

Let $y \in V$ be arbitrary and for $t \in (0, 1)$ define

$$z_t := (1 - t)x_0 + ty.$$

For t sufficiently small, $z_t \in B_\delta(x_0)$, hence $f(z_t) \geq f(x_0)$, and by convexity,

$$f(x_0) \leq f(z_t) \leq (1 - t)f(x_0) + tf(y).$$

Thus, we must have $0 \leq t(f(y) - f(x_0))$. Since $t > 0$, we conclude $f(y) \geq f(x_0)$.
(ii) If x_0 is a global minimum, then it is also local and by Theorem 4.4.3 we have $f'(x_0) = 0$. Now assume that $f'(x_0) = 0$. Lemma 4.4.2 yields $f(y) \geq f(x_0)$, for all $y \in V$, which leads to a global minimum. ∎

Essentials

You learned how differentiability is generalized to functions between normed vector spaces via the Fréchet derivative. This notion captures the idea of best linear approximation and provides a coordinate-free framework that unifies many results from single-variable calculus. For convex functions, every local minimum is a global minimum.

C A Completion adds structure in the form of Fréchet differentiability, which enables local linear approximation.

Fréchet derivative, differentiability, chain rule, convexity

R T A Fréchet differentiability shows that nonlinear mappings between normed spaces are locally governed by linear operators. In finite dimensions, the choice of bases represents these operators by matrices and reduces local analysis to linear algebra. Every concrete evaluation of such linearized models therefore requires representation, truncation, and finite precision. Consequently, differentiability provides the conceptual gateway through which analysis passes into numerical computation, where conditioning, stability, and algorithmic accuracy become the central questions.

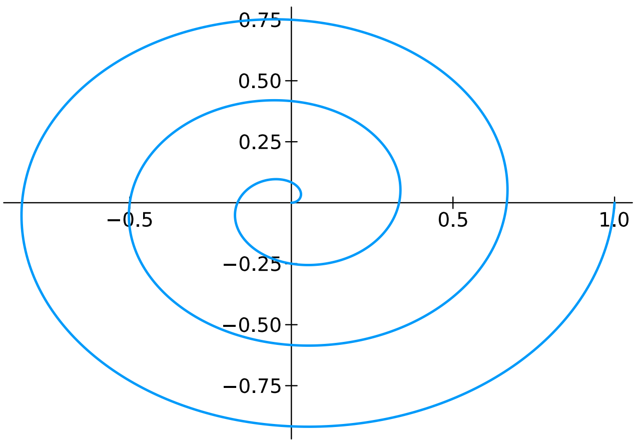
5 Analysis in Coordinates

We begin with a coordinate-free definition and then introduce suitable coordinates that allow for explicit computations. Curves provide a natural illustration of this principle: although defined abstractly in metric spaces, their length can be computed in coordinates by integrating the norm of the derivative.

Curves

Let X be a metric space. A map $f : [0,1] \to X$ is also called a curve. If $X = \mathbb{R}^n$, then f is called a space curve.

```
function myCurvePlot(f;n=1000)
    f1 = t->f(t)[1]
    f2 = t->f(t)[2]
    t = range(0,1,n)
    plot(f1.(t),f2.(t),lw=2)
end

f(t) = [t*cos(6*π*t), t*sin(6*π*t)]
myCurvePlot(f)
```



Definition 5.0.1 (Length of a curve in a metric space)**.** *Let (X,d) be a metric space and let $\gamma : [0,1] \to X$ be a curve. For a partition*

$$\mathcal{P} = \{0 = t_0 < t_1 < \cdots < t_n = 1\}$$

of $[0,1]$, define the polygonal length of γ along $\mathcal{P}$ by

$$\ell(\gamma,\mathcal{P}) := \sum_{j=1}^{n} d\big(\gamma(t_{j-1}),\gamma(t_j)\big).$$

The length *of* γ *is*

$$\ell(\gamma) := \sup_{\mathcal{P}} \ell(\gamma, \mathcal{P}),$$

where the supremum is taken over all partitions $\mathcal{P}$ *of* $[0,1]$. *The curve* γ *is called* rectifiable *if* $\ell(\gamma) < \infty$.

Definition 5.0.2. *Let* (X,d) *be a metric space and let* $\gamma : [0,1] \to X$ *be a curve. If the limit exists,*

$$|\dot{\gamma}|(t) = \lim_{h \to 0} \frac{d(\gamma(t+h), \gamma(t))}{|h|},$$

then $|\dot{\gamma}|(t)$ *is called the metric derivative of* γ *in* t.

For $X = \mathbb{R}^n$, we choose the standard metric

$$d(x,y) = \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2} = \|x - y\|_2.$$

A function $f : \mathbb{R} \to \mathbb{R}^n$ is written as

$$f(t) = \begin{pmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{pmatrix}.$$

We call f differentiable in t_0 if each $f_i : \mathbb{R} \to \mathbb{R}$, for $i = 1, \ldots, n$, is differentiable in t_0 . In that case we write

$$f'(t_0) = \begin{pmatrix} f_1'(t_0) \\ \vdots \\ f_n'(t_0) \end{pmatrix}.$$

This makes sense because the linear map

$$h \mapsto f'(t_0)h \in \mathbb{R}^n$$

approximates f locally in the sense

$$f(t_0 + h) = f(t_0) + f'(t_0)h + \varphi(h),$$

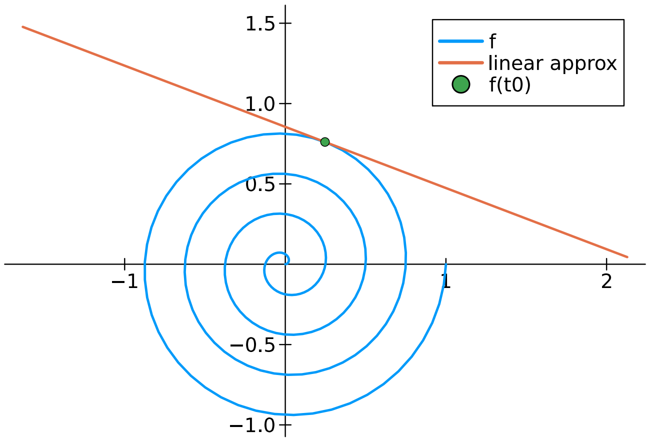
where $\varphi(h)/h \to 0$ for $h \to 0$. This is equivalent to

$$f_i(t_0 + h) = f_i(t_0) + f_i'(t_0)h + \varphi_i(h), \qquad i = 1, \ldots, n,$$

where $\varphi_i(h)/h \to 0$ for $h \to 0$.

```
using Calculus
f(t) = [t*cos(8*π*t), t*sin(8*π*t)]

function myAffineApprox(f, t0)
    n = length(f(0))
    df0 = [derivative(h -> f(h)[j], t0) for j in 1:n]
    return h -> f(t0) + h * df0
end

plot(t -> f(t)[1], t -> f(t)[2], 0, 1, label="f", lw=2, aspect_ratio=1)
```

```
t0 = .8
A = myAffineApprox(f, t0)
plot!(t -> A(t)[1], t -> A(t)[2], -0.1, 0.1, label="linear approx", lw=2)
scatter!([f(t0)[1]], [f(t0)[2]], label="f(t0)")
```



Definition 5.0.3. *Let (X, d) be a metric space and let $\gamma : [0,1] \to X$ be a curve. We say that γ is Lipschitz continuous if there is a constant $L > 0$ such that*

$$d(\gamma(t), \gamma(s)) \leq L|t - s|, \qquad t, s \in [0,1].$$

Lemma 5.0.4. *If the curve $\gamma : [0,1] \to \mathbb{R}^n$ is continuously differentiable, then it is Lipschitz continuous and its metric derivative $|\dot{\gamma}|$ satisfies*

$$|\dot{\gamma}|(t) = \|\gamma'(t)\|_2.$$

In particular, $|\dot{\gamma}|$ is continuous on $[0,1]$.

Proof. For $h \neq 0$, we calculate

$$\begin{aligned}\frac{\|\gamma(t+h) - \gamma(t)\|_2}{|h|} &= \left(\sum_{i=1}^{n} \frac{|\gamma_i(t+h) - \gamma_i(t)|^2}{|h|^2}\right)^{1/2} \\ &= \left(\sum_{i=1}^{n} \left|\frac{\gamma_i(t+h) - \gamma_i(t)}{h}\right|^2\right)^{1/2} \to \left(\sum_{i=1}^{n} |\gamma_i'(t)|^2\right)^{1/2} = \|\gamma'(t)\|_2,\end{aligned}$$

where $h \to 0$. Thus, metric derivative satisfies $|\dot{\gamma}|(t) = \|\gamma'(t)\|_2$.
To verify Lipschitz continuity, we first recall that $[0,1]$ is compact, hence $\gamma_i'([0,1]) \subseteq \mathbb{R}$ is compact since γ_i' is continuous. Taking the maximum over $i = 1, \ldots, n$, there is $C > 0$ such that

$$|\gamma_i'(t)| \leq C, \qquad t \in [0,1], \quad i = 1, \ldots, n.$$

For $s, t \in [0,1]$ the mean value theorem ensures that

$$|\gamma_i(s) - \gamma_i(t)| \leq C|s - t|.$$

Since $d_2 \leq \sqrt{n} d_\infty$, we deduce

$$\|\gamma(s) - \gamma(t)\|_2 \leq \underbrace{\sqrt{n}C}_{L} |s-t|.$$

$\square$

Theorem 5.0.5. *Let $\gamma : [0,1] \to \mathbb{R}^d$ be continuously differentiable. Then*

$$\ell(\gamma) = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_d'(t)|^2}\, dt = \int_0^1 \|\gamma'(t)\|_2\, dt.$$

Proof. The function $g(t) \mapsto \|\gamma'(t)\|_2$ is continuous on $[0,1]$.
Step 1: $\ell(\gamma) \leq \int_0^1 g(t)\, dt$. Let $\mathcal{P} = \{t_0, \ldots, t_m\}$ be a partition of $[0,1]$. For each j and each component, the mean value theorem gives

$$\gamma_k(t_j) - \gamma_k(t_{j-1}) = \gamma_k'(\xi_{j,k})(t_j - t_{j-1})$$

for some $\xi_{j,k} \in (t_{j-1}, t_j)$. Hence

$$|\gamma_k(t_j) - \gamma_k(t_{j-1})| \leq \sup_{t \in [t_{j-1}, t_j]} |\gamma_k'(t)|\, (t_j - t_{j-1}).$$

Therefore

$$\sqrt{\sum_{k=1}^n |\gamma_k(t_j) - \gamma_k(t_{j-1})|^2} \leq \sup_{t \in [t_{j-1}, t_j]} g(t)\, (t_j - t_{j-1}).$$

Summing over j and taking the supremum over all partitions yields

$$\ell(\gamma) \leq \int_0^1 \|\gamma'(t)\|_2\, dt.$$

Step 2: $\ell(\gamma) \geq \int_0^1 g(t)\, dt$. Since g is continuous, it is uniformly continuous. Let $\varepsilon > 0$ and choose a partition $\mathcal{P}$ with mesh sufficiently small so that

$$\sqrt{\sum_{k=1}^n |\gamma_k'(s) - \gamma_k'(t)|^2} < \varepsilon \quad \text{whenever } s, t \text{ lie in one subinterval.}$$

For each j, pick $\tau_j \in [t_{j-1}, t_j]$. Using the mean value theorem componentwise as above, we obtain

$$\left| \frac{\gamma_k(t_j) - \gamma_k(t_{j-1})}{t_j - t_{j-1}} - \gamma_k'(\tau_j) \right| \leq \varepsilon \qquad (k = 1, \ldots, n).$$

Hence

$$\sqrt{\sum_{k=1}^n \left| \frac{\gamma_k(t_j) - \gamma_k(t_{j-1})}{t_j - t_{j-1}} - \gamma_k'(\tau_j) \right|^2} \leq \sqrt{n}\, \varepsilon.$$

Thus

$$\sqrt{\sum_{k=1}^n |\gamma_k(t_j) - \gamma_k(t_{j-1})|^2} \geq \big(g(\tau_j) - \sqrt{n}\, \varepsilon\big)(t_j - t_{j-1}).$$

Summing gives

$$\ell(\gamma, \mathcal{P}) \geq \sum_{j=1}^{m} g(\tau_j)(t_j - t_{j-1}) - \sqrt{n}\,\varepsilon.$$

Passing to the limit (Riemann sums) and using $\ell(\gamma) \geq \ell(\gamma, \mathcal{P})$,

$$\ell(\gamma) \geq \int_0^1 g(t)\,dt.$$

Combining both inequalities proves

$$\ell(\gamma) = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\,dt.$$

□

Example 5.0.6 (circumference of the unit circle)**.** *The curve $\gamma : [0,1] \to \mathbb{R}^2$ given by*

$$\gamma(t) = \begin{pmatrix} \sin(2\pi t) \\ \cos(2\pi t) \end{pmatrix}$$

is a circle. Its length is

$$\begin{aligned} \ell(\gamma) &= \int_0^1 \sqrt{|2\pi\cos(2\pi t)|^2 + |-2\pi\sin(2\pi t)|^2}\,dt \\ &= 2\pi \int_0^1 \sqrt{\cos^2(2\pi t) + \sin^2(2\pi t)}\,dt \\ &= 2\pi \int_0^1 dt = 2\pi. \end{aligned}$$

```
function myPolygonLength(f;n=1000)
    t = range(0, 1, n)
    sum(norm(f(t[j+1]) - f(t[j])) for j in 1:n-1)
end

g(t) = [sin(2*π*t);cos(2*π*t)]
myPolygonLength(g), 2*π
```

```
(6.283174951057152, 6.283185307179586)
```

The length of a curve does not depend on the parametrization.

Lemma 5.0.7. *Let $f : [a,b] \to \mathbb{R}^n$ be a continuously differentiable function and let $\phi : [0,1] \to [a,b]$ be bijective and continuously differentiable. Then $\gamma := f \circ \phi : [0,1] \to \mathbb{R}^n$ satisfies*

$$\int_a^b \sqrt{|f_1'(x)|^2 + \cdots + |f_n'(x)|^2}\,dx = \int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\,dt.$$

Proof. The function ϕ is either strictly increasing or decreasing. Assume it is increasing, so that $\phi'(t) \geq 0$ with $\phi(0) = a$ and $\phi(1) = b$. Substituting $x = \phi(t)$ yields

$$\begin{aligned}
\int_0^1 \sqrt{|\gamma_1'(t)|^2 + \cdots + |\gamma_n'(t)|^2}\, dt &= \int_0^1 \sqrt{|f_1'(\phi(t))\phi'(t)|^2 + \cdots + |f_n'(\phi(t))\phi'(t)|^2}\, dt \\
&= \int_0^1 \sqrt{|f_1'(\phi(t))|^2 + \cdots + |f_n'(\phi(t))|^2}\, \phi'(t)\, dt \\
&= \int_a^b \sqrt{|f_1'(x)|^2 + \cdots + |f_n'(x)|^2}\, dx.
\end{aligned}$$

The case that ϕ is decreasing is analogous. □

5.1 Differentiability in coordinates

We begin in an infinite-dimensional, coordinate-free setting and then introduce a suitable representation together with a finite-dimensional truncation. This reduces the problem from the Hilbert space H to the finite-dimensional coordinate space $\mathbb{R}^N$, where explicit computations become possible.

Given a sequence $(\phi_n)_{n\in\mathbb{N}} \subseteq H$, we define the analysis (decomposition) operator by

$$\mathcal{D}_\phi x := (\langle x, \phi_n\rangle)_{n\in\mathbb{N}}.$$

If we suppose that $\mathcal{D}_\phi : H \to \ell^2(\mathbb{N})$ is bounded, then its adjoint operator is

$$\mathcal{D}_\phi^* : \ell^2(\mathbb{N}) \to H, \qquad \mathcal{D}_\phi^* c = \sum_{n=1}^{\infty} c_n \phi_n.$$

Definition 5.1.1. *Let H be a Hilbert space. We call $(\phi_n)_{n\in\mathbb{N}}, (\psi_n)_{n\in\mathbb{N}} \subseteq H$ a dual frame pair if both $\mathcal{D}_\phi, \mathcal{D}_\psi : H \to \ell^2(\mathbb{N})$ are bounded linear operators such that*

$$\mathcal{D}_\psi^* \mathcal{D}_\phi = \mathrm{id}_H\,.$$

Thus, a dual frame pair satisfies the reconstruction formula

$$\sum_{n\in\mathbb{N}} \langle x, \phi_n\rangle \psi_n = x, \qquad \forall\, x \in H.$$

Example 5.1.2. *Suppose that $(\phi_n)_{n\in\mathbb{N}} \subseteq H$ is an orthonormal basis. Then $\mathcal{D}_\phi : H \to \ell^2(\mathbb{N})$ is bounded and the expansion*

$$x = \sum_{n\in\mathbb{N}} \langle x, \phi_n\rangle \phi_n, \qquad \forall\, x \in H,$$

is exactly the dual frame pair condition $\mathcal{D}_\phi^*\mathcal{D}_\phi = \mathrm{id}_H$. Hence, every orthonormal basis with itself builds a pair of dual frames.

Dual frame pairs generalize the concept of orthonormal basis.

Let truncation and its adjoint be denoted by

$$\mathcal{T}_N : \ell^2(\mathbb{N}) \to \mathbb{R}^N, \qquad \mathcal{T}_N(c_1, c_2, \dots) = (c_1, \dots, c_N)^\top,$$
$$\mathcal{T}_N^* : \mathbb{R}^N \to \ell^2(\mathbb{N}), \qquad \mathcal{T}_N^*(c_1, \dots, c_N)^\top = (c_1, \dots, c_N, 0, \dots).$$

Define the truncated reconstruction maps on H by

$$Q_N : H \to \operatorname{span}\{\psi_n : n \le N\}, \qquad Q_N = \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi.$$

The explicit representation is

$$Q_N x = \sum_{n=1}^{N} \langle x, \phi_n \rangle \psi_n.$$

Lemma 5.1.3. *Let H be a Hilbert space and let $(\phi_n)_{n\in\mathbb{N}}$ and $(\psi_n)_{n\in\mathbb{N}}$ be a pair of dual frames. Then Q_N is uniformly bounded and converges pointwise to the identity,*

$$\|Q_N\| \le \|\mathcal{D}_\psi\| \, \|\mathcal{D}_\phi\|, \qquad Q_N \to \mathrm{id}_H, \quad \textit{pointwise.}$$

Proof. The pointwise convergence is just the dual frame expansion. The norm bound follows from $\|\mathcal{T}_N^* \circ \mathcal{T}_N\| \le 1$. □

Theorem 5.1.4 (R T A Representation, truncation for continuous maps I)**.** *Let H be a Hilbert space and let $(\phi_n)_{n\in\mathbb{N}}$ and $(\psi_n)_{n\in\mathbb{N}}$ be a pair of dual frames. Suppose that $f : H \to H$ is continuous. Define its representation by*

$$\tilde{f} : \ell^2(\mathbb{N}) \to \ell^2(\mathbb{N}), \qquad \tilde{f} := \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^*,$$
$$\tilde{f}_N : \mathbb{R}^N \longrightarrow \mathbb{R}^N, \qquad \tilde{f}_N := \mathcal{T}_N \circ \tilde{f} \circ \mathcal{T}_N^*.$$

We define the truncated approximation of f on H by using the truncated representation,

$$f_N : H \to \operatorname{span}\{\psi_n : n \le N\}, \qquad f_N := \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N \circ \mathcal{D}_\phi.$$

Then the following hold.

(i) Coordinate-free vs. coordinates:

$$f = \mathcal{D}_\psi^* \circ \tilde{f} \circ \mathcal{D}_\phi, \qquad \mathcal{D}_\phi \circ f = \tilde{f} \circ \mathcal{D}_\phi, \qquad f \circ \mathcal{D}_\psi^* = \mathcal{D}_\psi^* \circ \tilde{f}.$$

(ii) Truncated approximation: For every $N \in \mathbb{N}$,

$$f_N = Q_N \circ f \circ Q_N.$$

(iii) Coordinate-free approximation:

$$f_N \to f \quad \textit{pointwise for } N \to \infty.$$

(iv) Approximation in coordinates:

$$\mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N \to \tilde{f} \quad \textit{pointwise for } N \to \infty.$$

We may consider f_N as a finite-dimensional approximation of f.

Proof. (i) Since (ϕ_n) and (ψ_n) are dual, $\mathcal{D}_\psi^*\mathcal{D}_\phi = \mathrm{id}_H$. Thus

$$\mathcal{D}_\psi^* \circ \tilde{f} \circ \mathcal{D}_\phi = \mathcal{D}_\psi^* \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{D}_\phi = f,$$

which gives the first identity. The other two follow from

$$\begin{aligned} \tilde{f} \circ \mathcal{D}_\phi &= \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{D}_\phi = \mathcal{D}_\phi \circ f, \\ \mathcal{D}_\psi^* \circ \tilde{f} &= \mathcal{D}_\psi^* \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* = f \circ \mathcal{D}_\psi^*. \end{aligned}$$

(ii) Using $\tilde{f} = \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^*$ and $Q_N = \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi$, we compute

$$\begin{aligned} f_N &= \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ (\mathcal{T}_N \circ \tilde{f} \circ \mathcal{T}_N^*) \circ \mathcal{T}_N \circ \mathcal{D}_\phi \\ &= \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi \circ f \circ \mathcal{D}_\psi^* \circ \mathcal{T}_N^* \circ \mathcal{T}_N \circ \mathcal{D}_\phi \\ &= Q_N \circ f \circ Q_N. \end{aligned}$$

(iii) Fix $c \in H$ and estimate

$$\begin{aligned} \|Q_N \circ f \circ Q_N(c) - f(c)\| &\le \|Q_N \circ f \circ Q_N(c) - Q_N \circ f(c)\| + \|Q_N \circ f(c) - f(c)\| \\ &\le \|Q_N\| \, \|f \circ Q_N(c) - f(c)\| + \|Q_N \circ f(c) - f(c)\|. \end{aligned}$$

Since Q_N is uniformly bounded, $Q_N c \to c$ and f is continuous, both terms tend to zero for $N \to \infty$.
(iv) Let us now denote $P_N := \mathcal{T}_N^* \circ \mathcal{T}_N$, which simply keeps the first N coordinates. According to

$$\mathcal{T}_N^* \circ \tilde{f}_N \circ \mathcal{T}_N = P_N \circ \tilde{f} \circ P_N,$$

we now estimate

$$\begin{aligned} \|P_N \circ \tilde{f} \circ P_N\,(c) - \tilde{f}(c)\| &\le \|P_N \circ \tilde{f} \circ P_N\,(c) - P_N \circ \tilde{f}(c)\| + \|P_N \circ \tilde{f}(c) - \tilde{f}(c)\| \\ &\le \|\tilde{f} \circ \mathcal{T}_N^* \circ \mathcal{T}_N\,(c) - \tilde{f}(c)\| + \|P_N \circ \tilde{f}(c) - \tilde{f}(c)\| \end{aligned}$$

Since $P_N \to \mathrm{id}_{\ell^2(\mathbb{N})}$ pointwise and $\tilde{f}$ is continuous, both terms tend to zero. □

If we allow for two Hilbert spaces, $f : H_1 \to H_2$, then we need two pairs of dual frames and truncated reconstruction operators on H_i by

$$Q_{i,N_i} : H_i \to \mathrm{span}\{\psi_{i,n} : n \le N_i\}, \qquad Q_{i,N_i} := \mathcal{D}_{\psi_i}^* \circ \mathcal{T}_{N_i}^* \circ \mathcal{T}_{N_i} \circ \mathcal{D}_{\phi_i}, \qquad Q_{i,N_i} x = \sum_{n=1}^{N_i} \langle x, \phi_{i,n} \rangle \psi_{i,n}.$$

Theorem 5.1.5 (R T A Representation, truncation for continuous maps II). *Let H_1 and H_2 be Hilbert spaces and suppose that $f : H_1 \to H_2$ is continuous. Let $(\phi_{i,n})_{n\in\mathbb{N}}$ and $(\psi_{i,n})_{n\in\mathbb{N}}$ be a pair of dual frames for the Hilbert space H_i, for $i = 1, 2$. Define its representation by*

$$\begin{aligned} \tilde{f} &: \ell^2(\mathbb{N}) \to \ell^2(\mathbb{N}), & \tilde{f} &:= \mathcal{D}_{\phi_2} \circ f \circ \mathcal{D}_{\psi_1}^*, \\ \tilde{f}_{N_1,N_2} &: \mathbb{R}^{N_1} \longrightarrow \mathbb{R}^{N_2}, & \tilde{f}_{N_1,N_2} &:= \mathcal{T}_{N_2} \circ \tilde{f} \circ \mathcal{T}_{N_1}^*. \end{aligned}$$

We define the truncated approximation of f on H by using the truncated representation,

$$f_{N_1,N_2} : H \to \operatorname{span}\{\psi_n : n \le N_2\}, \qquad f_{N_1,N_2} := \mathcal{D}^*_{\psi_2} \circ \mathcal{T}^*_{N_2} \circ \tilde{f}_{N_1,N_2} \circ \mathcal{T}_{N_1} \circ \mathcal{D}_{\phi_1}.$$

Then the following hold.

(i) Coordinate-free vs. coordinates:

$$f = \mathcal{D}^*_{\psi_2} \circ \tilde{f} \circ \mathcal{D}_{\phi_1}, \qquad \mathcal{D}_{\phi_2} \circ f = \tilde{f} \circ \mathcal{D}_{\phi_1}, \qquad f \circ \mathcal{D}^*_{\psi_1} = \mathcal{D}^*_{\psi_2} \circ \tilde{f}.$$

(ii) Truncated approximation: For every $N \in \mathbb{N}$,

$$f_{N_1,N_2} = Q_{2,N_2} \circ f \circ Q_{1,N_1}.$$

(iii) Coordinate-free approximation:

$$f_{N_1,N_2} \to f \quad \textit{pointwise for } N_1, N_2 \to \infty.$$

(iv) Approximation in coordinates:

$$\mathcal{T}^*_{N_2} \circ \tilde{f}_{N_1,N_2} \circ \mathcal{T}_{N_1} \to \tilde{f} \quad \textit{pointwise for } N_1, N_2 \to \infty.$$

Proof. This is analogous to the proof of Theorem 5.1.4, hence omitted. □

The representation and truncation are compatible with the Fréchet derivative, so that

$$\tilde{f}' = \mathcal{D}_{\phi_2} \circ f' \circ \mathcal{D}^*_{\psi_1},$$
$$\tilde{f}'_{N_1,N_2} = \mathcal{T}_{N_2} \circ \tilde{f}' \circ \mathcal{T}^*_{N_1}.$$

Remark 5.1.6. *This representation-truncation procedure appears throughout analysis under different names. In numerical analysis it corresponds to Galerkin or Petrov-Galerkin projection methods; in harmonic analysis to matrix representations of operators with respect to frames; and in operator theory to the finite section method.*

For finite-dimensional vector spaces, we directly switch to coordinates without truncation.

Theorem 5.1.7 (reduction to coordinates)**.** *Let $\dim V = n$, $\dim W = m$, and $f : V \to W$. Choose bases $B_V \subseteq V$, $B_W \subseteq W$ and define $F : \mathbb{R}^n \to \mathbb{R}^m$ by*

$$F := \chi^{-1}_{B_W} \circ f \circ \chi_{B_V}, \qquad \textit{or equivalently by} \quad f = \chi_{B_W} \circ F \circ \chi^{-1}_{B_V}.$$

Then f is Fréchet differentiable if and only if F is Fréchet differentiable. In that case, for $x = \chi^{-1}_{B_V} v$, we have

$$f'(v) = \chi_{B_W} \, F'(x) \, \chi^{-1}_{B_V}.$$

In contrast to linear algebra, we do not restrict to linear maps: if f is nonlinear, then its coordinate representation $F : \mathbb{R}^n \to \mathbb{R}^m$ is nonlinear as well. Fréchet differentiable maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ are often simply called *differentiable.* Thus, by fixing bases, the analysis of f can be carried out entirely on the level of its coordinate representation $F : \mathbb{R}^n \to \mathbb{R}^m$. To study differentiability, the separation

$$F = \begin{pmatrix} F_1 \\ \vdots \\ F_m \end{pmatrix}$$

allows us to first restrict us to a single $F_j : \mathbb{R}^n \to \mathbb{R}$.

We now use directional derivatives, but restrict the directions $u \in \mathbb{R}^n$ to the standard basis vectors $e_1, \ldots, e_n$.

Definition 5.1.8 (Partial derivatives)**.** *The function $f : \mathbb{R}^n \to \mathbb{R}$ is called* partially differentiable *at $x \in \mathbb{R}^n$ if, for each $j = 1, \ldots, n$, the limit*

$$\partial_j f(x) := \lim_{h \to 0} \frac{f(x + he_j) - f(x)}{h}$$

exists. The row vector

$$\operatorname{grad} f(x) := (\partial_1 f(x), \ldots, \partial_n f(x))$$

is called the gradient *of f at x. Its transpose is a column vector and denoted by*

$$\nabla f(x) = \operatorname{grad} f(x)^\top.$$

The partial derivative $\partial_j f(x)$ is simply the derivative of the one-dimensional function

$$t \mapsto f(x_1, \ldots, x_{j-1}, t, x_{j+1}, \ldots, x_n)$$

at $t = x_j$, where all other coordinates are kept fixed.

Remark 5.1.9 (Feature sensitivity and gradient descent)**.** *Partial derivatives quantify how changes in individual coordinates affect the output of a multivariate model. The gradient describes the direction of strongest local change and underlies gradient-based optimization methods.*

Example 5.1.10. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x) = x_1^2 x_2 + x_2.$$

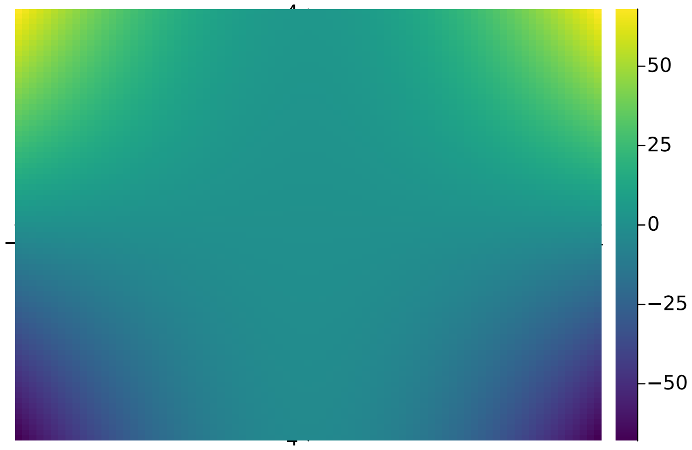
Its gradient is

$$\operatorname{grad} f(x) = \big(2x_1x_2,\ x_1^2 + 1\big).$$

As a row it represents a linear map from $\mathbb{R}^2$ to $\mathbb{R}$.
We now check the local approximation:

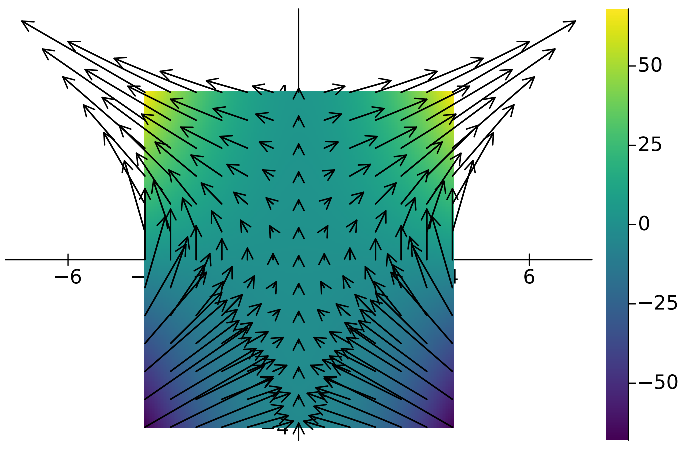
$$\begin{aligned} f(x+h) &= (x_1 + h_1)^2(x_2 + h_2) + x_2 + h_2 \\ &= x_1^2 x_2 + x_1^2 h_2 + 2x_1h_1x_2 + 2x_1h_1h_2 + h_1^2 x_2 + h_1^2 h_2 + x_2 + h_2 \\ &= \underbrace{x_1^2 x_2 + x_2}_{f(x)} + \underbrace{2x_1x_2h_1 + x_1^2 h_2 + h_2}_{\operatorname{grad} f(x)\, h} + \underbrace{2x_1h_1h_2 + x_2h_1^2 + h_1^2 h_2}_{\varphi(h)}. \end{aligned}$$

```
f(x,y) = x^2*y+y
x = -4:0.1:4
y = -4:0.1:4
heatmap(x, y, f)
```



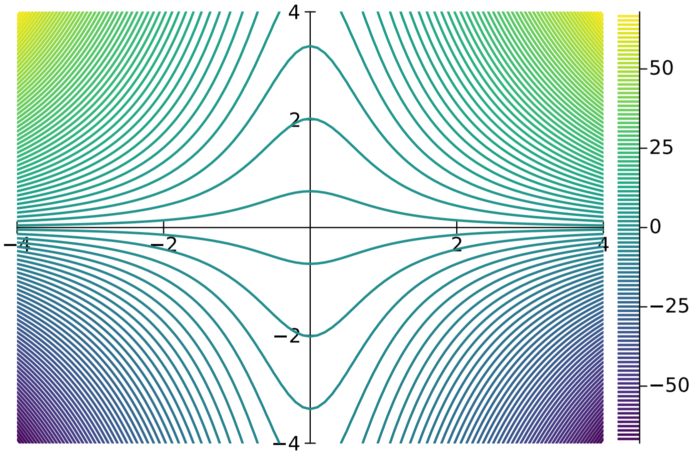
The gradient assigns a vector to each point (x, y).

```
gradf(x,y) = [2*x*y  x^2+1]

xq = range(-4, 4, length=13)
yq = range(-4, 4, length=13)

U = [gradf(x,y)[1] for y in yq, x in xq] *.1
V = [gradf(x,y)[2] for y in yq, x in xq] *.1

X = [x for y in yq, x in xq]
Y = [y for y in yq, x in xq]

quiver!(X, Y; quiver=(U, V), color=:black, linewidth=1.2)
```



We can also visualize the level sets of f.

```
contour(x, y, f,levels = 100)
```



Switching from the vector space setting to coordinates facilitates the actual computation of the derivative. The derivative of a differentiable function is the gradient.

Theorem 5.1.11 (Derivative and gradient in $\mathbb{R}^n$)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $x \in \mathbb{R}^n$. Then f is partially differentiable at x and*

$$f'(x) = \operatorname{grad} f(x).$$

In particular, $f'(x)e_j = \partial_j f(x)$, for $j = 1, \ldots, n$.

Proof. Since f is differentiable at x,

$$f(x+h) = f(x) + f'(x)h + o(\|h\|) \qquad (h \to 0).$$

Fix $j \in \{1, \ldots, n\}$ and set $h = te_j$. Then, as $t \to 0$,

$$\frac{f(x+te_j) - f(x)}{t} = \frac{f'(x)(te_j)}{t} + \frac{o(\|te_j\|)}{t} = f'(x)e_j + \frac{o(|t|)}{t}.$$

Hence the limit exists and equals $f'(x)e_j$, i.e.

$$\operatorname{grad} f(x)e_j = \partial_j f(x) = \lim_{t\to 0} \frac{f(x+te_j) - f(x)}{t} = f'(x)e_j.$$

Thus, the two linear maps $\operatorname{grad} f(x)$ and $f'(x)$ coincide, because they are equal on the basis $\{e_1, \ldots, e_n\}$. $\square$

The gradient points in the direction of "maximal change".

Example 5.1.12. *For differentiable $f : \mathbb{R}^n \to \mathbb{R}$, the gradient satisfies*

$$f(x+h) = f(x) + \operatorname{grad} f(x)h + o(\|h\|).$$

The Cauchy-Schwarz inequality implies

$$|\operatorname{grad} f(x)\, h| = |\langle \nabla f(x), h\rangle| \le \|\nabla f(x)\|\|h\|$$

with equality if and only if $h = \alpha \nabla f(x)$ *for some* $\alpha \in \mathbb{R}$*. In that case*

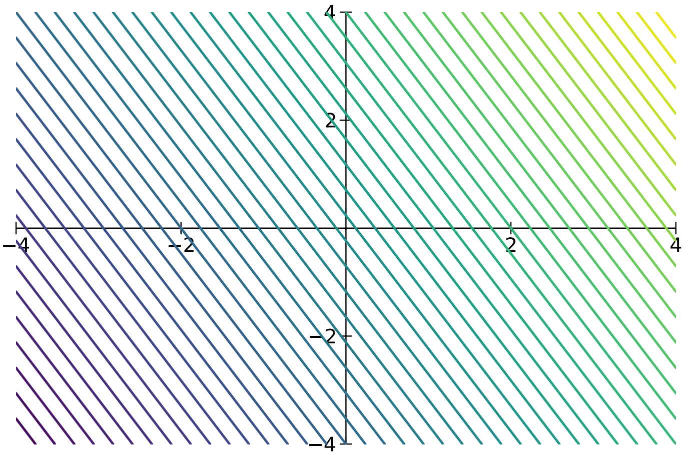
$$\operatorname{grad} f(x)\, h = \alpha \|\nabla f(x)\|^2.$$

For $\alpha > 0$ *the vector* h *points in the direction of the gradient and then* f *is increasing in this direction since* $\operatorname{grad} f(x)\, h > 0$*. For* $\alpha < 0$*, we have* $\operatorname{grad} f(x)\, h < 0$ *and, hence,* f *decreases in this direction.*

Thus, the local affine linear approximation of $f : \mathbb{R}^n \to \mathbb{R}$ in x_0 is

$$L_f(x) = f(x_0) + \operatorname{grad} f(x_0)(x - x_0).$$

```
(x0,y0) = (1,1)
Lf(x,y) = f(x0,y0)+only(gradf(x0,y0)*[x-x0 ; y-y0])    # only([x]) = x
contour(x, y, Lf,levels = 50, colorbar = false)
```



Alternatively, we may write the local affine linear approximation of f in x_0 as follows:

```
using ForwardDiff
f(x) = x[1]^3 - 3x[1] + x[1]*x[2]^2

function myFirstOrderApproximation(f,x0)
    ∇f = ForwardDiff.gradient(f,x0)
    return x -> f(x0)+∇f'*(x-x0)
end
```

Every differentiable function is partially differentiable. The reverse implication holds under an additional continuity assumption.

Theorem 5.1.13. *Let* $U \subseteq \mathbb{R}^n$ *be open and let* $f : U \to \mathbb{R}$*. If* f *is partially differentiable in* U *and all partial derivatives* $\partial_1 f, \ldots, \partial_n f$ *are continuous at* $x \in U$*, then* f *is differentiable at* x *and*

$$f'(x) = \operatorname{grad} f(x).$$

Proof. Let $x \in U$ and $h = (h_1, \dots, h_n) \in \mathbb{R}^n$ with $x + h \in U$. Define points

$$x^{(j)} := x + \sum_{i=1}^{j} h_i e_i, \qquad j = 0, \dots, n.$$

The telescope sum leads to

$$f(x+h) - f(x) = \sum_{j=1}^{n} \big(f(x^{(j)}) - f(x^{(j-1)}) \big).$$

Consider the one-dimensional functions

$$\varphi_j(t) := f(x^{(j-1)} + t e_j), \qquad j = 1, \dots, n,$$

defined for t near 0. The partial differentiability yields

$$\varphi_j'(t) = \lim_{\delta \to 0} \frac{f(x^{(j-1)} + (t+\delta) e_j) - f(x^{(j-1)} + t e_j)}{\delta} = \partial_j f(x^{(j-1)} + t e_j).$$

By the mean value theorem, there exists $\theta_j \in (0,1)$ such that

$$\begin{aligned} f(x^{(j)}) - f(x^{(j-1)}) = \varphi_j(h_j) - \varphi_j(0) &= \varphi_j'(\theta_j h_j) h_j \\ &= \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j. \end{aligned}$$

We now obtain

$$f(x+h) - f(x) = \sum_{j=1}^{n} \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j.$$

This implies

$$f(x+h) = f(x) + \sum_{j=1}^{n} \partial_j f(x)\, h_j + \varphi(h),$$

where the remainder $\varphi(h)$ is

$$\begin{aligned} \varphi(h) &= \sum_{j=1}^{n} \partial_j f(x^{(j-1)} + \theta_j h_j e_j)\, h_j - \sum_{j=1}^{n} \partial_j f(x)\, h_j \\ &= \sum_{j=1}^{n} \Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x) \Big)\, h_j. \end{aligned}$$

To verify $\varphi(h)/\|h\| \to 0$, we apply Cauchy-Schwarz in $\mathbb{R}^n$,

$$\begin{aligned} |\varphi(h)| &\leq \sum_{j=1}^{n} \Big| \Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x) \Big)\, h_j \Big| \\ &\leq \|h\| \Big(\sum_{j=1}^{n} \Big| \Big(\partial_j f(x^{(j-1)} + \theta_j h_j e_j) - \partial_j f(x) \Big) \Big|^2 \Big)^{1/2}. \end{aligned}$$

For $h \to 0$, we note that $x^{(j-1)} + \theta_j h_j e_j \to x$, so that continuity of $\partial_j f$ implies

$$\partial_j f(x^{(j-1)} + \theta_j h_j e_j) \to \partial_j f(x).$$

This leads to $\varphi(h)/\|h\| \to 0$ and hence f is differentiable in x with

$$f(x+h) = f(x) + \sum_{j=1}^{n} \partial_j f(x)\, h_j + \varphi(h) = f(x) + \operatorname{grad} f(x) h + \varphi(h).$$

□

Every function $f : \mathbb{R}^n \to \mathbb{R}^m$ can be written componentwise as

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_m(x) \end{pmatrix}, \qquad f_i : \mathbb{R}^n \to \mathbb{R}\,.$$

Remark 5.1.14 (Jacobian matrix)**.** *Since convergence in $\mathbb{R}^m$ means convergence in each component, the function $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if and only if all f_i, $i = 1, \dots, m$ are differentiable. In that case, the Jacobian matrix*

$$J_f(x) := \begin{pmatrix} \operatorname{grad} f_1(x) \\ \vdots \\ \operatorname{grad} f_m(x) \end{pmatrix} = \big(\partial_j f_i(x)\big)_{i,j} \in \mathbb{R}^{m \times n}\,.$$

is the derivative $f'(x) = J_f(x)$.

If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable on an open set U, then the mapping

$$x \mapsto f'(x) = J_f(x) \in \mathbb{R}^{m \times n}$$

is continuous if and only if all partial derivatives $x \mapsto \partial_j f_i(x)$ are continuous. In this case, f is called *continuously differentiable.*

We say that $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable if $\operatorname{grad} f : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. In other words, if the map $\partial_j f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable for all $j = 1, \dots, n$.

Lemma 5.1.15 (mixed partial derivatives commute)**.** *Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}$ be twice continuously partially differentiable. Then for all $i, j \in \{1, \dots, n\}$ and all $x \in U$,*

$$\partial_i \partial_j f(x) = \partial_j \partial_i f(x).$$

Proof. Fix $x \in U$ and indices i, j. Since U is open, there exists $\varepsilon > 0$ such that

$$x + se_i + te_j \in U \qquad \text{for all } |s|, |t| < \varepsilon.$$

Define the function $\varphi : \mathbb{R}^2 \to \mathbb{R}$ by

$$\varphi(s, t) := f(x + se_i + te_j).$$

Since f is twice continuously partially differentiable, the function φ has continuous second-order partial derivatives, and by the chain rule we have

$$\partial_1 \partial_2 \varphi(s, t) = \partial_i \partial_j f(x + se_i + te_j), \qquad \partial_2 \partial_1 \varphi(s, t) = \partial_j \partial_i f(x + se_i + te_j).$$

It therefore suffices to show that

$$\partial_1 \partial_2 \varphi(0, 0) = \partial_2 \partial_1 \varphi(0, 0).$$

Using the one-dimensional mean value theorem twice, there are $\xi_s \in (0, s)$ and $\xi_t \in (0, t)$, such that

$$\underbrace{\varphi(s,t)-\varphi(0,t)}_{p(t)}-\big(\underbrace{\varphi(s,0)-\varphi(0,0)}_{p(0)}\big)=p'(\xi_t)t=\partial_2\left(\varphi(s,\xi_t)-\varphi(0,\xi_t)\right)t$$
$$=\left(\partial_2\varphi(s,\xi_t)-\partial_2\varphi(0,\xi_t)\right)t$$
$$=\partial_1\partial_2\varphi(\xi_s,\xi_t)st.$$

We repeat this process with interchanged roles of s and t. There are $\eta_s \in (0,s)$ and $\eta_t \in (0,t)$, such that

$$\underbrace{\varphi(s,t)-\varphi(s,0)}_{q(s)}-\big(\underbrace{\varphi(0,t)-\varphi(0,0)}_{q(0)}\big)=q'(\eta_s)s=\partial_1\left(\varphi(\eta_s,t)-\varphi(\eta_s,0)\right)s$$
$$=\left(\partial_1\varphi(\eta_s,t)-\partial_1\varphi(\eta_s,0)\right)s$$
$$=\partial_2\partial_1\varphi(\eta_s,\eta_t)st.$$

We have shown that

$$\partial_1\partial_2\varphi(\xi_s,\xi_t)=\partial_2\partial_1\varphi(\eta_s,\eta_t).$$

The limit $(s,t)\to 0$ implies $(\xi_s,\xi_t),(\eta_s,\eta_t)\to 0$. The continuity of $\partial_i\partial_j f$ and $\partial_j\partial_i f$ yields continuity of $\partial_1\partial_2\varphi$ and $\partial_2\partial_1\varphi$ in 0, so that we arrive at

$$\partial_1\partial_2\varphi(0,0)=\partial_2\partial_1\varphi(0,0).$$

□

For $f:\mathbb{R}^n\to\mathbb{R}$, the gradient $\operatorname{grad} f$ collects first-order (partial) derivatives. Second-order differential information is collected by the Hessian.

Definition 5.1.16 (Hessian matrix)**.** *Let $U\subseteq\mathbb{R}^n$ be open and suppose that $f:U\to\mathbb{R}$ is twice continuously differentiable. The* Hessian matrix *of f at $x\in U$ is defined by*

$$H_f(x):=\big(\partial_i\partial_j f(x)\big)_{i,j=1}^n.$$

Since f is twice continuously differentiable, the mixed partial derivatives commute, and hence the Hessian matrix is symmetric. When it comes to local extrema in the next section, it plays the role of the second derivative of univariate functions to decide if a local extremum is a local maximum or minimum.

Example 5.1.17 (Hessian)**.** *Consider the function $f:\mathbb{R}^2\to\mathbb{R}$ defined by*

$$f(x)=x_1^3+x_2^2\exp(x_1).$$

We compute

$$\begin{aligned}\partial_1 f(x)&=3x_1^2+x_2^2\exp(x_1), & \partial_2 f(x)&=2x_2\exp(x_1)\\ \partial_1\partial_1 f(x)&=6x_1+x_2^2\exp(x_1), & \partial_2\partial_2 f(x)&=2\exp(x_1)\\ \partial_2\partial_1 f(x)&=2x_2\exp(x_1), & \partial_1\partial_2 f(x)&=2x_2\exp(x_1).\end{aligned}$$

The Hessian matrix is

$$H_f(x) = \begin{pmatrix} 6x_1 + x_2^2 \exp(x_1) & 2x_2 \exp(x_1) \\ 2x_2 \exp(x_1) & 2\exp(x_1) \end{pmatrix}.$$

Essentials

Pairs of dual frames extend the concept of orthonormal bases and provide structured coordinate representations. Through these representations, continuous maps on Hilbert spaces can be discretized. Truncating the discretized representation yields finite-dimensional approximations of the original map. If f is defined on finite-dimensional coordinate-free normed spaces, then, after choosing bases, the Fréchet derivative of f corresponds to the Fréchet derivative of its coordinate representation. Fréchet differentiability in coordinates reduces to the existence of partial derivatives. You thus learned that differentiability in $\mathbb{R}^n$ is expressed in terms of partial derivatives, gradients, and Jacobian matrices. Moreover, mixed partial derivatives commute under suitable regularity assumptions.

RTA Continuous maps are represented in coordinates via dual frames. Truncation provides finite-dimensional approximations. Representation by coordinates turns differentiability into partial derivatives. Gradients and Jacobians lead to local linear approximations.

Hilbert space, Riesz representation theorem, adjoint operator, orthonormal basis, pairs of dual frames, partial derivatives, gradient, Jacobian matrix, Hessian matrix

5.2 Taylor's formula and local extrema

This section develops Taylor's formula for multivariate functions as a local polynomial approximation and uses it to study local extrema. We derive practical criteria for local extrema based on gradients and Hessians.

First we introduce higher-order partial derivatives and derive Taylor's formula in coordinate spaces. This will allow us to classify local extrema using second-order information.

For $k \in \mathbb{N}$ and $j \in \{1, \ldots, n\}$, we define the k-th order partial derivative by

$$\partial_j^k := \underbrace{\partial_j \cdots \partial_j}_{k \text{ times}}.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, we define

$$\partial^\alpha := \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}, \qquad |\alpha| := \alpha_1 + \cdots + \alpha_n, \qquad \alpha! := \alpha_1! \cdots \alpha_n!.$$

For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $y = (y_1, \ldots, y_n) \in \mathbb{R}^n$, we write

$$(x - y)^\alpha := (x_1 - y_1)^{\alpha_1} \cdots (x_n - y_n)^{\alpha_n}.$$

Theorem 5.2.1 (Taylor formula in $\mathbb{R}^n$). *Let $U \subseteq \mathbb{R}^n$ be open, $k \in \mathbb{N}$, and let $f : U \to \mathbb{R}$ be*

k times continuously differentiable. Then for every $x_0 \in U$,

$$f(x) = \sum_{|\alpha| \le k} \frac{\partial^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha + o(\|x - x_0\|^k) \qquad (x \to x_0).$$

The Taylor polynomial,

$$x \mapsto \sum_{|\alpha| \le k} \frac{\partial^\alpha f(x_0)}{\alpha!} (x - x_0)^\alpha,$$

provides a polynomial approximation of f near x_0.

Remark 5.2.2 (Local approximation). *In data analysis, Taylor expansions are used to approximate complex models locally by simpler ones.*

Proof. Set $h := x - x_0$ and assume $\|h\|$ is small enough so that the line segment $\{x_0 + th : t \in [0, 1]\} \subseteq U$. Define the one-dimensional function

$$g : [0, 1] \to \mathbb{R}, \qquad g(t) := f(x_0 + th).$$

Step 1: compute $g^{(j)}(0)$ in multi-index form. For $j \in \{0, \dots, k\}$ one proves by induction (using the chain rule and the commutation of mixed partial derivatives) that g is k-times continuously differentiable and

$$g^{(j)}(t) = \sum_{|\alpha| = j} \frac{j!}{\alpha!} (\partial^\alpha f)(x_0 + th)\, h^\alpha, \qquad t \in [0, 1]. \tag{5.1}$$

In particular,

$$\frac{g^{(j)}(0)}{j!} = \sum_{|\alpha| = j} \frac{(\partial^\alpha f)(x_0)}{\alpha!} h^\alpha.$$

Summing over $j = 0, \dots, k$ yields

$$\sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} = \sum_{|\alpha| \le k} \frac{(\partial^\alpha f)(x_0)}{\alpha!} h^\alpha,$$

which is exactly the Taylor polynomial of order k.

Step 2: The univariate Taylor formula with integral remainder. The univariate Taylor formula for g yields

$$g(1) = \sum_{j=0}^{k-1} \frac{g^{(j)}(0)}{j!} + \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} g^{(k)}(t)\, dt.$$

We add and subtract $g^{(k)}(0)$ inside the integral and use $\int_0^1 (1-t)^{k-1}\, dt = 1/k$ to obtain

$$g(1) = \sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} + \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} \big(g^{(k)}(t) - g^{(k)}(0)\big)\, dt.$$

Since $g(1) = f(x)$, the remainder

$$R(h) := \int_0^1 \frac{(1-t)^{k-1}}{(k-1)!} \big(g^{(k)}(t) - g^{(k)}(0)\big)\, dt.$$

leads to the formula

$$f(x) = \sum_{j=0}^{k} \frac{g^{(j)}(0)}{j!} + R(h).$$

Step 3: estimate the remainder. We use the representation (5.1) for $j = k$ and derive

$$g^{(k)}(t) - g^{(k)}(0) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} \Big((\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big) h^\alpha.$$

Hence, with $|h^\alpha| \leq \|h\|^{|\alpha|} = \|h\|^k$,

$$|g^{(k)}(t) - g^{(k)}(0)| \leq \|h\|^k \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big| \sum_{|\alpha|=k} \frac{k!}{\alpha!}.$$

Thus, there is a constant $C > 0$ that depends only on k and n such that

$$|R(h)| \leq C \, \|h\|^k \sup_{t \in [0,1]} \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big|.$$

Since each $\partial^\alpha f$ with $|\alpha| = k$ is continuous at x_0, the function

$$\omega(h) := \sup_{t \in [0,1]} \max_{|\alpha|=k} \Big| (\partial^\alpha f)(x_0 + th) - (\partial^\alpha f)(x_0) \Big| \to 0$$

as $h \to 0$. Hence $R(h) = o(\|h\|^k)$. $\square$

Example 5.2.3 (second-order Taylor expansion)**.** *For $k = 2$, Taylor's formula can be written in the compact form*

$$f(x) = f(x_0) + \operatorname{grad} f(x_0)(x - x_0) + \frac{1}{2}(x - x_0)^\top H_f(x_0)(x - x_0) + o(\|x - x_0\|^2).$$

```julia
using ForwardDiff
f(x) = x[1]^2 +x[1]*x[2]+2*x[2]^2

function mySecondOrderApproximation(f,x0)
    ∇f = ForwardDiff.gradient(f,x0)
    Hf = ForwardDiff.hessian(f, x0)
    return x -> f(x0)+∇f'*(x-x0)+1/2*(x-x0)'*Hf*(x-x0)
end
```

Example 5.2.4. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x) = x_1^2 + x_1 x_2 + 2x_2^2.$$

At the point $x_0 = (0,0)$ we compute $f(0,0) = 0$ and

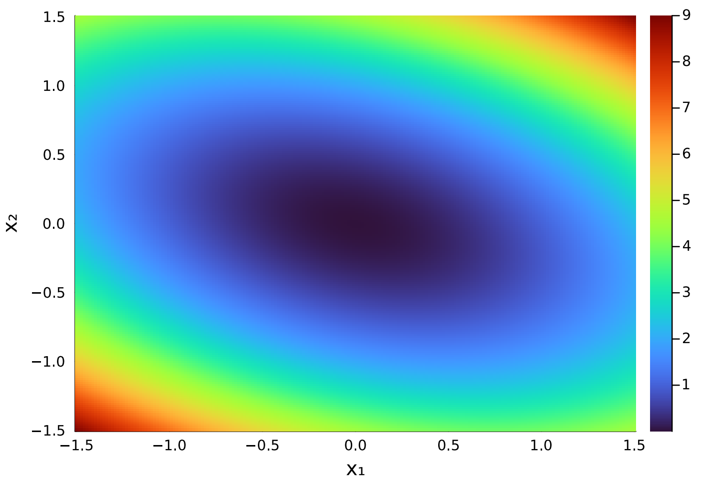
$$\operatorname{grad} f(0,0) = (0,0), \qquad H_f(0,0) = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

Thus the second-order Taylor polynomial of f at $(0,0)$ is given by

$$\begin{aligned} f(x) = \frac{1}{2}(x_1, x_2)\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \frac{1}{2}(x_1, x_2)\begin{pmatrix} 2x_1 + x2 \\ x_1 + 4x_2 \end{pmatrix} \\ &= \frac{1}{2}\left(2x_1^2 + x_1x_2 + x_1x_2 + 4x_2^2\right) \\ &= x_1^2 + x_1x_2 + 2x_2^2. \end{aligned}$$

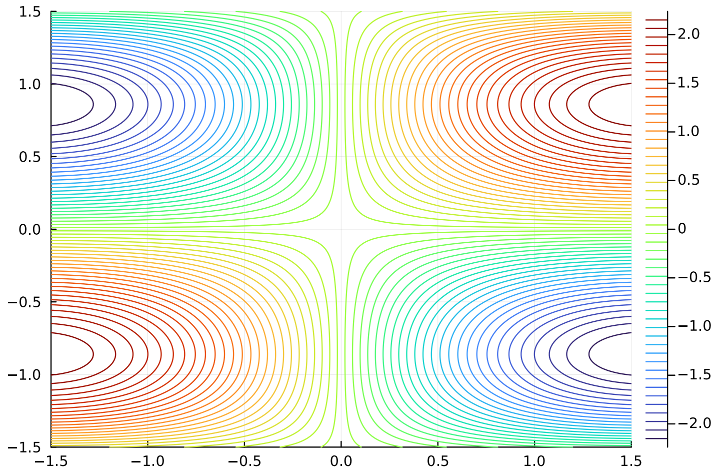
It coincides with f, so that the polynomial approximation is exact in this case.

```
x1 = range(-1.5, 1.5, length=200)
x2 = range(-1.5, 1.5, length=200)

F2(x) = mySecondOrderApproximation(f,[0;0])(x)

Z = [F2([x;y]) for y in x2, x in x1]
heatmap(x1, x2, Z, xlabel="x₁", ylabel="x₂")
```

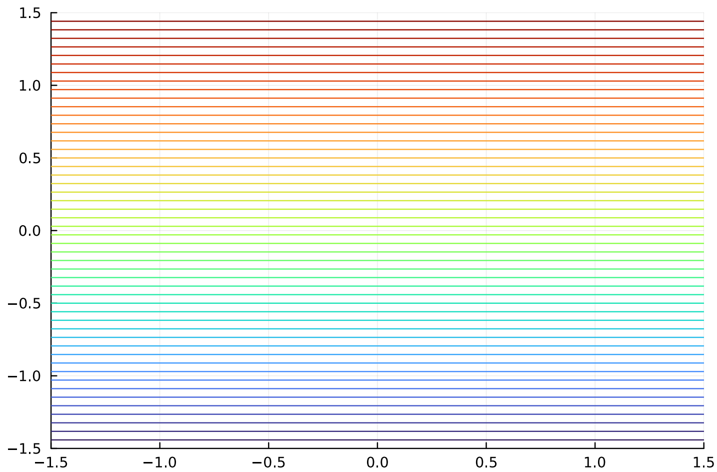


We also consider $f(x) = 4\sin(x_1)\cos(x_2)x_2$ and compute the first and second order approximation in $x_0 = (1,0)^\top$.

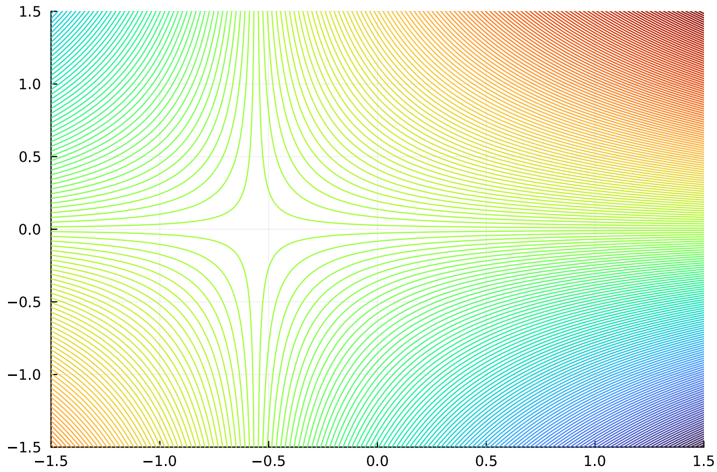
```
f(x) = 4*sin(x[1])*cos(x[2])*x[2]
x1 = range(-1.5, 1.5, length=200)
x2 = range(-1.5, 1.5, length=200)

Z = [f([x;y]) for y in x2, x in x1]

contour(x1, x2, Z,levels = 50)
```




```
x0 = [1;0]

F1(x) = myFirstOrderApproximation(f,x0)(x)

Z = [F1([x;y]) for y in x2, x in x1]
contour(x1, x2, Z,levels = 50,colorbar=false)
```




```
F2(x) = mySecondOrderApproximation(f,x0)(x)

Z = [F2([x;y]) for y in x2, x in x1]

contour(x1, x2, Z,levels = 200, colorbar=false)
```



Recall that a matrix $A \in \mathbb{R}^{n\times n}$ is called positive definite if

$$x^\top Ax > 0, \qquad \forall\, x \in \mathbb{R}^n \setminus \{0\}.$$

It is called negative definite if $-A$ is positive definite and indefinite if there are $x, y \in \mathbb{R}^n$ such that

$$x^\top Ax < 0, \qquad y^\top Ay > 0.$$

The gradient and the Hessian combined provide a sufficient condition for local extrema.

Theorem 5.2.5 (Second-order sufficient conditions for extrema)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable and let $x_0 \in \mathbb{R}^n$ be a critical point, i.e.* $\operatorname{grad} f(x_0) = 0$.

(i) If $H_f(x_0)$ is positive definite, then x_0 is a strict local minimum.

(ii) If $H_f(x_0)$ is negative definite, then x_0 is a strict local maximum.

(iii) If $H_f(x_0)$ is indefinite, then x_0 is a saddle point.

Remark 5.2.6. *If the Hessian $H_f(x_0)$ is positive semidefinite or negative semidefinite but not definite, the second-order test is inconclusive.*

Remark 5.2.7 (Local extrema)**.** *In data-driven models, local extrema correspond to locally optimal parameter values where small changes do not improve the objective function.*

Proof. Since f is twice continuously differentiable, Taylor's formula of second order yields, for h sufficiently small,

$$f(x_0 + h) = f(x_0) + \operatorname{grad} f(x_0) \cdot h + \frac{1}{2} h^\top H_f(x_0) h + o(\|h\|^2).$$

Because x_0 is a critical point, $\operatorname{grad} f(x_0) = 0$, and hence

$$f(x_0 + h) = f(x_0) + \frac{1}{2} h^\top H_f(x_0) h + \varphi(h),$$

where $\varphi(h)/\|h\|^2 \to 0$ for $h \to 0$.

(i) The idea is to verify that $\frac{1}{2}h^\top H_f(x_0)h$ is positive and larger than $\varphi(h)$. We first define

$$\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}.$$

As the pre-image of the closed set $\{1\}$ of a continuous map $\|\cdot\|$, the set $\mathbb{S}^{n-1}$ is closed. It is also bounded. Heine-Borel extends to from $\mathbb{R}$ to $\mathbb{R}^n$ and therefore $\mathbb{S}^{n-1}$ is compact. Since $h \mapsto \frac{1}{2}h^\top H_f(x_0)h$ is continuous, its image of $\mathbb{S}^{n-1}$ is compact. Therefore, it assumes its minimum, which must be a positive number since $H_f(x_0)$ is positive definite. We define

$$\alpha := \min_{h \in \mathbb{S}^{n-1}} \frac{1}{2}h^\top H_f(x_0)h.$$

For $h \in \mathbb{R}^n \setminus \{0\}$, we calculate

$$\frac{1}{2}h^\top H_f(x_0)h = \frac{1}{2}\|h\|\frac{h^\top}{\|h\|}H_f(x_0)\frac{h}{\|h\|}\|h\| \geq \frac{1}{2}\alpha\|h\|^2.$$

Since $\varphi(h)/\|h\|^2 \to 0$ for $h \to 0$, there is $\delta > 0$ such that

$$\frac{|\varphi(h)|}{\|h\|^2} \leq \frac{\alpha}{4}, \qquad \forall \|h\| \leq \delta.$$

For $\|h\| \leq \delta$, we obtain

$$\begin{aligned} f(x_0 + h) &= f(x_0) + \frac{1}{2}h^\top H_f(x_0)h + \varphi(h) \\ &\geq f(x_0) + \frac{\alpha}{2}\|h\|^2 - \frac{\alpha}{4}\|h\|^2 \\ &= f(x_0) + \frac{\alpha}{4}\|h\|^2. \end{aligned}$$

This shows that x_0 is a strict local minimum.
(ii) If $H_f(x_0)$ is negative definite, the same argument applied to $-f$ shows that x_0 is a strict local maximum.
(iii) Finally, if $H_f(x_0)$ is indefinite, there exist vectors $u, v \in \mathbb{R}^n$ such that

$$u^\top H_f(x_0)u > 0 \quad \text{and} \quad v^\top H_f(x_0)v < 0.$$

Choosing $x = x_0 + tu$ and $x = x_0 + tv$ with t sufficiently small and using the Taylor expansion shows that $f(x) - f(x_0)$ takes both positive and negative values arbitrarily close to x_0. Hence x_0 is a saddle point. □

Example 5.2.8. *Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by*

$$f(x_1, x_2) = x_1^2 + x_2^2.$$

Then $\operatorname{grad}(f)(x) = (2x_1, 2x_2)$, *hence the only critical point is $x_0 = (0,0)$. Moreover,*

$$H_f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

which is positive definite. Thus $(0,0)$ is a strict local minimum.

Let us revisit Example 4.4.7.

Example 5.2.9. *The function* $f : \mathbb{R}^2 \to \mathbb{R}$*,* $f(x_1, x_2) = x_1 x_2$ *has the gradient and Hessian*

$$\operatorname{grad} f(x) = (x_2\,,\, x_1)\,, \qquad H_f(x) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The gradient vanishes for $x = 0$*, and the Hessian is indefinite due to*

$$(x_1, x_2)\, H_f(x) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2x_1x_2.$$

Thus, $x = 0$ *is a saddle point as already stated in Example 4.4.7.*

Computing extrema by gradient descent

Suppose we want to minimize a convex function $f : \mathbb{R}^n \to \mathbb{R}$. We obtain its minimum by a fixed point iteration.

Convexity plays a key role, and we first characterize convexity by the Hessian matrix. Recall that a matrix $A \in \mathbb{R}^{n\times n}$ is called positive semi-definite if

$$x^\top A x \geq 0, \qquad \forall\, x \in \mathbb{R}^n\,.$$

Lemma 5.2.1 (Second-order characterization of convexity)**.** *Let* $f \in \mathcal{C}^2(\mathbb{R}^n)$*. Then* f *is convex if and only if* $H_f(x)$ *is positive semi-definite for all* $x \in \mathbb{R}^n$*.*

The first order characterization of convexity in Lemma 4.4.2 yields

$$f(x) - f(y) \leq f'(x)\,(x - y), \qquad x, y \in \mathbb{R}^n\,.$$

Moreover, we rewrite convexity as follows.

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2), \qquad x_1, x_2 \in \mathbb{R}^n, \quad \lambda_1, \lambda_2 \in [0,1], \quad \lambda_1 + \lambda_2 = 1.$$

The extension to convex linear combinations of length m is known as Jensen's inequality.

Lemma 5.2.2 (Jensen's inequality)**.** *Let* $f : \mathbb{R}^n \to \mathbb{R}$ *be convex and* $m \in \mathbb{N}$ *with* $m \geq 1$*. Then*

$$f(\sum_{i=1}^m \lambda_i x_i) \leq \sum_{i=1}^m \lambda_i f(x_i), \qquad x_i \in \mathbb{R}^n, \quad \lambda_i \in [0,1], \quad \sum_{i=1}^m \lambda_i = 1.$$

Proof. The case $m = 1$ is obvious and $m = 2$ is the definition of convexity. Assume that the claim holds for m. To verify the inequality for $m + 1$, we consider $\lambda_i \in [0,1]$ and $\sum_{i=1}^{m+1} \lambda_i = 1$. There must be one index i such $\lambda_i \neq 0$. Without loss of generality, we may assume that $i = m+1$, otherwise we may reorder. For $\lambda_{m+1} \neq 0$, we obtain

$$\begin{aligned} f(\sum_{i=1}^{m+1} \lambda_i x_i) &= f((1-\lambda_{m+1}) \underbrace{\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i}_{\tilde{x}_1} + \lambda_{m+1} x_{m+1}) \\ &\leq (1-\lambda_{m+1}) f(\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i) + \lambda_{m+1} f(x_{m+1}). \end{aligned}$$

The induction hypothesis for m leads to

$$f(\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} x_i) \leq \sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} f(x_i),$$

because $\sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} = \frac{1}{1-\lambda_{m+1}} \sum_{i=1}^{m} \lambda_i = 1$. We therefore derive

$$\begin{aligned} f(\sum_{i=1}^{m+1} \lambda_i x_i) &\leq (1-\lambda_{m+1}) \sum_{i=1}^{m} \frac{\lambda_i}{1-\lambda_{m+1}} f(x_i) + \lambda_{m+1} f(x_{m+1}) \\ &= \sum_{i=1}^{m+1} \lambda_i f(x_i). \end{aligned}$$

□

Gradient descent

Theorem 5.2.3 (Gradient descent). *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable and assume that*

$$\|\nabla f(x)\| \leq C \qquad \text{for all } x \in \mathbb{R}^d$$

for some $C > 0$. Let x^* be a minimizer of f. For an initial $x_0 \in \mathbb{R}^d$, consider gradient descent

$$x_{k+1} := x_k - \eta \nabla f(x_k)$$

with step size $\eta > 0$. Define the average iterate

$$\bar{x}_N := \frac{1}{N} \sum_{k=0}^{N-1} x_k.$$

Then for every $N \geq 1$,

$$|f(\bar{x}_N) - f(x^*)| \leq \frac{\|x_0 - x^*\|^2}{2\eta N} + \frac{\eta C^2}{2}.$$

In particular, with $\eta = \|x_0 - x^*\|/(C\sqrt{N})$,

$$|f(\bar{x}_N) - f(x^*)| \leq N^{-\frac{1}{2}} C \|x_0 - x^*\|.$$

Thus, the error is bounded by $|f(\bar{x}_N) - f(x^*)|$ decays as $N^{-1/2}$.

Proof. Let $e_k := x_k - x^*$. Expanding the update gives

$$\begin{aligned}\|e_{k+1}\|^2 &= \|x_k - \eta\nabla f(x_k) - x^*\|^2 \\ &= \|e_k - \eta\nabla f(x_k)\|^2 \\ &= \|e_k\|^2 - 2\eta\langle\nabla f(x_k), e_k\rangle + \eta^2\|\nabla f(x_k)\|^2,\end{aligned}$$

and rearranging terms leads to

$$2\eta\langle\nabla f(x_k), e_k\rangle = \|e_k\|^2 - \|e_{k+1}\|^2 + \eta^2\|\nabla f(x_k)\|^2.$$

The first-order characterization of convexity in Lemma 4.4.2 yields

$$f(x_k) - f(x^*) \le \operatorname{grad} f(x_k)\,(x_k - x^*) = \langle\nabla f(x_k), e_k\rangle.$$

The gradient bound, $\|\nabla f(x_k)\|^2 \le C^2$, leads to

$$2\eta\big(f(x_k) - f(x^*)\big) \le \|e_k\|^2 - \|e_{k+1}\|^2 + \eta^2 C^2.$$

Summing from $k = 0$ to $N - 1$ telescopes:

$$\begin{aligned}2\eta\sum_{k=0}^{N-1}\big(f(x_k) - f(x^*)\big) &\le \sum_{k=0}^{N-1}(\|e_k\|^2 - \|e_{k+1}\|^2) + N\eta^2C^2 \\ &= \|e_0\|^2 - \|e_{N+1}\|^2 + N\eta^2C^2 \\ &\le \|e_0\|^2 + N\eta^2C^2.\end{aligned}$$

Divide by $2\eta N$ and use convexity (Jensen) $f(\bar{x}_N) \le \frac{1}{N}\sum_{k=0}^{N-1} f(x_k)$ to conclude the claim. $\square$

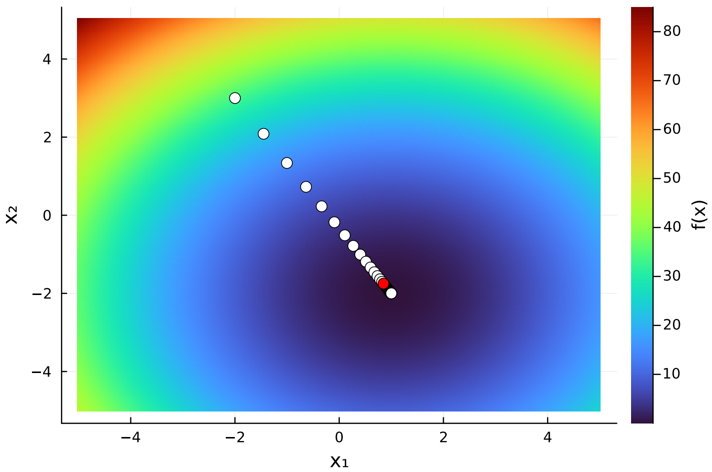
To implement gradient descent, we store all iterates and return both, the iterates and the average $\bar{x}_N$.

```
function myGradientDescent(f, x0; η = 0.01, N = 1000)
    d = length(x0)
    x_vec = zeros(d,N)
    x_vec[:,1] = x0

    for k in 1:N-1
        x_vec[:,k+1] = x_vec[:,k] - η * ForwardDiff.gradient(f,x_vec[:,k])
    end

    x_avg = sum(x_vec[:,k] for k=1:N)/N

    return x_vec , x_avg
end
```

The function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x_1, x_2) = (x_1 - 1)^2 + (x_2 + 2)^2$ has the minimum in $x^* = (1, -2)^\top$.

```
f(x::Vector) = (x[1] - 1)^2 +(x[2] + 2)^2

x1 = range(-5, 5, length=200)
x2 = range(-5, 5, length=200)
Z = [f([x, y]) for y in x2, x in x1]
heatmap(x1, x2, Z, xlabel="x₁", ylabel="x₂", colorbar_title="f(x)")
```

```
x0 = [-2;3]
x_vec , x_avg = myGradientDescent(f, x0; η = 0.01)

heatmap(x1, x2, Z, xlabel="x₁", ylabel="x₂", colorbar_title="f(x)", c = :turbo)
scatter!(x_vec[1,1:10:end], x_vec[2,1:10:end], color="white", markersize=5)
scatter!(x_avg[1,:], x_avg[2,:], color="red", markersize=5)
```



If f is not convex, then gradient descent converges towards a critical point. To see this, we still need some preparation.

Definition 5.2.4. *A function $g : \mathbb{R}^d \to \mathbb{R}$ is called L-Lipschitz continuous if there exists a constant $L > 0$ such that*

$$\|g(x) - g(y)\| \le L\|x - y\|, \qquad \forall x, y \in \mathbb{R}^d.$$

Lemma 5.2.5. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and suppose that ∇f is L-Lipschitz continuous. Then*

$$f(y) - f(x) \le \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|^2, \qquad \forall\, x, y \in \mathbb{R}^d.$$

Proof. Define the path $\phi : [0, 1] \to \mathbb{R}$ by

$$\phi(t) = f(x + t(y - x)).$$

The chain rule leads to

$$\phi'(t) = \nabla f(x + t(y - x))^\top (y - x).$$

The fundamental theorem of calculus and Cauchy-Schwarz yield

$$\begin{aligned}
f(y) - f(x) &= \int_0^1 \phi'(t)dt \\
&= \int_0^1 \nabla f(x + t(y-x))^\top (y-x)dt \\
&= \nabla f(x)^\top (y-x) + \int_0^1 \left(\nabla f(x + t(y-x) - \nabla f(x)^\top\right)^\top (y-x)dt \\
&\le \nabla f(x)^\top (y-x) + \int_0^1 \|\nabla f(x + t(y-x) - \nabla f(x)\| \|y-x\| dt \\
&\le \nabla f(x)^\top (y-x) + \int_0^1 L\|t(y-x)\| \|y-x\| dt \\
&= \nabla f(x)^\top (y-x) + L\|y-x\|^2 \int_0^1 t dt \\
&= \nabla f(x)^\top (y-x) + \frac{L}{2}\|y-x\|^2.
\end{aligned}$$

□

Theorem 5.2.6. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable, bounded from below, so that $f(x) \ge r_0$ for all $x \in \mathbb{R}^d$. Suppose that ∇f is L-Lipschitz continuous. Then gradient descent*

$$x_{k+1} = x_k - \eta \nabla f(x_k), \qquad 0 < \eta < \frac{2}{L},$$

satisfies

$$\nabla f(x_k) \to 0.$$

Proof. By Lemma 5.2.5,

$$f(x_{k+1}) - f(x_k) \le \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{L}{2}\|x_{k+1} - x_k\|^2.$$

According to $x_{k+1} = x_k - \eta \nabla f(x_k)$, we derive

$$f(x_{k+1}) - f(x_k) \le -\eta \|\nabla f(x_k)\|^2 + \frac{L\eta^2}{2}\|\nabla f(x_k)\|^2.$$

We define $c := \eta(1 - \frac{L\eta}{2}) > 0$ and obtain

$$f(x_{k+1}) - f(x_k) \le -\eta(1 - \frac{L\eta}{2})\|\nabla f(x_k)\|^2 = -c\|\nabla f(x_k)\|^2.$$

This implies $c\|\nabla f(x_k)\|^2 \le f(x_k) - f(x_{k+1})$ and summing from $k = 0$ to N yields

$$c\sum_{k=0}^N \|\nabla f(x_k)\|^2 \le \sum_{k=0}^N (f(x_k) - f(x_{k+1})) = f(x_0) - f(x_{N+1}) \le f(x_0) - r_0.$$

Since this inequality holds for all $N \in \mathbb{N}$, we deduce that the series converges with

$$c\sum_{k=0}^\infty \|\nabla f(x_k)\|^2 \le f(x_0) - r_0.$$

This implies that $\|\nabla f(x_k)\|^2 \to 0$. $\square$

Stochastic gradient descent

If $F : \mathbb{R}^d \to \mathbb{R}$ is an average over simpler components,

$$F(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x),$$

then we can use alternatives to the gradient of F. Instead of computing

$$\operatorname{grad} F(x) = \frac{1}{n} \sum_{i=1}^{n} \operatorname{grad} f_i(x)$$

in each iteration, we restrict us to only one index $i \in \{1, \dots, n\}$. That is we replace $\operatorname{grad} F(x)$ by $\operatorname{grad} f_i(x)$, where $i \in \{1, \dots, n\}$ is random in each iteration.

Remark 5.2.7. *This makes gradient descent significantly more efficient and turns out to be a game changer in large scale machine learning scenarios.*

Theorem 5.2.8 (Stochastic gradient descent for finite-sum problems)**.** *Let $f_1, \dots, f_n : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable, and define*

$$F(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x).$$

Let x^* be a minimizer of f. Assume that

$$\|\nabla f_i(x)\| \leq C \quad \textit{for all } x \in \mathbb{R}^d,\ i = 1, \dots, n,$$

for some $C > 0$.
Let $(i_k)_{k \in \mathbb{N}}$ be independent uniformly distributed on $\{1, \dots, n\}$, and define

$$x_{k+1} = x_k - \eta \nabla f_{i_k}(x_k),$$

with step size $\eta > 0$. Then for $\bar{x}_N := \frac{1}{N} \sum_{k=0}^{N-1} x_k$,

$$\mathbb{E}[F(\bar{x}_N)] - F(x^*) \leq \frac{\|x_0 - x^*\|^2}{2\eta N} + \frac{\eta C^2}{2}.$$

In particular, choosing $\eta = \|x_0 - x^*\|/(C\sqrt{N})$ yields

$$\mathbb{E}[F(\bar{x}_N)] - F(x^*) \leq N^{-\frac{1}{2}}\, C \|x_0 - x^*\|.$$

Proof idea. The only probabilistic input we use is the following elementary fact: if i is independently, uniformly distributed on $\{1, \dots, n\}$ and $\psi : \{1, \dots, n\} \to \mathbb{R}^m$ is any function, then

$$\mathbb{E}_i[\psi(i)] = \frac{1}{n}\sum_{j=1}^{n}\psi(j).$$

In our setting, this implies

$$\mathbb{E}_{i_k}\left[\nabla f_{i_k}(x)\right] = \frac{1}{N}\sum_{i=1}^{N}\nabla f_i(x) = \nabla F(x). \tag{5.2}$$

We now follow the proof of gradient descent and use the expectation at the appropriate locations. We omit the details. □

```
function myStochasticGradientDescent(F, x0; η = 0.01, N = 10_000)
    d = length(x0)
    n = length(F)
    x_vec = zeros(d,N)
    x_vec[:,1] = x0

    for k in 1:N-1
        i = rand(1:n)
        x_vec[:,k+1] = x_vec[:,k] - η * ForwardDiff.gradient(F[i],x_vec[:,k])
    end

    x_avg = sum(x_vec[:,k] for k=1:N)/N

    return x_vec , x_avg
end

F = [ x -> (x[1] - 1)^2 + (x[2] - 2)^2,
      x -> 2(x[1] - 1)^2 + (x[2] - 2)^2,
      x -> (x[1] - 1)^2 + 3(x[2] - 2)^2 ]

x0 = [10.0; -5.0]

x_vec , x_avg = myStochasticGradientDescent(F,x0;η=0.01, N=100_000)
x_avg
```

```
2-element Vector{Float64}:
 1.003158302447018
 1.997624054388961
```

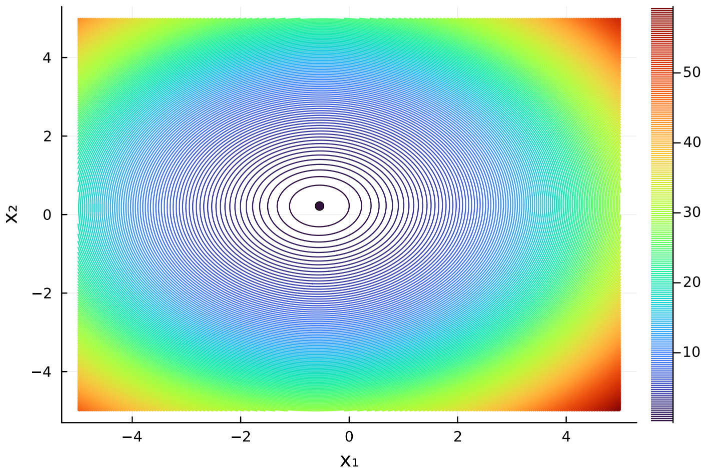
We now provide an example for large n.

```
using Random, ForwardDiff, Statistics
n = 1_000
d = 2

# model setup
Random.seed!(42)
A = randn(n, d)
x_true = randn(d)
y = A * x_true .+ 0.1*randn(n)

F = [x -> (dot(A[i, :],x) - y[i])^2 for i in 1:n]
```

```
x0 = zeros(d)
x_vec, x_avg = myStochasticGradientDescent(F, x0; η=0.01, N=10_000)

println("x_true = ", x_true)
println("x_avg  = ", x_avg)
```

```
x_true = [-0.5445681210073285, 0.2208132624522603]
x_avg = [-0.5467396476613867, 0.21803998455760248]
```

```
g(x) = sum(F[i](x) for i=1:n)/n
Z = [g([x, y]) for y in x2, x in x1]
contour(x1, x2, Z, xlabel="x₁", ylabel="x₂", levels = 200)
scatter!(x_avg[1,:],x_avg[2,:])
```



Mini-batch SGD

Mini-batch stochastic gradient descent uses a small subset of gradients instead of a single i_k for more robust outcomes. For $F : \mathbb{R}^d \to \mathbb{R}$ given by

$$F(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x),$$

the gradient

$$\operatorname{grad} F(x) = \frac{1}{n} \sum_{i=1}^{n} \operatorname{grad} f_i(x)$$

is replaced in each iteration by a smaller subset (batch) of gradients over f_i. This means at iteration k, we choose a batch

$$B_k \subseteq \{1, \dots, n\}$$

and use the iteration (update) rule

$$x_{k+1} = x_k - \eta \sum_{i \in B_k} \nabla f_i(x_k).$$

C R T A We started from a finite-dimensional normed vector space described in coordinate-free form. By completion, this space extended to a Banach space and, in the presence of an inner product, to a Hilbert space. The structure gained in this process enabled differential calculus in the form of Fréchet derivatives defined without reference to coordinates.
For computation, coordinates were introduced, enabling partial derivatives. Truncation then replaced the full Taylor expansion by a polynomial of finite degree. Approximation quantified the resulting error via bounds for the Taylor remainder in the underlying norm. Thus, this chapter followed the entire computational cycle: from finite-dimensional coordinate-free structure through completion to infinite-dimensional analysis, and back through coordinates and truncation to finite representation together with controlled approximation.

Essentials

Taylor's formula extends to multivariate functions, providing local polynomial approximations. Local extrema occur at critical points and definiteness of the Hessian matrix enables us to distinguish minima, maxima, and saddle points. Convexity is characterized by positive semi-definiteness of the Hessian. Gradient and stochastic gradient descent are fixed point iterations to numerically determine minimizers.

T A Truncating the Taylor expansion provides low order function representations. The approximation error is controlled by the order of the Taylor polynomial.

Taylor expansion, Taylor polynomial, local extrema, second-order conditions, saddle points, convex function, gradient descent, stochastic gradient descent

T A Multivariate analysis provides local linear models of nonlinear maps through Fréchet derivatives, Jacobians, and Taylor expansions. Once such linearizations are available, the central question is how perturbations of the input influence the computed value. This sensitivity is quantified by the concept of conditioning and forms the starting point of numerical analysis and finite-precision computation.

5.3 Newton's method in $\mathbb{R}^n$

To formulate a fixed point iteration to determine zeros of multivariate functions $f : \mathbb{R}^n \to \mathbb{R}^n$, we need some preparation.

Theorem 5.3.1 (Implicit Function Theorem). *Let $U_1 \subset \mathbb{R}^n$ and $U_2 \subset \mathbb{R}^m$ be open sets, and let*

$$f : U_1 \times U_2 \to \mathbb{R}^m, \qquad (x, y) \mapsto f(x, y),$$

be continuously differentiable. Suppose that

$$f(a, b) = 0$$

for some $(a,b) \in U_1 \times U_2$*, and that the Jacobian matrix*

$$\frac{\partial f}{\partial y}(a,b) = \begin{pmatrix} \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial y_1} & \cdots & \frac{\partial f_m}{\partial y_m} \end{pmatrix}$$

is invertible.
Then there exist neighborhoods $V_1 \subset U_1$ *of* a *and* $V_2 \subset U_2$ *of* b*, together with a continuously differentiable map*

$$g : V_1 \to V_2,$$

such that

(i) $f(x, g(x)) = 0$ *for all* $x \in V_1$*,*

(ii) Moreover, for all $(x,y) \in V_1 \times V_2$*,*

$$f(x,y) = 0 \quad \Longleftrightarrow \quad y = g(x).$$

Theorem 5.3.2 (Inverse Function Theorem)**.** *Let* $U \subset \mathbb{R}^n$ *be open and let* $f : U \to \mathbb{R}^n$ *be continuously differentiable. Suppose that* $a \in U$ *and that* $f'(a)$ *is invertible.*
Then there exist neighborhoods $U_0 \subset U$ *of* a *and* $V_0 \subset \mathbb{R}^n$ *of* $b = f(a)$ *such that* $f : U_0 \to V_0$ *is bijective and* $f^{-1} : V_0 \to U_0$ *is continuously differentiable with*

$$(f^{-1})'(b) = (f'(a))^{-1}.$$

Proof. Define $F : \mathbb{R}^n \times U \to \mathbb{R}^n$ according to

$$F(x,y) = x - f(y).$$

Then $F(f(a), a) = 0$ and $\frac{\partial F}{\partial y}(x,y) = -f'(y)$. Since $Df(a)$ is invertible, so is $\frac{\partial F}{\partial y}(f(a), a)$. By the Implicit Function Theorem, there exist neighborhoods

$$V_1 \ni f(a), \qquad V_2 \ni a,$$

and a continuously differentiable map $g : V_1 \to V_2$ such that (according to Part (i))

$$F(x, g(x)) = 0.$$

Hence $x - f(g(x)) = 0$, so that $f(g(x)) = x$ for all $x \in V_1$.
We use Part (ii) to see that $(x,y) \in V_1 \times V_2$ with $F(x,y) = 0$ implies $y = g(x)$. Since then $x = f(y)$, we obtain $y = g(f(y))$. Hence, g is the inverse of f on suitable neighborhoods contained in V_1 and V_2. □

Lemma 5.3.3. *Suppose* $f : \mathbb{R}^n \to \mathbb{R}^m$ *is continuously differentiable and* $B_r \subseteq \mathbb{R}^n$ *a closed ball. Then* f *is L-Lipschitz continuous on* B_r *with Lipschitz constant*

$$L = \sup_{z \in B_r} \|f'(z)\|$$

Proof. For $x, y \in B_r$, define $g : [0,1] \to \mathbb{R}^m$ by $g(t) = f(y + t(x-y))$. The chain rule yields

$$g'(t) = f'(y + t(x-y))(x-y).$$

The fundamental theorem of calculus implies

$$f(x) - f(y) = g(1) - g(0) = \int_0^1 f'(y + t(x-y))(x-y)dt.$$

Taking norms on both sides leads to

$$\|f(x) - f(y)\| \leq \int_0^1 \|f'(y + t(x-y))\| \|(x-y)\| dt \leq L\|x-y\|.$$

□

We formulate Newton's method on $\mathbb{R}^n$.

Theorem 5.3.4. *Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is twice continuously differentiable and $x^* \in \mathbb{R}^n$ such that $f(x^*) = 0$ and $f'(x^*)$ is invertible. Then there is $r > 0$ such that for all $x_0 \in B_r(x^*)$ the iterative sequence*

$$x_{k+1} = x_k - f'(x_k)^{-1} f(x_k) \tag{5.3}$$

converges towards x^* and there is $C > 0$ such that

$$\|x_{k+1} - x^*\| \leq C\|x_k - x^*\|^2.$$

Proof. The inverse function theorem (5.3.2) implies that there is a radius $r > 0$ and a constant $M > 0$ such that

$$\|f'(x)^{-1}\| \leq M, \qquad \forall\, x \in B_r(x^*).$$

Put $e_k := x_k - x^*$ and again observe

$$\int_0^1 f'(x^* + te_k)e_k dt = f(x_k) - f(x^*) = f(x_k).$$

We may add and subtract $f'(x_k)e_k$ to obtain

$$f(x_k) = f'(x_k)e_k + \int_0^1 (f'(x^* + te_k - f'(x_k))e_k dt.$$

By subtracting x^* from both sides of Newton's iteration, we get

$$\begin{aligned} e_{k+1} &= e_k - f'(x_k)^{-1}\Big(f'(x_k)e_k + \int_0^1 (f'(x^* + te_k) - f'(x_k))e_k dt\Big) \\ &= -f'(x_k)^{-1} \int_0^1 (f'(x^* + te_k) - f'(x_k))e_k dt. \end{aligned}$$

We may take norms on both sides and the triangle inequality still holds for the integration (as a limit of summations),

$$\begin{aligned}
\|e_{k+1}\| &\leq \|f'(x_k)^{-1}\| \int_0^1 \|f'(x^* + te_k) - f'(x_k)\| \|e_k\| dt \\
&\leq \|f'(x_k)^{-1}\| \int_0^1 L\|x^* + te_k - x_k\| \|e_k\| dt \\
&= \|f'(x_k)^{-1}\| L \int_0^1 \|(t-1)e_k\| \|e_k\| dt \\
&= \|f'(x_k)^{-1}\| L \|e_k\|^2 \int_0^1 (1-t) dt \\
&= \|f'(x_k)^{-1}\| L \|e_k\|^2 \frac{1}{2} \\
&\leq \frac{ML}{2} \|e_k\|^2.
\end{aligned}$$

This is the quadratic error estimate with $C := \frac{ML}{2}$. To check for convergence, we decrease the radius of the ball centered at x^*. Choose a radius $\rho > 0$ such that $\rho \leq r$ and $C\rho \leq \theta < 1$. For $x_0 \in B_\rho(x^*)$, we have $\|e_0\| \leq \rho < 1$ and induction yields $\|e_k\| \leq \rho$, so that the chain of inequalities leads to

$$\|e_{k+1}\| \leq C\|e_k\|^2 \leq C\rho\|e_k\| = \theta\|e_k\| \leq \ldots \theta^{k+1}\|e_0\| \to 0.$$

Thus, $x_k \to x^*$. $\square$

To compute $f'(x_k)^{-1} f(x_k)$, we usually do NOT invert the matrix $f'(x_k)$ but instead solve the linear system of equations

$$f'(x_k)z = f(x_k)$$

for $z \in \mathbb{R}^n$.

Example 5.3.5. *Consider $f : \mathbb{R}^n \to \mathbb{R}^n$ given by*

$$f(x) = \begin{pmatrix} x_1^2 - a_1 \\ \vdots \\ x_n^2 - a_n \end{pmatrix}, \qquad a_1, \ldots, a_n > 0.$$

To find $x^* \in \mathbb{R}^n$ such that $f(x^*) = 0$, we first determine the Jacobian

$$f'(x) = \operatorname{diag}(2x_1, \ldots, 2x_n)$$

As opposed to the numerical implementation, we here determine the inverse matrix

$$f'(x)^{-1} = \operatorname{diag}(\frac{1}{2x_1}, \ldots, \frac{1}{2x_n}).$$

The iteration is

$$\begin{aligned}
x^{(k+1)} &= x^{(k)} - \operatorname{diag}(\frac{1}{2x_1}, \ldots, \frac{1}{2x_n}) f(x^{(k)}) \\
&= x^{(k)} - \begin{pmatrix} \frac{(x_1^{(k)})^2 - a_1}{2x_1^{(k)}} \\ \vdots \\ \frac{(x_n^{(k)})^2 - a_n}{2x_n^{(k)}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}(x_1^{(k)} + \frac{a_1}{x_1^{(k)}}) \\ \vdots \\ \frac{1}{2}(x_n^{(k)} + \frac{a_n}{x_n^{(k)}}) \end{pmatrix}.
\end{aligned}$$

The iteration splits into each separate component. This is a familiar iteration in one dimension.

Example 5.3.6. *Consider $g(y) = y^2 - b$ for $b > 0$. The Newton iteration for the 1-dimensional function is*

$$y^{(k+1)} = y^{(k)} - \frac{(y^{(k)})^2 - b}{2y^{(k)}} = \frac{1}{2}(y^{(k)} + \frac{b}{y^{(k)}}),$$

and $y^{(k)} \to \sqrt{b}$ for suitable choice of $y^{(0)}$.

Thus, for suitable choices of $x^{(0)} \in \mathbb{R}^n$, we have $x^{(k)} \to \begin{pmatrix} \sqrt{a_1} \\ \vdots \\ \sqrt{a_n} \end{pmatrix}$.

The Newton iteration is a fixed point iteration by

$$\Phi(x) = x - f'(x)^{-1} f(x),$$

since $x_{k+1} = \Phi(x_k)$ becomes (5.3) with

$$f(x^*) = 0 \quad \Leftrightarrow \quad \Phi(x^*) = x^*.$$

Since $f(x^*) = 0$, the derivative of Φ is

$$\Phi'(x^*) = I_n - \big(f'(\cdot)f(\cdot)\big)'(x^*) = I_n - f'(x^*)^{-1} f'(x^*) = I_n - I_n = 0.$$

Thus, we have $\|\Phi'(x^*)\| = 0$. Banach-fixed-point theorem only needs $\|\Phi'(x^*)\| < 1$ to establish linear convergence, cf. Theorem 4.4.12. The Newton iteration leads to $\|\Phi'(x^*)\| = 0$ and then derives quadratic convergence.

Example 5.3.7. *Fix an integer $m \geq 2$. By identifying $\mathbb{C}$ with $\mathbb{R}^2$, the complex function $\mathbb{C} \to \mathbb{C}$, $z \mapsto z^m - 1$, is written as $f : \mathbb{R}^2 \to \mathbb{R}^2$,*

$$f(x, y) = \begin{pmatrix} \mathrm{Re}((x + iy)^m - 1) \\ \mathrm{Im}((x + iy)^m - 1) \end{pmatrix}.$$

Its zeros are the m-th roots of unity $\omega_j = e^{2\pi i j/m}$, $j = 0, \ldots, m - 1$.
To determine the Newton iteration in terms of z, we need to determine the Jacobian of f denoted by $f'(z_n) = f'(x_n, y_n)$. Let $h \in \mathbb{C}$ be a small increment. The binomial theorem,

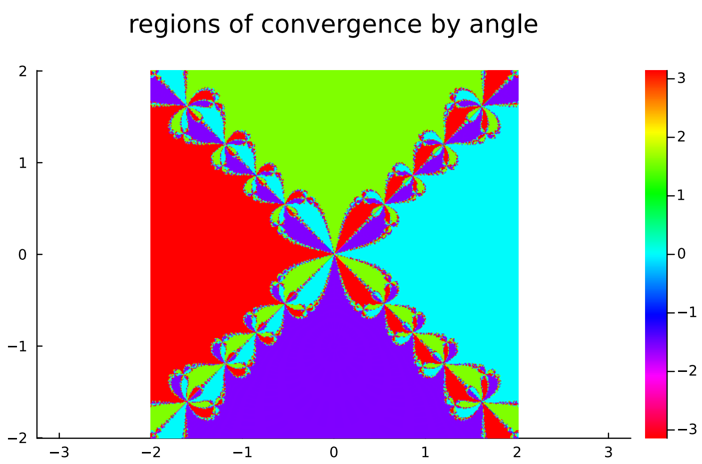
$$(z + h)^m = z^m + mz^{m-1}h + \sum_{j=2}^{m} \binom{m}{j} z^{m-j} h^j,$$

implies that $f'(z)h = mz^{m-1}h$. The inverse is multiplication by $\frac{1}{mz^{m-1}}$, so that the Newton iteration becomes

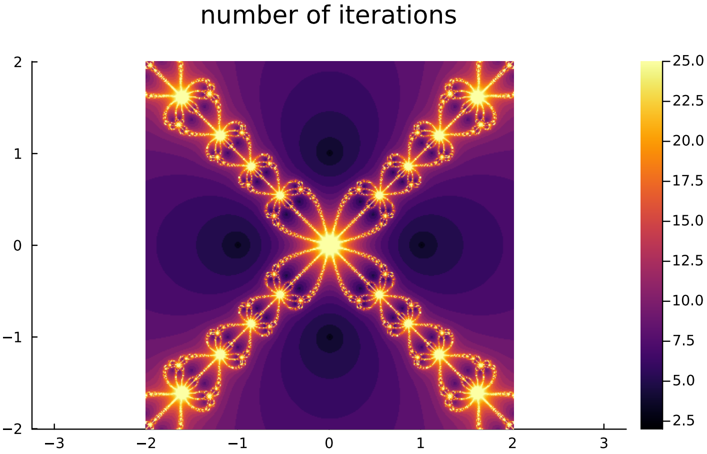
$$z_{k+1} = z_k - \frac{f(z_k)}{f'(z_k)} = z_k - \frac{z_k^m - 1}{mz_k^{m-1}}.$$

We investigate numerically for which initializations $z_0 \in \mathbb{C}$ the Newton iterations converges and if so then to which root.

```
function myNewton(f, ∂f, z0; tol=1e-8, maxiter=100)
    z = copy(z0)
    for k in 1:maxiter
        z = z-f(z)/∂f(z)
        if abs(f(z))<tol
            return z, k
        end
    end
    return z, maxiter
end

function newtonFractal(f,∂f,s)
    step = 0.1e-1 # 10^-2
    interval = -s:step:s
    L = length(interval)

    Zgrid = [a+b*im for b in interval, a in interval]

    Y = myNewton.(f,∂f,Zgrid;tol=1e-10,maxiter=100)
    Z = [Y[k,l][1] for k in 1:L, l in 1:L]
    K = [Y[k,l][2] for k in 1:L, l in 1:L]

    return Z, K, interval
end

m = 4
f(z) = z^m-1
fprime(x) = m*x^(m-1)

X, K, interval = newtonFractal(f,fprime,2);

myColor = range(HSV(0,1,1), stop = HSV(-360,1,1), length = 90)
heatmap(interval, interval, angle.(X), color = myColor, clim = (-π,π), aspect_ratio
    =:1, title = "regions of convergence by angle")
```




```
heatmap(interval,interval,K, clim = (2,25),  aspect_ratio=:equal,title = "number of
    iterations")
```



Where on the real axis is no convergence? The Newton iteration breaks when there is some $z_k = 0$. Obviously, there cannot be any convergence for $z_0 = 0$.

Assume $m = 3$. The Newton iteration then is

$$z_{k+1} = z_k - \frac{z_z^3 - 1}{3z_k^2} = \frac{2}{3} z_k + \frac{1}{3z_k^2} \tag{5.4}$$

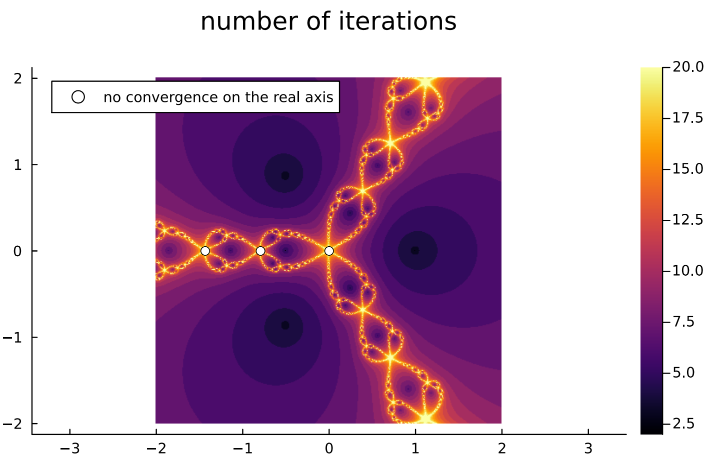
If $z_1 = 0$, then we have

$$0 = \frac{2}{3} z_0 + \frac{1}{3z_0^2} \qquad \Rightarrow \qquad z_0 = -\left(\frac{1}{2}\right)^{1/3}. \tag{5.5}$$

Next, we consider $z_2 = 0$. This yields $z_1 = -\left(\frac{1}{2}\right)^{1/3}$, which leads to

$$-\left(\frac{1}{2}\right)^{1/3} = \frac{2}{3} z_0 + \frac{1}{3z_0^2} \qquad \Rightarrow \qquad \cdots \qquad \Rightarrow \qquad z_0 \approx -1.4338 \tag{5.6}$$

In that case, we obtain:



Essentials

The inverse function theorem is a consequence of the implicit function theorem and ensures that an invertible Jacobi matrix implies local invertibility. It guarantees that Newton's method in d dimensions converges locally, which solves nonlinear systems of equations by a fixed point iteration.

A The Newton iterations approximate solutions to nonlinear systems of equations.

implicit function theorem, inverse function theorem, Newton iteration

5.4 Local extrema under constraints

Minimization of a function $f : \mathbb{R}^n \to \mathbb{R}$ is often constraint to complicated subsets of $\mathbb{R}^n$. We derive the tools to compute local extrema under constraints.

Definition 5.4.1 (submanifolds)**.** *A subset $M \subset \mathbb{R}^n$ is called a k-dimensional submanifold if for every $x_0 \in M$, there exist*

- *an open neighborhood $U \subset \mathbb{R}^n$ of x_0,*
- *a continuously differentiable function $g : U \to \mathbb{R}^r$ with $r = n - k$,*

such that

$$M \cap U = \{x \in U : g(x) = 0\}, \qquad \operatorname{rank} g'(x) = r, \quad \forall x \in M \cap U.$$

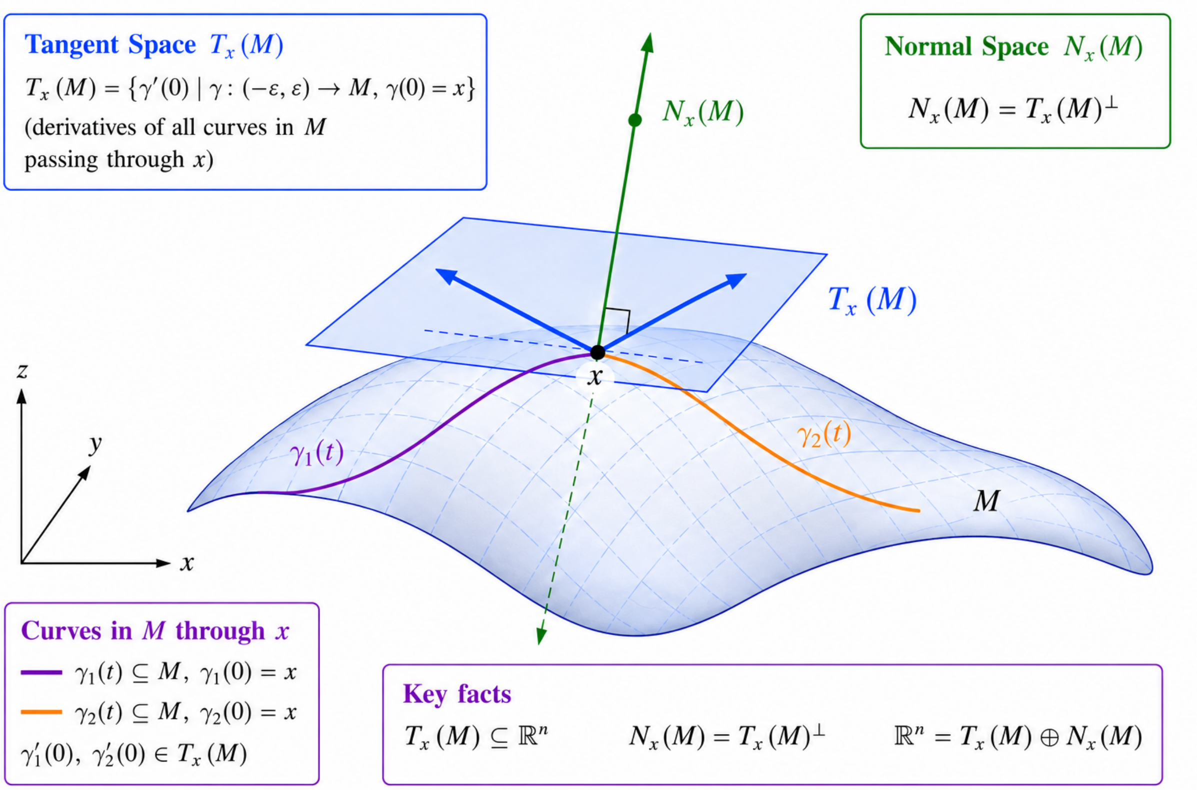
Example 5.4.2. *Consider*

$$M = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 - 1 = 0\}.$$

Then M is a 1-dimensional submanifold of $\mathbb{R}^2$.
Indeed, for $U = \mathbb{R}^2$, the function $g : \mathbb{R}^2 \to \mathbb{R}$ given by

$$g(x) = x_1^2 + x_2^2 - 1,$$

yields $g'(x) = (2x_1, 2x_2)$, which has rank one for $x \in M \cap U = M$ since $(0,0)^\top \notin M$.



Definition 5.4.3 (tangent space). *Let $M \subset \mathbb{R}^n$ be a k-dimensional submanifold and let $x \in M$.*

(a) The tangent space of M at x is

$$T_x(M) = \left\{\gamma'(0) : \gamma : (-\varepsilon, \varepsilon) \to M \text{ continuously differentiable, } \gamma(0) = x\right\}.$$

(b) The normal space of M at x is

$$N_x(M) = T_x(M)^\perp.$$

By definition, the normal space $N_x(M)$ is a linear subspaces of $\mathbb{R}^n$.

Definition 5.4.4. *Let $\Omega \subseteq \mathbb{R}^n$ be open and $f : \Omega \to \mathbb{R}$. Suppose $M \subseteq \mathbb{R}^n$ is a k-dimensional submanifold. We say that $x \in M \cap \Omega$ is a local maximum of f on M if there is an open neighborhood $U \subseteq \mathbb{R}^n$ of x such that*

$$f(y) \leq f(x), \qquad \forall y \in U \cap M \cap \Omega.$$

A local minimum of f on M is defined analogously.

Theorem 5.4.5 (extrema under constraints). *Let $\Omega \subset \mathbb{R}^n$ be open, $f : \Omega \to \mathbb{R}$ be continuously differentiable, and suppose that M is a k-dimensional submanifold. If $x_0 \in \Omega \cap M$ is* *a local extremum of f on M, then*

$$\nabla f(x_0) \in N_{x_0}(M).$$

Proof. Let $\gamma : (-\varepsilon, \varepsilon) \to M$ be a continuously differentiable curve such that $\gamma(0) = x_0$. Define $F : (-\varepsilon, \varepsilon) \to \mathbb{R}$ by $F(t) = f(\gamma(t))$. Since f has a local extremum at x_0 on M, the function F has a local extremum at 0 and hence $F'(0) = 0$. The chain rule implies

$$0 = F'(0) = f'(x_0)\gamma'(0) = \langle \nabla f(x_0), \gamma'(0) \rangle.$$

Since $\gamma'(0) \in T_{x_0}(M)$ was arbitrary,

$$\nabla f(x_0) \perp T_{x_0}(M).$$

Therefore, $\nabla f(x_0) \in N_{x_0}(M)$. □

Theorem 5.4.6. *Let $M \subset \mathbb{R}^n$ be a k-dimensional submanifold and $x \in M$ with a neighborhood $U \subseteq \mathbb{R}^n$ of x such that $g : U \to \mathbb{R}^r$ with $r = n - k$ is continuously differentiable with $M \cap U = \{y \in U : g(y) = 0\}$ and $\operatorname{rank} g'(y) = r$ for all $y \in M \cap U$. Then*

$$T_x(M) = \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^{\perp},$$
$$N_x(M) = \operatorname{span}\{\nabla g_1(x), \ldots, \nabla g_r(x)\}.$$

Since the Jacobian of g is

$$g'(x) = \begin{pmatrix} \operatorname{grad} g_1(x) \\ \vdots \\ \operatorname{grad} g_r(x) \end{pmatrix} = \begin{pmatrix} (\nabla g_1(x))^{\top} \\ \vdots \\ (\nabla g_r(x))^{\top} \end{pmatrix},$$

the rank requirement ensures that $\{\nabla g_1(x), \ldots, \nabla g_r(x)\}$ are linearly independent. The theorem further says that $T_x(M) = \operatorname{null} g'(x) = \{v \in \mathbb{R}^n : g'(x)v = 0\}$ is a linear subspace of dimension k, $N_x(M)$ is a $n-k$-dimensional linear subspace, and

$$\mathbb{R}^n = T_x(M) \oplus N_x(M).$$

Proof. To show $T_x(M) \subseteq \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^{\perp}$, let $v \in T_x(M)$. By definition of the tangent space, there exists a continuously differentiable curve $\gamma : (-\varepsilon, \varepsilon) \to M$ such that $\gamma(0) = x$ and $\gamma'(0) = v$. Since $\gamma(t) \in M$ for all $t \in (-\varepsilon, \varepsilon)$, we have

$$g_i(\gamma(t)) = 0, \qquad i = 1, \ldots, r.$$

Differentiating and applying the chain rule yields

$$0 = \frac{d}{dt} g_i(\gamma(t))\Big|_{t=0} = g_i'(x)\gamma'(0) = g_i'(x)v = \langle \nabla g_i(x), v \rangle, \qquad i = 1, \ldots, r.$$

Thus, we have verified $T_x(M) \subseteq \{\nabla g_1(x), \ldots, \nabla g_r(x)\}^{\perp}$.
To verify the reverse inclusion, take $v \in \mathbb{R}^n$ such that $v \perp \nabla g_i(x) = 0$, for $i = 1, \ldots, r$. Define $F : \mathbb{R} \times \mathbb{R}^r \to \mathbb{R}^r$ by

$$F(t, y) = g(x + tv + (g'(x))^{\top} y).$$

Then we have $F(0,0) = g(x) = 0$ and $\frac{\partial F}{\partial y}(0,0) = g'(x)(g'(x))^\top$. Since $g'(x)(g'(x))^\top \in \mathbb{R}^{r\times r}$ is invertible, the implicit function theorem implies that there is $\varepsilon > 0$ and a continuously differentiable $\eta : (-\varepsilon, \varepsilon) \to \mathbb{R}^r$ such that $\eta(0) = 0$ and

$$F(t, \eta(t)) = 0.$$

Thus, the curve $\gamma : (-\varepsilon, \varepsilon) \to M$,

$$\gamma(t) = x + tv + (g'(x))^\top \eta(t)$$

is continuously differentiable and indeed maps into M because

$$g(\gamma(t)) = F(t, \eta(t)) = 0.$$

Differentiating at $t = 0$ yields

$$0 = g'(x)\gamma'(0) = g'(x)(v + (g'(x))^\top \eta'(0)) = g'(x)v + g'(x)(g'(x))^\top \eta'(0).$$

The assumption on v implies $g'(x)v = 0$, so that we derive

$$g'(x)(g'(x))^\top \eta'(0) = 0.$$

Since $g'(x)(g'(x))^\top$ is invertible, we must have $\eta'(0) = 0$. Therefore

$$\gamma'(0) = v + (g'(x))^\top \eta'(0) = v.$$

Thus $v \in T_x(M)$, and we have verified that $\{\nabla g_1(x), \dots, \nabla g_r(x)\}^\perp \subseteq T_x(M)$. □

Corollary 5.4.7. *Let $U \subset \mathbb{R}^n$ be open and let*

$$M = \{x \in U : g_1(x) = \cdots = g_r(x) = 0\}$$

be a k-dimensional submanifold and $\operatorname{rank} g'(y) = r$ for all $y \in M \cap U$. Let

$$f : U \to \mathbb{R}$$

be continuously differentiable and suppose that $f|_M$ has a local extremum at $x_0 \in M$. Then there are coefficients $\lambda_1, \dots, \lambda_r \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^{r} \lambda_i \nabla g_i(x_0).$$

Example 5.4.8. *Find the extrema of $f(x) = x_1 x_2$ subject to the constraint $g(x) = x_1^2 + x_2^2 - 1 = 0$.*
We compute

$$\nabla f(x) = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}, \qquad \nabla g(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}.$$

At a constrained extremum, $\nabla f(x) = \lambda \nabla g(x)$*, so*

$$\begin{aligned} x_2 &= 2\lambda x_1, \\ x_1 &= 2\lambda x_2, \\ x_1^2 + x_2^2 - 1 &= 0. \end{aligned}$$

Multiplying the first two equations by x_1 *and* x_2*, respectively makes the right hand sides equal and we derive* $x_1^2 = x_2^2$*, hence,* $x_1 = \pm x_2$*.*
Case 1: $x_1 = x_2$
Using the third equations, we obtain $2x_1^2 = 1$*, thus* $x_1 = x_2 = \pm\frac{1}{\sqrt{2}}$*. The function value is* $f(x) = \frac{1}{2}$*.*
Case 2: $x_1 = -x_2$
Again, $2x_1^2 = 1$*, so that* $x_1 = \pm\frac{1}{\sqrt{2}}$ *and* $x_2 = \mp\frac{1}{\sqrt{2}}$*. The function value is* $f(x) = -\frac{1}{2}$*.*
Since the circle $\{x \in \mathbb{R}^2 : g(x) = 0\}$ *is compact and* f *is continuous, it must attain its maximum and its minimum. Those are* $1/2$ *ad* $= -1/2$*, respectively.*

Example 5.4.9. *Consider* $f : \mathbb{R}^3 \to \mathbb{R}$ *given by* $f(x) = \|x\|^2$ *and find the minimizer on* $M = \{x \in \mathbb{R}^3 : g(x) = 0\}$*, where*

$$\begin{aligned} g_1(x) &= x_1^2 + x_2^2 - 1 = 0 \\ g_2(x) &= x_3 - x_1 = 0. \end{aligned}$$

The Jacobian of g *is*

$$g(x) = \begin{pmatrix} 2x_1 & 2x_2 & 0 \\ -1 & 0 & 1 \end{pmatrix},$$

and has rank 2 for all $x \in M$*. By Corollary 5.4.7, any constrained extremum satisfies* $\nabla f(x) = \lambda_1 \nabla g_1(x) + \lambda_2 \nabla g_2(x)$*. Since*

$$\nabla f = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{pmatrix}, \qquad \nabla g_1 = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 0 \end{pmatrix}, \qquad \nabla g_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

we obtain

$$\begin{aligned} 2x_1 &= 2\lambda_1 x_1 - \lambda_2 \\ 2x_2 &= 2\lambda_1 x_2 \\ 2x_3 &= \lambda_2. \end{aligned}$$

Using the constraint $x_3 = x_1$*, the last equation gives* $\lambda_2 = 2x_1$*. Substituting into the first equation yields* $(2 - \lambda_1)x_1 = 0$*, while the second equation becomes* $(1 - \lambda_1)x_2 = 0$*. If* $x_1 = 0$*, then* $x_2^2 = 1$ *and therefore* $x = (0, \pm 1, 0)^\top$ *with* $\lambda = (1, 0)^\top$*.*
If $x_1 \neq 0$*, then* $\lambda_1 = 2$*, so that* $x_2 = 0$*, and hence* $x_1^2 = 1$ *and* $x = (\pm 1, 0, \pm 1)^\top$ *with* $\lambda = (2, \pm 2)^\top$*.*
Evaluating the objective,

$$f(0, \pm 1, 0) = 1, \qquad f(\pm 1, 0, \pm 1) = 2.$$

Therefore the constrained minimizers are $(0, \pm 1, 0)^\top$ *with minimal value* $f(0, \pm 1, 0) = 1$*.*

Newton's method for constrained extrema

Let
$$f : \mathbb{R}^n \to \mathbb{R}$$
and let
$$M = \{x \in \mathbb{R}^n : g(x) = 0\}, \qquad g : \mathbb{R}^n \to \mathbb{R}^r,$$
be a k-dimensional submanifold. We define the Lagrangian by
$$L : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}, \qquad L(x, \lambda) = f(x) - \lambda^\top g(x).$$
To identify local extrema, we must find zeros of the function
$$F : \mathbb{R}^{n+r} \to \mathbb{R}^{n+r}, \qquad F(x, \lambda) = \begin{pmatrix} \nabla_x L(x, \lambda) \\ g(x) \end{pmatrix} = \begin{pmatrix} \nabla f(x) - \sum_{i=1}^r \lambda_i \nabla g_i(x) \\ g(x) \end{pmatrix}.$$
Newton's method applied to F yields the iteration
$$(x_{m+1}, \lambda_{m+1}) = (x_m, \lambda_m) - F'(x_m, \lambda_m)^{-1} F(x_m, \lambda_m),$$
provided that the Jacobian matrix $F'(x_m, \lambda_m)$ is invertible. The derivative of F uses the Hessian of L with respect to x, $H_x L$, and is given by
$$F'(x, \lambda) = \begin{pmatrix} H_x L(x, \lambda) & -J_g(x)^\top \\ J_g(x) & 0 \end{pmatrix}.$$

```
using LinearAlgebra
using ForwardDiff

function myNewtonConstrained(f, g, x0, λ0; tol=1e-10, maxiter=30)
    x = copy(x0)
    n = length(x)
    λ = copy(λ0)
    r = length(λ)
    for k in 1:maxiter
        ∇f = ForwardDiff.gradient(f, x)
        G  = ForwardDiff.jacobian(g, x)
        L(z) = f(z) - dot(λ, g(z))
        H = ForwardDiff.hessian(L, x)
        F = vcat(∇f - G'λ, g(x))
        if norm(F) < tol return x, λ, k end
        K = [H   -G'; G   zeros(r, r)]
        Δ = -K \ F
        x += Δ[1:n]
        λ += Δ[n+1:end]
    end
    return x, λ, maxiter
end

f(x) = x[1] * x[2]
g(x) = [x[1]^2 + x[2]^2 - 1]
x0 = [4.1, 1.4]
λ0 = [0.0]
x, λ, it = myNewtonConstrained(f, g, x0, λ0)
println("iterations = ", it), println("λ = ", λ), println("x = ", x)
println("f(x) = ", f(x)), println("g(x) = ", g(x));
```

```
iterations = 8
λ= [0.4999999999999491]
x = [0.707106781186652, 0.707106781186652]
f(x) = 0.5000000000001478
g(x) = [2.9554136915521667e-13]
```

```
f(x) = x[1]^2+x[2]^2+x[3]^2
g(x) = [x[1]^2 + x[2]^2 - 1; x[3] - x[1]]
x0 = [1, 1, 1]
λ0 = [0.0, 0.0]
x, λ, it = myNewtonConstrained(f, g, x0, λ0)
println("iterations = ", it), println("λ = ", λ), println("x = ", x)
println("f(x) = ", f(x)), println("g(x) = ", g(x));
```

```
iterations = 7
λ= [1.0, 0.0]
x = [0.0, 1.0, 0.0]
f(x) = 1.0
g(x) = [0.0, 0.0]
```

Theorem 5.4.1 (Second-order sufficient conditions under constraints). *Let $U \subset \mathbb{R}^n$ be open, let $f : U \to \mathbb{R}$ be twice continuously differentiable, and suppose that $g : U \to \mathbb{R}^r$ is twice continuously differentiable such that $M = \{x \in U : g(x) = 0\}$ is a k-dimensional submanifold with* $\operatorname{rank} g'(x) = r = n - k$ *for all $x \in M$. Define the Lagrangian by*

$$L : U \times \mathbb{R}^r \to \mathbb{R}, \qquad L(x, \lambda) = f(x) - \sum_{j=1}^{r} \lambda_j g_j(x).$$

Suppose that there exist $x_0 \in M$ and $\lambda \in \mathbb{R}^r$ such that $\nabla_x L(x_0, \lambda) = 0$. Then the following hold.

1. *If $H_x L(x_0, \lambda)$ is positive definite on the tangent space $T_{x_0}(M)$, i.e.*

$$v^\top H_x L(x_0, \lambda) v > 0, \qquad \forall v \in T_{x_0}(M) \setminus \{0\},$$

then x_0 is a strict local minimum of f on M.

2. *If $H_x L(x_0, \lambda)$ is negative definite on the tangent space $T_{x_0}(M)$, i.e.*

$$v^\top H_x L(x_0, \lambda) v < 0, \qquad \forall v \in T_{x_0}(M) \setminus \{0\},$$

then x_0 is a strict local maximum of f on M.

Proof. We prove the minimum case. Let $\gamma : (-\varepsilon, \varepsilon) \to M$ be a twice continuously differentiable curve with $\gamma(t) \in M$ and $\gamma(0) = x_0$. Set $v := \gamma'(0) \in T_{x_0}(M)$ and define $\varphi : (-\varepsilon, \varepsilon) \to \mathbb{R}$ by

$$\varphi(t) = f(\gamma(t)).$$

We aim to investigate its Taylor formula in $t = 0$,

$$f(\gamma(t)) = f(x_0) + \varphi'(0)t + \frac{1}{2}\varphi''(0)t^2 + o(t^2)$$

to check that $f(\gamma(t)) - f(x_0) > 0$ for sufficiently small $|t|$.
The chain rule yields

$$\varphi'(0) = \nabla f(x_0)^\top v = \sum_{i=1}^{r} \lambda_i \nabla g_i(x_0)^\top v = 0,$$
$$\varphi''(0) = v^\top Hf(x_0)v + \nabla f(x_0)^\top \gamma''(0).$$

Since $g_i(\gamma(t)) = 0$, differentiating twice yields

$$0 = v^\top Hg_i(x_0)v + \nabla g_i(x_0)^\top \gamma''(0).$$

Multiplying by λ_i, summing over i, and using $\nabla f(x_0) = \sum_{i=1}^{r} \lambda_i \nabla g_i(x_0)$, we obtain

$$\nabla f(x_0)^\top \gamma''(0) = -v^\top \left(\sum_{i=1}^{r} \lambda_i Hg_i(x_0) \right) v.$$

The observations $\varphi'(0) = 0$ and

$$\begin{aligned} \varphi''(0) &= v^\top Hf(x_0)v - v^\top \left(\sum_{i=1}^{r} \lambda_i Hg_i(x_0) \right) v \\ &= v^\top (Hf(x_0) - \sum_{i=1}^{r} \lambda_i Hg_i(x_0))v \\ &= v^\top (H_x L(x_0, \lambda))v \end{aligned}$$

imply that Taylor's formula of φ is

$$\begin{aligned} f(\gamma(t)) - f(x_0) &= \varphi'(0)t + \frac{1}{2}\varphi''(0)t^2 + o(t^2) \\ &= \frac{1}{2} v^\top (H_x L(x_0, \lambda))vt^2 + o(t^2). \end{aligned}$$

If $H_x L(x_0, \lambda)$ is positive definite on $T_{x_0}(M)$, then the right-hand side is positive for all sufficiently small $t \neq 0$. Hence x_0 is a strict local minimum of f under the constraint.
The maximum case follows analogously from negative definiteness. □

We continue Example 5.4.8 and check the Hessian of the Lagrangian on the tangent space.

Example 5.4.2. *The Lagrangian for $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x) = x_1 x_2$ with the constraint $g(x) = x_1^2 + x_2^2 - 1$ is the map*

$$L(x, \lambda) = x_1 x_2 - \lambda x_1^2 - \lambda_2^2 + \lambda.$$

The Hessian $H_x L$ is

$$H_x L(x, \lambda) = \begin{pmatrix} -2\lambda & 1 \\ 1 & -2\lambda \end{pmatrix}.$$

According to Example 5.4.8, candidate local extrema are

$$\pm(1/\sqrt{2}, 1/\sqrt{2}), \qquad \lambda = 1/2,$$
$$\pm(1/\sqrt{2}, -1/\sqrt{2}), \qquad \lambda = -1/2,$$

For $\lambda = 1/2$, *the Hessian is* $H_x L(x, 1/2) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. *The normal space is the span of* $\nabla g(\pm 1/\sqrt{2}, \pm 1/\sqrt{2}) = \pm(\sqrt{2}, \sqrt{2})^\top$. *The tangent space is the orthogonal complement, hence, the span of* $(1, -1)^\top$. *We observe*

$$(1, -1) \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = -4.$$

It is therefore negative definite on the tangent space and the points $\pm(1/\sqrt{2}, 1/\sqrt{2})$ *are local constraint maxima.*

For the candidates $(\pm 1/\sqrt{2}, \mp 1/\sqrt{2})^\top$, *we similarly compute* $\nabla g(\pm 1/\sqrt{2}, \mp 1/\sqrt{2}) = (\pm\sqrt{2}, \mp\sqrt{2})^\top$. *The tangent space is the span of* $(1, 1)^\top$ *and the Hessian is* $H_x L(x, -1/2) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. *We compute*

$$(1, 1) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 4.$$

Thus, the points $\pm(1/\sqrt{2}, -1/\sqrt{2})$ *are local constraint minima.*

We now complete Example 5.4.9.

Example 5.4.3. *Consider* $f : \mathbb{R}^3 \to \mathbb{R}$ *given by* $f(x) = \|x\|^2 = x_1^2 + x_2^2 + x_3^2$, *and find the constrained extrema on* $M = \{x \in \mathbb{R}^3 : g(x) = 0\}$, *where* $g_1(x) = x_1^2 + x_2^2 - 1 = 0$ *and* $g_2(x) = x_3 - x_1 = 0$.
The tangent space is

$$T_x M = \{v \in \mathbb{R}^3 : \nabla g_i(x) \perp v = 0,\ i = 1, 2\}.$$

The gradients of g_i *are*

$$\nabla g_1(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 0 \end{pmatrix}, \qquad \nabla g_2(x) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

The Lagrangian is

$$L(x, \lambda) = x_1^2 + x_2^2 + x_3^2 - \lambda_1(x_1^2 + x_2^2 - 1) - \lambda_2(x_3 - x_1).$$

The Hessian with respect to the x*-variables is*

$$H_x L(x, \lambda) = \begin{pmatrix} 2 - 2\lambda_1 & 0 & 0 \\ 0 & 2 - 2\lambda_1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

According to Example 5.4.9, the extrema candidates are

$$(0, \pm 1, 0)^\top, \qquad \lambda = (1, 0)^\top,$$
$$(\pm 1, 0, \pm 1)^\top, \qquad \lambda = (2, \pm 2)^\top.$$

At these points, we have $f(0, \pm 1, 0) = 1$ *and* $f(\pm 1, 0, \pm 1) = 2$.
Starting with $(0, \pm 1, 0)^\top$, *we compute*

$$H_x L(0, \pm 1, 0, 1, 0) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad T_{(0,\pm 1,0)^\top} M = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

For $\alpha \in \mathbb{R} \setminus \{0\}$, *we check definiteness by*

$$\alpha(1, 0, 1)^\top \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \alpha^2\, 2 > 0$$

Thus, $(0, \pm 1, 0)^\top$ *are local constraint minima.*
For the candidates $(\pm 1, 0, \pm 1)^\top$, *the gradients of* g_i *are*

$$\nabla g_1(\pm 1, 0, \pm 1) = \begin{pmatrix} \pm 2 \\ 0 \\ 0 \end{pmatrix}, \qquad \nabla g_2(\pm 1, 0, \pm 1) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Hence, the Hessian of the Lagrangian and the tangent space are

$$H_x L(\pm 1, 0, \pm 1, 2, -2) = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad T_{(\pm 1,0,\pm 1)^\top} M = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

The Hessian is negative definite on the tangent space because, for $\alpha \in \mathbb{R} \setminus \{0\}$,

$$\alpha(0, 1, 0)^\top \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = -\alpha^2\, 2 < 0.$$

Thus, the points $(\pm 1, 0, \pm 1)^\top$ *are local constraint maxima.*

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